AsseSync: Assessment Management System for Naawan Central School

A Capstone Project

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ABSTRACT

The traditional manual assessment process at Naawan Central School presents significant challenges, including inefficiencies in test creation, vulnerability of physical documents to damage, and inconsistencies in data management. To address these issues, this study developed "AsseSync," an Assessment Management System designed to streamline the creation, administration, and analysis of student assessments. The system was developed using the Agile Software Development Model, utilizing Vue.js, Inertia.js, and Laravel for the web-based interface, and PostgreSQL for database management.

AsseSync integrates essential features such as a centralized Test Bank for storing and reusing questions, automated bubble sheet scanning for efficient grading, and real-time Item Analysis to compute difficulty and discrimination indices. Furthermore, the system incorporates Gemini AI to assist in question generation and provide data-driven insights. System testing involved Black Box Testing and User Acceptance Testing (UAT) with 30 teachers from Naawan Central School, evaluated using the System Usability Scale (SUS). The results indicate that AsseSync successfully reduces the manual workload for teachers, secures assessment data, and improves the reliability of student evaluations, marking a significant improvement over traditional methods.

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CHAPTER 1
INTRODUCTION

This chapter provides an overview of the Assessment Management System for Naawan Central School, focusing on its purpose, objectives, and significance. It also introduces the scope of the project, contextualizes the need for the AsseSync within the challenges identified at Naawan Central School, and explains the rationale behind adopting the Agile Software Development Model.

1.1. Background of the Project

Assessment is traditionally defined as a process of gathering evidence what each student actually knows, understands, and can do. The word assess comes from the Latin root assidere, which means to sit beside. If we take this definition of assessment to heart, we begin to see it as an opportunity to glean information about what students know and are able to do, so educators can facilitate and guide their next steps (Hanson, 2024). According to the Department of Education et al. (2019), assessment plays an important role in determining the quality of basic education. In the context of the K to 12 Program, assessment results shall be used to look into the learners' performance so that relevant and responsive policies/programs/reforms can be introduced to further improve teaching and learning quality.

Test items that are not analyzed can be a factor that makes the results of the assessment low. Quality of items is not yet known because the teacher has not carried out the analysis. The assessment process mandated by the Department of Education, particularly through item analysis conducted by the teachers. Item analysis is an effective method for assessing not only the test but can also revel important issues of the whole educational process from curriculum design, implementation, assessment and evaluation (Sowdani et al., 2018). Given that teachers play a vital role in shaping learners' learning, with support from administrators, parents, and the wider school community. Primary teachers carry the main responsibility for helping students build strong foundations in knowledge and skills (Allen, 2019, as cited in Bravo & Paceño, 2023). In this situation, Assessment Management Systems should be implemented as well to assist teachers in performing their tasks efficiently.

The Assessment Management System (AsseSync) is essential as it assists teachers in efficiently performing their tasks. At Naawan Central School, the system provides significant help since manual management of assessments is time-consuming and prone to data loss. AsseSync enables teachers to automatically check students' assessments through the use of bubble sheets and also offers a Test Bank feature for storing and reusing assessment items for future use. Moreover, the system integrates Gemini AI through API implementation, enhancing its capabilities in automated data processing, intelligent

assessment analysis, and generating insightful feedback to support teachers in decision-making.

1.2. Context of the Project

This project is conceived to address the pressing challenges faced by teachers at Naawan Central School in managing student assessments. The project team, composed of Jaymei Rey A. Cortez as the Project Manager, Lloyd S. Gonzaga as the Programmer, and John Robert G. Adlawan as the System Analyst, conducted a detailed interview with Zairine H. Tanque, a teacher at Naawan Central School, on October 2, 2024, to gather insights into the difficulties teachers encounter in the current system.

During the interview, several key issues are highlighted. Firstly, teachers are unable to save and reuse the questions they create for assessments, leading to inefficiencies when preparing future tests. This results in a repetitive cycle where the assessment process begins with the teacher checking whether the Department of Education (DepEd) provides an assessment. If an assessment is not provided, the teacher creates an assessment test for a specific subject and proceeds to print it. Once the assessment is prepared, the teacher distributes it to the students. The students then answer the assessment, after which the teacher collects the completed assessments for evaluation.

This project organizes the assessment process, provides valuable insights into question effectiveness, and enables teachers to focus on helping students who need assistance in contrast to their manual processes, which will be further discussed in our narrative listing.

1.3. Narrative Listing

These are the processes followed in the manual assessment process (see Figure 1) and in AsseSync assessment process (see Figure 2).

1.3.1. Manual Assessment process

In figure 1. The manual assessment process begins with the teacher checking if the Department of Education (DepEd) provides a standardized assessment test. If none is provided, the teacher creates a multiple-choice questionnaire (MCQ) for the assessment. Once created, the teacher prints the test and distributes it to the students. The students answer the test, and the teacher collects the completed papers. After collection, the teacher checks the assessments, arranges the students' scores in either ascending or descending order, and identifies the upper and lower 27% of students. The teacher then records the frequencies of correct answers from both groups for each item. Next, the item difficulty is computed, followed by the distractor quality and the discrimination index. The teacher then

checks the reliability and validity to determine whether certain items need revision or removal. ; if not, the evaluated assessment is kept for future use.

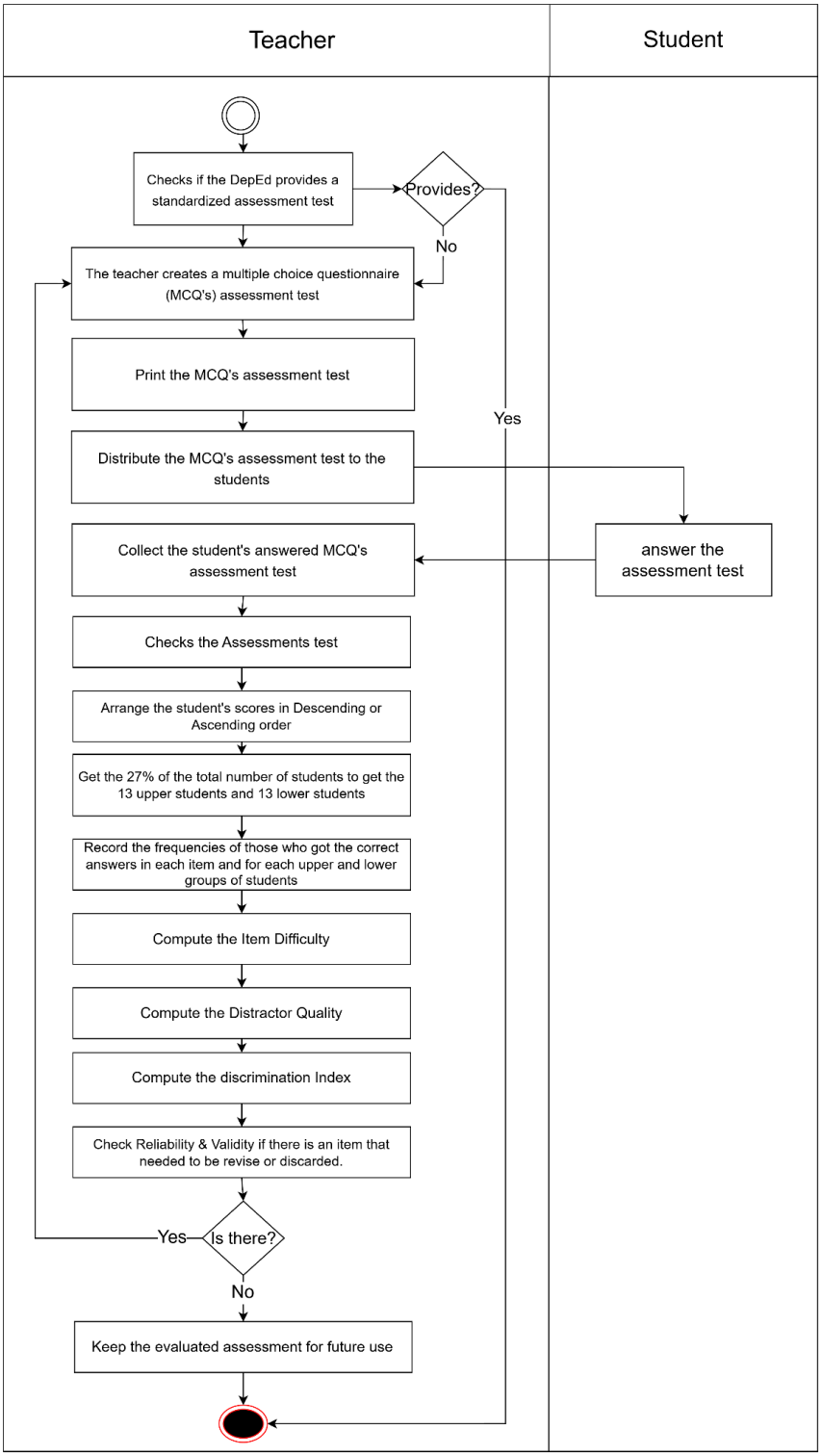


Figure 1. Manual Assessment process

1.3.1. AsseSync Assessment process

Figure 2 shows that in the system-assisted process, the admin begins by checking if DepEd provides a standardized assessment test. If available, the standardized test is imported into the system's test bank; if not, the teacher creates a multiple-choice assessment test. The teacher inputs the questions, answer choices, and answer keys, then saves them in the system. The teacher may then print the assessment, which includes a bubble sheet, and conduct the assessment with the students. Students answer the test using the bubble sheet, which is later collected by the teacher. Afterward, the teacher imports the students' bubble sheets into the system and clicks the evaluate button. The system automatically checks the bubble sheets and computes the item difficulty, distractor quality, and discrimination index. The evaluation results are then displayed for the teacher to review, allowing them to decide if certain items need revision or removal. Once the review is complete, the finalized and evaluated assessment is saved into the test bank as a standardized test for future use. If not , the assessment test will need to be reconducted using the new version of the assessment test with the revised questions and replaced until the evaluation results will be all retained.

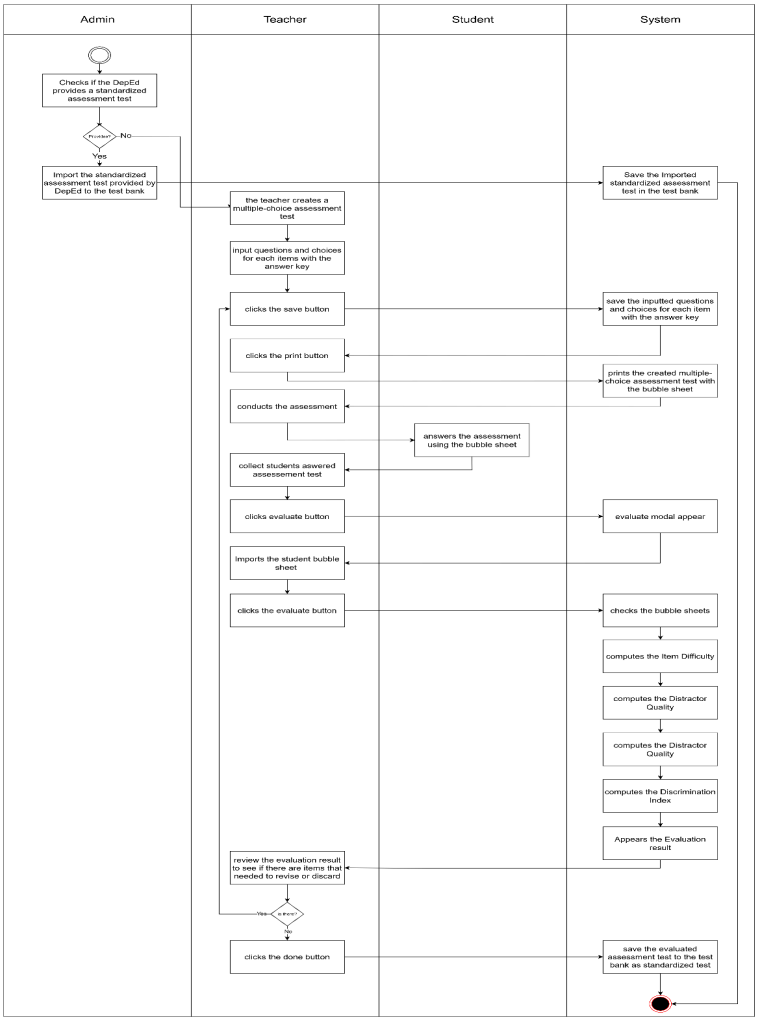


Figure 2. AsseSync System process

1.4. Issues and Problems

1.4.1. Document Vulnerability

Physical questionnaires that are provided by DepEd were stored in paper form. There are instances of damage from water, fire, or wear and tear over time. The loss of questionnaires can significantly disrupt the assessment process, affecting both teachers and students. If original questionnaires are lost or damaged, educators may need to recreate assessments, which can result in inconsistency and extra workload.

1.4.2. Data Inconsistency

The test questionnaires provided by DepEd, which are often downloaded from the internet, sometimes lead to downloading the wrong questionnaire or an unstandardized assessment test, which can cause confusion, inconsistencies in evaluation, and potential misalignment with the intended learning outcomes.

1.5. Objective of the Project

1.5.1. General Objective

This project aims to develop an Assessment Management System (AMS) that addresses the specific challenges faced by Naawan Central School in managing assessments. The AsseSync will highlight the process of creating, managing, and analyzing assessments, which is currently a time-consuming and error-prone manual task.

1.5.2. Specific Objectives

- to design an Assessment Management System (AMS) based on the business processes and policies of NCS;
- to develop the system based on the design; and
- to test the functionality and usability of the developed system using the System Usability Scale (SUS) and the Adjective Rating Scale (ARS) for the interpretation of the SUS score.

1.6. Significance of the Project

This study is significant because it addresses the challenges in the traditional assessment process by introducing an automated AsseSync Management System (AMS) that features test creation, test banking, and item analysis. Conducting this research contributes to both practice and knowledge. it provides a practical solution for schools in improving assessment efficiency, accuracy, and reliability, while also adding to the growing body of studies on educational technology and assessment systems. By evaluating the usability and effectiveness of the system, the research provide insights that can guide future system enhancements and inform other institutions aiming to adopt similar innovations.

1.6.1. Teachers

For teachers, this research demonstrates how technology can reduce the time spent on repetitive administrative tasks toward teaching and learning. The study highlights the potential of AsseSync to improve test creation, ensure alignment with learning objectives, and make item analysis more accessible.

1.6.2. Admin

For school administrators, particularly IT staffs, the study provides evidence on how the system supports quality assurance in assessments. The findings emphasize the role of AsseSync in minimizing inconsistencies, improving record management, and providing reliable data for decision-making.

1.6.3. students

For students, this research is significant because it showcases how a systematic approach to assessments can provide more accurate feedback on performance. By identifying strengths and weaknesses, the system can help learners focus on areas needing improvement and foster better academic outcomes.

1.6.4. Naawan Central School

For Naawan Central School, the study is valuable as it introduces an innovation that enhances the overall assessment process within the school. The system's implementation strengthens the school's capacity to deliver quality education and promotes evidence-based evaluation practices. The system's hybrid architecture supports offline paper-based testing while utilizing a centralized web database to prevent data loss through automated redundancy.

1.6.5. Mindanao State University at Naawan

For Mindanao State University at Naawan, the research contributes to the institution's mission of advancing technology-driven solutions in education. The study also serves as a reference point for future research projects, particularly those aimed at integrating technology into teaching, learning, and assessment.

1.6.6. Researchers

For the researchers, this study is significant because it provides an opportunity to apply theoretical knowledge into practice by developing a real-world solution to an existing educational challenge. It enhances their skills in system development, project management, and research writing. Moreover, the study contributes to their academic growth and serves as a foundation for future studies on system development and educational technologies.

1.7. Scope and Limitation of the Project

1.7.1. Scope

The Assessment Management System (AsseSync) is developed to assist teachers of Naawan Central School in efficiently managing, evaluating, and recording student assessments. The system provides a comprehensive set of features designed to automate and automate the assessment process.

It includes an Assessment Banking module that allows teachers to store, organize, and reuse test items for future assessments. The system also supports the importing of questionnaires provided by the Department of Education (DepEd) and automatically converts them into text (`.txt`) format, making them compatible for digital assessment creation within the system. Imported questionnaires are automatically stored in the Assessment Bank for easy retrieval and reuse.

The system offers Item Analysis that can be conducted through bubble sheet scanning or manual data input, with automatic computation of analysis results. Furthermore, AsseSync enables automatic checking of student assessments and automatic recording of scores per student, minimizing manual effort and reducing the risk of data loss.

The system also provides an exporting feature that allows teachers to generate and download analysis reports in CSV format, facilitating submission and reporting requirements for DepEd.

In addition, AsseSync integrates Gemini AI through an API to provide intelligent assistance such as generating insights, interpreting results, and offering data-driven recommendations for teachers. Assessment creation is designed with a Google Forms-inspired interface, ensuring a familiar, intuitive, and user-friendly experience. Overall, AsseSync enhances efficiency, accuracy, and data management in the assessment process.

1.7.2. Limitation

The system will only support multiple choices exam questionnaire and will not support hands-on activities, or alternative assessment formats such as portfolios, presentations, or projects, as DepEd only requires Multiple Choice Questionnaires (MCQs). The system parses PDF and DOCX files to extract only the questions and choices. Some of the test questionnaires have a picture, the scanning of pictures is excluded in the system so it should be put manually.

For this reason, NCS students are unable to take online exams or quizzes because the school lacks the devices to lend to the students. Therefore, the AsseSync will be exclusively employed by the teachers and admin to develop questions, and perform item analysis only, while the formulated questions shall be exported and printed, for the

face-to-face examinations for these students. Also, the Item analysis only covers from grade 4 to grade 6 due to the exam format. The exam format needs bubble answer sheets so that the Item Analysis check would be automated by just taking pictures and scanned.

The system's functionality is limited to providing raw scores derived from assessment inputs. Grading functionality (transmutation and report card generation) is excluded as it belongs to the Grading Module of the centralized ecosystem. AsseSync focuses exclusively on Assessment Quality Assurance and providing raw score data via export.

1.8. Software Development Model of the Project

This project will adopt the Agile Software Development Model, a flexible and iterative approach that emphasizes incremental progress, collaboration, and adaptability (Laoyan, 2025). The Agile model is particularly suitable for the development of the Assessment Management System (AMS) as it accommodates the evolving nature of system requirements and ensures continuous feedback from stakeholders, particularly the teachers and administrators of Naawan Central School.

1.9. Summary of the Chapter

This chapter introduced the foundation of the study, beginning with the background and context of the project that highlighted the challenges in the traditional assessment process. It presented the manual and automated processes of assessment through narrative listing and diagrams, showing the clear distinction between existing practices and the proposed solution, AsseSync. The issues and problems, such as document vulnerability and data inconsistency, were identified to justify the need for the system.

The chapter also presented the objectives of the project, both general and specific, emphasizing the system’s aim to assessment creation, test banking, and evaluation. The significance of the project was discussed, detailing the benefits for teachers, administrators, students, Naawan Central School, Mindanao State University at Naawan, and the researchers themselves. Furthermore, the scope and limitations of the study were clarified to define the system’s limit and boundaries. Finally, the software development methodology to be used in the study was introduced, providing the basis for how the system will be designed and implemented.

Overall, Chapter 1 established the importance and relevance of developing an Assessment Management System, setting the direction for the succeeding chapters, particularly the review of related literature and systems in Chapter 2.

CHAPTER 2
REVIEW OF RELATED LITERATURE

2.1. Related Literature

2.1.1. Evaluating Online Assessment Strategies: A Systematic Review of Reliability and Validity in ELearning Environments

From classroom settings to online education, e-learning has redefined teaching methodologies which have been made possible through technology. This development has led to the democratization of learning. Finally, it should be noted that the application of online assessment methods is crucial in assessing and improving student performance in distance education. As of the 1960s, e-learning can be credited with its inception with the work of early computer-assisted instruction which significantly expanded with the internet during the 1990s (Bates, 2019; Clark & Mayer, 2016; Mohammed et al., 2023).

2.1.2. Assessment, Student Learning and Classroom Practice: A Review

Assessment in its various forms has always been a central part of educational practice. Evidence gleaned from the empirical literature suggests that assessment, especially high stakes external assessment has an effect on how teachers teach and consequently, how students learn. Through focus group discussion, the findings show that although teachers are expected to nurture evaluative thinking skills in their students/students this is not reflected in the assessment and teaching and learning practices of student-teachers. The paper argues that for teachers to be effective in promoting the desired goals of the basic school curriculum, greater recognition must be accorded to the influence of assessment on teaching and learning, the understanding of which could arguably play an important role in introducing changes that will promote the cognitive processes and thinking skills desired in our schools and classrooms (Amua-Sekyi, 2016).

2.1.3. Student and Staff Perceptions of a Learning Management System for Blended Learning in Teacher Education

In a study by Holmes et al. (2018), the use of online quizzes as a tool for student assessment revealed diverse responses among staff. Of the six staff members interviewed, only one actively used online quizzes as a regular part of their assessment strategy. This teacher integrated quizzes into the learning process by having students complete them after reading assigned materials and attending in-person classes. The responses to the quizzes are then brought into the subsequent face-to-face sessions as discussion prompts. This method ensured that the online assessments are meaningfully connected with in-class learning, promoting a blended learning environment where digital interactivity complemented traditional classroom discussions.

However, the majority of the staff are less enthusiastic about using online quizzes. The main concerns revolved around the nature of the course content, which some believed is not well-suited for such assessments. Additionally, there is apprehension about potential technical issues, such as students facing technological failures during quizzes, which could unfairly impact their performance and cause unnecessary stress.

2.1.4. Item Analysis of Multiple-choice Questions (MCQs): Assessment Tool For Quality Assurance Measures

The article by (Elgadal & Mariod, 2021) highlights the significance of well-constructed Multiple-Choice Questions (MCQs) in educational assessment, particularly in evaluating higher cognitive skills aligned with Bloom's taxonomy. As outlined in their work, MCQs serve not only to assess basic knowledge recall but also to gauge more complex cognitive abilities, such as data interpretation, synthesis, and application of knowledge. The stem of an MCQ, often presented as a clinical case scenario, effectively measures core competencies and intended learning outcomes (ILOs), thereby providing educators with reliable feedback and insights for curriculum reform. Item analysis is a crucial process for evaluating the reliability and validity of test items after examinations. It examines the effectiveness of both the stem questions and their distractors, enabling educators to refine or eliminate poorly performing items before creating a comprehensive exam bank for future assessments. Key metrics in item analysis include the difficulty index (DIF-I), which indicates the level of challenge posed by each question, and the discrimination index (DIS-I), which assesses a question's ability to differentiate between high and low-performing students. These analyses are vital for maintaining the quality and effectiveness of assessments.

Furthermore, the system's item analysis algorithms are aligned with the quality assurance standards mandated by DepEd Order No. 8, s. 2015 (Policy Guidelines on Classroom Assessment for the K to 12 Basic Education Program). This policy emphasizes the necessity of establishing the validity and reliability of assessment tools. To comply with this mandate, AsseSync utilizes standard Classical Test Theory (CTT) formulas to automate the computation of Difficulty Index (p) and Discrimination Index (D), providing teachers with the statistical evidence required to validate their test questionnaires.

2.1.5. Banking on Item Banks

Banerjee and Das (2021) explore the growing significance of item banks in enhancing educational assessments. The article discusses how organizing and categorizing test items help teachers craft reliable, aligned assessments while ensuring a smoother test creation process. The authors highlight the role of software platforms like Moodle in managing these item banks, improving teachers' ability to assess student performance effectively. The integration of data-driven insights supports instructional adjustments, leading to better educational outcomes (Banerjee & Das, 2021).

2.2. Related Systems

This section presents the related studies that relate to the proposed project and its characteristics, which are shown below:

2.2.1. Iteman

Iteman is a software platform for automated psychometric reporting (test and item analysis) with classical test theory. It means that the software doesn't just calculate statistics – it automatically builds you a tech report in MS Word, potentially hundreds of pages, with embedded narratives, graphs, and tables like you see here. You can take the item analysis pages and share them on your screen to discuss with the content experts (Assessment Systems, 2024).

The features of the Iteman are listed below:

a. Test Bank Page

- **Item Management**: View, add, modify, and remove test items and categories as needed.
- **Item Metadata Customization**: Add or update metadata (e.g., difficulty level, cognitive level, item type) for individual items.
- **Batch Import/Export**: Import items from external files or export items for backup and sharing with other systems.
- **Item Status Indicators**: Mark items as active, inactive, or under review to keep track of items available for analysis.
- **Search and Filter Tools**: Quickly locate items based on specific criteria like topic, difficulty, or other tags.

- **Visualization Tools**: Display graphs and charts, such as histograms or item characteristic curves, to visualize data.
- **Custom Date Filtering**: Select date ranges for specific analyses, comparing item and test performance across periods.
- **Export Data**: Save data in various formats (e.g., CSV, Excel) for further analysis or integration into other software.

b. History Page

- **Transaction History**: Access previous test runs, including details on test form, time, and score distribution.
- **Editable Logs**: Make notes or add comments to specific test runs or item analyses for future reference.
- **Re-run Analyses**: Re-analyze past tests with updated settings or metrics without re-importing data.
- **Review Item Changes**: Track changes in item performance and flag items that have shown variability.
- **Detailed Transaction Records**: Access comprehensive details for each transaction, including individual item statistics.

c. Profile Page

- **User Management**: Add, update, and manage profiles for test administrators, analysts, or content experts.
- **Organization Profile Settings**: Update details like organization name, logo, and contact information for reporting purposes.
- **Access Permissions**: Assign permissions to users based on roles to limit access to specific pages or features.
- **Activity Logs**: Track user actions within the system for auditing and accountability.
- **Personalized Dashboard**: Customize the home view for each user role to prioritize the most relevant pages and data.

These features simplify the item and test management process, enhancing test quality, user collaboration, and reporting capabilities within Iteman.

d. Item Analysis Page

- **Statistical Overview**: Display key metrics for each item, including difficulty, discrimination, and point-biserial correlations.
- **Reliability Scores**: View reliability coefficients (e.g., Cronbach's alpha) for the overall test and subtests.
- **Item Flagging**: Automatically flag items with low discrimination or reliability concerns for further review.
- **Group Comparison**: Compare item performance across different demographic groups to check for bias.
- **Exportable Reports**: Generate analysis reports in Word or PDF formats, ready for distribution to stakeholders.

e. Test Setup Page

- **Test Form Management**: Add, rename, modify, or delete test forms as needed within a test bank.
- **Scoring Adjustments**: Set scoring rules, such as partial credit or correct response scoring, for each test form.
- **Time-Based Comparisons**: Compare test performance across different test sessions to track improvements.
- **Subtest Configuration**: Divide tests into subtests for analysis of specific skills or topic areas.
- **Randomization Options**: Enable or disable randomization options for test items or sections as needed.

f. Statistics Page

- **Test-Level Summary Statistics**: View overall test statistics, such as mean score, standard deviation, and score distribution.
- **Item Performance Trends**: Track how item performance has changed over time or across multiple test forms.

- **Visualization Tools**: Display graphs and charts, such as histograms or item characteristic curves, to visualize data.
- **Custom Date Filtering**: Select date ranges for specific analyses, comparing item and test performance across periods.
- **Export Data**: Save data in various formats (e.g., CSV, Excel) for further analysis or integration into other software.

g. History Page

- **Transaction History**: Access previous test runs, including details on test form, time, and score distribution.
- **Editable Logs**: Make notes or add comments to specific test runs or item analyses for future reference.
- **Re-run Analyses**: Re-analyze past tests with updated settings or metrics without re-importing data.
- **Review Item Changes**: Track changes in item performance and flag items that have shown variability.
- **Detailed Transaction Records**: Access comprehensive details for each transaction, including individual item statistics.

h. Profile Page

- **User Management**: Add, update, and manage profiles for test administrators, analysts, or content experts.
- **Organization Profile Settings**: Update details like organization name, logo, and contact information for reporting purposes.
- **Access Permissions**: Assign permissions to users based on roles to limit access to specific pages or features.
- **Activity Logs**: Track user actions within the system for auditing and accountability.
- **Personalized Dashboard**: Customize the home view for each user role to prioritize the most relevant pages and data.

These features simplify the item and test management process, enhancing test quality, user collaboration, and reporting capabilities within Iteman.

2.2.2. Questionmark

Questionmark is a comprehensive assessment management platform that allows educators to create and manage assessments with ease, enabling a diverse range of question types and assessment formats. This system supports both formative and summative assessments, helping educators align their evaluations with learning outcomes (Questionmark, 2024).

The features of the Iteman are listed below:

a. Test Bank Page

- **Item Repository Management**: Store, categorize, and manage a central repository of test items, organized by topic, difficulty, and item type.

- **Flexible Question Types**: Support for various question formats, including multiple-choice, true/false, essay, drag-and-drop, and more.
- **Metadata and Tagging**: Attach metadata (e.g., difficulty level, cognitive level, keywords) to each item for easier organization and retrieval.
- **Question Import/Export**: Import items from external files or export questions for use in other platforms.
- **Search and Filter Capabilities**: Quickly locate items with advanced search and filter options based on metadata, question type, or status.

b. Item Analysis Page

- **Detailed Item Statistics**: View key metrics such as difficulty level, discrimination index, and item response frequency.
- **Reliability Analysis**: Calculate reliability metrics for each item and overall assessments to gauge consistency.
- **Item Flagging and Alerts**: Automatically flag items with low discrimination or poor performance indicators for review.
- **Subgroup Performance Comparison**: Analyze item performance across different student subgroups to identify potential bias.
- **Graphical Analysis Tools**: Visualize item data with charts, such as item characteristic curves and response distribution graphs, to aid decision-making.

c. Assessment Setup Page

- **Assessment Configuration**: Create and manage multiple assessment types, including quizzes, exams, and surveys, with customizable settings.
- **Scoring and Feedback Options**: Set scoring rules, partial credit, and automated feedback to provide learners with meaningful results.
- **Randomization and Shuffling**: Configure randomization settings for questions and answer choices to enhance assessment security.
- **Adaptive Testing Options**: Implement adaptive testing configurations that adjust item difficulty based on learner responses.
- **Timed Assessments**: Set time limits for each assessment, with options for time extensions as needed.

d. Statistics Page

- **Assessment-Level Summary Statistics**: Access comprehensive statistics on overall performance, such as mean score, median, and standard deviation.
- **Trend Analysis**: Track changes in assessment performance over time to identify improvement or decline in student outcomes.
- **Visualization Tools**: Use graphs and charts, such as histograms and scatter plots, to visualize score distributions and other data trends.

- **Custom Date Filtering**: Filter assessment data by specific date ranges for comparative analysis across terms or semesters.
- **Data Export Options**: Export assessment data in multiple formats (e.g., CSV, Excel, PDF) for in-depth analysis or record-keeping.

e. History Page

- **Assessment Attempt Logs**: View a log of all attempts for each assessment, including participant details, score, and completion time.
- **Detailed Item Review**: Access individual responses and detailed item-level data from past attempts for insights into learner understanding.
- **Feedback Tracking**: Review automated feedback provided to participants and track any custom feedback added by instructors.
- **Re-analysis Capabilities**: Re-evaluate past assessment attempts with updated scoring or analytical settings without re-importing data.
- **Session Notes**: Attach notes or observations to specific assessment attempts for reference in future reviews.

f. Profile Page

- **User Role Management**: Create and update profiles for instructors, administrators, and content creators with customizable permissions.
- **Organization Profile Settings**: Set up and manage institutional details such as logo, name, and contact information for reports.
- **Role-Based Access Control**: Define and manage access controls based on user roles, ensuring secure and appropriate data access.
- **User Activity Log**: Track user actions within the platform for transparency and accountability in assessment management.
- **Customized Dashboard Views**: Tailor the homepage view for each user based on role, highlighting relevant tools and data insights.

2.2.3. Respondus

A tool that integrates with various LMSs to provide secure assessment creation and management, focusing on test integrity and analytics. Respondus offers a range of features that significantly enhance item analysis and test bank management for educational assessments (Solutions for Higher Ed - Respondus, 2021).

The features of the Respondus are listed below:

a. Test Bank Page

- **Secure Question Repository**: Store and manage questions in a centralized, secure test bank that integrates directly with LMSs.
- **Diverse Question Types**: Supports multiple question formats, including multiple-choice, true/false, essay, fill-in-the-blank, and matching.

- **Metadata Tagging**: Add tags and metadata such as difficulty level, topic, and cognitive level for efficient organization and retrieval.
- **Import/Export Options**: Import test banks from files (Word, Excel, or Respondus format) and export to compatible LMSs.
- **Search and Filter Tools**: Easily search and filter questions based on criteria like question type, metadata tags, or item status.

b. Item Analysis Page

- **Detailed Item Metrics**: Access analytics such as difficulty index, discrimination index, and average time spent on each item.
- **Reliability and Consistency Scores**: Calculate test reliability (e.g., Cronbach's alpha) to assess the internal consistency of the assessment.
- **Item Performance Alerts**: Automatically flag items that perform below specified thresholds for review.
- **Subgroup Analysis**: View item performance by subgroup, helping identify potential bias or disparities in question difficulty.
- **Visualization Options**: Visualize data with charts like item characteristic curves and response frequency distributions for clear analysis.

c. Assessment Setup Page

- **Secure Assessment Configuration**: Create exams with security settings, such as browser lockdown and proctoring, to prevent cheating.
- **Scoring and Feedback Customization**: Configure automatic scoring, partial credit, and detailed feedback for each question.
- **Randomization Settings**: Set up randomization of questions and answer choices to maintain test integrity across test-takers.
- **Adaptive Release Options**: Enable conditional release of questions or sections based on student responses.
- **Timed Exam Options**: Configure time limits for assessments, with customizable grace periods for students if needed.

d. Statistics Page

- **Comprehensive Assessment Analytics**: Access assessment-wide statistics, including mean scores, standard deviation, and completion rates.
- **Time-Based Analysis**: Track assessment performance over specific time periods, comparing results across different student groups or terms.
- **Visual Analytics**: Utilize visual tools, such as score histograms and scatter plots, for an intuitive understanding of assessment results.
- **Data Filtering and Exporting**: Filter data by date range or student subgroup, and export results in formats like CSV, Excel, or PDF for reporting.

- **Reliability and Validity Insights**: Generate insights into assessment validity and reliability, aiding in quality assurance.

e. History Page

- **Attempt Logs and Detailed Records**: View a history of assessment attempts, including individual item responses and overall scores.
- **Audit Trails and Logs**: Access a complete audit trail of student activity during assessments, useful for verifying test integrity.
- **Re-analysis Options**: Re-run analysis on past exams to incorporate new scoring criteria or updated analytics settings.
- **Detailed Feedback Review**: Review student responses and instructor feedback provided during past assessments.
- **Session Comments**: Add notes or comments to specific assessment attempts for contextual reference or follow-up.

f. Profile Page

- **User Access Control**: Set up user profiles with role-based access for instructors, proctors, and administrators to secure sensitive data.
- **Institutional Profile Management**: Update and manage organizational information, such as branding, for consistency across assessments.
- **Permission Management**: Define custom permissions for each user role, ensuring compliance with privacy and security protocols.
- **Activity Monitoring**: Track and monitor user activity within the platform, enhancing security and accountability.
- **Dashboard Customization**: Customize dashboard views by role, allowing users to quickly access the most relevant data and features.

2.2.4. ExamSoft

ExamSoft provides a comprehensive assessment management solution that includes test banks and detailed item analysis tools. Educators can create various question types, analyze student performance, and assess the reliability and validity of each question (ExamSoft, n.d.).

The features of the ExamSoft are listed below:

a. Test Bank Page

- **Centralized Question Repository**: Store and manage a comprehensive database of questions with centralized access for ease of organization.
- **Variety of Question Types**: Support for multiple question formats, such as multiple-choice, essay, fill-in-the-blank, true/false, and hotspot questions.
- **Metadata Tagging and Custom Fields**: Attach tags and metadata like difficulty level, Bloom's taxonomy level, or specific learning objectives to each item.

- **Bulk Import and Export Options**: Import question banks from various file formats and export them for use in other systems or backup.
- **Advanced Search and Filter Functions**: Easily locate items using search and filter options, based on criteria like topic, type, or metadata tags.

b. Item Analysis Page

- **In-depth Item Metrics**: View performance metrics for each item, such as difficulty index, discrimination index, and item response analysis.
- **Reliability and Consistency Scores**: Assess the reliability of each question and the overall assessment (e.g., Cronbach's alpha) for consistency.
- **Automatic Item Flagging**: Automatically flag low-performing items or those that do not meet reliability standards for review.
- **Bias and Fairness Analysis**: Analyze question performance across subgroups to identify any biases or issues with question fairness.
- **Graphical Data Representation**: Utilize visual aids like item characteristic curves, frequency charts, and distribution graphs for clear data insights.

c. Assessment Setup Page

- **Customizable Assessment Design**: Configure a wide range of assessment types, including exams, quizzes, and formative assessments, with customizable settings.
- **Scoring and Feedback Control**: Set up automatic scoring rules, define partial credit options, and create customized feedback for each question.
- **Randomization and Shuffling**: Apply randomization to questions and answer choices to reduce the chance of cheating.
- **Adaptive Testing Capabilities**: Implement adaptive testing configurations that adjust question difficulty based on the student's responses.
- **Timing and Accessibility Settings**: Set time limits, allow for extra time or accessibility accommodations, and set assessment retake rules.

d. Statistics Page

- **Comprehensive Assessment Data**: Access summary statistics, such as mean, median, and standard deviation, to evaluate overall assessment performance.
- **Time-Based Comparison Analysis**: Compare performance over time across terms or semesters to identify trends in student learning.
- **Visual Analytics Tools**: Leverage visual tools like histograms, score distribution graphs, and trend lines for easy interpretation of data.
- **Customizable Data Filters**: Filter assessment data by date, section, or specific student groups for focused analyses.
- **Downloadable Reports**: Export data in formats like CSV, Excel, or PDF for detailed analysis or record-keeping.

e. History Page

- **Session and Attempt Logs**: Access logs of past assessment attempts, including individual responses, scores, and time spent per question.
- **Audit and Review Trails**: Review detailed records of all student activity during the assessment for security and integrity purposes.
- **Historical Item Performance Review**: Access past item statistics to assess stability and identify items that may need refinement.
- **Feedback Review and Tracking**: Review student feedback responses to gain insights into their understanding and experience.
- **Session Notes and Annotations**: Add notes or observations for specific attempts, helping instructors keep contextual information on hand.

f. Profile Page

- **User Role and Access Control**: Manage user roles and permissions, assigning tailored access for instructors, admins, and proctors.
- **Institutional Profile Management**: Set up and manage institutional branding, including name, logo, and other details for consistency across reports.
- **Custom Permission Settings**: Define and manage access controls for each role to ensure data privacy and security compliance.
- **Activity Monitoring and Tracking**: Track user actions within the platform for accountability and auditing purposes.
- **Dashboard Customization**: Allow users to customize dashboard views according to their role, prioritizing relevant tools and information.

2.2.5. Moodle

Moodle, as an open-source learning management system, includes robust quiz and test features that facilitate the creation of question banks and comprehensive item analysis. Educators can gather data on question effectiveness based on student performance metrics (*Home | Moodle.org*, n.d.).

The features of the Moodle are listed below:

a. Question Bank Page

- **Centralized Question Storage**: Organize and manage a comprehensive question bank within the LMS, accessible across courses.
- **Diverse Question Formats**: Supports various question types, including multiple-choice, true/false, short answer, matching, calculated, and essay questions.
- **Question Metadata and Tagging:** Add metadata such as category, difficulty level, or tags to questions for better organization and retrieval.
- **Bulk Import/Export Functionality**: Import questions in multiple formats (e.g., GIFT, Aiken, XML) and export to use in other Moodle courses or systems.

- **Search and Filter Options**: Use search and filter tools to find questions based on specific tags, question type, or difficulty level.

b. Item Analysis Page

- **Comprehensive Item Metrics**: View essential metrics, including facility index (difficulty), discrimination index, and response frequency for each option.
- **Reliability Statistics**: Access overall quiz reliability metrics, such as Cronbach's alpha, to assess the consistency of the assessment.
- **Automatic Question Flagging**: Identify items with low discrimination or high guess rates for potential revision.
- **Group Comparison Analysis**: Compare question performance across different groups to identify any disparities or areas of bias.
- **Data Visualization**: Display item analysis results with visual aids like bar charts for quick assessment of question performance.

c. Quiz Setup Page

- **Flexible Quiz Configuration**: Set up quizzes with options for graded quizzes, self-assessments, and formative assessments, customizable by course requirements.
- **Scoring and Feedback Options**: Configure scoring rules, partial credit, and feedback messages to enhance the learning experience.
- **Question and Answer Randomization**: Enable randomization of questions and answer options to maintain test security.
- **Conditional Release Options**: Control question release or adapt question pathways based on student responses.
- **Timing and Accessibility Settings**: Set time limits, provide accommodations, and enable multiple attempts based on course needs.

d. Statistics Page

- **Quiz Performance Overview**: Access detailed statistics on quiz performance, including average score, median, standard deviation, and time taken.
- **Trend Analysis Over Time**: Track quiz performance across multiple cohorts or terms for insights into course effectiveness.
- **Visual Analysis Tools**: Use histograms, score distribution graphs, and trend lines to understand data trends and insights visually.
- **Customizable Filters**: Filter data by date range, student group, or course section to focus on specific areas for analysis.
- **Exportable Reports**: Export quiz statistics and item analysis data in various formats (e.g., CSV, Excel, PDF) for record-keeping and reporting.

e. History Page

- **Attempt Logs and Records**: Access records of all quiz attempts, including detailed response data and scores for each student.
- **Detailed Activity Logs**: Track student activity within quizzes, useful for understanding time taken per question or reviewing behavior during the test.
- **Historical Item Performance Review**: Review past data on question performance to identify trends or items needing updates.
- **Feedback Tracking**: View automated feedback provided to students and manually add additional feedback if necessary.
- **Attempt Notes**: Add instructor notes on specific attempts for context or follow-up with students.

f. Profile Page

- **Role-Based Access and Permissions**: Manage roles and permissions for instructors, course designers, and administrators to secure quiz and test data.
- **Institution Profile Management**: Customize course and quiz settings to align with institutional standards, including course branding.
- **Permissions Control**: Assign access levels to each user role, ensuring privacy and control over sensitive assessment data.
- **User Activity Monitoring**: Track user interactions within the platform for transparency and security.
- **Customizable Dashboard**: Tailor the dashboard to highlight relevant tools, statistics, and activities based on user roles.

2.2.6. Google Forms

Google Forms is a go-to tool for fast, free, and efficient data collection. It's used by marketers, educators, HR professionals, and SaaS teams alike, thanks to its simplicity, customizability, and seamless integration with other Google Workspace tools. Whether you're collecting feedback, running a quick poll, organizing an event, or building a simple quiz (Zafar, 2025).

The features of the Google Forms are listed below:

a. Form Creation Page

- **Customizable Questions:** Add various question types (e.g., multiple-choice, dropdown, linear scale, short answer).
- **Themes and Branding:** Customize colors, fonts, and add logos for a professional look.

b. Question Bank Page

- **Reusable Questions:** Store frequently used questions for reuse in other forms.
- **Topic Tagging:** Organize questions by topics or tags for quick retrieval.

c. Response Analysis Page

- **Real-Time Results:** View responses as they come in with summary statistics and visualizations.
- **Export Data:** Export responses to Google Sheets or download as CSV for further analysis.

These features make Google Forms a versatile and easy-to-use platform for collecting and managing data in a wide range of applications, from education to business workflows.

2.2.7. Google Drive

Google Drive is a cloud-based storage service that enables users to store, access, and share files and folders online. The web service provides a centralized cloud storage hub where changes to content are automatically saved and synchronized across all connected devices. This allows multiple users to collaborate on the same content in real time and always see the latest version of each file (Hanna et al., 2025).

The features of Google Drive are listed below:

Storage and File Management Page

- **Flexible Storage Options:** Store files of any type with free storage up to 15GB and expandable paid plans.
- **Organization Tools:** Create folders, add color codes, and use advanced search for efficient file management.

Collaboration and Sharing Page

- **Real-Time Collaboration:** Work on shared files simultaneously with others in Docs, Sheets, and Slides.
- **Customizable Sharing Settings:** Control access with options for view, comment, or edit permissions.

Integration and Accessibility Page

- **Cross-Platform Accessibility:** Access files from web browsers, mobile apps, or desktop backups.
- **Seamless Integration:** Directly attach files to Gmail, insert them into Google Docs, or use them in other Google Workspace tools.

These features make Google Drive a robust and user-friendly platform suitable for personal, educational, and professional use, ensuring files are accessible, organized, and secure.

2.2.8. EvalBee

EvalBee is a versatile grading app for educators that allows teachers to design MCQ answer sheets, scan them using a device camera, and generate detailed performance reports (EvalBee, 2025).

Key Features:

- **Exam Templates:** Supports the creation of custom templates for a wide range of exams, including competitive tests like JEE, NEET, SAT, and more. These templates can accommodate up to 300 questions with multiple pages.
- **Answer Formats:** Offers flexibility in answer options such as true/false, matrix, numerical inputs, and 2 to 10 choices per question.
- **Automated Grading:** Teachers can use their phone's camera to quickly scan and grade answer sheets, saving significant time.
- **Performance Analytics:** Provides detailed reports with insights into student performance. These can be exported to Excel or shared via email.
- **Multilingual and Multi-device Support:** Available in various languages and compatible with iOS devices, including iPhones and iPads.

2.2.9. Test Item Analysis Calculator

Teachers often rely on automated tools to compute item difficulty, discrimination, and distractor performance, particularly in schools without statistical software. One of the earlier tools commonly used is the Test Item Analysis Calculator, which provides a simple spreadsheet-based system for analyzing test items. The purpose of item analysis is to assess and improve the reliability of your tests. If test reliability is low, test validity is also low. This is why we do test item analysis to improve the validity of a test by improving its reliability. (DepEd Tambayan, 2020).

2.3. Summary of the Chapter

In summary, this chapter provides an overview of existing literature and assessment systems relevant to developing the Assessment Management System (AMS) for Naawan Central School. It highlights the positive aspects of data-driven assessments, such as improved efficiency and enhanced learning outcomes. However, challenges, such as potential technical limitations and adaptation issues, are also acknowledged. Factors affecting the project include the specific needs of the school, the adaptability of teachers to new technology, and the reliability of the existing infrastructure. These insights are crucial for tailoring an effective Assessment Management system that addresses both the strengths and limitations identified in current systems.

CHAPTER 3
CONCEPTS, THEORIES AND TECHNICAL BACKGROUND

This chapter describes the project guidelines, theories, and technologies used in developing the system to provide a better understanding of its development process.

3.1. Agile Model

The Agile model was chosen as the Software Development Life Cycle (SDLC) approach for this project." (Original was a fragment). Agile was first conceptualized with the publication of the Agile Manifesto in 2001 by Kent Beck, Martin Fowler, and others, emphasizing iterative development, flexibility, and continuous improvement (Hohl et al., 2018).

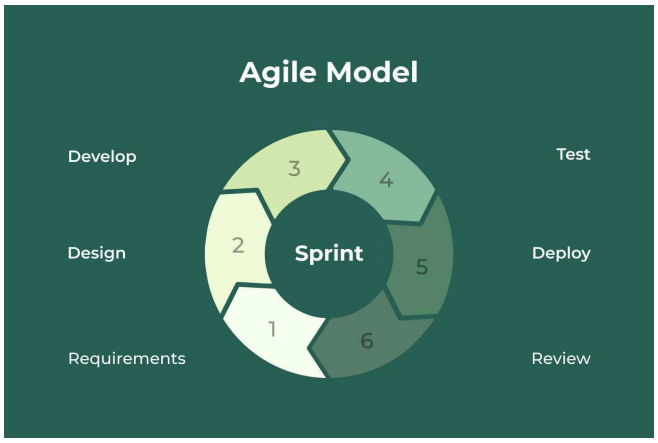


Figure 3. The Agile Model

Compared to traditional SDLC models like Waterfall and V-Model, Agile offers unique advantages. While these linear models generally follow a sequential path, Agile allows development teams to remain highly responsive to change through frequent evaluations and adjustments. Key benefits include enhanced flexibility, which enables teams to adjust requirements or adapt to business needs mid-project without major disruptions. Agile also promotes continuous feedback and improvement through regular client collaboration, allowing for real-time feedback and fast issue resolution. Furthermore, Agile's incremental value delivery ensures that usable components can be released even before the complete system is finalized, which supports early user engagement and validation. For this project, the team chose Agile due to its suitability for dynamic projects, as it enables seamless incorporation of client feedback and timely adjustment to new requirements, ensuring that the final product is both functional and closely aligned with client expectations. In practice, Agile divides development into short, iterative segments called sprints, each of a defined duration. At the conclusion of each sprint, the team reviews its progress, gathers feedback, and makes

any necessary adjustments for the next sprint, thus ensuring a flexible and adaptive project progression.

The team selected the Agile model for this project due to its collaborative nature and its ability to incorporate ongoing feedback from the client. Regular meetings with Ma'am Zairine H. Tanque, a teacher at Naawan Central School, allowed the developers to adapt their work based on real-time input, ensuring the system aligned with the school's specific needs. Agile's adaptability and focus on stakeholder involvement made it the ideal choice for this project's methodology. The following will cover the six stages of the Agile Model.

3.1.1. Phases of the Agile Model

The first phase of the Agile Model is the planning phase. In this stage, the needs and requirements are gathered through interviews and consultations with the respective client (Ma'am Zairine H. Tanque, the teacher) of Naawan Central School for the developers to develop the project. The developers used their smartphones to record the important discussion during the interview with the teacher of Naawan Central School , and they also used pens and notebooks to draft the idea on how to develop the system based on the client's input. After the developers gathered the important data, they started to analyze what to design for the system.

The second phase of the project is the design phase. In this stage, the developers focused on designing for the system, such as the Test bank and Item analysis. Entity-Relationship Diagram (ERD) for the system's database design and Unified Modeling Language (UML) for the project. Using these tools, developers are able to translate requirements into visual representations that effectively express the project's design characteristics. These designs served as a model for the succeeding stages of development, assisting in the implementation of the Assessment Management System.

The third phase of the project is the development phase. In this stage, the developers began the actual implementation and coding process. The system is developed using a variety of languages, such as HTML, CSS, and others.

In the fourth phase, the developed system is tested to determine whether it met the client's needs and whether using it left the client satisfied. The developers tested the functionality and usability of the system by coordinating with the client, the teachers of Naawan Central School. The System Usability Scale (SUS) standard formula is used to assess the system's usability and results. The fifth phase is deployment. The assessment management system prototype is developed and implemented by the proponents at this stage. An orientation and a user manual are included in the prototype to help users learn how to operate the system efficiently. After getting the client's initial input on the prototype, the proponents performed the required adjustments and modifications in response to the input.

This completes the cycle of Agile development. The proponents presented the client with the final outcome and explained how the system satisfies the specifications after finishing all earlier phases of development. The client is given the chance to evaluate the system and offer comments on its general satisfaction, usability, and usefulness. Further iterations or 15 improvements may be made in response to the client's feedback, or the project may move on to the next phase while adhering to Agile principles and expanding the development process.The system architecture chosen for this project is centralized, which consolidates all data and processes in a single location. This setup enhances data access and management, ensuring efficient and secure data handling. The centralized architecture also simplifies maintenance and supports the Agile model’s iterative approach, making it ideal for projects that benefit from regular updates and improvements.

3.2. Unified Modelling Language (UML)

The Unified Modeling Language (UML) is a general-purpose, standardized modeling language used in software engineering to specify, visualize, construct, and document the artifacts of a system. As Donald Bell (2023) explains, UML was created by software engineers Grady Booch, Ivar Jacobson, and James Rumbaugh often called the “Three Amigos”. In the mid-1990s, and it was adopted as a standard by the Object Management Group (OMG) in 1997. Since its standardization, UML has evolved through versions such as UML 1.x and UML 2.x to support both structural and behavioral aspects of system design, making it a key tool for modern software architects and developers (Bell, 2023).

As a comprehensive modeling language, UML provides a variety of diagram types that allow developers to capture both the structural and behavioral aspects of a system. According to Bell (2023), UML diagrams are generally categorized into structural diagrams, which describe the static architecture of a system, and behavioral diagrams, which illustrate dynamic interactions and workflows. These diagrams include use case diagrams, class diagrams, activity diagrams, and sequence diagrams offer different overviews that help clarify system requirements and design decisions.

In this project, UML plays a central role during the design phase by providing a standardized and visual way of representing the system. Using UML diagrams ensures that both the development team and stakeholders share a common understanding of the system’s functionality, architecture, and boundaries. This not only strengthens communication but also allows potential issues to be detected early, contributing to more efficient and accurate system development (Bell, 2023). In the case of the Assessment Management System, the diagrams produced by the developers served as a blueprint for both the front-end and back-end components, acting as the structural foundation upon which the project was conceptualized and implemented.

3.2.1. Use Case Diagram

Originally introduced by Ivar Jacobson as part of object-oriented modeling, the Use Case Diagram was later integrated into UML to show the interactions between actors and the system. Rather than focusing on the internal details of implementation, it provides an illustration view of system functionality by illustrating relationships among actors, use cases, and system boundaries. This visualization helps developers and stakeholders identify essential system interactions and clear requirements. In this project, the Use Case Diagram served to define the primary user roles and their corresponding actions within the AsseSync Management System (AMS), ensuring that the system scope aligns with user needs and expectations (Becker, Kuhn, & Laue, 2019).

3.2.2. Activity Diagram

Activity diagrams provide a dynamic view of a system by modeling workflows of steps of activities and actions, including choice, repetition, and simultaneous. It helps illustrate how processes progress from one activity to another, making them particularly effective for capturing the logic of complex use cases. As shown in the empirical study by Hashim, Ibrahim, Rejab, Romli, and Mohd (2018), activity diagrams are easier to comprehend than other behavioral UML diagrams, such as state charts, when applied to tasks like test case generation. In the AsseSync project, activity diagrams supported the visualization of processes and decision flows, ensuring that the functional behavior of the system was well understood by both developers and stakeholders.

3.2.3. Context Diagram

Provides a high-level view of the system's boundaries and interactions with external entities such as stakeholders without detailing internal implementation specifics. According to UML Diagrams in Software Engineering Research: A Systematic Literature Review (Koç, Erdoğan, Barjakly & Peker, 2021), UML diagrams are frequently used in software engineering to identify system requirements and scope, helping distinguish which entities are external and which are within system boundaries. In the AMS project, a Context Diagram plays a role by mapping out external actors and interfaces, defining system inputs and outputs clearly. That overview enabled the developers to set system scope appropriately and ensure alignment with stakeholder expectations, helping to guide system construction and improvements.

3.2.4. Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) is a widely used modeling technique for representing how information moves through a system, highlighting the processes, data stores, and external entities involved. Rather than focusing on implementation details, it emphasizes the logical flow of data, making it valuable in both analysis and design phases. According to

Cheema, Tariq, and Pires (2023), DFDs serve as a foundational tool in software engineering for capturing system requirements and ensuring that data dependencies and transformations are clearly understood. In the AMS project, the DFD provided a structured representation of how assessment-related data is processed, stored, and shared across different actors and components. This clarity allowed developers to align system workflows with user expectations and ensure efficient handling of inputs and outputs.

The Entity-Relationship Model (ERM), introduced by Peter P. S. Chen in his 1976 paper “The Entity-Relationship Model — Toward a Unified View of Data”, provides a visual representation of data within a system and how data elements relate to each other (Chen, 1976). In this project, ERM will be used to design the database for the Assessment Management System through an Entity-Relationship Diagram (ERD). The ERD helps define entities (such as teachers, students, admin and assessments) and the relationships between them, supporting a structured approach to data management.

Also, recent research titled “An Improved Generic ER Schema for Conceptual Modeling of Information Systems” (Pieris, Wijegunesekera & Dias, 2020) highlights the importance of preserving all constraints, relationship types, and attribute information when converting ER models into relational schemas. This ensures that the database structure for AsseSync will facilitate efficient data storage and retrieval while maintaining integrity and supporting complex queries. The ERD provides a clear foundation for the database, ensuring that data relationships are well-defined and consistent, which is critical for accurately representing real-world relationships within the system.

3.3. Functional Decomposition Diagram

Functional decomposition is a fundamental concept in software development that allows breaking down complex systems into smaller, more manageable and understandable components. It is a process of dividing a system into smaller functional units or modules, each responsible for a specific task or functionality (Investopedia, 2022).

3.4. Software Development Tools

This section discusses the frameworks, programming languages, and other tools used by the developers in designing and implementing the system.

3.4.1. JavaScript

According to MacManus (2020), JavaScript was invented in May 1995 by Brendan Eich. This tool is used to bring interactivity to web pages, enabling real-time data updates and dynamic user engagement .

3.4.2. PHP (Hypertext Preprocessor)

PHP was originally developed by Rasmus Lerdorf in 1994 (WPEExperts, 2025), is a server-side scripting language that supports backend operations like data storage, retrieval, and processing.

3.4.3. HyperText Markup Language (HTML)

HTML was first introduced by Tim Berners-Lee in 1991 as a structural framework for web content (EBSCO, 2023). This tool organizes the layout and content structure.

3.4.4. Cascading Style Sheets (CSS)

CSS was introduced in 1996 to style HTML content, a concept originally proposed by Håkon Wium Lie (GeeksforGeeks, 2025).It styles HTML elements for a visually consistent and accessible presentation across web pages.

3.4.5. Tailwind CSS Framework

Tailwind CSS was created by Adam Wathan, Steve Schoger, Jonathan Reinink, and David Hemphill and first released in 2017 (Telerik, 2025), is a CSS framework that allows quick, utility-based styling for custom user interfaces.

3.4.6. Laravel

Laravel, an open-source PHP framework created by Taylor Otwell in 2011, simplifies backend tasks like routing, authentication, and database management (Gonzalez, 2024).

3.4.7. Inertia

Inertia.js, created by Jonathan Reinink in 2018, is a framework designed to build single-page applications using classic server-side routing and controllers (Redmond, 2019).

3.4.8. Vue.js

Vue.js, created by Evan You in 2014, is a progressive JavaScript framework for building user interfaces (Whitfield, 2025). Developed in China, it is created to provide a lightweight and flexible alternative to other JavaScript frameworks, enabling developers to build reactive and component-driven applications with ease.

3.4.9. PostgreSQL

PostgreSQL, initially developed by Michael Stonebraker and his team at the University of California, Berkeley later renamed in 1996 to reflect its support for SQL is a relational database system known for reliability, scalability, and data integrity (PostgreSQL Global Development Group, 2024).

3.5. Software Test Tools

In developing the Assessment Management System (AMS) for Naawan Central School, it is crucial to implement a comprehensive software testing approach to ensure that the system meets both functional requirements and user expectations. A range of testing methods will be applied to verify and validate the AMS's performance, usability, and reliability. Each testing type contributes to the goal of producing a robust, user-friendly platform that enhances the assessment process for teachers and provides an engaging experience for students. Table 2 outlines the sequence of testing steps, and this section further discusses each testing method in detail, including Black Box Testing, Functional and Non-Functional Testing, Unit Testing, Integration Testing, System Testing, Acceptance Testing, and Usability Testing.

3.5.1. Black Box Test

Black Box Testing is a software testing technique that assesses the functionality of a system without requiring knowledge of its internal structure or code, and as software complexity grew in the 1970s, the need for user-focused validation made this approach more formally recognized (Robinson & Heusser, 2024). This approach is particularly beneficial to the Assessment Management System (AMS) because it enables testers to verify that all components work according to requirements, ensuring that teachers and students interact with a reliable, functional interface.

3.5.1.1. Functional Testing

Functional Testing is a method for verifying that software functionalities align with specified requirements by supplying inputs and evaluating outputs, and while no single inventor is credited, it evolved during the 1960s and 1970s in response to increasing software complexity (Robinson & Heusser, 2024). Functional Testing is essential for the AMS, as it helps validate that core features like assessment creation, student response tracking, and item analysis are operational and meet the system's specific requirements. This testing approach ensures that the AMS’s functionalities are reliable and meet user expectations.

3.5.1.2. Non-Functional Testing

Non-Functional Testing focuses on attributes such as performance, usability, reliability, and scalability evaluating how a system operates under various conditions rather than what it does and emerged as a distinct testing concern in the late 20th century as systems increased in complexity (Salahirad et al., 2023). Non-Functional Testing is crucial for AMS, as it verifies that the system can handle data-intensive processes like real-time updates, student assessments, and progress tracking without lag or instability. This testing ensures that AsseSync is optimized for speed, usability, and reliability, contributing to a smoother user experience.

3.5.1.3. Unit Testing

Unit Testing is a method where individual components or modules of a system are tested separately to ensure each one functions as intended. While variations of component-level testing date back to the 1950s and 1960s, unit testing as a more formalized practice became more structured during the 1970s (McDonough, 2021). Unit Testing is valuable in AsseSync development because it allows for the isolated verification of core features, such as assessment modules, student records, and progress analysis components. Testing each unit individually enables early identification of issues, ensuring that the basic building blocks of the AsseSync are robust before they are integrated into the larger system.

3.5.1.4. System Testing

System Testing is a comprehensive method that examines the complete, integrated system to verify it meets specified requirements. Its formal recognition as a distinct level in the testing life cycle became more widespread during the 1980s with the maturation of software development methodologies and greater emphasis on system quality (CertLibrary, 2024). This testing is essential for AsseSync as it validates the system's functionality as a whole, ensuring that each part, from assessment management to data analysis, integrates seamlessly. By conducting System Testing, developers ensure that AsseSync operates as intended in a real-world setting, which is critical for delivering a reliable product.

3.5.1.5. User Acceptance Test

User Acceptance Testing (UAT) is the final phase in the testing lifecycle where real users validate whether a system meets their needs and requirements. While no single inventor is credited, UAT became formalized as a distinct testing phase in software engineering practices during the late 20th century (Gordon et al., 2022; Leung & Wong, 1997). UAT is vital for the AMS, as it involves teachers and administrators at Naawan Central School, allowing them to assess whether the system is easy to use, meets functional requirements, and addresses specific needs in assessment management. This feedback-driven testing phase helps ensure that AsseSync is fully aligned with the expectations and practical needs of its end-users.

Table 1. Software Test.

Black Box Test			
Functional Testing			
Steps	**Testing**	**Deliverables/Tools**	**Participant(s)**
1	Unit Testing	Test Cases and Test Case Results	Software Tester(s)
2	System Testing	Task Scenarios (Scenario 1, Scenario 2, etc. per End-User Module) and Task Scenario Results	Software Tester(s)
3	Acceptance Testing using the UAT Model	Task List for testing the System and the Results	End-Users
Non-Functional Testing			
4	Usability Testing	Task List for testing the System, System Usability Scale, and Adjective Rating Scale for Result Interpretation	At least 30 End-Users

3.6. Summary of the Chapter

This chapter presented the essential tools, development model, and testing strategies for building the Assessment Management System (AMS). The Agile Development Model supports flexibility and collaboration, while tools like JavaScript, PHP, and PostgreSQL ensure efficient performance and data management. Testing methods, including Black Box and User Acceptance Testing, are discussed to confirm the system's reliability and usability.

CHAPTER 4
METHODOLOGY

This chapter discusses the methods and procedures by which the project is developed. It includes a discussion of the methods by the developers to develop and design the system and test its functionality and usability.

4.1. Software Design

The project was managed through the Agile approach, which divides the development process into several key phases. Each phase will focus on specific aspects of the project, ensuring that progress is controlled while allowing for ongoing stakeholder involvement. This structured yet flexible approach will ensure that the project remains aligned with stakeholder goals and accommodates any necessary adjustments as the system develops.

Phase 1: Project Planning

The initial phase involved close collaboration with stakeholders to define the system's overall requirements and establish a clear development plan. During this phase, the project team has:

- Collaborated with teachers and administrators to gather system requirements
- Prioritized key functionalities, including test bank management and item analysis
- Established a detailed project timeline, including key milestones, deliverables, and sprint cycles

This phase ensured that the project is aligned with the educational goals of Naawan Central School and that the system addresses the specific challenges faced by educators in managing assessments.

During this phase, Requirement Analysis was involved, which focused on collecting and evaluating data to define the system's needs accurately. The developers conducted comprehensive interviews and brainstorming sessions with teachers at Naawan Central School. A key meeting took place on October 2, 2024, during which Ma'am Zairine H. Tanque provided valuable insights into the school's assessment processes and challenges faced by the teachers. These discussions are crucial for understanding the procedural requirements and business rules specific to the school's context. The developers meticulously reviewed and analyzed each requirement, ensuring that the gathered information captured the core functionalities needed for the AMS. This involved understanding the teachers' workflow for preparing and managing assessments, conducting item analysis, and identifying the need for an efficient system to handle these academic tasks. The analysis phase led to a detailed

specification of system requirements aimed at improving efficiency and effectiveness in assessment management, specifically tailored for teachers at Naawan Central School.

Phase 2: Sprint-Based Development

The core of the Agile methodology lies in its sprint-based development structure. Each sprint will last approximately three weeks, followed by a one-week review and planning period. This allows for focused development while leaving room for adjustments between sprints. The project team will incrementally develop the system, with each sprint dedicated to a specific subset of the AMS's functionality. By dividing the development into sprints, the team will be able to deliver working components of the system early, enabling stakeholders to test and review features as they are completed.

Phase 3: Feedback and Iteration

One of the core principles of Agile is continuous feedback and iteration. After each sprint, the team will:

- Present the current progress to stakeholders for review and feedback
- Gather feedback on both the functionality and usability of the system
- Refine the system based on the feedback received, making necessary adjustments in subsequent iterations

This iterative process ensures that the system continues to align with the evolving needs of its users. It also allows the team to address any issues or concerns before moving on to the next sprint, ensuring the highest level of satisfaction among stakeholders.

Phase 4: Deployment and Evaluation

Once the core functionalities of the AsseSynchave been developed, tested, and refined, the system will be deployed in a real-world classroom setting for evaluation. During this phase, the team will:

- Monitor the system's performance in a live environment to ensure that it functions as intended
- Collect feedback from teachers on the system's effectiveness and ease of use
- Implement any necessary refinements based on the feedback collected during the deployment phase

This real-world evaluation will ensure that the system not only meets the technical requirements but also fulfills its intended purpose of improving assessment management at Naawan Central School.

4.1.1. System Diagrams

- **Use Case Diagram**: This diagram mapped out the main interactions between users (teachers) and the AMS, highlighting core functions like question preparation, administering assessments, and performing item analysis.
- **Activity Diagram**: It illustrates the sequence of actions within each significant function, ensuring a logical and intuitive flow of operations, such as creating assessments or running analytics.
- **Context Diagram**: This high-level diagram depicted the AsseSync interactions with external entities, defining the system's boundaries and the data exchanged.
- **Data Flow Diagram (DFD)**: The DFD provided a detailed visualization of data movement within the system, identifying processes, data storage locations, and key information exchanges.

4.1.2. Database Design

To support efficient data management and retrieval, the system's database is designed using an Entity-Relationship Diagram (ERD). The ERD defined key entities such as Teachers, Assessments, and Questions, as well as the relationships between them. This robust database structure facilitated real-time analytics and ensure data integrity, making the AsseSynca reliable tool for teachers to manage assessments and analyze student performance efficiently.

4.2. Software Development

The development team, alongside key stakeholders, initiated the creation of a sprint iteration process, beginning with the sprint backlog derived from the overall product backlog. This iterative approach followed the Agile methodology and continued on a bi-weekly basis until the Assessment Management System (AMS) was completed. During each sprint iteration, the developers conducted meetings twice a month. The first meeting is a sprint planning session to define goals, prioritize tasks, and allocate responsibilities, while the second meeting is a sprint retrospective to review progress, identify obstacles, and discuss improvements.

Throughout the sprint cycle, team members worked individually on assigned tasks and provided asynchronous daily progress updates. These updates ensured efficient communication and allowed team members to stay aligned with the overall project timeline and address any challenges promptly. The AMS, a web-based system, is developed using Vue.js for the front-end, Inertia.js for seamless single-page application behavior, Laravel for the back-end framework, and PostgreSQL for secure and efficient data management.

The front-end and back-end leads collaborated continuously to ensure that all system components are seamlessly integrated. Communication is facilitated through regular code

reviews and collaborative problem-solving sessions, which helps maintain high-quality code and efficient data flow between the client and server.

Frequent sprint reviews and retrospectives are conducted to ensure that the development of the AsseSyncis is aligned with the requirements of Naawan Central School. These sessions provided opportunities to gather feedback from stakeholders, particularly the teachers, and to make iterative improvements. Bi-weekly communication with key stakeholders ensures that updates on development milestones are communicated effectively and that any feedback regarding system functionality or usability is addressed promptly.

This structured and collaborative approach, centered on Agile principles, enabled the development team to quickly respond to challenges, incrementally improve the system, and deliver a robust and user-friendly AsseSync that fulfilled all specified functional requirements.

4.3. Functional Requirements

The following are the expected functionalities of the Assessment Management System (AMS), as specified by Naawan Central School, based on their academic requirements and assessment processes. AsseSync is designed to simplify assessment management, focusing on the essential functions teachers need to manage classes and questions efficiently.

4.3.1. Web-Based System

4.3.1.1. Teacher

The Teacher module includes four primary functionalities, each dedicated to effective assessment management and database interaction:

- **Log-in**: Teacher users must have a valid login ID to securely access the system and perform their respective functions. The system ensures that user data is protected and only authorized teachers can manage assessments.
- **Create Question**: Teachers can create and organize questions for various assessments. The system supports different question types, such as multiple-choice, true/false, and short answer, allowing teachers to customize content to meet academic standards.
- **Questions/Item Analysis**: The AsseSync includes a robust item analysis feature that evaluates questions based on student responses. It includes important metrics, such as checking test score reliability, item difficulty, analyzing Item Distractor Quality and discrimination index, providing teachers with insights to refine and improve assessment quality. This analysis helps ensure that questions are fair and effective in measuring student understanding.

- **TestBank**: A comprehensive test bank feature enables teachers to store, categorize, and retrieve questions efficiently. Questions can be tagged based on subjects, difficulty levels, and learning objectives, making it easy to reuse and organize content for future assessments. This functionality saves time and enhances the consistency of assessments.

4.3.1.2. Admin

The Admin module includes four primary functionalities, each dedicated to affective assessment management and database interaction:

- **Log-in**: Admin users must have a valid login ID to securely access the system and perform their respective functions. The system ensures that user data is protected and only authorized teachers can manage assessments.
- **Import Questions:** Admin can Import Test Questions from DepEd.
- **Edit Question:** Admin can Edit questions from the Test banks.
- **Delete Question:** Admin can delete questions from the Test banks.

4.3.1.3. Web-Base Log Functions

- Log-in: Registered users must have a valid login id to perform their respective user functions.
- Log-out: Users can log-out from the site.

4.4. Web-Base Functional and Technical Requirements

4.4.1. Technical Requirements

These are the technical expectations for the reliability, security, and performance of the AsseSyncweb application.

4.4.1.1. Performance Requirements:

- Data Response Time: Responses to load and display data, such as class details, test questions, or assessment results, should take no more than 5 seconds.
- Scalability: The system should handle data efficiently as more assessments and questions are added to the test bank.
- Real-Time Analysis: The system must compute and display item analysis metrics (item difficulty, discrimination index) for assessments in real-time.

4.4.1.2. Security Requirements:

- Data Protection:The AsseSync will use a centralized, secure database to safeguard sensitive assessment data.

- User Authentication: The system will require a secure login for teachers to ensure that only authorized users can access and manage assessments.
- Data Backup and Recovery: Automated backups ensure data is protected in case of system failure.

4.4.1.3. Error Handling:

- User-Friendly Error Messages: When an error occurs, the system will display informative messages to guide users without revealing system internals.
- System Downtime: AsseSync should handle unexpected errors to prevent loss of information and minimize downtime, ensuring reliable access for teachers during peak usage.

4.4.1.4. Logging:

- User Activity Logs: Logs will capture successful user logins, along with user names, dates, and times.
- Change Tracking: Logs will record any changes made to classes, assessments, or test banks, with timestamps and user details.
- Error Logs: Logs will capture error events, including failed login attempts, database issues, or other operational disruptions.

4.4.2. System Design Diagrams

To ensure the AsseSync meets functional and technical requirements, the development team used the following UML diagrams:

- System Architecture Diagram
- Use Case Diagram
- Functional Decomposition Diagram
- Context Diagram
- Data Flow Diagram (DFD)

4.4.3. Database Design

To support AsseSync functions, a centralized database structure is chosen for reliable data management. Database design steps included:

- Conceptual Data Modeling: Identified entities such as Teacher, Class,Test Questionnaire.
- Logical Data Modeling: Defined relationships, ensuring proper links between entities to allow for easy retrieval and updates.

- Physical Data Modeling: Implemented the database using PostgreSQL, focusing on performance and scalability to handle multiple assessments and student records.
- Database Implementation: Tested database integrity and optimized queries to ensure the AsseSync can handle complex queries without significant delays.

4.4. Software Testing

To ensure the functionality and usability of the proposed system, a structured black box testing approach shown in Chapter 3 will be followed. The testing process will include:

4.4.1. Functional Testing

4.4.1.1. Unit Testing:

The IT Faculty of Mindanao State University at Naawan will validate individual features, such as creating assessments, saving questions, and conducting item analysis, with results documented for each test case.

4.4.1.2. System Testing:

Test scenarios will cover tasks like creating full assessments, saving tests, and generating reports, ensuring that the system behaves as expected in various user scenarios.

4.4.1.3. Acceptance Testing (User Acceptance Testing - UAT):

Teachers from Naawan Central School will participate in UAT. Feedback from teachers will guide final adjustments, confirming that the system aligns with educational needs and usability expectations.

4.4.2. Non-Functional Testing

4.4.2.1. Usability Testing:

At least 30 teachers from Naawan Central School will assess system usability. The IT Department will conduct this testing using a task list, System Usability Scale (SUS), and an Adjective Rating Scale for interpretation. The goal is to ensure that the interface is intuitive and accessible for daily use.

4.5. Software Usability

The System Usability Scale (SUS) is employed to provide a reliable and valid measure of the usability and acceptance of the AsseSync: Assessment Management System among its users. The SUS questionnaire is designed to assess the system’s functions, design, and overall user satisfaction. To maximize feedback, The developers integrated both positive and negative statements, ensuring a balanced evaluation of the system's usability.

Additionally, three open-ended prompts are implemented to capture users' positive and negative impressions, as well as any suggestions for improvement, further enhancing the feedback process.

The SUS questionnaire for the AsseSync consisted of ten questions, five of which are positively worded, and five are negatively worded. The questions aimed to assess various aspects of the AsseSync, including usability, learnability, ease of use, and user satisfaction.

The following positively worded questions are designed to assess how users viewed the system in terms of ease of use, integration, and learnability:

- “I think that I would like to use this system frequently."
- “I thought the system was easy to use."
- "I found the various functions in this system were well integrated."
- "I would imagine that most people would learn to use this system very quickly"
- "I felt very confident using the system."

The first question, "I think that I would like to use this system frequently," is designed to measure users' interest in regular use of AsseSync. The second question, "I thought the system was easy to use," assessed users' perception of the effortlessness required to operate the system. The third question, "I found the various functions in this system were well integrated," measured how effectively different features of AsseSync worked together cohesively. The fourth question, "I would imagine that most people would learn to use this system very quickly," evaluated the learnability and intuitiveness of the system. The fifth question, "I felt very confident using the system," assessed the users' sense of control and self-assurance while interacting with AsseSync.

The following negatively worded questions are intended to assess any difficulties users experienced with the system, including the need for technical support, consistency, and intuitiveness:

- “I found the system unnecessarily complex.”
- “I think that I would need the support of a technical person to be able to use this system.”
- “I thought there was too much inconsistency in this system.”
- “I found the system very cumbersome/awkward to use.”
- “I needed to learn a lot of things before I could get going with this system.”

These questions focused on users’ perceptions of the system’s potential flaws, specifically assessing its intuitiveness, complexity, and the necessity of support or significant learning. The first question, "I found the system unnecessarily complex," is designed to measure the extent to which users perceive AsseSync as overly complicated. The second

question, "I think that I would need the support of a technical person to be able to use this system," assessed users' perceived dependence on technical assistance to operate AsseSync. The third question, "I thought there was too much inconsistency in this system," aimed to evaluate the lack of uniformity in the system's interface and behavior. The fourth question, "I found the system very cumbersome/awkward to use," assessed whether users found the system difficult or unwieldy in practice. The fifth question, "I needed to learn a lot of things before I could get going with this system," is aimed at determining the perceived steepness of the learning curve associated with AsseSync's interface and features.

By integrating both positively and negatively worded questions, the developers aimed to balance user responses and reduce response bias. This approach provided a more comprehensive and actionable assessment of the AsseSync's usability.

4.5.1. The Test Participants

The objective is to conduct the test to the teachers from Naawan Central School, with a minimum of 30 participants to ensure that the system is evaluated from a different perspective. The developer of this project expected to achieve insights from teachers with different age groups and whether there is a specific area where our system needed improvements based on different preferences.

4.5.2. The Usability Test Setup

The usability test is scheduled to be conducted at Naawan Central School. An orientation script (See Appendix A) is used to introduce the system to the selected teachers participating in the test. Test participants are provided with a consent form (See Appendix B) to ensure their voluntary participation and adherence to ethical considerations. The functionality of the system is thoroughly tested, and participants' subjective evaluations are gathered using the System Usability Scale (SUS) form (See Appendix C). The SUS scores are then computed to assess overall usability, and the Adjective Rating Scale (ARS) is applied for interpretation. This structured approach allowed for a comprehensive evaluation of the system's usability.

4.6. Summary of the Chapter

This chapter describes the methodology used in developing the AsseSync for Naawan Central School. It describes the Agile approach to system design. Key components such as use case, activity, context, and data flow diagrams with visualization of system functions, while providing a database design ensured efficient data handling. The chapter also covers the development process by integrating Vue.js, Inertia.js, Laravel, and PostgreSQL to build a web-based platform. And lastly, structured testing methods unit, system, and user acceptance testing together with a usability assessment by teachers to validate the system's functionality and user-friendliness.

CHAPTER 5
RESULTS AND DISCUSSION

This chapter provides an overview of the design, development, and testing results of the Assessment Management System created for the teachers of Naawan Central School. It includes information on the project's implementation and testing phases, highlighting how the system addresses the specific needs of the assessment management system.

5.1. Design Results

5.1.1. System Architecture

This section presents a system architecture diagram of the Assessment Management System.

Figure 4 shows the system architecture of the Assessment Management System, designed to facilitate efficient interaction between the teacher and admin's user interface and the central database. Teachers access the system remotely via the internet, using the web application to manage test banks and perform item analysis. On the other hand, the admin manages the Test questionnaire provided by the DepEd. The system server interfaces with pgAdmin 4, a management tool that communicates directly with the PostgreSQL database. This architecture ensures seamless data flow by allowing teachers to create, store, and retrieve assessment data efficiently, with secure and organized access to test questions and analysis tools. This setup offers a reliable and user-friendly experience tailored to the needs of teachers.

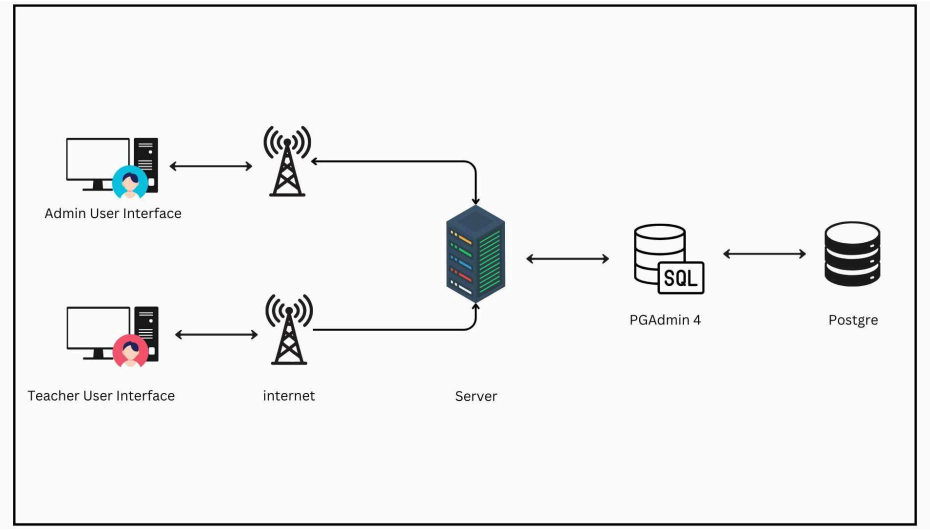


Figure 4. System Architecture of the Assessment Management System

5.1.2. Hardware and Software Design Components

Table 2 shows the hardware components used in developing the Assessment Management System. The workstation includes a Ryzen 3 3200G processor for reliable performance, 8GB of memory to support development tasks, and a 500GB hard drive for storing project files and tools. This setup provides a sufficient and stable environment for system development.

Table 2. Hardware Design Component used in the development of Assessment Management System.

Workstation	**Specifications**	
	Components	**Details**
Unit for Web Development	Processor	Ryzen 3 3200G
	Memory	8GB
	Hard Drive	500GB

Table 3 presents the software used in the development of the system. HTML5, Tailwind CSS, Vue.js, and Inertia.js were used for building the front-end interface, while Laravel (PHP8) and PostgreSQL handled the back-end processes. Windows 10 served as the operating system, providing a stable platform for all development activities.

Table 3. Software Design Component used in the development of Assessment Management System

Software Component	**Usage**
HTML5, Tailwind CSS, Vue.js (JavaScript), Inertia.js	Front End
Laravel (PHP8 Framework), PostgreSQL	Back End
Windows 10	Operating System

5.1.3. Use Case Diagram

Figure 5 presents the use case diagram of the Assessment Management System, where the teacher is the primary actor responsible for creating, managing, evaluating, and analyzing assessments, while the system administrator supports by managing questionnaires and test banks.

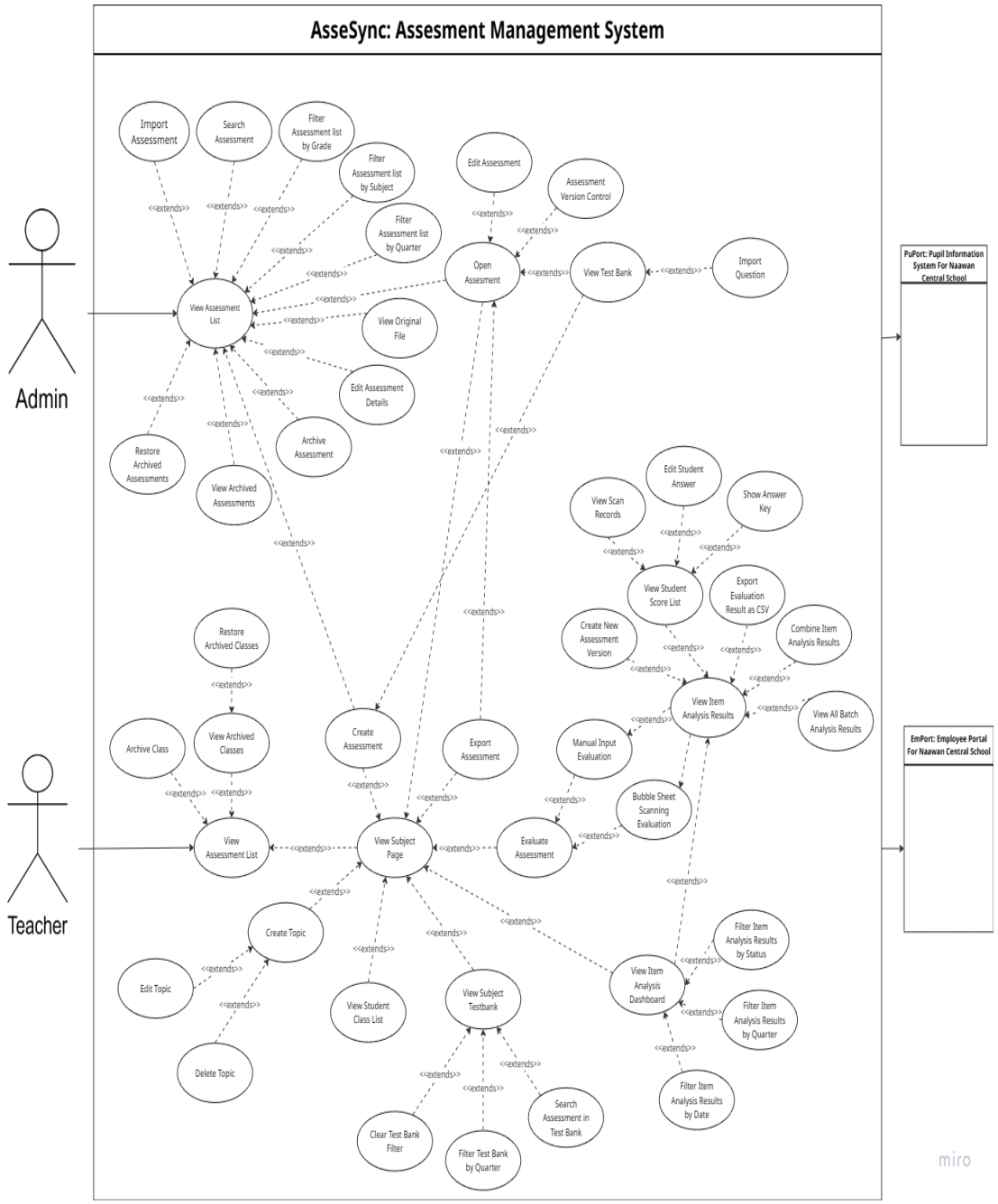


Figure 5. Use Case Diagram of Assessment Manage itement System

5.1.4. Activity Diagram

This part presents the activity diagram of the Assessment Management system per use case. In addition to defining business processes and use cases. This is used to document how system procedures have been implemented.

Figure 6 illustrates the process of viewing the assessment list in the system. The admin initiates the action by clicking the "Dashboard" button. The system responds by displaying the dashboard page, which includes a comprehensive list of available assessments.

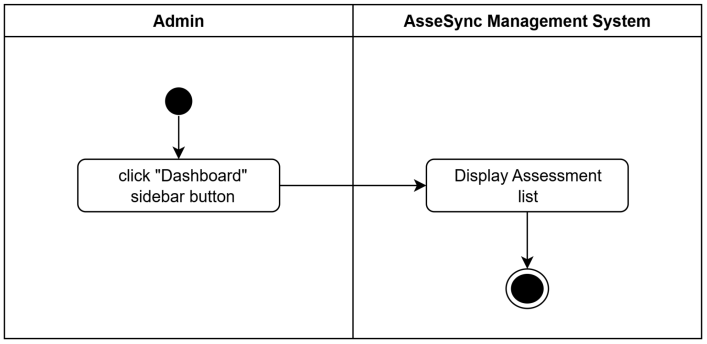


Figure 6. Activity Diagram of Admin View Assessment List

Figure 7 illustrates the Admin Import Assessment process in the AsseSync Management System. The process begins when the Admin clicks the “Import Assessment” button, prompting the system to display the Import Assessment modal. The Admin then inputs the required details and uploads the assessment file, either in DOCX or PDF format, by clicking or using drag-and-drop functionality. At this stage, the Admin may choose to cancel the process, which terminates the workflow, or click “Save” to proceed.

If the Admin clicks “Save”, the system validates the input and asks whether to save. If “No” is selected, the process ends. If “Yes”, the system displays the “Extracted Questions” modal for the Admin to review. At this point, the Admin can either click “Cancel” to stop the process or “Confirm” to proceed. Once confirmed, the system displays a success message: “Questionnaire uploaded successfully”, completing the import process.

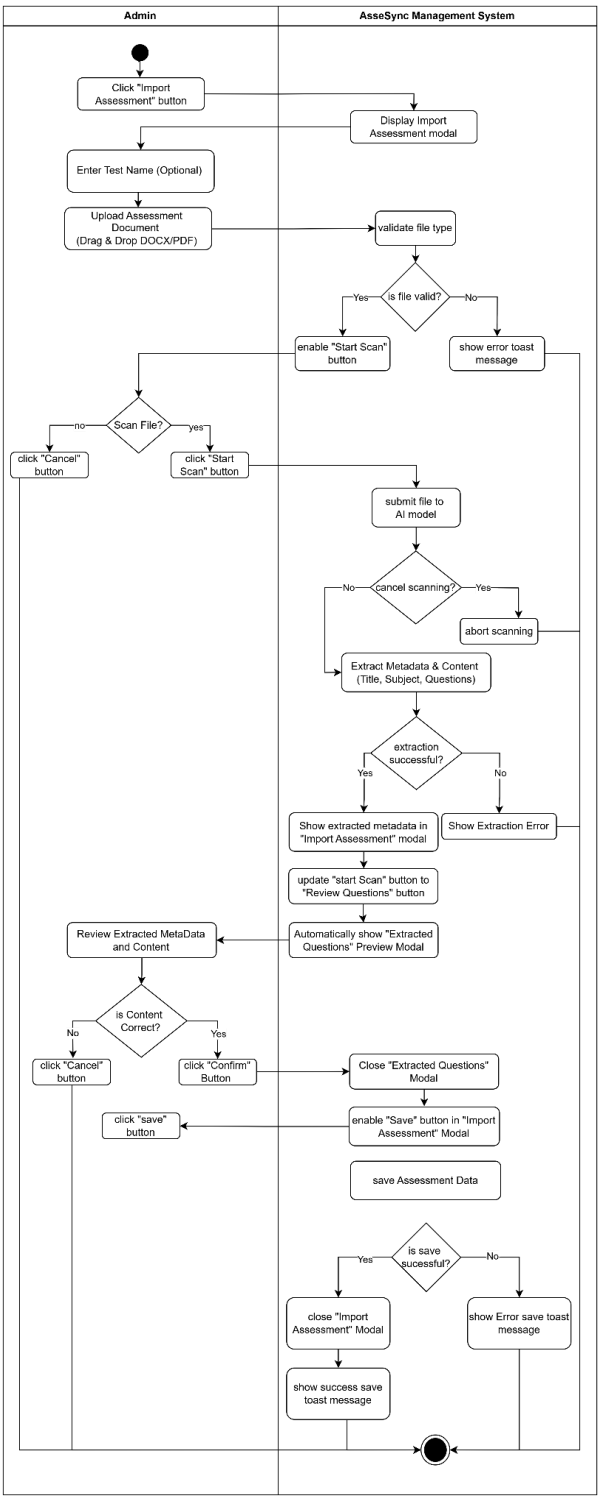


Figure 7. Activity Diagram of Admin Import Assessment

Figure 8 illustrates the overall process involves the Admin or Teacher initiating the creation/editing of an assessment, navigating through multiple steps to select assessment type, input details, choose question sources (uploading files, selecting from the test bank, or generating new questions), confirming question inclusions, setting up assessment configurations and distribution, and finally, either saving the assessment or generating a printable template, concluding the multi-stage creation workflow.

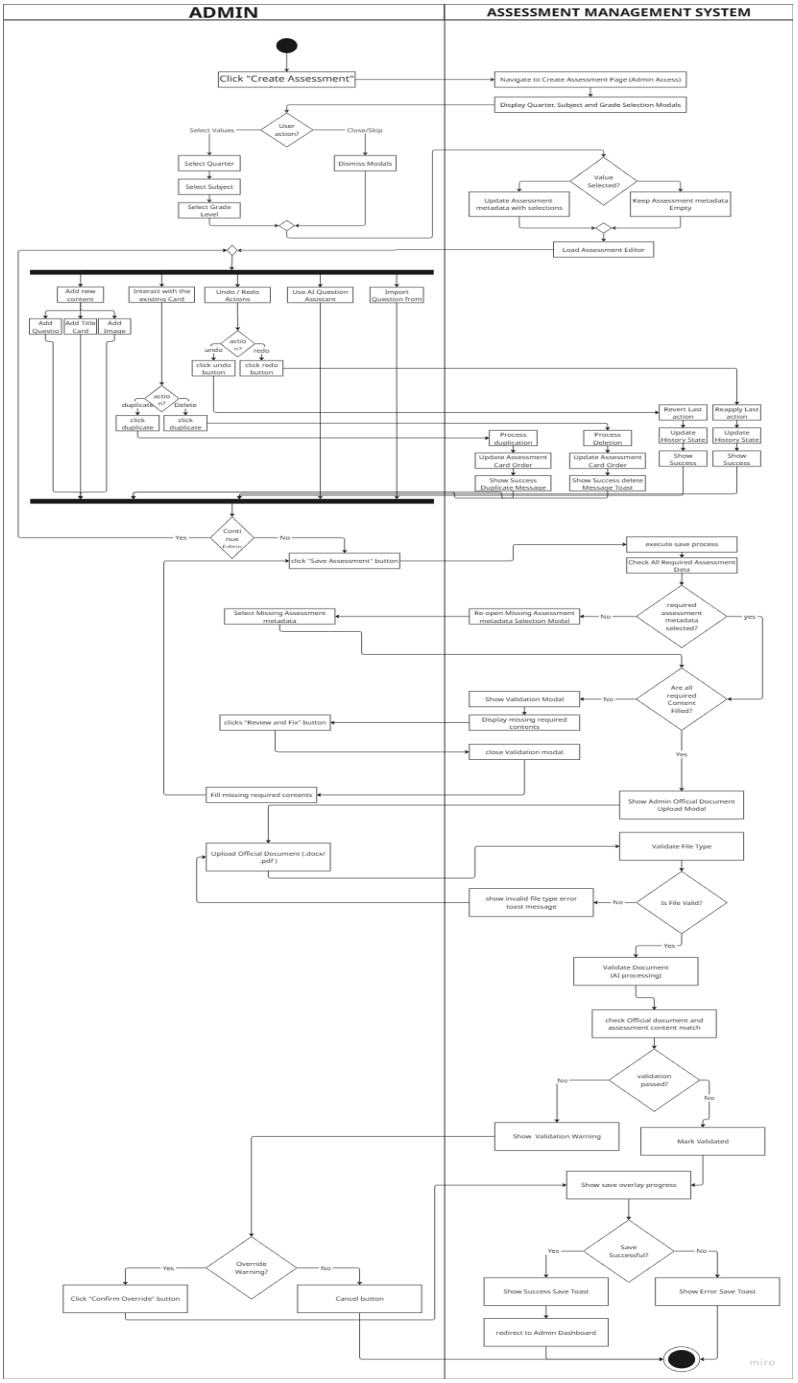


Figure 8. Activity Diagram of Admin Create Assessment

Figure 9 illustrates the process by which an Admin or Teacher uses the Assessment Management System to create assessment questions, leveraging both an AI feature and a Test Bank.

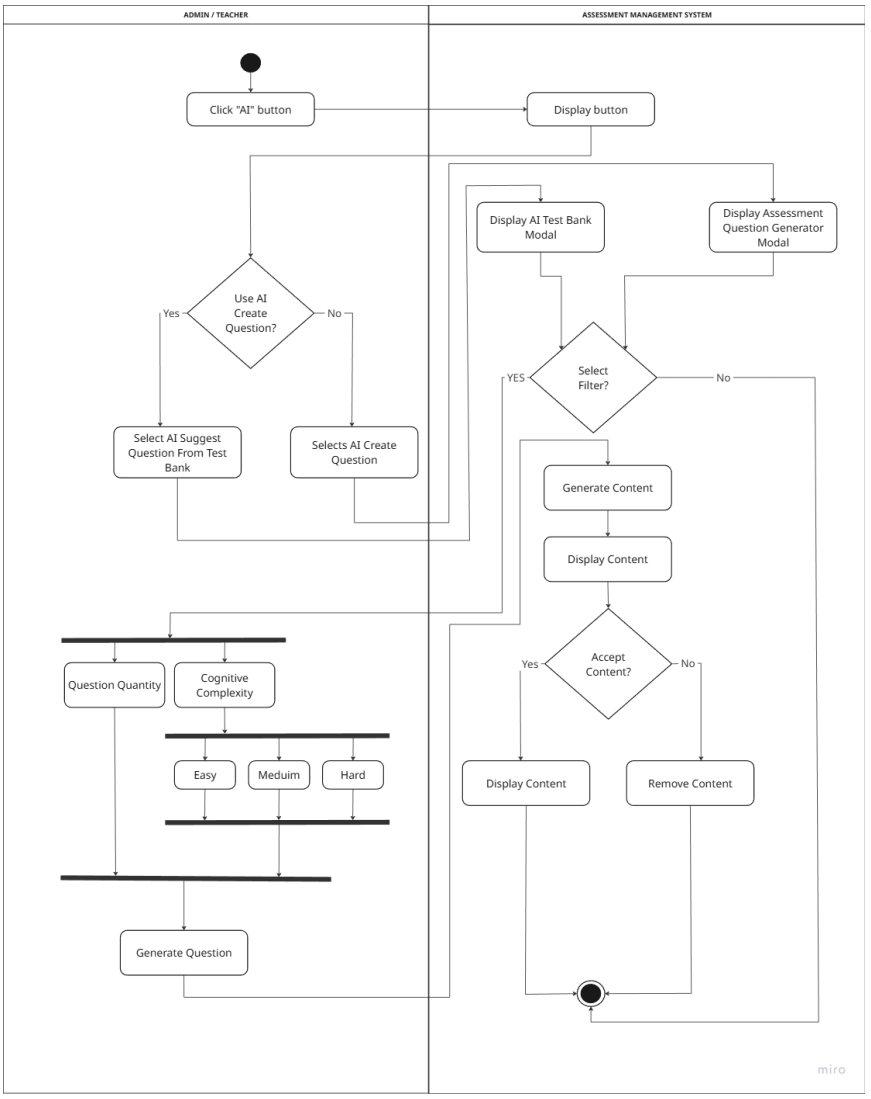


Figure 9. Activity Diagram of Admin and Create Assessment using AI

Figure 10 illustrates the process of searching for assessments using the search bar on the dashboard page. The user inputs a keyword into the search bar, and the system performs a clustering operation to match the keyword with relevant assessments. The search results are then displayed, enabling the user to quickly access the desired information based on their query.

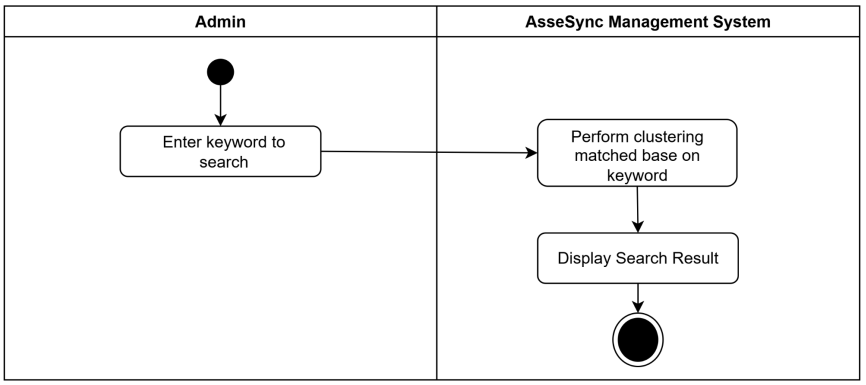


Figure 10. Activity Diagram of Admin Search Assessment

This diagram illustrates the process of filtering assessments by grade in the AsseSync Management System. When the admin clicks the “Grade” dropdown button, the system displays a list of available grades. After the admin selects a grade from the dropdown, the system displays the corresponding assessment list based on the selected grade.

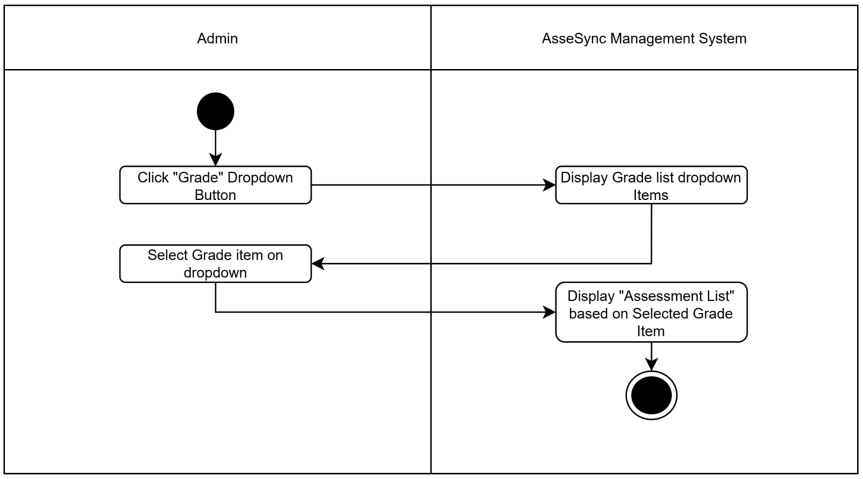


Figure 11. Activity Diagram of Admin Filter Assessment list by Grade

Figure 12 illustrates the process of filtering assessments by subject in the AsseSync Management System. When the admin clicks the “Subject” dropdown button, the system displays a list of available subjects. After the admin selects a subject from the dropdown, the system updates and displays the corresponding assessment list based on the selected subject.

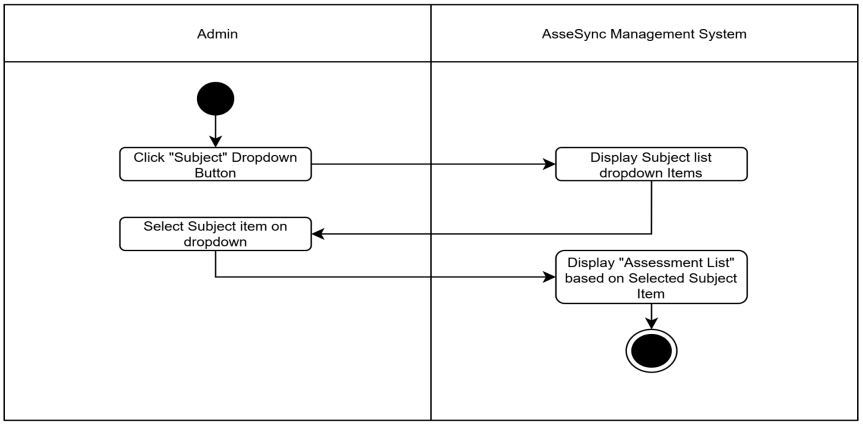


Figure 12. Activity Diagram of Admin Filter Assessment list by Subject

Figure 13 illustrates the process of filtering assessments by quarter in the AsseSync Management System. When the admin clicks the “Quarter” dropdown button, the system displays a list of available quarters. After the admin selects a quarter from the dropdown, the system displays the corresponding assessment list based on the selected quarter.

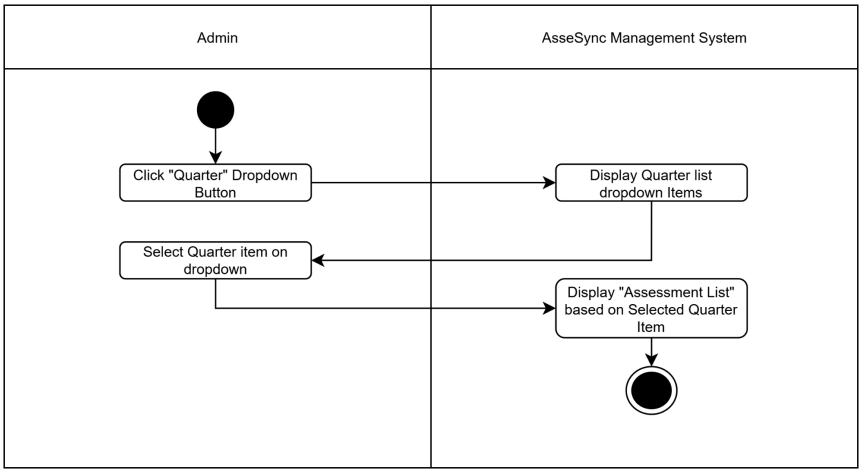


Figure 13. Activity Diagram of Admin Filter Assessment list by Quarter

Figure 14 illustrates the process of clearing filters in the AsseSync Management System. When the admin clicks the “Clear” button, the system removes all selected filters and resets the assessment list to its default state. This ensures that all assessments are displayed without any applied filtering criteria.

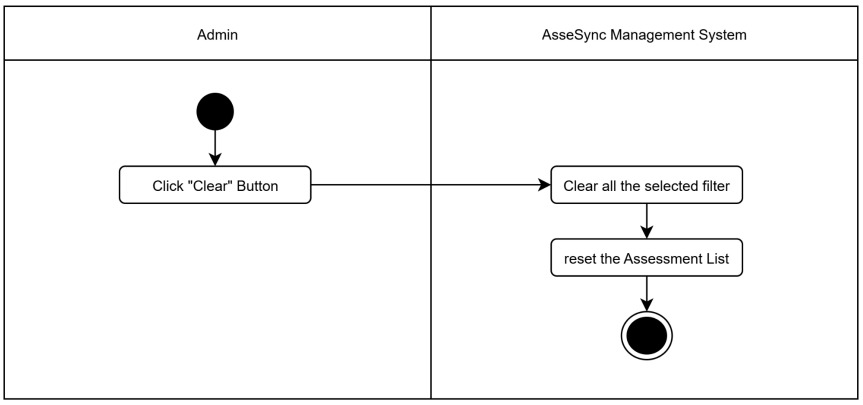


Figure 14. Activity Diagram of Admin Clear Filter

Figure 15 illustrates the Admin Open Assessment process in the AsseSync Management System, where the Admin clicks the “Open Assessment” icon, prompting the system to display the assessment form along with the assessment library (test bank) for review or further action.

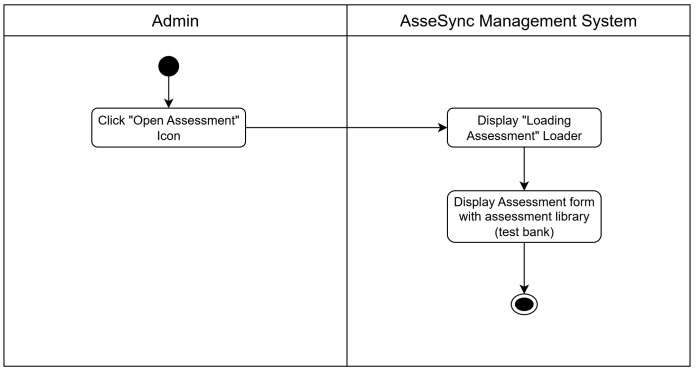


Figure 15. Activity Diagram of Admin Open Assessment

Figure 16 illustrates the process of managing an assessment in the AsseSync Management System. When the admin clicks the “Open Assessment” icon, the system displays a loading screen and then presents the assessment form with the assessment library (test bank). The admin can perform various actions such as adding a title, questions, or images; uploading or duplicating questions; and saving, exporting, or exiting the assessment. Depending on the chosen action, the system executes the corresponding operation and completes the process.

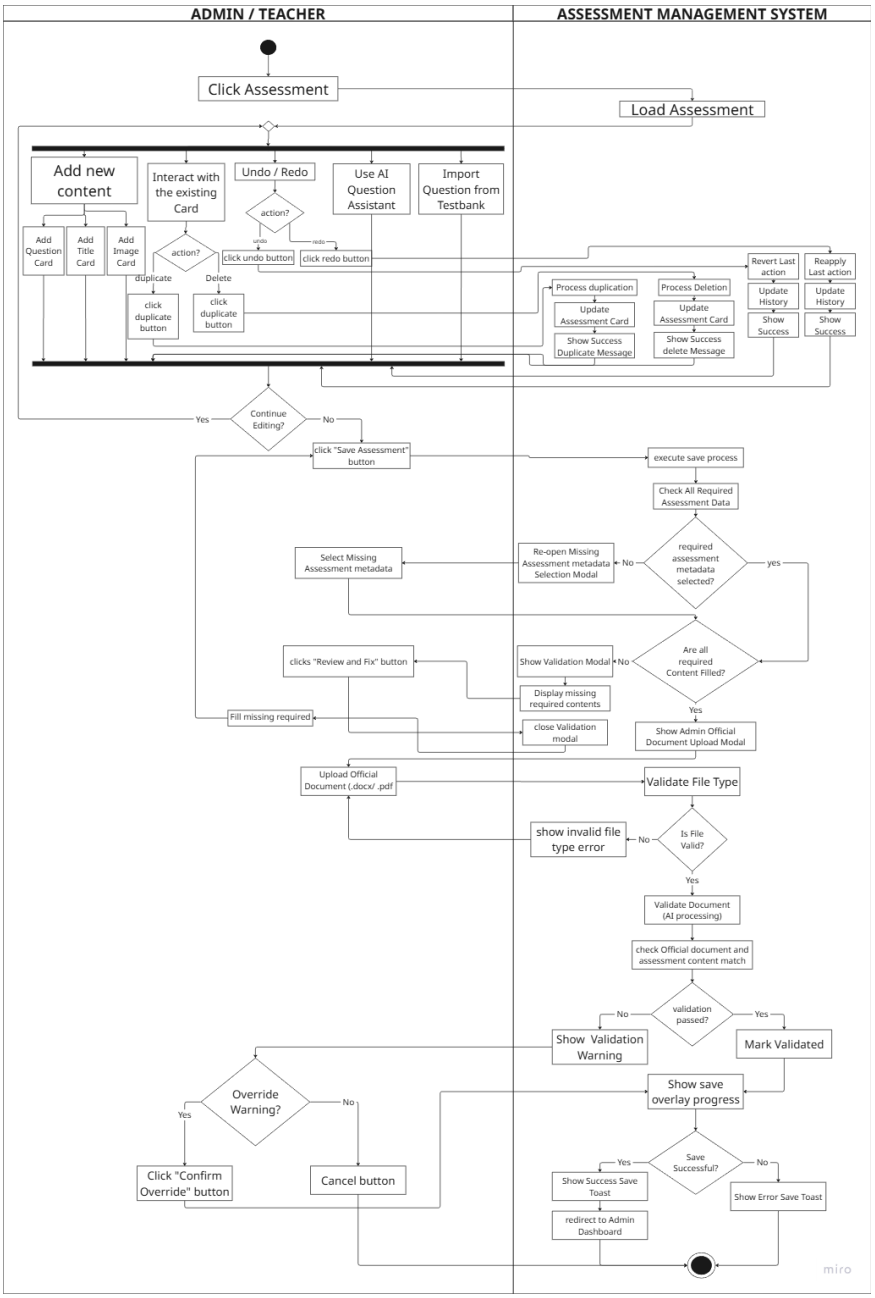


Figure 16. Activity Diagram of Admin Edit Assessment

Figure 17 illustrates the process of opening an assessment in the AsseSync Management System. When the admin clicks the “Open Assessment” icon, the system displays a loading screen while retrieving the necessary data. Once the process is complete, the system presents the assessment form along with the associated assessment library.

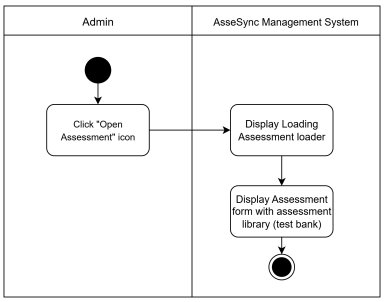


Figure 17. Activity Diagram of Admin View Assessment Library

Figure 18 illustrates the process of importing questions from the test bank into an assessment within the AsseSync Management System. The process begins when the admin clicks the “Open Assessment” icon. The admin then clicks the “Show Test Bank” button to open the test bank panel. The admin can then choose to “Select All” or pick specific questions to import. Once the desired questions are chosen, clicking “Import Selected” imports them into the current assessment form.

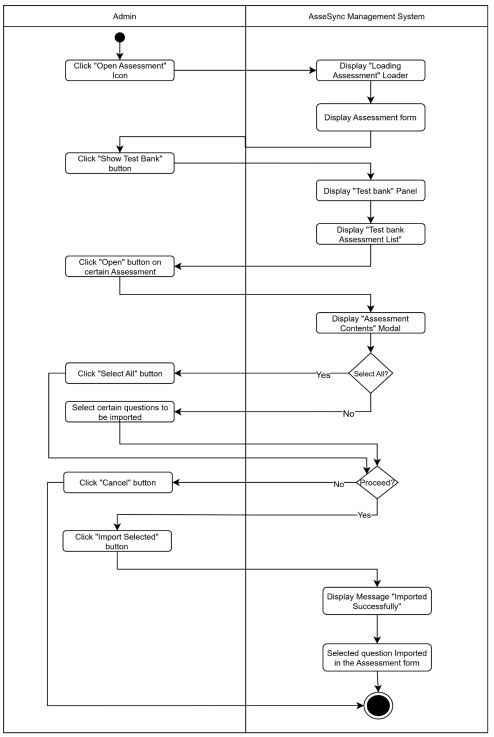


Figure 18. Activity Diagram of Admin Import Question

Figure 19 outlines the steps for a user to view the history of an assessment, select a specific version, and then choose whether to set that selected version as the new Main Version or simply switch to viewing/editing that version.

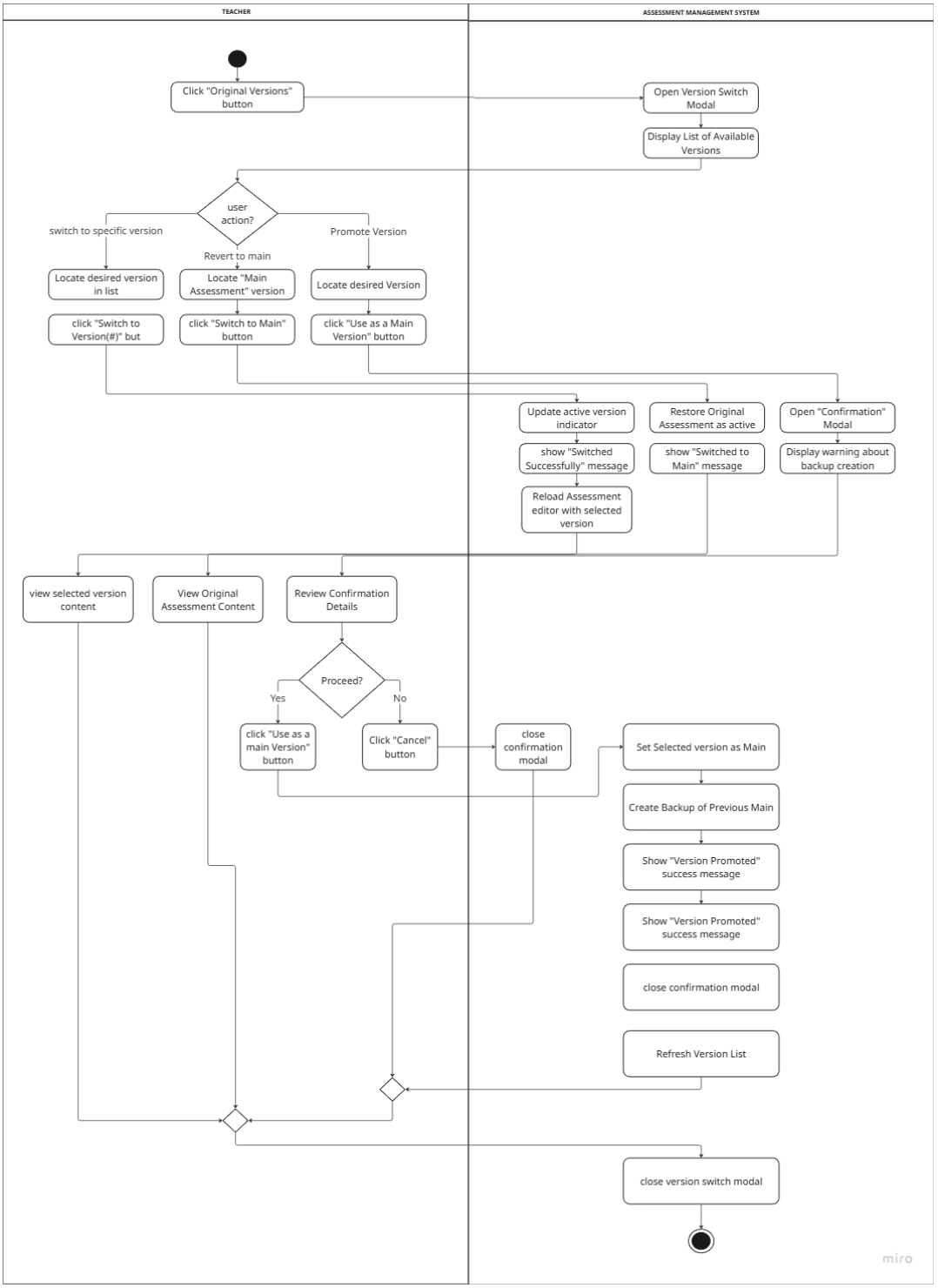


Figure 19. Activity Diagram of Admin/Teacher Switch Assessment Version Control

Figure 20 shows the workflow for accessing the original file associated with an assessment. The process starts when the user clicks the "View Original File" button. Following this action, the system immediately proceeds to Display Loading Modal to indicate that the file retrieval is in progress. Upon successful loading, the system completes the process by transitioning to and displaying The Assessment Original File Interface, allowing the user to view the source document.

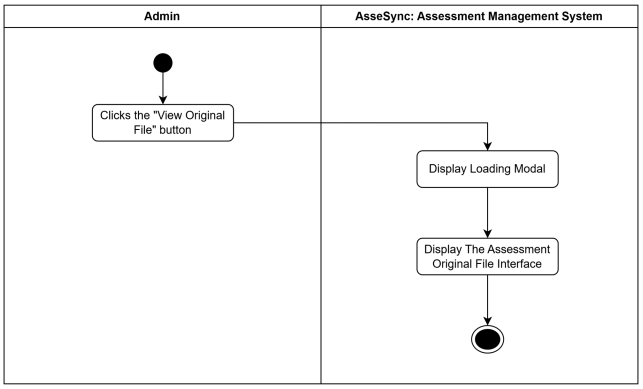


Figure 20. Activity Diagram of Admin View Original File

Figure 21 shows the workflow of editing assessment details in the AsseSync Management System, highlighting the interaction between the Admin and the System. The process starts when the admin clicks the “edit” assessment icon, after which the system displays an edit form modal. The admin then inputs the necessary changes to the assessment details. At this point, the admin can either cancel the action, which ends the process without saving, or click the “save” button to confirm. If saved, the system updates the assessment and displays a confirmation message “Edited successfully”, completing the process.

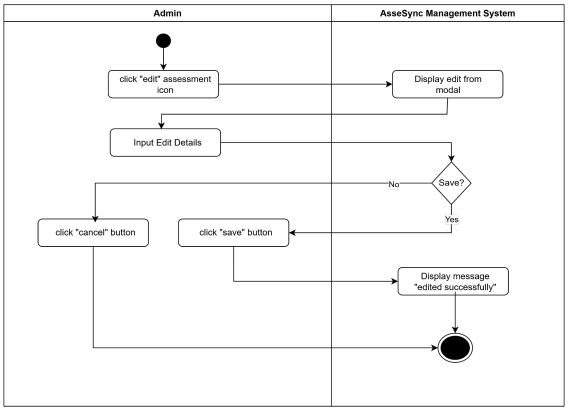


Figure 21. Activity Diagram of Edit Assessment Details

Figure 22 illustrates the workflow of archiving an assessment in the AsseSync Management System, showing the interaction between the Admin and the System. The admin starts by clicking the “archive” assessment icon, which prompts the system to display a confirmation dialog. At this point, the admin has two options: cancel the action, which ends the process, or confirm the archive by clicking the “archive” button. If confirmed, the system archives the assessment and displays a confirmation message “Archive successfully”, marking the completion of the process.

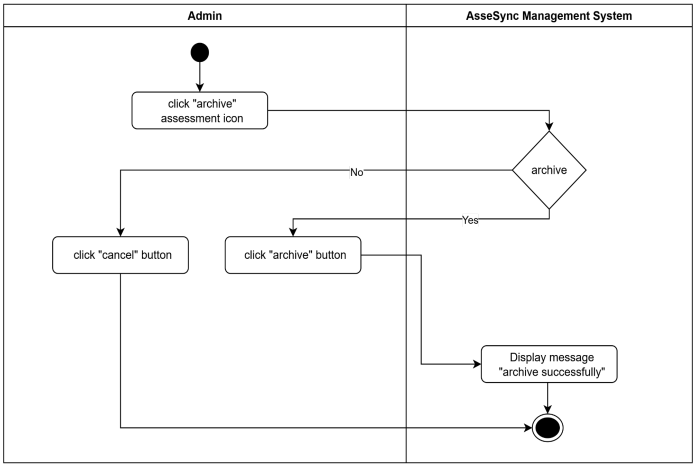


Figure 22. Activity Diagram for Admin Archive Assessment

Figure 23 illustrates the process of accessing the archive assessment feature in the AsseSync Management System. When the admin clicks the “Archive Assessment” button, the system responds by displaying the archive assessment interface, allowing the admin to manage archived assessment records.

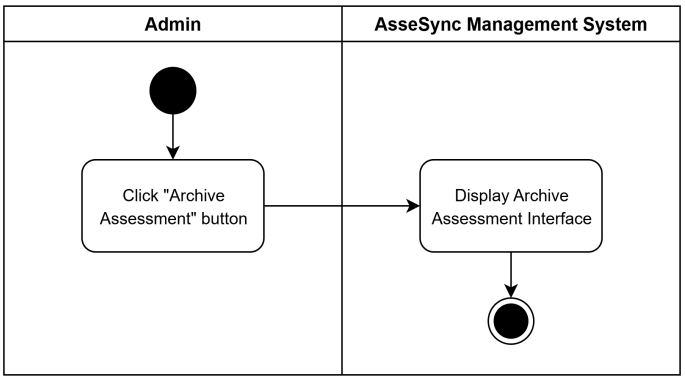


Figure 23. Activity Diagram of Admin View Archive Assessment

Figure 24 illustrates the process of restoring archived assessments in the AsseSync Management System. The admin begins by clicking the “Archive Assessment” button, prompting the system to display the archive assessment interface. The admin then selects an assessment to unarchive and clicks the “Restore Selected” button. The system confirms the action by displaying a success message indicating that the assessment has been restored.

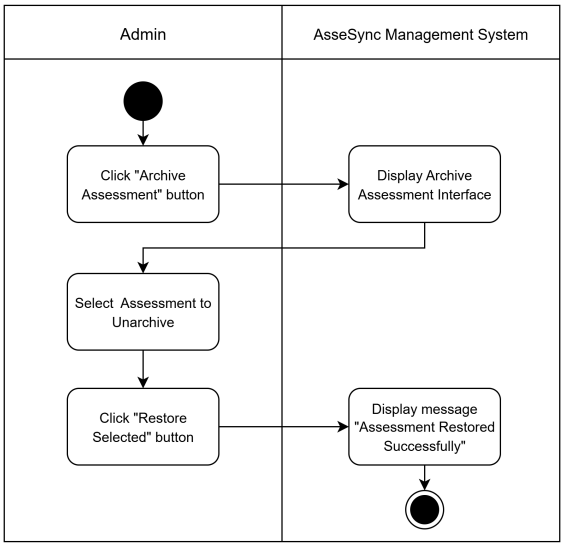


Figure 24. Activity Diagram of Admin Restore Archived Assessment

Figure 25 illustrates the process of navigating to the home section in the AsseSync Management System. When the teacher clicks the “Home” sidebar button, the system displays the subject list, allowing the teacher to view and access available subjects.

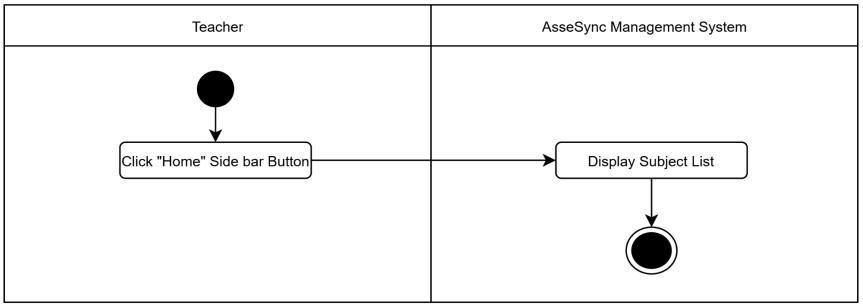


Figure 25. Activity Diagram of Teacher View Subject List

Figure 26 illustrates the workflow for archiving a class list. The Teacher initiates an action by clicking the "more options/vertical ellipsis," selecting the "Archive class" button, and confirming with "Continue," which results in the system displaying a "Restore Successfully" message and resetting the Archive Class Dashboard.

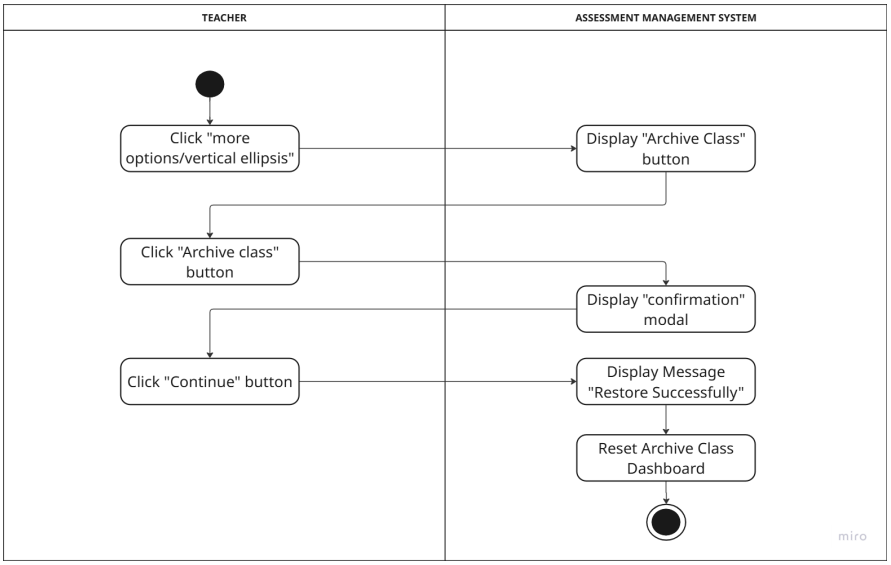


Figure 26. Activity Diagram of Teacher Archive Class

Figure 27 illustrates the workflow for viewing an archived class by clicking the "Archive Class" option in the side bar; the user navigates directly to the system's "Archived Class" Dashboard.

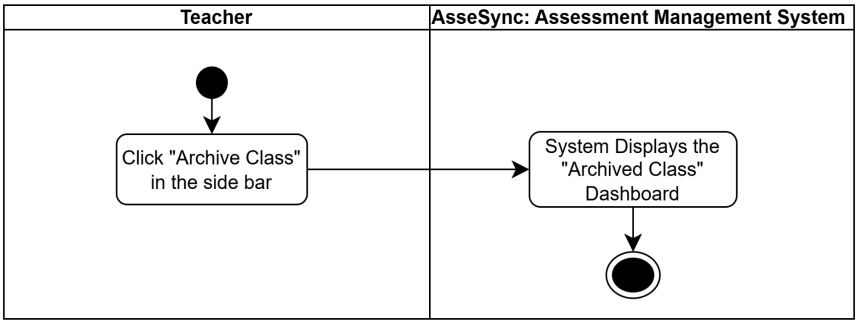


Figure 27. Activity Diagram of Teacher View Archived Class

Figure 28 illustrates the workflow for Teacher Restore Archive Class, by clicking the Subject Class Card, confirming the action in the Confirmation Modal by clicking "Confirm," after which the system displays a "Restore Successfully" message and resets the Archive Class Dashboard.

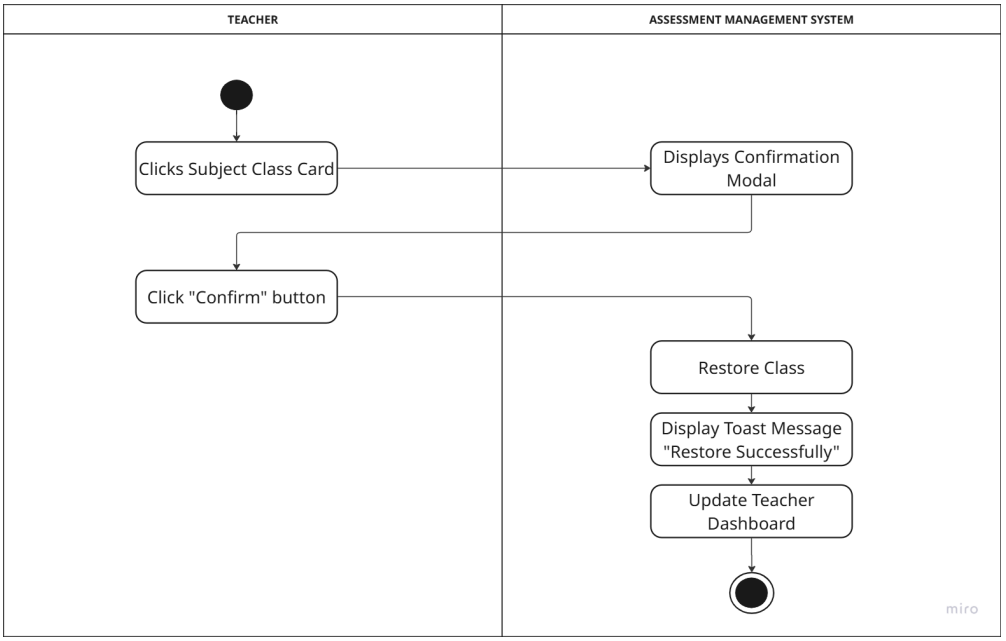


Figure 28. Activity Diagram of Teacher Restore Archive Class

Figure 29 illustrates the process of accessing a specific subject page in the AsseSync Management System. When the teacher selects a subject grade, the system responds by displaying the corresponding subject page, providing access to subject-specific assessments and resources

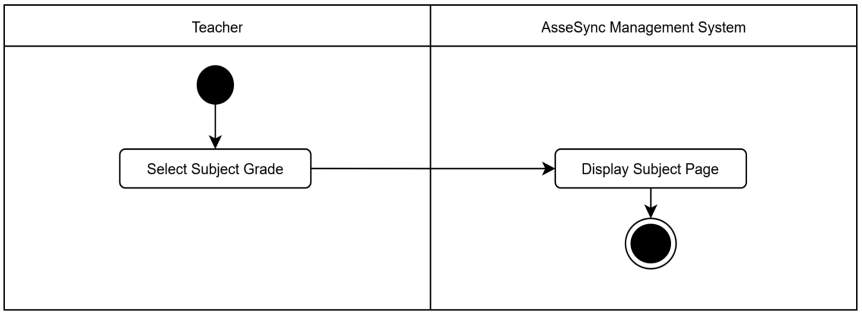


Figure 29. Activity Diagram of Teacher View Subject Page

Figure 30 illustrates the Teacher initiates topic creation by Clicking the "Create Topic" button, which displays a modal where they Input Required Fields; if the Teacher chooses to save and all required fields are filled, the system saves the Topic Data to database, shows a Success Create Topic Toast Message, and finally Display Topic as Card in Subject Page, otherwise the process highlights missing fields or ends if the Teacher clicks "Cancel."

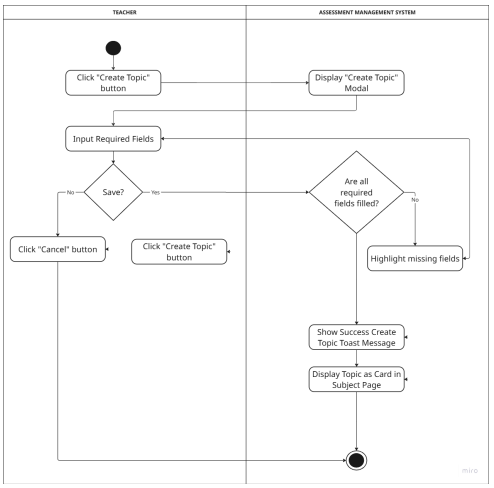


Figure 30. Activity Diagram of Teacher Create Topic

Figure 31 illustrates the Teacher initiates topic editing by Clicking "Edit Topic", which displays the modal allowing them to Edit Topic Details; if the Teacher chooses to save changes and all required fields are complete, the system displays a "Topic Updated Successfully" Toast Message and Show Updated Topic Detail in Subject Page, otherwise the system highlights missing fields or cancels the action if the "Cancel" button is clicked.

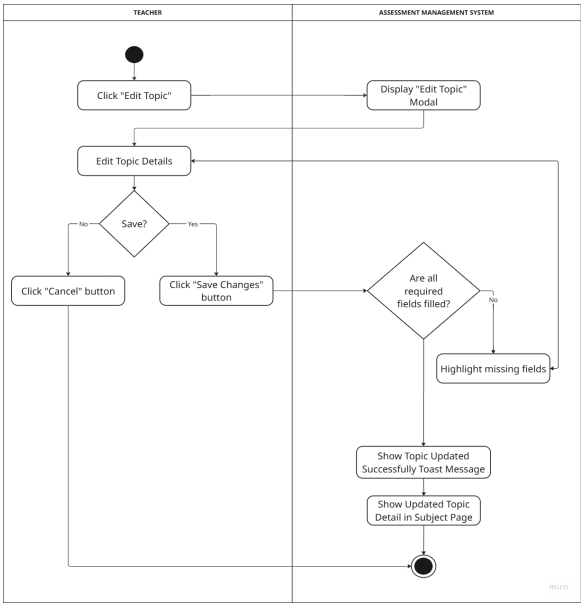


Figure 31. Activity Diagram of Teacher Edit Topic

Figure 32 illustrates the Teacher initiates topic deletion by Clicking "Delete Topic", prompting the system to check if topic has assessments; if assessments exist, an error message appears ("Cannot Delete Topic"), but if none exist, the system displays a confirmation modal, and if the Teacher clicks "Delete Topic" button, the system shows deletion progress, a "Topic Deleted Successfully" toast message, and completes the process, otherwise the Teacher can cancel the action.

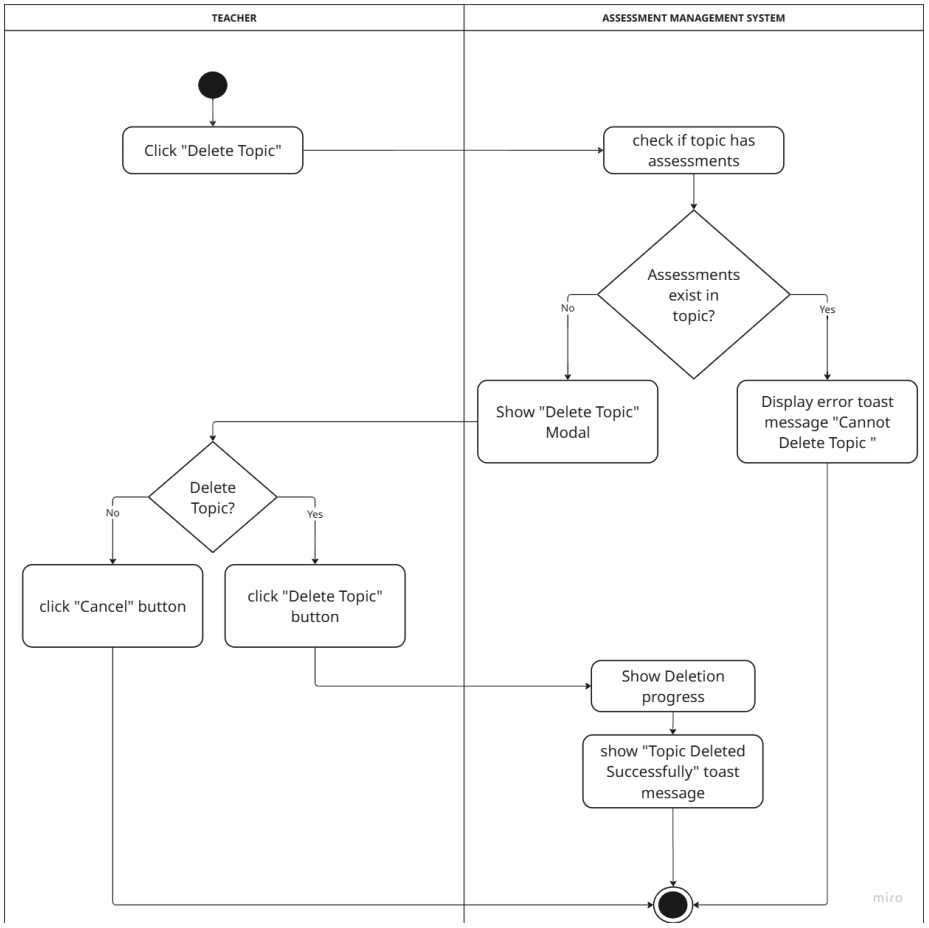


Figure 32. Activity Diagram of Teacher Edit Topic

Figure 33 illustrates the process of creating an assessment in the AsseSync Management System. The process begins when the teacher or admin clicks the “Create Assessment” button. The system then checks whether a topic has been created. If no topic exists, a warning message is displayed; otherwise, the system proceeds to show the “Pick a Topic” or “Select Quarter for Exam” modal. Within the form, the user can add titles, questions, images, or upload and manage questions. Upon saving, a loader and confirmation message indicating that the assessment has been successfully saved.

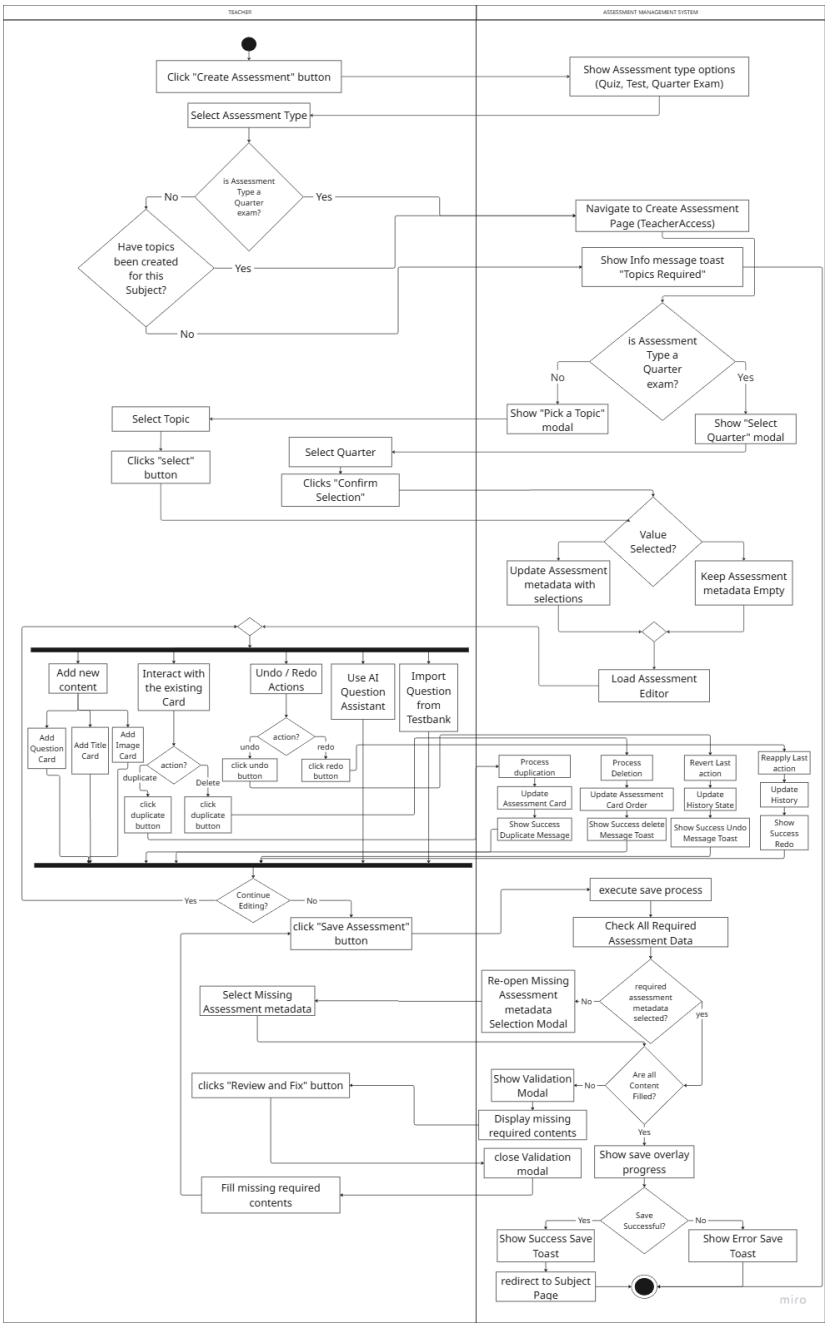


Figure 33. Activity Diagram of Teacher Create Assessment

Figure 34 illustrates the process of export via the "Export" button, configuring the file format (PDF/Word) and content inclusions (Bubble Sheet/Answer Keys), which triggers the system to process the export, display a success message, and finally prompt the Admin to name and save the file locally.

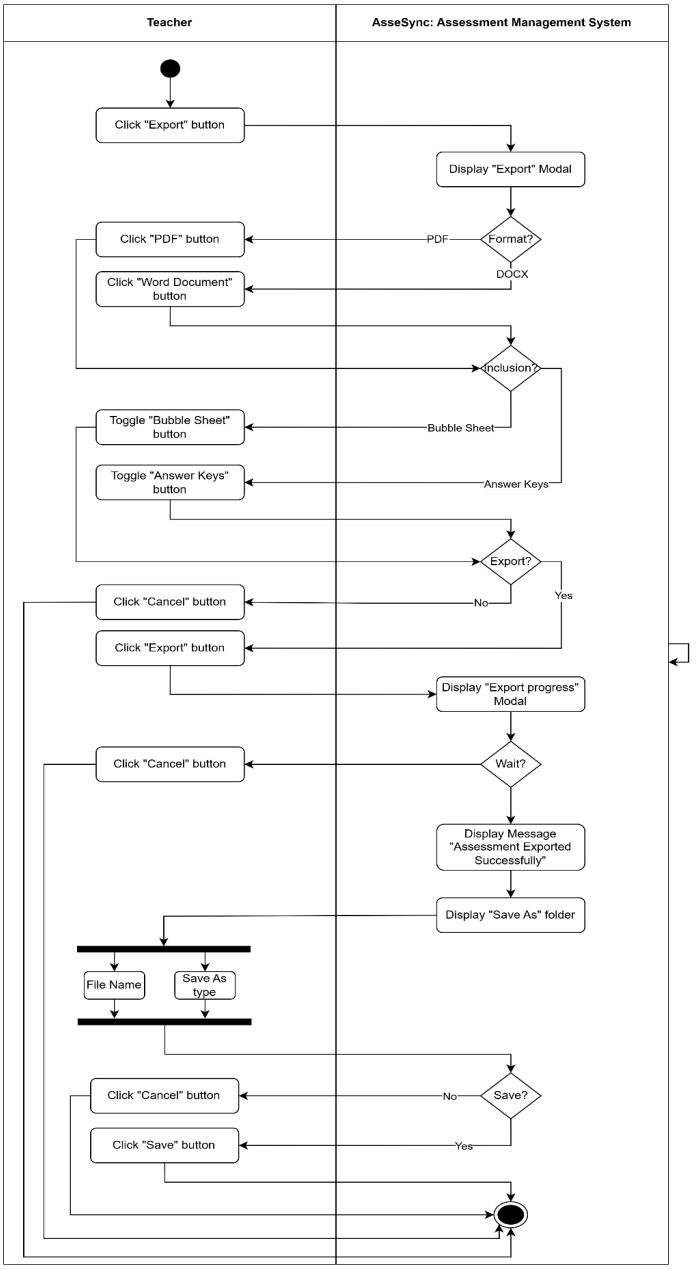


Figure 34. Activity Diagram of Teacher Export Assessment

Figure 35 illustrates the process flow for how a teacher evaluates student assessments in the AsseSync Management System.

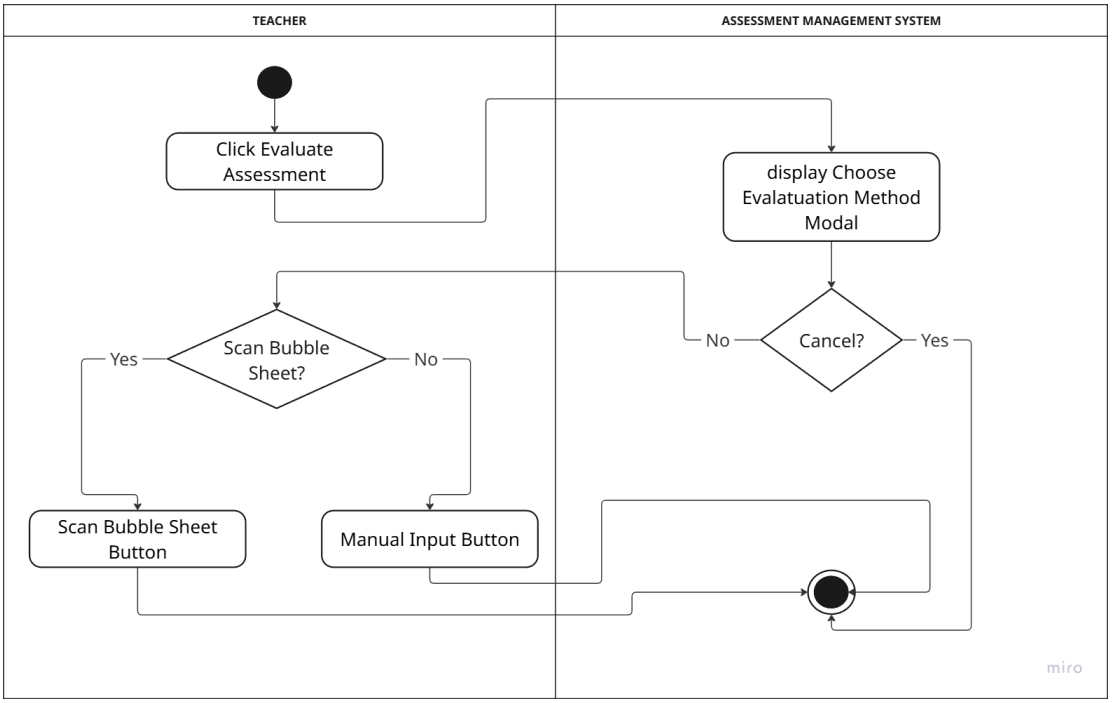


Figure 35. Activity Diagram of Teacher Evaluate Assessment

Figure 36 illustrates the process of the Teacher using Bubble sheet scanning as the main feature of the system for uploading and processing bubble sheets, likely for automated scoring.

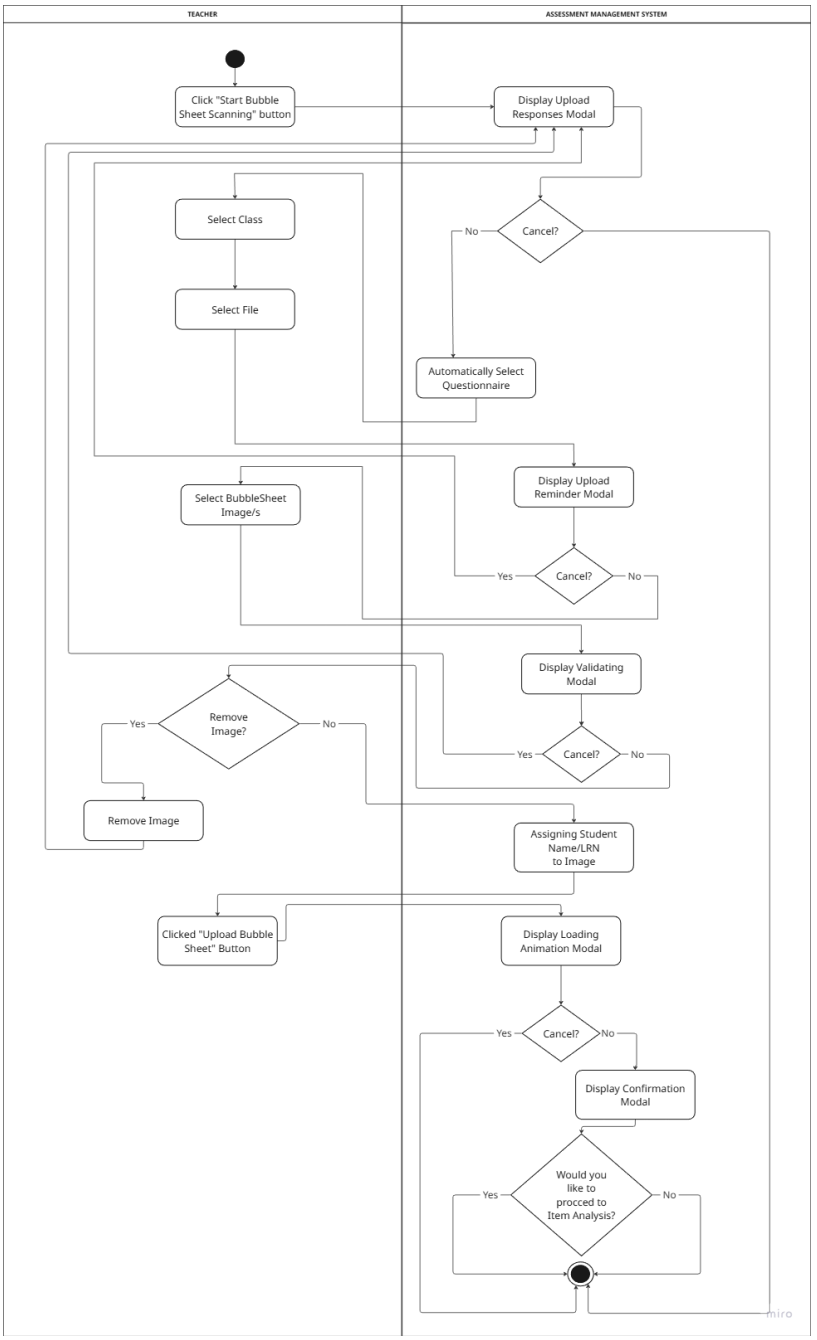


Figure 36. Activity Diagram of Teacher Bubble Sheet Scanning Evaluation

Figure 37 illustrates the manual input workflow in the Assessment Management System, where the user may load saved progress or start fresh, select a class, and input answers for each student while navigating between entries. After completing all students, the system shows all recorded responses for review, clearing, or editing. Finally, the user confirms submission through review and submit modals, completing the manual input process.

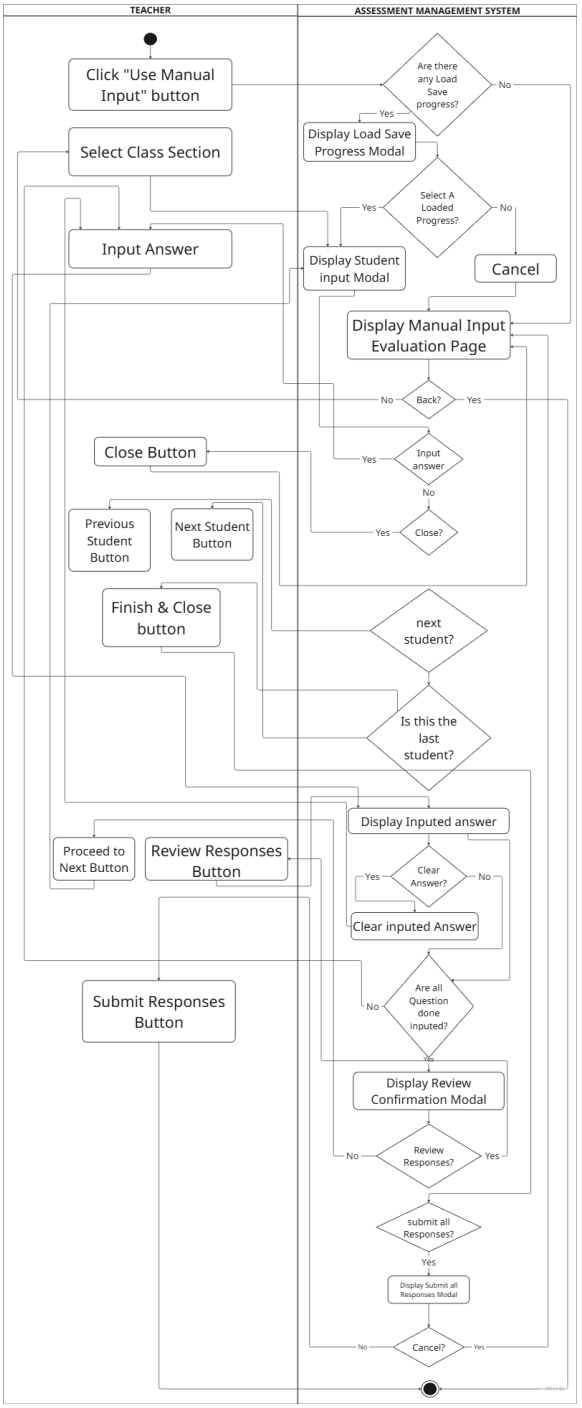


Figure 37. Activity Diagram of Teacher Manual Input Evaluation

Figure 38 demonstrates how the teacher interacts with the AsseSync Management System to view assessment Evaluation Result. The process begins when the teacher clicks the Item Analysis Result tab, prompting the system to display the Item Analysis Dashboard. The teacher then selects an assessment to view if it is marked as complete, the system displays the Item Analysis Result; otherwise, the teacher may click the Continue Analysis icon to proceed with the analysis.

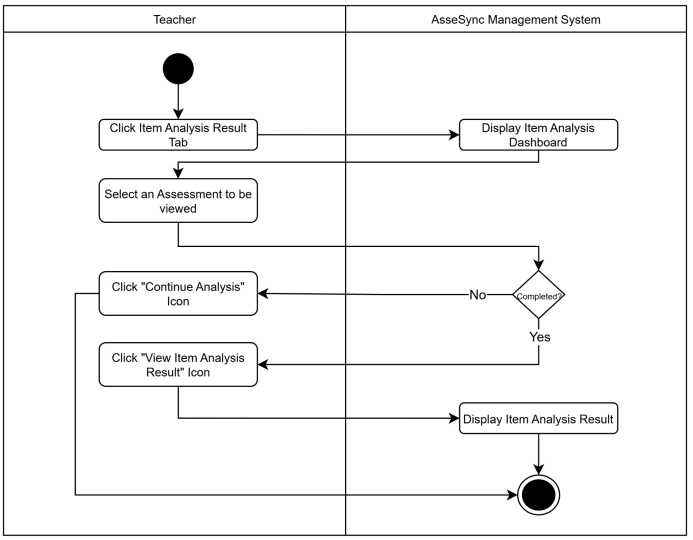


Figure 38. Activity Diagram of Teacher View Assessment Evaluation Result

Figure 39 illustrates the content revision workflow in the Assessment Management System, with the teacher initiating the process and the system checking for any saved progress before proceeding. If progress exists, a save-progress modal appears; otherwise, the teacher continues directly to viewing and revising items.

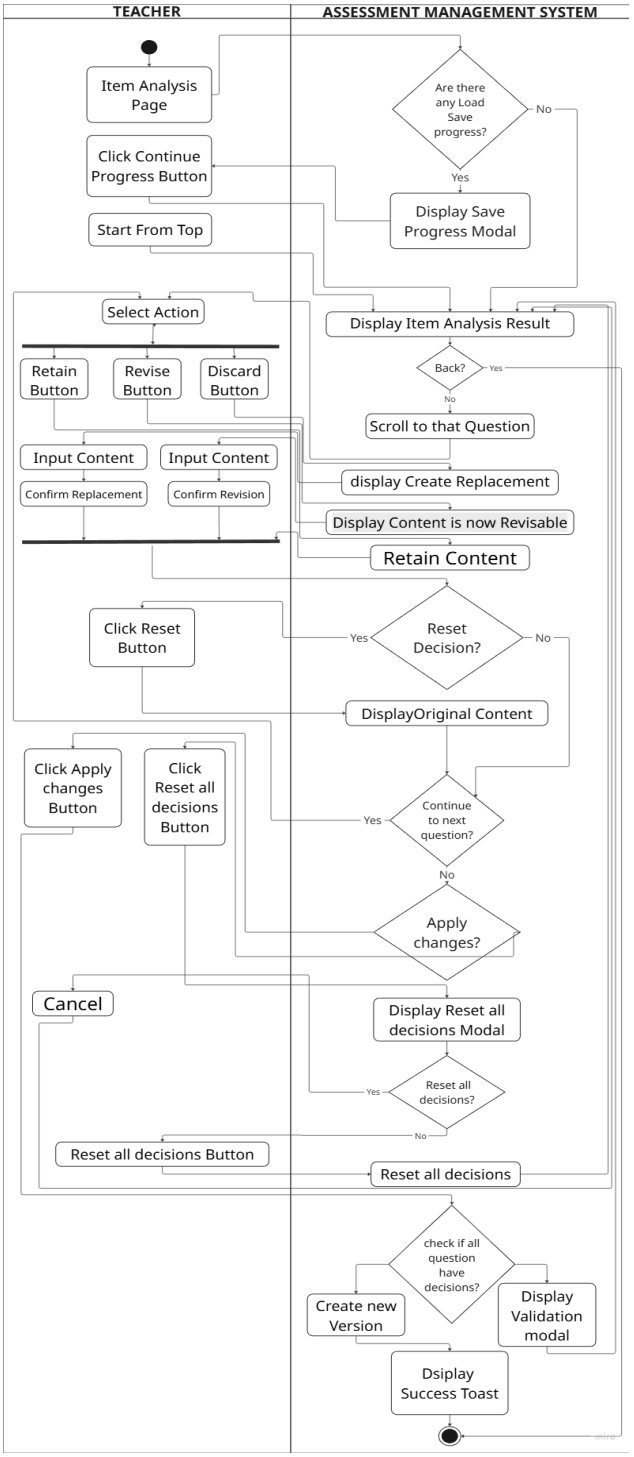


Figure 39. Activity Diagram of Teacher Create New Assessment Test Version

Figure 40 demonstrates how the Teacher initiates the process by clicking the "Student Score" Tab, which causes the Assessment Management System to immediately Display "Student Score list" Page, concluding the navigational workflow.

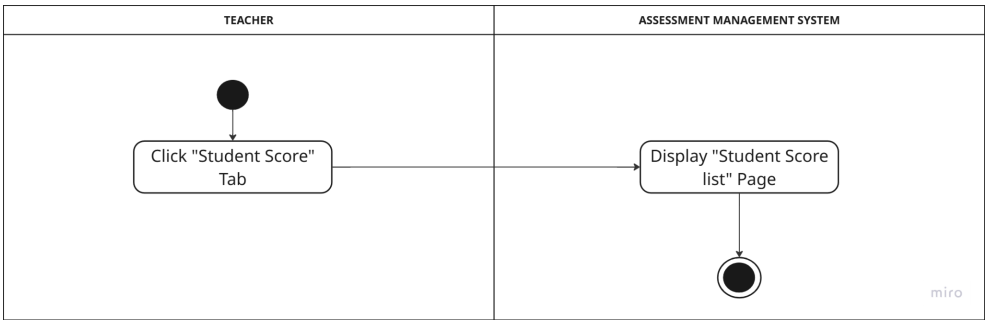


Figure 40. Activity Diagram of Teacher View Student Score List

Figure 41 illustrates View Scan Records. The flow begins with the teacher clicking the Records button and the system loads the records and displays them per student records.

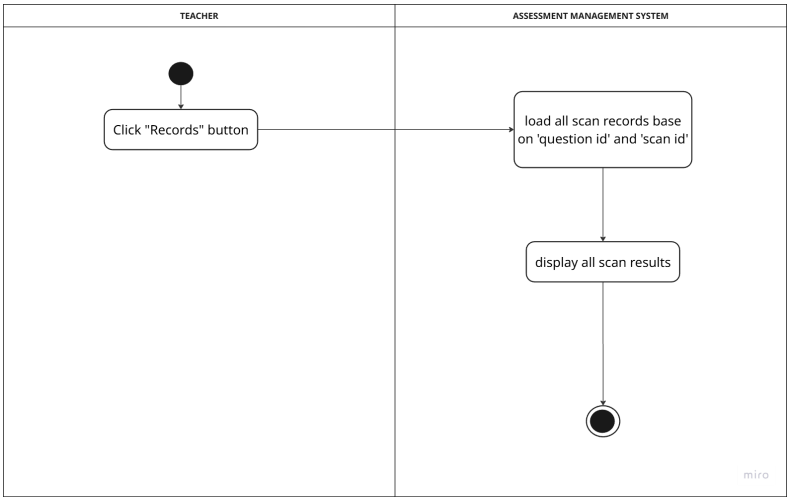


Figure 41. Activity Diagram of Teacher View Scan Records

Figure 42 illustrates the editing workflow in the Assessment Management System, beginning when the teacher clicks the “Edit” button, prompting the system to display an editable version of the recorded responses. After making changes, the teacher can either save or cancel the modifications. If the teacher chooses to save, the system applies the updated responses, updates the student records and item analysis results, and then displays a success message. If the teacher cancels, no changes are applied, and the process ends.

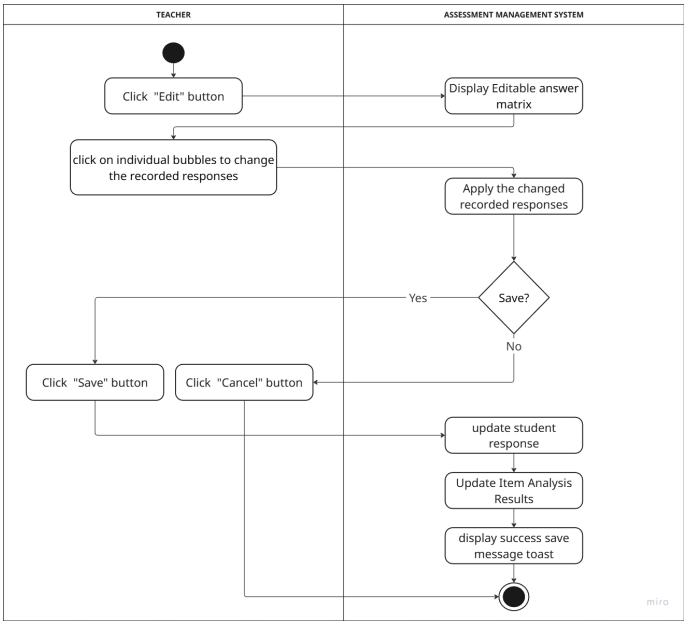


Figure 42. Activity Diagram of Teacher Edit Student Answer

Figure 43 illustrates the process of Exporting Evaluation Result in CSV by clicking the "more options/vertical ellipsis" to reveal and click the "Import in CSV" button, which prompts the system to display the "Download" progress, then the Save As Folder, allowing the Teacher to input the File Name and Save As type, and finally either Click "Cancel" button or Click "Save" button to conclude the operation.

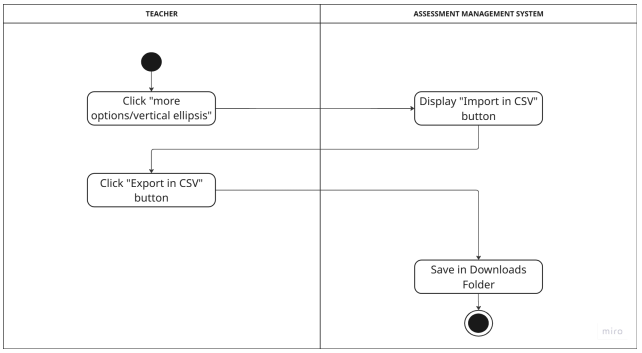


Figure 43. Activity Diagram of Teacher Export Evaluation Result in CSV

Figure 44 illustrates how the teacher initiates the combine item analysis process, makes selections or clears them, while the system displays the modal, handles decision points such as cancel, select all, and manual select, and ultimately updates the item analysis results and displays a success confirmation.

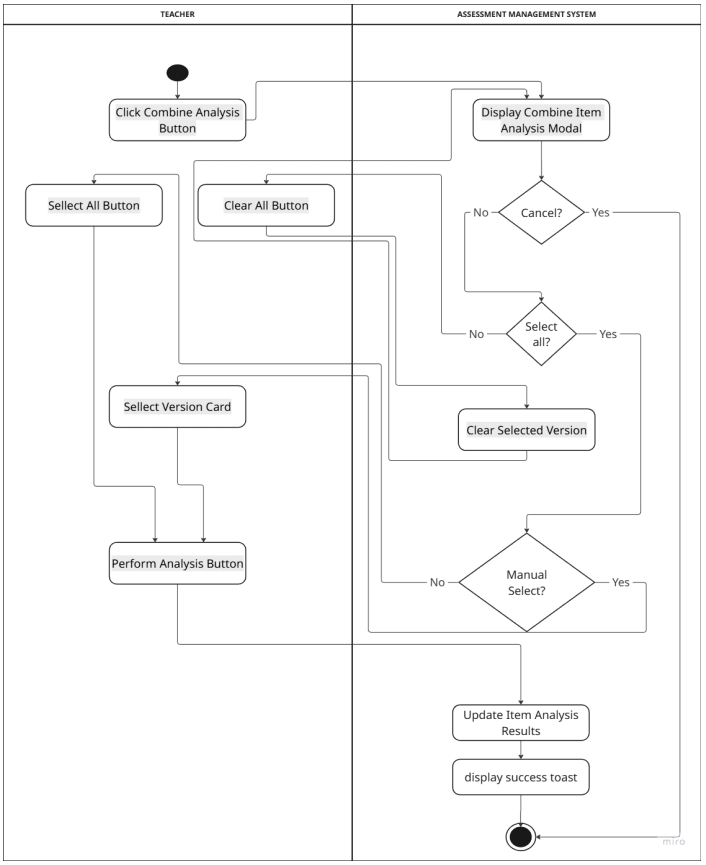


Figure 44. Activity Diagram of Teacher Combine Item analysis Result

Figure 45 illustrates the process of accessing the Item Analysis Dashboard in the AsseSync Management System. When the teacher clicks the “Item Analysis Result” tab, the system displays the Item Analysis Dashboard, providing performance insights and metrics.

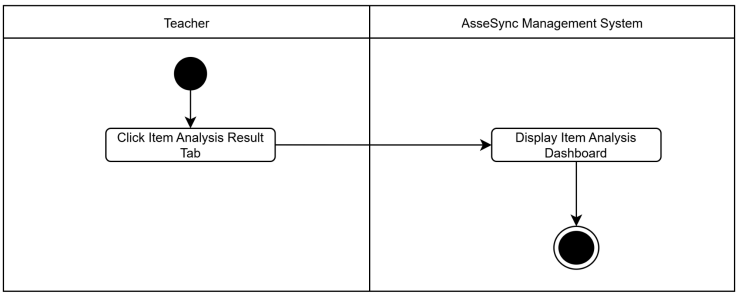


Figure 45. Activity Diagram of Teacher View Item Analysis Dashboard

Figure 46 shows the process for filtering assessments by status in the AsseSync Management System. The teacher clicks the “All Status” dropdown button, after which the system displays the list of status options. Upon selecting a status, the system displays the “Analysis Assessment” based on the selected status item.

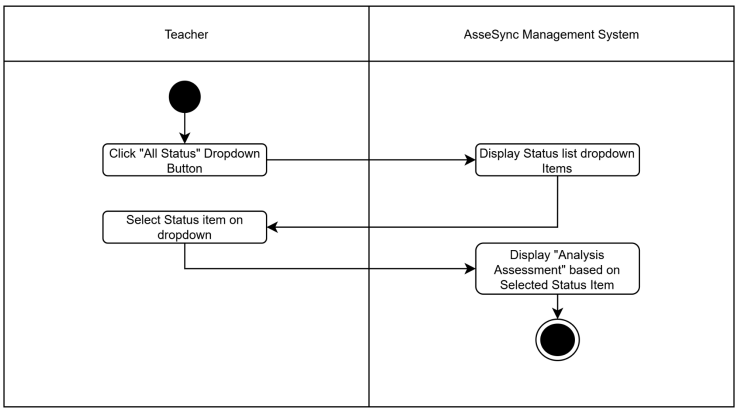


Figure 46. Activity Diagram of Teacher Item Analysis Filter by Status

Figure 47 illustrates the process for filtering analysis assessments by quarter in the AsseSync Management System. The teacher clicks the “Quarter” dropdown button, prompting the system to display the available quarter list. After the teacher selects a quarter, the system displays the “Analysis Assessment” corresponding to the selected quarter item.

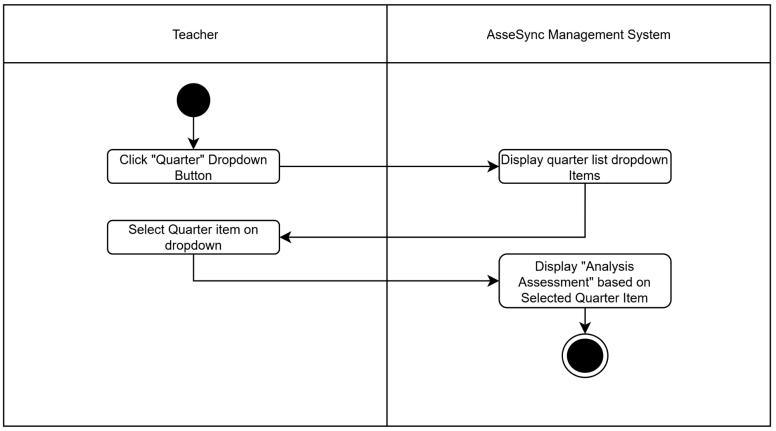


Figure 47. Activity Diagram of Teacher Item Analysis Quarter Filter

Figure 48 shows the process for sorting analysis assessments by date in the AsseSync Management System. The teacher clicks the “Date” dropdown button, prompting the system to display a list of available date options. Once the teacher selects a date item, the system displays the sorted “Analysis Assessment” results based on the chosen date.

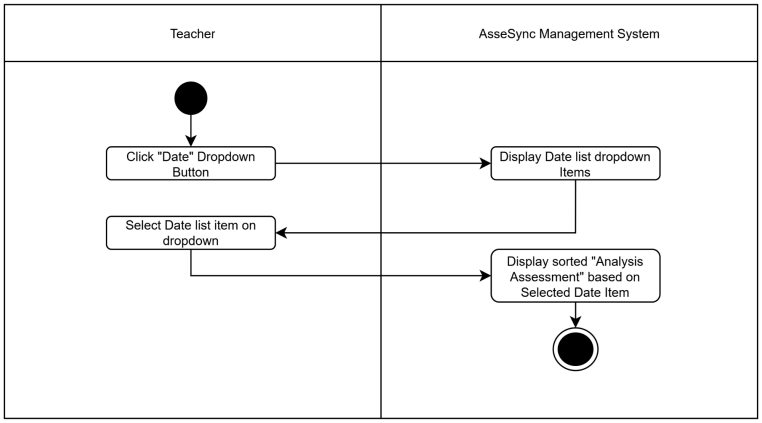


Figure 48. Activity Diagram of Teacher Item Analysis Sort by Date

Figure 49 illustrates the process of accessing the test bank in the AsseSync Management System. When the teacher clicks the “Test Bank” tab, the system displays the test bank page, allowing the teacher to view, manage, and organize assessment questions for future use.

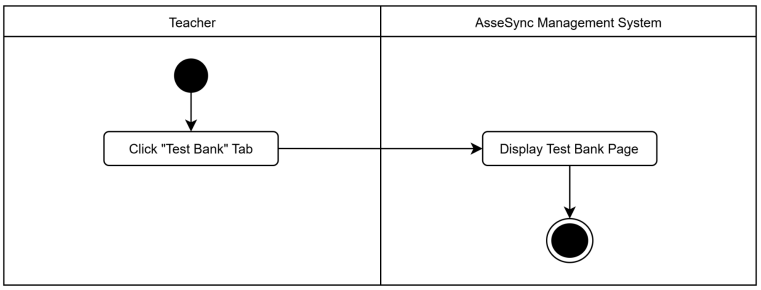


Figure 49. Activity Diagram of Teacher View Subject Test Bank

Figure 50 illustrates the process of searching for assessments in the AsseSync Management System. The teacher enters a keyword into the “Search Assessment” input field, prompting the system to perform a clustering operation to identify and match relevant assessments based on the entered keyword. The system then displays the search results, allowing the teacher to quickly locate and access the desired assessments.

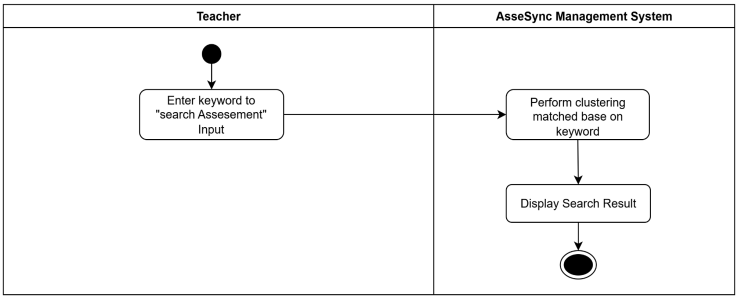


Figure 50. Activity Diagram of Teacher Search Subject Test Bank

Figure 51 illustrates the process of filtering the test bank by Quarter, the Teacher begins by Clicking the "Quarter" dropdown button, which displays the list of items, and after the Teacher Selects Quarter item on dropdown items, the Assessment Management System proceeds to Reset Test Bank List based on the selected quarter, completing the filtering process.

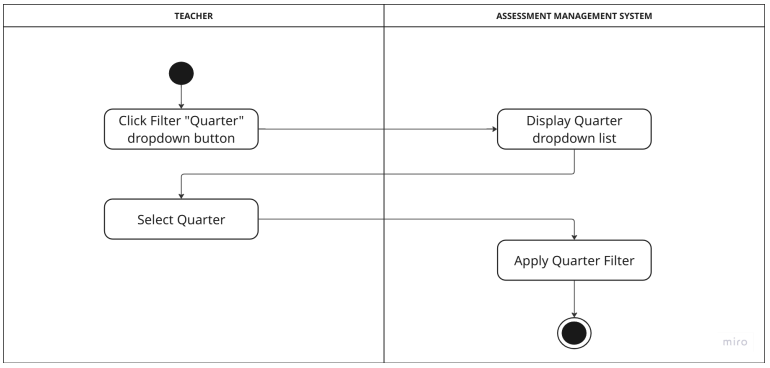


Figure 51. Activity Diagram of Teacher Filter Test Bank by Quarter

Figure 52 illustrates the process of filtering test bank items by difficulty level in the AsseSync Management System. The teacher clicks the “Difficulty” dropdown button, prompting the system to display available difficulty level options. Once the teacher selects a level, the system displays the “Test Bank List” based on the selected difficulty level.

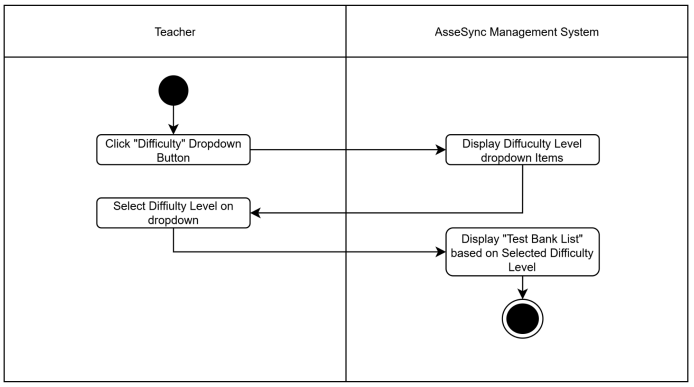


Figure 52. Activity Diagram of Teacher Filter Test Bank by Difficulty

Figure 53 illustrates the process of clearing all selected filters in the AsseSync Management System. When the teacher clicks the “Clear” button, the system removes all active filters and resets the Test Bank List to its default state.

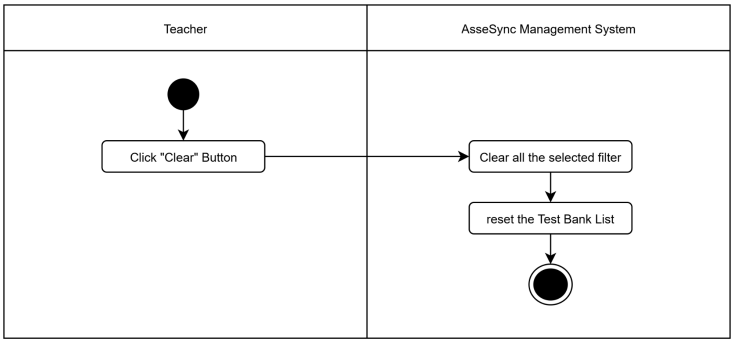


Figure 53. Activity Diagram of Teacher Clear Test Bank Filter

Figure 54 illustrates how the teacher navigates between subject and class list pages within the AsseSync Management System. The teacher first selects a subject grade, prompting the system to display the corresponding subject page. When the teacher clicks the “Class List” tab, the system transitions to show the class list page, completing the interaction.

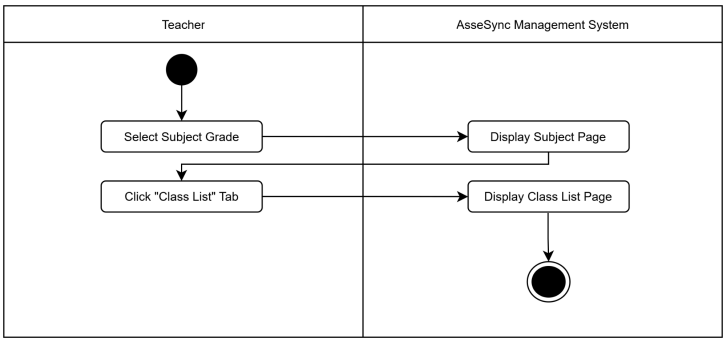


Figure 54. Activity Diagram of Teacher View Student Class List

5.1.5. Functional Decomposition Diagram

This section presents the Functional Decomposition Diagram of the project. The developers detail the interaction between different components that leads to the overall functionality.

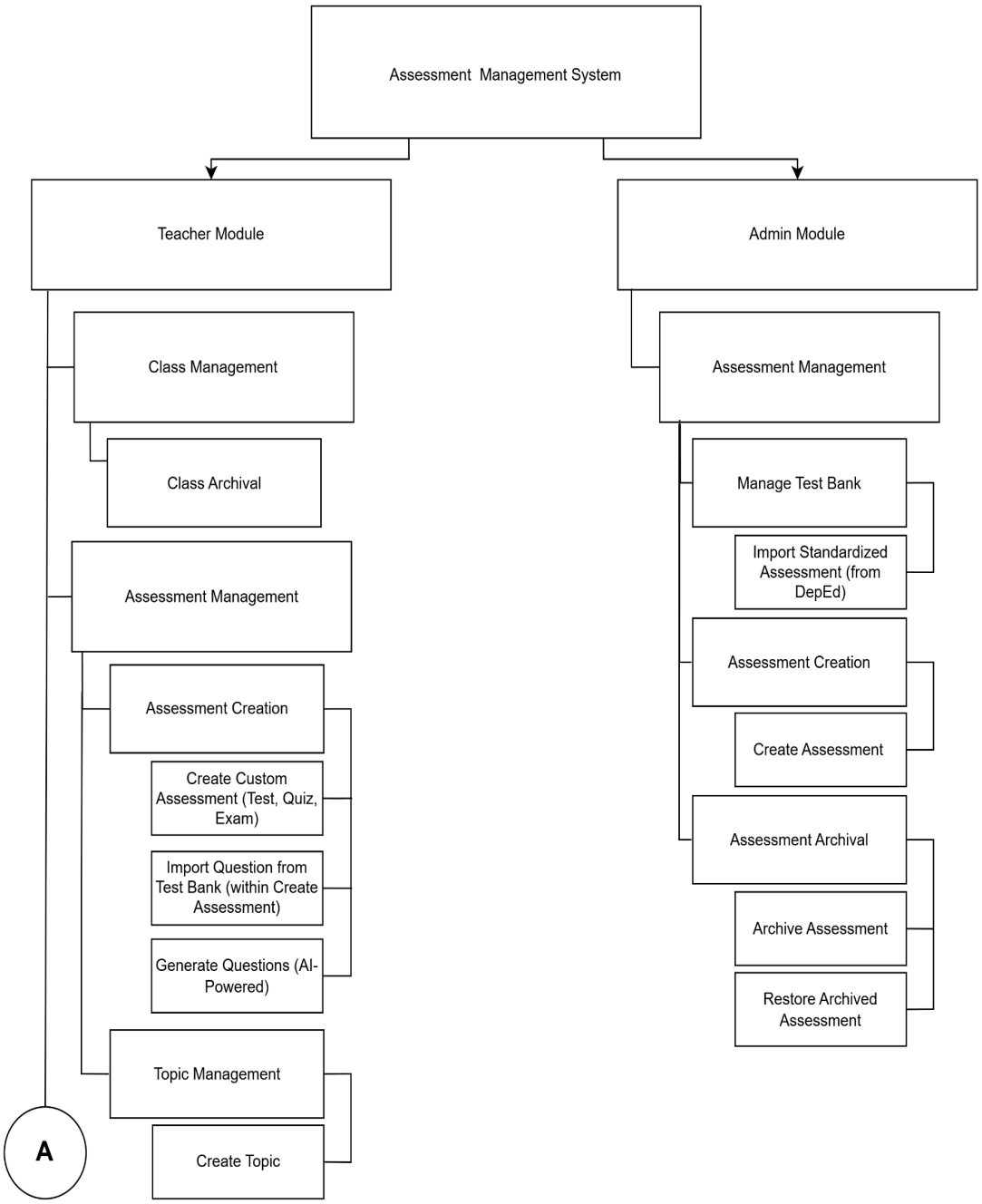


Figure 55. Functional Decomposition Diagram of Assessment Management System

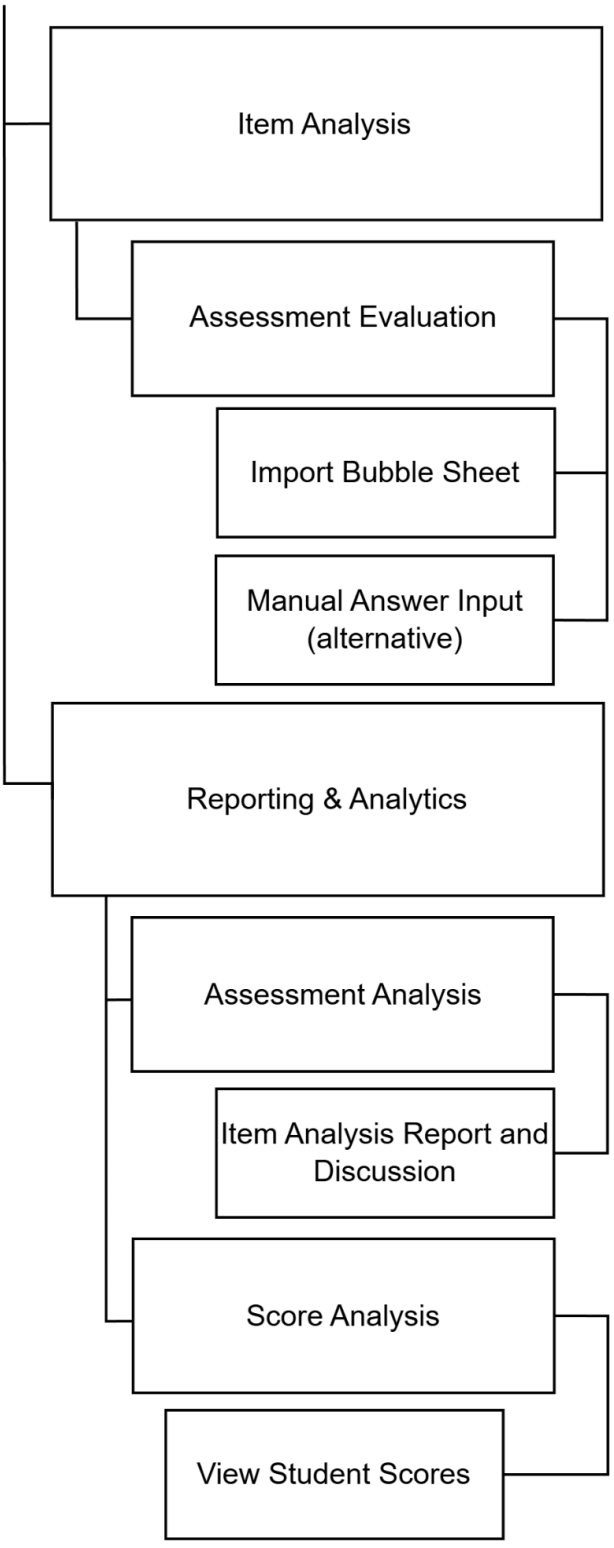


Figure 55 cont. Functional Decomposition Diagram of Assessment Management System

5.1.6. Context Diagram

This section outlines the functional requirements for the system, including how it should respond to inputs and behave in various scenarios. Figure 12 provides a summary of the system in the form of a context diagram. A context diagram is a high level view of a system that serves as a visual representation of the entities that interact with a system, a component of a system, and its environment. It displays the specifications and scope of the created system, as well as the project's implications.

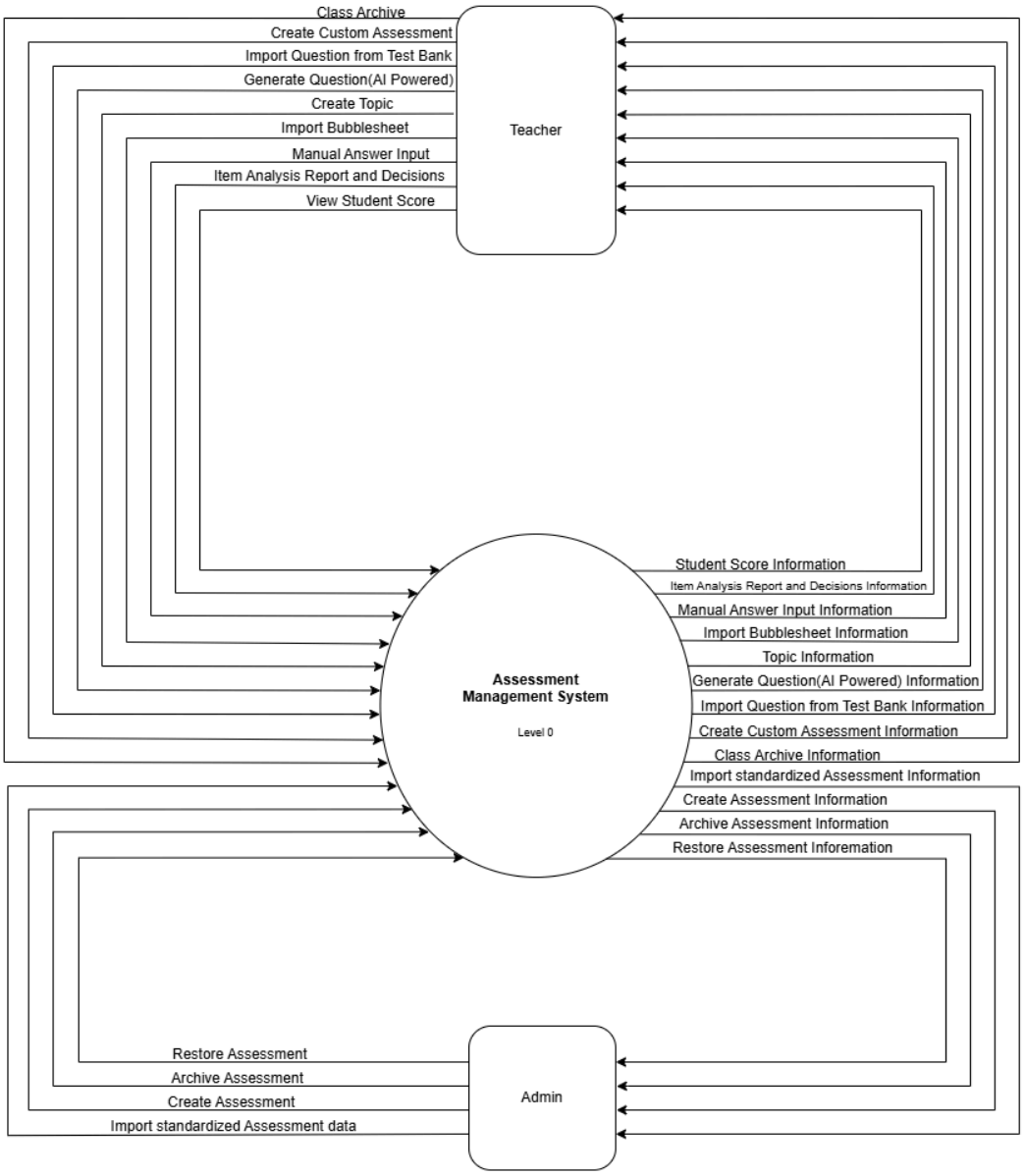
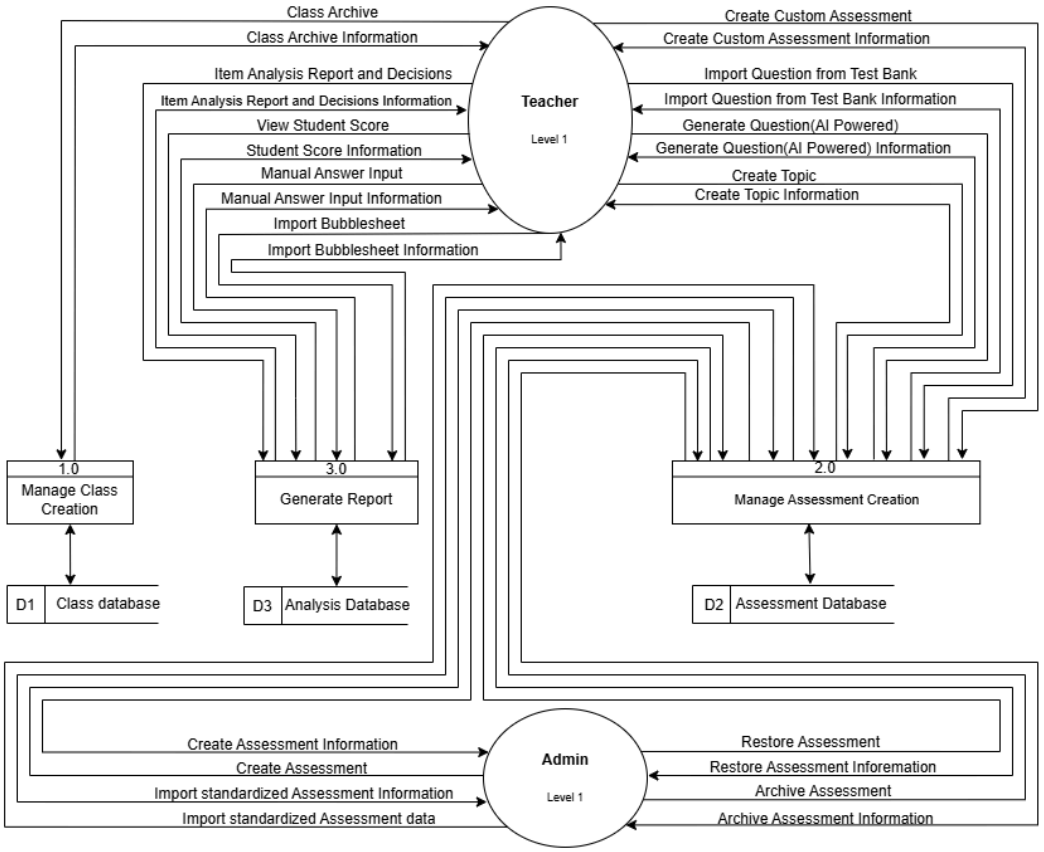


Figure 56. Context Diagram of Assessment Management System

5.1.7. Data Flow Diagram



This section depicts the data flow of a diagram of the project. It gives details about the flow of data in the Assessment Management System as shown in Figure 54.

Figure 57. Data Flow Diagram of Assessment Management System

5.1.8. Database Design using Entity Relationship Model

This section depicts the database design of the system using Entity Relationship Model. The developers describe the connections between entity sets that are kept in a database in Figure56. This diagram assisted them in describing the logical structure of the Assessment Management System database.

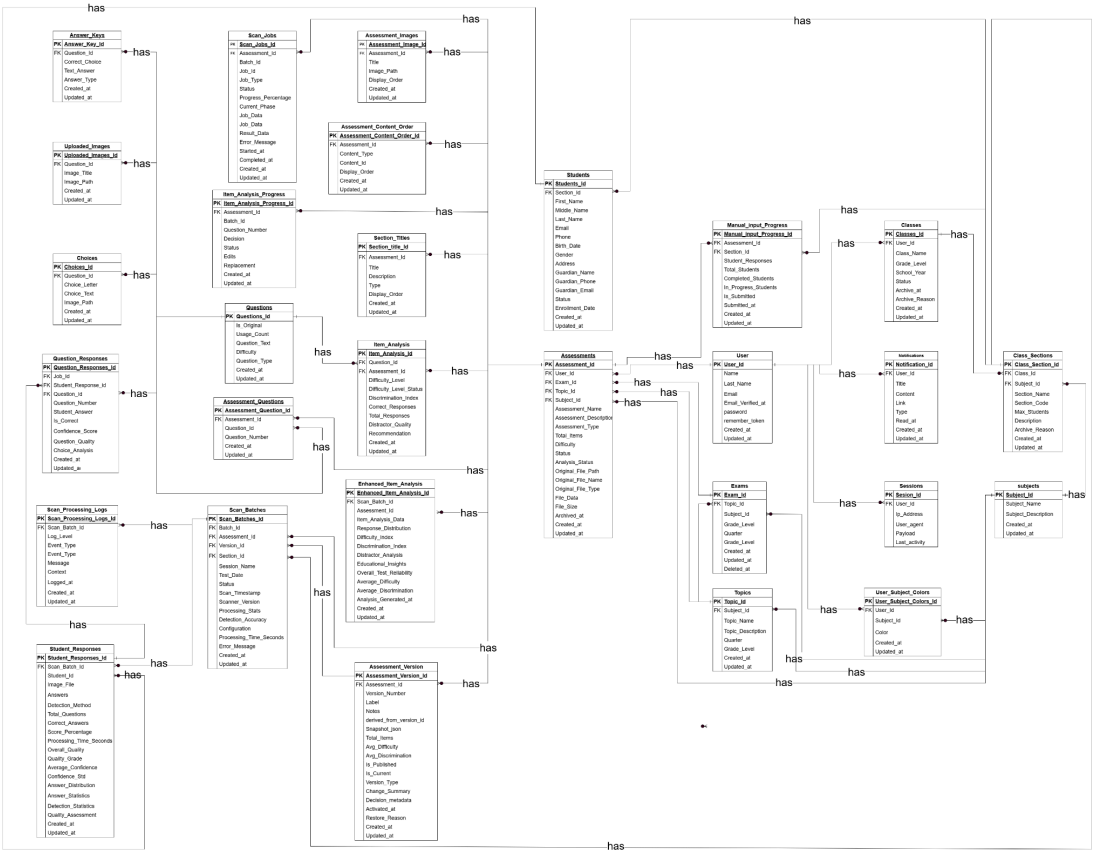


Figure 58. Entity Relationship Diagram of Assessment Management System

5.1.9. Data Dictionary of the Designed Database

Table 4 displays the data dictionary of the database of the system to explain the definition and purpose of data elements within the system while giving each named data object a relevant description

Table 4. Data Dictionary of Assessment Management System

schema_name	table_name	obj_ty	order	is_key	column_name	data_type	nullable	column_descr
public	answer_keys	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	answer_keys	TBL	2	FK,UK	question_id	bigint(64)	NOT NULL	NULL
public	answer_keys	TBL	3		correct_choice	character varying(1)	NULL	NULL
public	answer_keys	TBL	4		text_answer	text	NULL	NULL
public	answer_keys	TBL	5		answer_type	character varying(50)	NULL	NULL
public	answer_keys	TBL	6		created_at	timestamp without time zone(6)	NULL	NULL
public	answer_keys	TBL	7		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessment_content_order	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	assessment_content_order	TBL	2	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	assessment_content_order	TBL	3		content_type	character varying(50)	NULL	NULL
public	assessment_content_order	TBL	4		content_id	bigint(64)	NOT NULL	NULL
public	assessment_content_order	TBL	5		display_order	integer(32)	NOT NULL	NULL

public	assessment_content_order	TBL	6		created_at	timestamp without time zone(6)	NULL	NULL
public	assessment_content_order	TBL	7		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessment_images	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	assessment_images	TBL	2	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	assessment_images	TBL	3		title	character varying(255)	NULL	NULL
public	assessment_images	TBL	4		image_path	character varying(500)	NOT NULL	NULL
public	assessment_images	TBL	5		display_order	integer(32)	NULL	NULL
public	assessment_images	TBL	6		created_at	timestamp without time zone(6)	NULL	NULL
public	assessment_images	TBL	7		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessment_questions	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	assessment_questions	TBL	2	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	assessment_questions	TBL	3	FK	question_id	bigint(64)	NOT NULL	NULL
public	assessment_questions	TBL	4		question_number	integer(32)	NOT NULL	NULL
public	assessment_questions	TBL	5		created_at	timestamp without time zone(6)	NULL	NULL

public	assessment_questions	TBL	6		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessment_versions	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	assessment_versions	TBL	2	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	assessment_versions	TBL	3	UK	version_number	integer(32)	NOT NULL	NULL
public	assessment_versions	TBL	4		label	character varying(255)	NULL	NULL
public	assessment_versions	TBL	5		notes	text	NULL	NULL
public	assessment_versions	TBL	6		derived_from_version_id	bigint(64)	NULL	NULL
public	assessment_versions	TBL	7		snapshot_json	jsonb	NULL	NULL
public	assessment_versions	TBL	8		total_items	integer(32)	NULL	NULL
public	assessment_versions	TBL	9		avg_difficulty	numeric(5,2)	NULL	NULL
public	assessment_versions	TBL	10		avg_discrimination	numeric(5,2)	NULL	NULL
public	assessment_versions	TBL	11		is_published	boolean	NULL	NULL
public	assessment_versions	TBL	12		is_current	boolean	NULL	NULL
public	assessment_versions	TBL	13		version_type	character varying(50)	NULL	NULL
public	assessment_versions	TBL	14		change_summary	text	NULL	NULL

public	assessment_versions	TBL	15		decision_metadata	jsonb	NULL	NULL
public	assessment_versions	TBL	16		activated_at	timestamp without time zone(6)	NULL	NULL
public	assessment_versions	TBL	17		restore_reason	text	NULL	NULL
public	assessment_versions	TBL	18		created_at	timestamp without time zone(6)	NULL	NULL
public	assessment_versions	TBL	19		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessments	TBL	1	PK	assessment_id	bigint(64)	NOT NULL	Formerly questionnaire_id - primary key
public	assessments	TBL	2	FK	user_id	bigint(64)	NOT NULL	NULL
public	assessments	TBL	3	FK	subject_id	bigint(64)	NOT NULL	NULL
public	assessments	TBL	4		assessment_name	character varying(255)	NOT NULL	NULL
public	assessments	TBL	5		assessment_description	text	NULL	NULL
public	assessments	TBL	6		assessment_type	character varying(100)	NULL	NULL
public	assessments	TBL	7	FK	topic_id	bigint(64)	NULL	NULL
public	assessments	TBL	8	FK	exam_id	bigint(64)	NULL	NULL
public	assessments	TBL	9		total_items	integer(32)	NULL	NULL
public	assessments	TBL	10		difficulty	character varying(50)	NULL	NULL
public	assessments	TBL	11		status	character varying(50)	NULL	NULL

public	assessments	TBL	12		analysis_status	character varying(50)	NULL	NULL
public	assessments	TBL	13		original_file_path	character varying(500)	NULL	NULL
public	assessments	TBL	14		original_file_name	character varying(255)	NULL	NULL
public	assessments	TBL	15		original_file_type	character varying(50)	NULL	NULL
public	assessments	TBL	16		file_data	bytea	NULL	NULL
public	assessments	TBL	17		file_size	bigint(64)	NULL	NULL
public	assessments	TBL	18		archived_at	timestamp without time zone(6)	NULL	NULL
public	assessments	TBL	19		created_at	timestamp without time zone(6)	NULL	NULL
public	assessments	TBL	20		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessments	TBL	21		deleted_at	timestamp without time zone(6)	NULL	NULL
public	choices	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	choices	TBL	2	FK	question_id	bigint(64)	NOT NULL	NULL
public	choices	TBL	3		choice_letter	character varying(1)	NULL	NULL
public	choices	TBL	4		choice_text	text	NOT NULL	NULL
public	choices	TBL	5		image_path	character varying(500)	NULL	NULL
public	choices	TBL	6		created_at	timestamp without time zone(6)	NULL	NULL

public	choices	TBL	7		updated_at	timestamp without time zone(6)	NULL	NULL
public	class_sections	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	class_sections	TBL	2	FK,UK	class_id	bigint(64)	NOT NULL	NULL
public	class_sections	TBL	3	FK,UK	subject_id	bigint(64)	NOT NULL	NULL
public	class_sections	TBL	4	UK	section_name	character varying(255)	NOT NULL	NULL
public	class_sections	TBL	5		section_code	character varying(50)	NULL	NULL
public	class_sections	TBL	6		max_students	integer(32)	NULL	NULL
public	class_sections	TBL	7		description	text	NULL	NULL
public	class_sections	TBL	8		status	character varying(50)	NULL	NULL
public	class_sections	TBL	9		created_at	timestamp without time zone(6)	NULL	NULL
public	class_sections	TBL	10		updated_at	timestamp without time zone(6)	NULL	NULL
public	classes	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	classes	TBL	2	FK	user_id	bigint(64)	NOT NULL	NULL
public	classes	TBL	3		class_name	character varying(255)	NOT NULL	NULL
public	classes	TBL	4		grade_level	character varying(50)	NULL	NULL
public	classes	TBL	5		school_year	character varying(50)	NULL	NULL
public	classes	TBL	6		status	character varying(50)	NULL	NULL

public	classes	TBL	7		archived_at	timestamp without time zone(6)	NULL	NULL
public	classes	TBL	8		archived_reason	text	NULL	NULL
public	classes	TBL	9		created_at	timestamp without time zone(6)	NULL	NULL
public	classes	TBL	10		updated_at	timestamp without time zone(6)	NULL	NULL
public	enhanced_item_analysis	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	enhanced_item_analysis	TBL	2	FK	scan_batch_id	bigint(64)	NOT NULL	NULL
public	enhanced_item_analysis	TBL	3	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	enhanced_item_analysis	TBL	4		item_analysis_data	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	5		response_distribution	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	6		difficulty_index	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	7		discrimination_index	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	8		distractor_analysis	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	9		educational_insights	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	10		overall_test_reliability	numeric(5,2)	NULL	NULL

public	enhanced_item_analysis	TBL	11		average_difficulty	numeric(5,2)	NULL	NULL
public	enhanced_item_analysis	TBL	12		average_discrimination	numeric(5,2)	NULL	NULL
public	enhanced_item_analysis	TBL	13		analysis_generated_at	timestamp without time zone(6)	NULL	NULL
public	enhanced_item_analysis	TBL	14		created_at	timestamp without time zone(6)	NULL	NULL
public	enhanced_item_analysis	TBL	15		updated_at	timestamp without time zone(6)	NULL	NULL
public	exams	TBL	1	PK	exam_id	bigint(64)	NOT NULL	NULL
public	exams	TBL	2	FK,UK	topic_id	bigint(64)	NULL	NULL
public	exams	TBL	3	FK,UK	subject_id	bigint(64)	NOT NULL	NULL
public	exams	TBL	4	UK	grade_level	character varying(50)	NULL	NULL
public	exams	TBL	5	UK	quarter	character varying(50)	NULL	NULL
public	exams	TBL	6	UK	exam_type	character varying(50)	NULL	NULL
public	exams	TBL	7		created_at	timestamp without time zone(6)	NULL	NULL
public	exams	TBL	8		updated_at	timestamp without time zone(6)	NULL	NULL
public	exams	TBL	9		deleted_at	timestamp without time zone(6)	NULL	NULL
public	item_analysis	TBL	1	PK	id	bigint(64)	NOT NULL	NULL

public	item_analysis	TBL	2	FK	question_id	bigint(64)	NOT NULL	NULL
public	item_analysis	TBL	3	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	item_analysis	TBL	4		difficulty_level	numeric(5,2)	NULL	NULL
public	item_analysis	TBL	5		difficulty_level_status	character varying(50)	NULL	NULL
public	item_analysis	TBL	6		discrimination_index	numeric(5,2)	NULL	NULL
public	item_analysis	TBL	7		correct_responses	integer(32)	NULL	NULL
public	item_analysis	TBL	8		total_responses	integer(32)	NULL	NULL
public	item_analysis	TBL	9		distractor_quality	numeric(5,2)	NULL	NULL
public	item_analysis	TBL	10		recommendation	character varying(50)	NULL	NULL
public	item_analysis	TBL	11		created_at	timestamp without time zone(6)	NULL	NULL
public	item_analysis	TBL	12		updated_at	timestamp without time zone(6)	NULL	NULL
public	item_analysis_progress	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	item_analysis_progress	TBL	2	FK,UK	assessment_id	bigint(64)	NOT NULL	NULL
public	item_analysis_progress	TBL	3	UK	batch_id	character varying(255)	NOT NULL	NULL
public	item_analysis_progress	TBL	4	UK	question_number	integer(32)	NOT NULL	NULL
public	item_analysis_progress	TBL	5		decision	character varying(50)	NULL	NULL

public	item_analysis_progress	TBL	6		status	character varying(50)	NULL	NULL
public	item_analysis_progress	TBL	7		edits	jsonb	NULL	NULL
public	item_analysis_progress	TBL	8		replacement	jsonb	NULL	NULL
public	item_analysis_progress	TBL	9		created_at	timestamp without time zone(6)	NULL	NULL
public	item_analysis_progress	TBL	10		updated_at	timestamp without time zone(6)	NULL	NULL
public	manual_input_progress	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	manual_input_progress	TBL	2	FK,UK	assessment_id	bigint(64)	NOT NULL	NULL
public	manual_input_progress	TBL	3	FK,UK	section_id	bigint(64)	NOT NULL	NULL
public	manual_input_progress	TBL	4		student_responses	jsonb	NULL	NULL
public	manual_input_progress	TBL	5		total_students	integer(32)	NULL	NULL
public	manual_input_progress	TBL	6		completed_students	integer(32)	NULL	NULL
public	manual_input_progress	TBL	7		in_progress_students	integer(32)	NULL	NULL
public	manual_input_progress	TBL	8	UK	is_submitted	boolean	NULL	NULL
public	manual_input_progress	TBL	9		submitted_at	timestamp without time zone(6)	NULL	NULL

public	manual_input_progress	TBL	10		created_at	timestamp without time zone(6)	NULL	NULL
public	manual_input_progress	TBL	11		updated_at	timestamp without time zone(6)	NULL	NULL
public	notifications	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	notifications	TBL	2	FK	user_id	bigint(64)	NOT NULL	NULL
public	notifications	TBL	3		title	character varying(255)	NOT NULL	NULL
public	notifications	TBL	4		content	text	NOT NULL	NULL
public	notifications	TBL	5		link	character varying(255)	NULL	NULL
public	notifications	TBL	6		type	character varying(50)	NULL	NULL
public	notifications	TBL	7		read_at	timestamp without time zone(6)	NULL	NULL
public	notifications	TBL	8		created_at	timestamp without time zone(6)	NULL	NULL
public	notifications	TBL	9		updated_at	timestamp without time zone(6)	NULL	NULL
public	question_responses	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	question_responses	TBL	2	FK	student_response_id	bigint(64)	NOT NULL	NULL
public	question_responses	TBL	3	FK	question_id	bigint(64)	NOT NULL	NULL
public	question_responses	TBL	4		question_number	integer(32)	NOT NULL	NULL

public	question_responses	TBL	5		student_answer	character varying(10)	NULL	NULL
public	question_responses	TBL	6		is_correct	boolean	NULL	NULL
public	question_responses	TBL	7		confidence_score	numeric(5,2)	NULL	NULL
public	question_responses	TBL	8		question_quality	character varying(50)	NULL	NULL
public	question_responses	TBL	9		choice_analysis	jsonb	NULL	NULL
public	question_responses	TBL	10		created_at	timestamp without time zone(6)	NULL	NULL
public	question_responses	TBL	11		updated_at	timestamp without time zone(6)	NULL	NULL
public	questions	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	questions	TBL	2		is_original	boolean	NULL	NULL
public	questions	TBL	3		usage_count	integer(32)	NULL	NULL
public	questions	TBL	4		question_text	text	NOT NULL	NULL
public	questions	TBL	5		difficulty	character varying(50)	NULL	NULL
public	questions	TBL	6		question_type	character varying(50)	NULL	NULL
public	questions	TBL	7		created_at	timestamp without time zone(6)	NULL	NULL
public	questions	TBL	8		updated_at	timestamp without time zone(6)	NULL	NULL
public	scan_batches	TBL	1	PK	id	bigint(64)	NOT NULL	NULL

public	scan_batches	TBL	2	UK	batch_id	character varying(255)	NOT NULL	NULL
public	scan_batches	TBL	3	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	scan_batches	TBL	4	FK	version_id	bigint(64)	NULL	NULL
public	scan_batches	TBL	5	FK	section_id	bigint(64)	NULL	NULL
public	scan_batches	TBL	6		session_name	character varying(255)	NULL	NULL
public	scan_batches	TBL	7		test_date	date(3)	NULL	NULL
public	scan_batches	TBL	8		status	character varying(50)	NULL	NULL
public	scan_batches	TBL	9		scan_timestamp	timestamp without time zone(6)	NULL	NULL
public	scan_batches	TBL	10		scanner_version	character varying(50)	NULL	NULL
public	scan_batches	TBL	11		processing_stats	jsonb	NULL	NULL
public	scan_batches	TBL	12		detection_accuracy	jsonb	NULL	NULL
public	scan_batches	TBL	13		configuration	jsonb	NULL	NULL
public	scan_batches	TBL	14		processing_time_seconds	integer(32)	NULL	NULL
public	scan_batches	TBL	15		error_message	text	NULL	NULL
public	scan_batches	TBL	16		created_at	timestamp without time zone(6)	NULL	NULL
public	scan_batches	TBL	17		updated_at	timestamp without time zone(6)	NULL	NULL
public	scan_jobs	TBL	1	PK	id	bigint(64)	NOT NULL	NULL

public	scan_jobs	TBL	2	UK	job_id	uuid	NOT NULL	NULL
public	scan_jobs	TBL	3		batch_id	character varying(255)	NULL	NULL
public	scan_jobs	TBL	4	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	scan_jobs	TBL	5		job_type	character varying(50)	NULL	NULL
public	scan_jobs	TBL	6		status	character varying(50)	NULL	NULL
public	scan_jobs	TBL	7		progress_percentage	integer(32)	NULL	NULL
public	scan_jobs	TBL	8		current_phase	character varying(255)	NULL	NULL
public	scan_jobs	TBL	9		job_data	jsonb	NULL	NULL
public	scan_jobs	TBL	10		result_data	jsonb	NULL	NULL
public	scan_jobs	TBL	11		error_message	text	NULL	NULL
public	scan_jobs	TBL	12		started_at	timestamp without time zone(6)	NULL	NULL
public	scan_jobs	TBL	13		completed_at	timestamp without time zone(6)	NULL	NULL
public	scan_jobs	TBL	14		created_at	timestamp without time zone(6)	NULL	NULL
public	scan_jobs	TBL	15		updated_at	timestamp without time zone(6)	NULL	NULL
public	scan_processing_logs	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	scan_processing_logs	TBL	2	FK	scan_batch_id	bigint(64)	NOT NULL	NULL

public	scan_processing_logs	TBL	3		log_level	character varying(50)	NULL	NULL
public	scan_processing_logs	TBL	4		event_type	character varying(100)	NULL	NULL
public	scan_processing_logs	TBL	5		message	text	NULL	NULL
public	scan_processing_logs	TBL	6		context	jsonb	NULL	NULL
public	scan_processing_logs	TBL	7		logged_at	timestamp without time zone(6)	NULL	NULL
public	scan_processing_logs	TBL	8		created_at	timestamp without time zone(6)	NULL	NULL
public	scan_processing_logs	TBL	9		updated_at	timestamp without time zone(6)	NULL	NULL
public	section_titles	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	section_titles	TBL	2	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	section_titles	TBL	3		title	character varying(255)	NOT NULL	NULL
public	section_titles	TBL	4		description	text	NULL	NULL
public	section_titles	TBL	5		type	character varying(50)	NULL	NULL
public	section_titles	TBL	6		display_order	integer(32)	NULL	NULL
public	section_titles	TBL	7		created_at	timestamp without time zone(6)	NULL	NULL
public	section_titles	TBL	8		updated_at	timestamp without time zone(6)	NULL	NULL

public	sessions	TBL	1	PK	id	character varying(255)	NOT NULL	NULL
public	sessions	TBL	2	FK	user_id	bigint(64)	NULL	NULL
public	sessions	TBL	3		ip_address	character varying(45)	NULL	NULL
public	sessions	TBL	4		user_agent	text	NULL	NULL
public	sessions	TBL	5		payload	text	NOT NULL	NULL
public	sessions	TBL	6		last_activity	integer(32)	NOT NULL	NULL
public	student_responses	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	student_responses	TBL	2	FK	scan_batch_id	bigint(64)	NOT NULL	NULL
public	student_responses	TBL	3	FK	student_id	bigint(64)	NOT NULL	NULL
public	student_responses	TBL	4		image_file	character varying(500)	NULL	NULL
public	student_responses	TBL	5		answers	jsonb	NULL	NULL
public	student_responses	TBL	6		detection_method	character varying(50)	NULL	NULL
public	student_responses	TBL	7		total_questions	integer(32)	NULL	NULL
public	student_responses	TBL	8		correct_answers	integer(32)	NULL	NULL
public	student_responses	TBL	9		score_percentage	numeric(5,2)	NULL	NULL
public	student_responses	TBL	10		processing_time_seconds	integer(32)	NULL	NULL
public	student_responses	TBL	11		overall_quality	character varying(50)	NULL	NULL

public	student_responses	TBL	12		quality_grade	character varying(10)	NULL	NULL
public	student_responses	TBL	13		average_confidence	numeric(5,2)	NULL	NULL
public	student_responses	TBL	14		confidence_std	numeric(5,2)	NULL	NULL
public	student_responses	TBL	15		answer_distribution	jsonb	NULL	NULL
public	student_responses	TBL	16		answer_statistics	jsonb	NULL	NULL
public	student_responses	TBL	17		detection_statistics	jsonb	NULL	NULL
public	student_responses	TBL	18		quality_assessment	jsonb	NULL	NULL
public	student_responses	TBL	19		created_at	timestamp without time zone(6)	NULL	NULL
public	student_responses	TBL	20		updated_at	timestamp without time zone(6)	NULL	NULL
public	students	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	students	TBL	2	FK	section_id	bigint(64)	NOT NULL	NULL
public	students	TBL	3	UK	student_id	character varying(255)	NOT NULL	NULL
public	students	TBL	4		first_name	character varying(255)	NOT NULL	NULL
public	students	TBL	5		middle_name	character varying(255)	NULL	NULL
public	students	TBL	6		last_name	character varying(255)	NOT NULL	NULL
public	students	TBL	7		email	character varying(255)	NULL	NULL

public	students	TBL	8		phone	character varying(50)	NULL	NULL
public	students	TBL	9		birth_date	date(3)	NULL	NULL
public	students	TBL	10		gender	character varying(50)	NULL	NULL
public	students	TBL	11		address	text	NULL	NULL
public	students	TBL	12		guardian_name	character varying(255)	NULL	NULL
public	students	TBL	13		guardian_phone	character varying(50)	NULL	NULL
public	students	TBL	14		guardian_email	character varying(255)	NULL	NULL
public	students	TBL	15		status	character varying(50)	NULL	NULL
public	students	TBL	16		enrollment_date	date(3)	NULL	NULL
public	students	TBL	17		created_at	timestamp without time zone(6)	NULL	NULL
public	students	TBL	18		updated_at	timestamp without time zone(6)	NULL	NULL
public	subjects	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	subjects	TBL	2		subject_name	character varying(255)	NOT NULL	NULL
public	subjects	TBL	3		subject_description	text	NULL	NULL
public	subjects	TBL	4		created_at	timestamp without time zone(6)	NULL	NULL

public	subjects	TBL	5		updated_at	timestamp without time zone(6)	NULL	NULL
public	topics	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	topics	TBL	2	FK	subject_id	bigint(64)	NOT NULL	NULL
public	topics	TBL	3		topic_name	character varying(255)	NOT NULL	NULL
public	topics	TBL	4		topic_description	text	NULL	NULL
public	topics	TBL	5		quarter	character varying(50)	NULL	NULL
public	topics	TBL	6		grade_level	character varying(50)	NULL	NULL
public	topics	TBL	7		created_at	timestamp without time zone(6)	NULL	NULL
public	topics	TBL	8		updated_at	timestamp without time zone(6)	NULL	NULL
public	uploaded_images	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	uploaded_images	TBL	2	FK	question_id	bigint(64)	NOT NULL	NULL
public	uploaded_images	TBL	3		image_title	character varying(255)	NULL	NULL
public	uploaded_images	TBL	4		image_path	character varying(500)	NOT NULL	NULL
public	uploaded_images	TBL	5		created_at	timestamp without time zone(6)	NULL	NULL

public	uploaded_images	TBL	6		updated_at	timestamp without time zone(6)	NULL	NULL
public	user_subject_colors	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	user_subject_colors	TBL	2	FK,UK	user_id	bigint(64)	NOT NULL	NULL
public	user_subject_colors	TBL	3	FK,UK	subject_id	bigint(64)	NOT NULL	NULL
public	user_subject_colors	TBL	4		color	character varying(50)	NOT NULL	NULL
public	user_subject_colors	TBL	5		created_at	timestamp without time zone(6)	NULL	NULL
public	user_subject_colors	TBL	6		updated_at	timestamp without time zone(6)	NULL	NULL
public	users	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	users	TBL	2		name	character varying(255)	NOT NULL	NULL
public	users	TBL	3		last_name	character varying(255)	NULL	NULL
public	users	TBL	4	UK	email	character varying(255)	NOT NULL	NULL
public	users	TBL	5		email_verified_at	timestamp without time zone(6)	NULL	NULL
public	users	TBL	6		password	character varying(255)	NOT NULL	NULL
public	users	TBL	7		remember_token	character varying(100)	NULL	NULL

public	users	TBL	8		created_at	timestamp without time zone(6)	NULL	NULL
public	users	TBL	9		updated_at	timestamp without time zone(6)	NULL	NULL

5.1.10. Summary of the Design and Development of the Assessment Management System

This section highlights the different diagrams and tables used in providing comprehension on how the Assessment Management system is designed and developed. Furthermore, this section also presents the system requirements for the developed system. Lastly, after the development of the system, the GUI of the system is then shown.

5.2. Development Results

5.2.1. Graphical User Interface of Assessment Management System

A Graphical User Interface (GUI) is a type of user interface that enables users to interact with electronic devices using graphical icons and visual indicators, rather than text-based commands. GUIs are commonly used in modern systems to improve usability and provide an intuitive experience.

The Assessment Management System is designed specifically for teachers and admins, with primary functions focused on managing a test bank and conducting item analysis. The test bank allows users to create, store, and organize assessment questions efficiently, while the item analysis function helps evaluate question effectiveness. The GUI provides an intuitive interface that enables users to simplify assessment creation and analysis tasks, enhancing the overall management of assessments.

5.2.2. Graphical User Interface of Assessment Management System's Desktop Application

5.2.2.1. A Teacher and Admin User Interface

The figures present the graphical user interface (GUI) pages for the user interfaces of the Assessment Management System for both teachers and admins. The teacher's homepage is organized by classes, allowing teachers to manage assessments. Key features

include access to the test bank for each class, item analysis for each questionnaire, and tools for creating, organizing questions, and generating item analysis. The admin interface provides access to the test bank for managing DepEd questionnaires and importing them into the system.

Figure 59 shows the graphical user interface of the teacher and admin user login form.

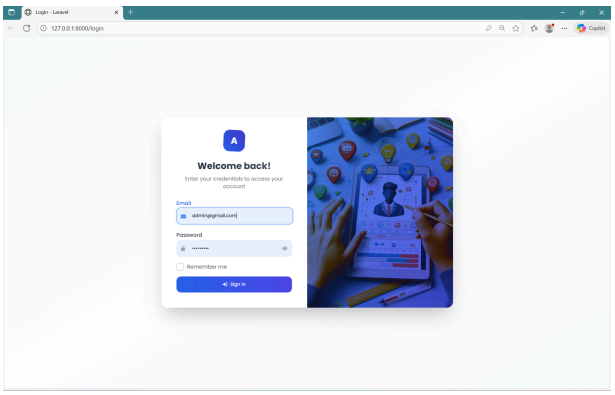


Figure 59. AsseSync Teacher and Admin User Interface Login Form

Figure 60 displays the Admin Dashboard of the ASSESYNC system. This interface allows administrators to manage, track, and analyze assessments by providing features for importing questionnaires, creating new assessments, exporting data, applying filters, and viewing comprehensive assessment information within a structured table.

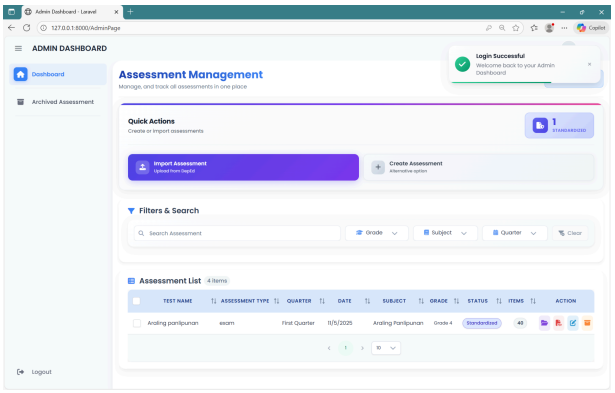


Figure 60. AsseSync Admin Dashboard.

Figure 61 displays the Import Assessment Modal of the ASSESYNC system, which is accessed from the Admin Dashboard. This modal allows admin to upload existing questionnaires (supporting PDF, DOC, and DOCX files) and automatically extract questions. Before uploading the document, the administrator must categorize the assessment by specifying the Test Name, Quarter Type, Subject, and Grade using the required input file.

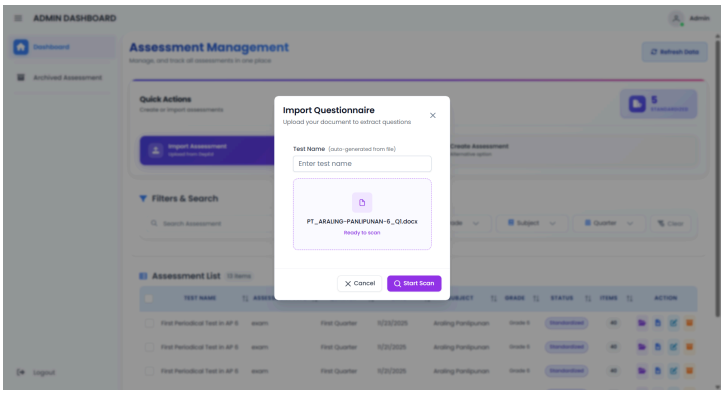


Figure 61. AsseSync Admin Import Assessment Modal

Figure 62 illustrates the Extracted Questions Modal that appears after an administrator uploads a document via the Import Assessment feature. The purpose of this component is to allow the administrator to review and confirm the content automatically extracted by the system. The modal presents the detected questions and multiple-choice options, including the identified correct answer (highlighted in green, e.g., Choice B, "Ekwador"). It also displays system warnings or relevant metadata extracted from the document. The process concludes when the administrator clicks "Confirm" to finalize the assessment's creation.

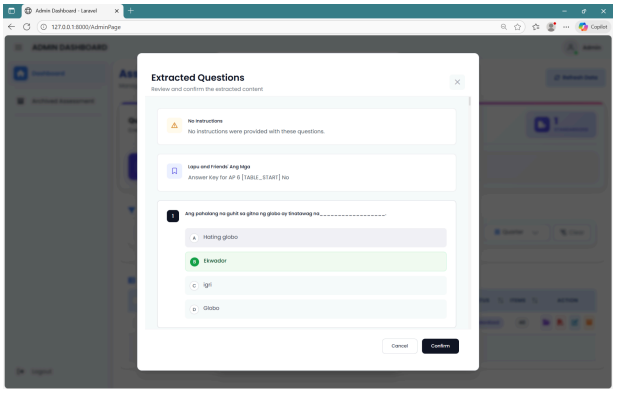


Figure 62. AsseSync Admin Extracted Questions Modal

Figure 63 illustrates the final stage of the Import Assessment process, appearing after the document has been successfully uploaded, scanned, and questions have been extracted. The modal displays a clear confirmation message ("Document scanned successfully") along with the number of extracted questions (e.g., "40 questions extracted"). The administrator can review the filled assessment details (Test Name, Quarter Type, Subject, Grade) and utilize the "Preview" option to examine the extracted questions before final commitment. To complete the creation of the assessment, the administrator clicks the "Save" button, or selects "Cancel" to discard the operation.

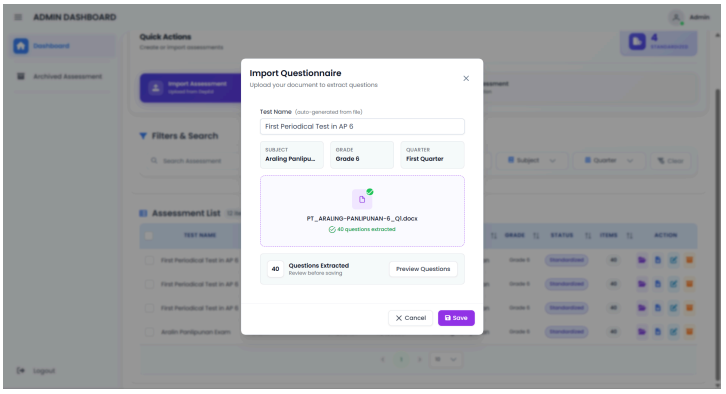


Figure 63. AsseSync Admin Save Assessment Modal

Figure 64 depicts the Saving Assessment loading state that appears immediately after the administrator clicks the "Save" button in the Import Assessment modal. This component provides visual feedback to the user that the system is actively processing and saving the data. A message, "Saving Assessment... Please wait while we process your questionnaire," is displayed prominently. The input fields and the previous content of the modal become temporarily disabled and dimmed, indicating that the system is currently performing a background operation to finalize the creation of the new assessment record.

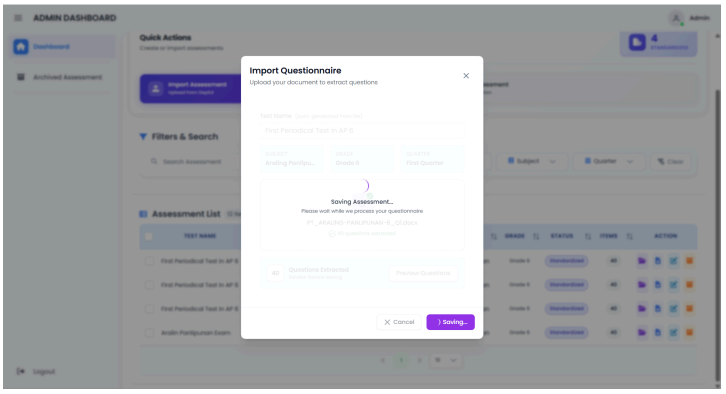


Figure 64. AsseSync Admin Saving Assessment loading Modal

Figure 65 illustrates the final outcome of the assessment import process, showing the administrator returned to the Admin Dashboard. A toast notification appears in the top right corner, providing confirmation with the message: "Questionnaire Uploaded Successfully. Your questionnaire has been saved." Concurrently, the newly saved assessment is immediately visible within the Assessment List table. The table now reflects the updated inventory, allowing the administrator to verify the successful creation and access the assessment for further management actions.

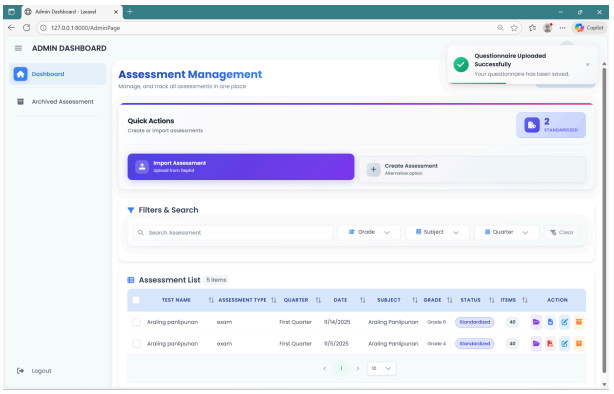


Figure 65. AsseSync Admin Assessment Saved Confirmation and Dashboard Update

Figure 66 displays the initial step in the manual Create Assessment process, presenting the Select Quarter for Exam modal. This step requires the administrator to categorize the new assessment by selecting the appropriate grading period. The modal lists four distinct options: First Quarter, Second Quarter, Third Quarter, and Fourth Quarter, each clearly defined by its corresponding grading period description. Once the administrator selects the desired quarter, they must click "Confirm Selection" to proceed to the next step of defining the assessment (such as selecting the subject).

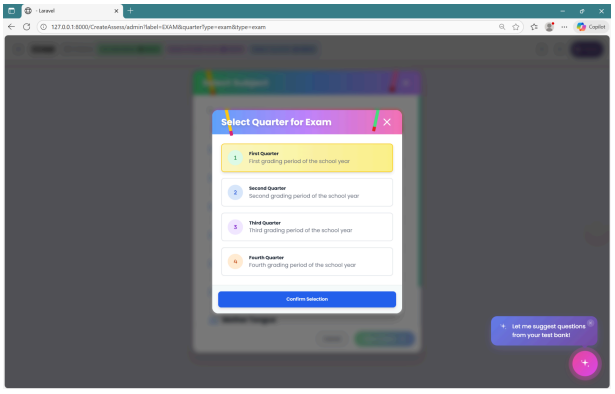


Figure 66. AsseSync Admin Select Quarter for Exam modal

Figure 67 shows the Select Grade Level Modal, which is the second step in the manual assessment creation process, following the quarter selection. This modal requires the administrator to define the target grade level for the new assessment. It presents several clearly labeled grade options (e.g., Grade 4, Grade 5, Grade 6), each accompanied by descriptive text specifying the corresponding grade level and typical age range. Once the appropriate grade is selected, the administrator must click "Confirm Selection" to proceed to the subsequent step, which typically involves selecting the specific subject for the assessment.

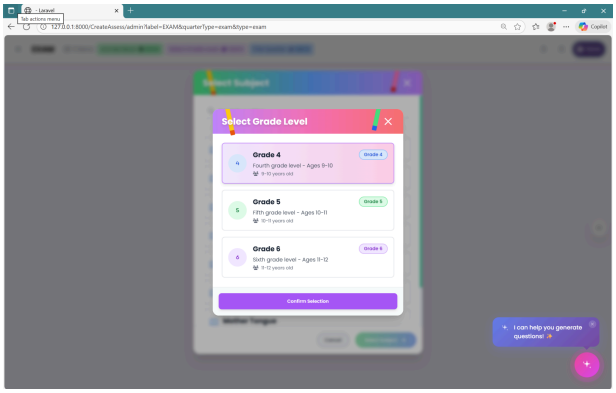


Figure 67. AsseSync Admin Select Grade Level Modal

Figure 68 shows the Select Subject Modal, the third and final categorization step in the manual assessment creation process. This modal prompts the administrator to choose the specific academic subject for the new assessment. It features a Search bar to quickly locate a subject within the extensive list (e.g., Mathematics, Science, English, Araling Panlipunan, etc.). After selecting the desired subject, the administrator clicks the "Select Subject" button to finalize the assessment's basic categorization and proceed to the interface for building or populating the questionnaire content.

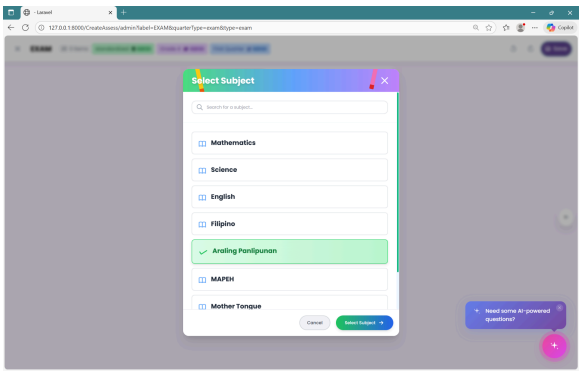


Figure 68. AsseSync Admin Select Subject Modal

Figure 69 shows the Select Grade Level Modal, which is the second step in the manual assessment creation process, following the quarter selection. This modal requires the administrator to define the target grade level for the new assessment. It presents several clearly labeled grade options (e.g., Grade 4, Grade 5, Grade 6), each accompanied by descriptive text specifying the corresponding grade level and typical age range. Once the appropriate grade is selected, the administrator must click "Confirm Selection" to proceed to the subsequent step, which typically involves selecting the specific subject for the assessment.

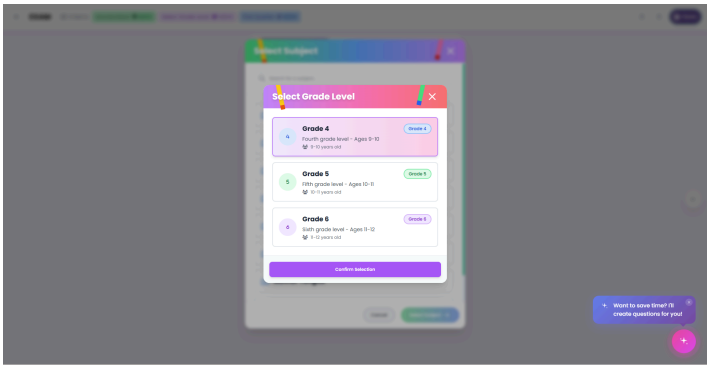


Figure 69. AsseSync Admin Select Grade Level Modal

Figure 70 displays the dedicated Assessment Creation and Editing Interface, where the administrator manually builds the questionnaire after selecting the quarter, grade, and subject. The interface clearly shows the categorization tags (e.g., Exam, Araling Panlipunan, Grade 4, First Quarter) at the top for context. The main toolbar provides options to Add Question, set the Title and Description, insert an Image, or Import questions. The central area is used for defining the assessment details, and the top-right Save button is used to finalize the work.

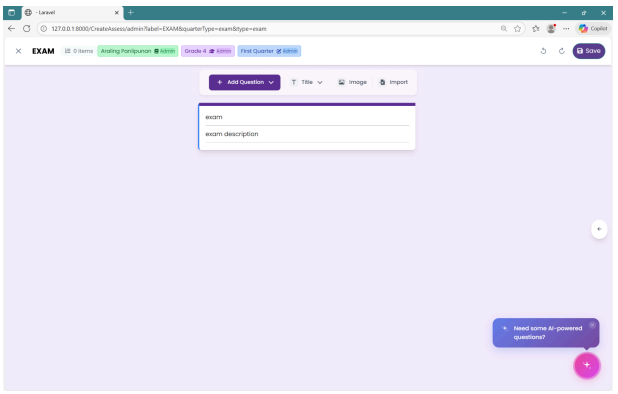


Figure 70. AsseSync Admin Assessment Creation and Editing Interface

Figure 71 illustrates the options available when the administrator clicks the "Title" button within the Assessment Creation and Editing Interface. This feature allows the administrator to structure the assessment content by inserting different levels of textual dividers. The dropdown menu provides two distinct options: "Title" (for the main assessment title or sections) and "Section Title" (for sub-headings or smaller organizational units). This functionality is crucial for administrators who wish to organize a lengthy assessment into logical and clearly defined sections before adding questions

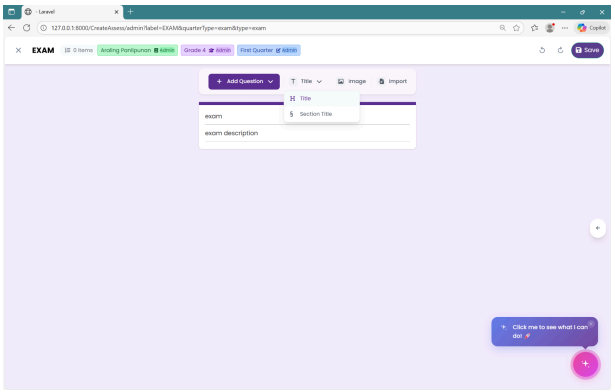


Figure 71. AsseSync Admin Assessment Editing Interface - Title Selection

. Figure 72 illustrates the result of inserting a Title element into the Assessment Creation and Editing Interface. Once inserted, the element provides two dedicated input fields: one for the primary Title and another for the accompanying Title Description. This functionality allows the administrator to clearly delineate different sections within the examination, such as Part I or specific instructions for a group of questions, thereby enhancing the structure and clarity of the overall assessment document.

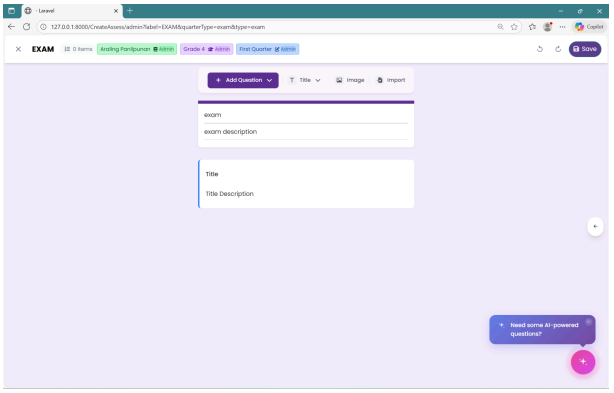


Figure 72. AsseSync Admin Assessment Editing Interface - Title Input

Figure 73 illustrates the functionality available when the administrator clicks the "Add Question" button within the Assessment Creation and Editing Interface. This dropdown menu presents the available question formats that can be added to the examination. The primary option shown is "Multiple Choice", allowing the administrator to insert a standard question element requiring a single correct answer from a list of options. This feature is fundamental for building the actual content of the questionnaire based on the selected question type.

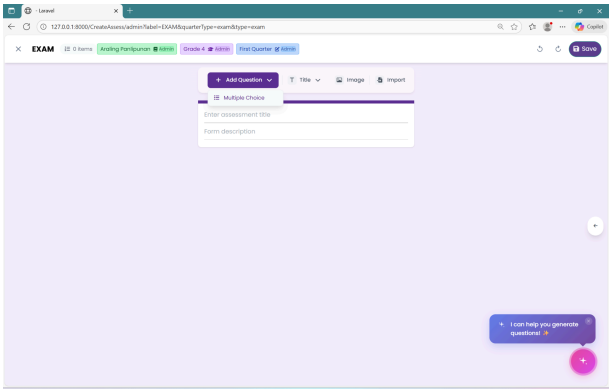


Figure 73. AsseSync Admin Assessment Editing Interface - Add Question Menu

Figure 74 displays the fully structured interface for a manually created Multiple Choice Question element. The question block allows the administrator to input the question text and define up to five answer options (A to E). Key features include the ability to insert an image within the question or an option, and the option to delete an answer option. Crucially, the system allows the administrator to designate the correct answer by selecting the radio button next to the appropriate option (as seen with Option C), which defines the scoring key for the assessment.

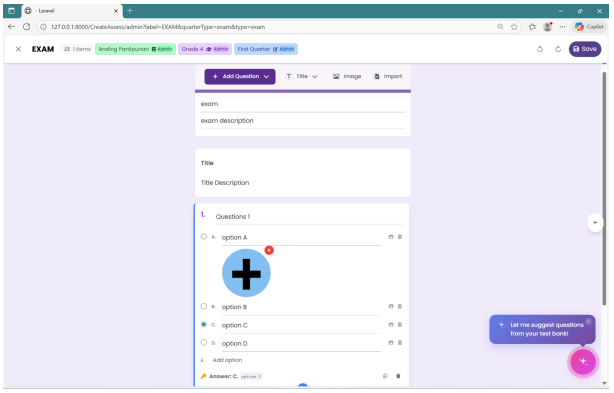


Figure 74. AsseSync Admin Assessment Editing Interface - Multiple Choice Question Creation Interface

Figure 75 illustrates the result of inserting a Section Title element into the Assessment Creation and Editing Interface. This component serves as a second-level header, useful for creating sub-sections within a larger title, and includes input fields for the Section Title itself and an optional Description. The Section Title appears below existing questions and is clearly labeled, in this example, as "Section 1 of 1." This feature is essential for administrators who require granular structural organization within the assessment document, improving the readability and flow for the test-takers.

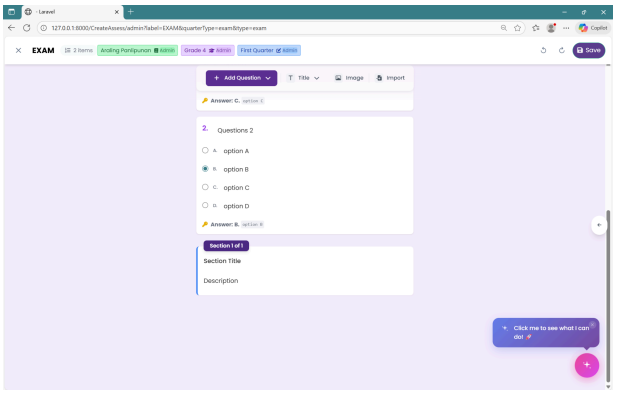


Figure 75. AsseSync Admin Assessment Editing Interface - Section Title Input

Figure 76 illustrates the Add Image feature within the Assessment Creation and Editing Interface, accessed by clicking the "Image" dropdown button and selecting "Add Image." This functionality inserts a dedicated block into the assessment, providing an Image Title input field and a drag-and-drop or click-to-upload area. The system supports common image formats (PNG & JPG) and specifies a file size limit of up to 10MB. This feature allows administrators to enhance questions or sections with visual aids, charts, or diagrams to improve test clarity and comprehension.

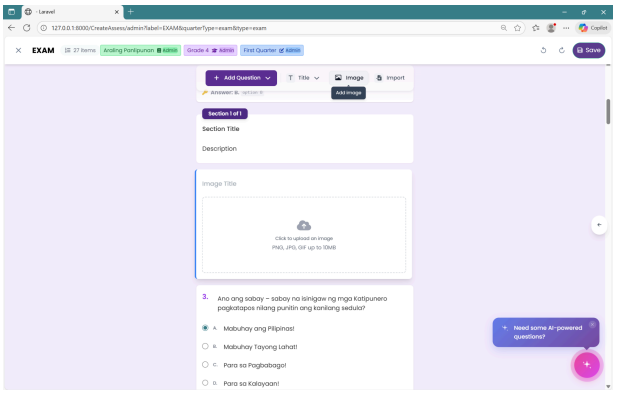


Figure 76. AsseSync Admin Assessment Editing Interface - Add Image Feature

Figure 77 illustrates the Import feature within the Assessment Creation and Editing Interface, which opens a sidebar containing the Test Bank. This functionality allows the administrator to quickly reuse questions from existing standardized assessments within the system. The Test Bank sidebar enables the user to search assessments and view a list of available questionnaires, showing key details like the number of questions, creation date, and subject (Araling Panlipunan in this example). By selecting an assessment from this bank, the administrator can efficiently import its questions into the current document, saving significant time during the creation process.

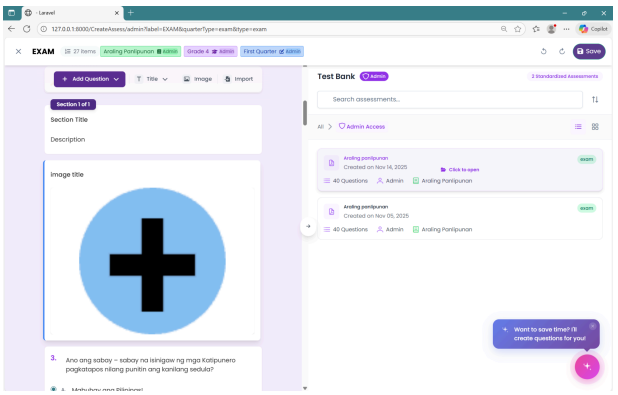


Figure 77. AsseSync Admin Assessment Editing Interface - Import Assessment from Test Bank

Figure 78 displays the Question Selection Modal, which appears when the administrator chooses an assessment from the Test Bank. This modal allows for selective importing of questions. The administrator can individually select questions or use the "Select All" checkbox. The process is finalized by clicking the "Import Selected" button, which adds the chosen questions to the current assessment being created.

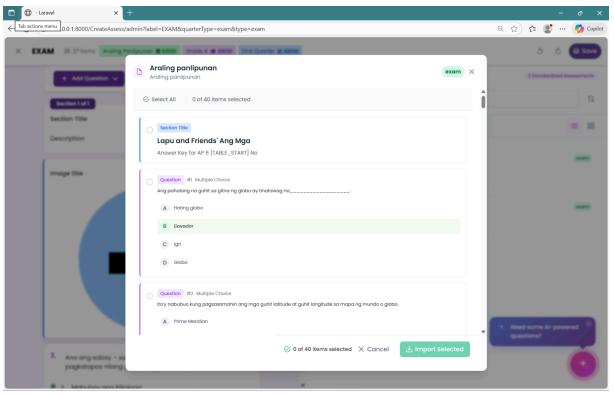


Figure 78. AsseSync Admin Assessment Editing Interface - Import Assessment Question Selection Modal

Figure 79 illustrates the state of the Question Selection Modal after the administrator has chosen to select all available questions for import. Upon clicking "Select All," the interface updates to display a count of "41 of 41 items selected" (in this example) and replaces the "Select All" option with "Deselect All." All questions are marked with a checkmark, confirming their inclusion. This feature provides a rapid method for administrators to duplicate or fully reuse an existing assessment by importing its entire content into the currently created examination. The selected questions are then added by clicking "Import Selected."

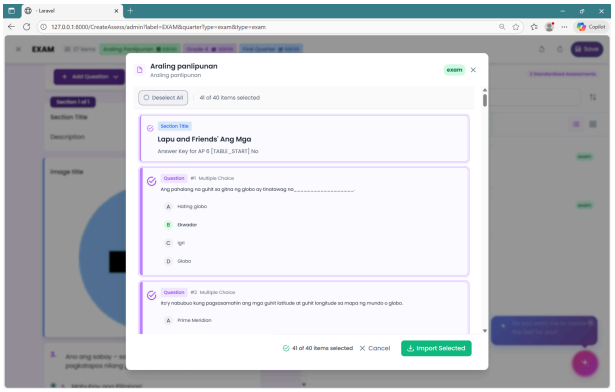


Figure 79. AsseSync Admin Assessment Editing Interface - Import Assessment Modal - Select All State

Figure 80 illustrates the Assessment Creation and Editing Interface immediately following a successful import of questions from the Test Bank. The newly imported questions are appended to the assessment document, as evidenced by the increase in the question count at the top (e.g., "40 Items"). A small success notification appears at the bottom-left of the screen, confirming the operation: "Imported 41 questions successfully!" This notification provides an "Undo" option, allowing the administrator to quickly revert the import if necessary. The administrator can now review, edit, or continue building the newly populated assessment.

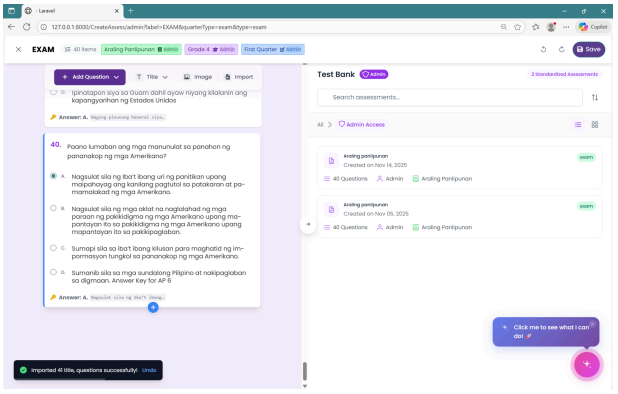


Figure 80. AsseSync Admin Assessment Editing Interface - Successful Question Import

Figure 81 showcases the use of the Artificial Intelligence (AI) features accessible via the floating action button in the Assessment Creation and Editing Interface. This feature provides two powerful options for quickly populating the assessment content: "AI Create Questions" and "AI Suggest Questions from Test Bank." The former utilizes generative AI to create new questions based on the assessment's subject and grade level, while the latter employs AI to intelligently suggest relevant questions from the existing Test Bank. These tools significantly accelerate the question-writing process for the administrator.

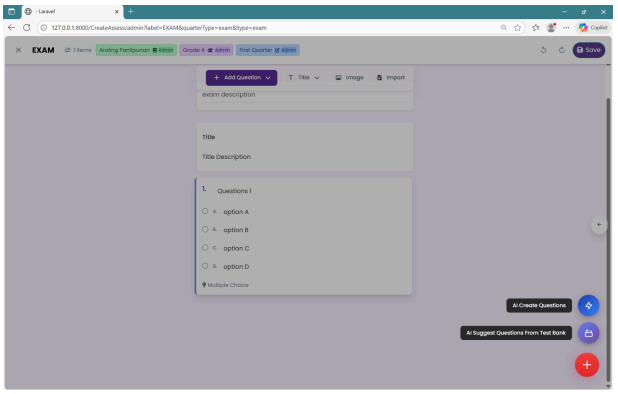


Figure 81. AsseSync Admin Assessment Editing Interface - AI Question Generation Options

Figure 82 displays the AI Test Bank Search Modal, which is activated when the administrator selects the "AI Suggest Questions from Test Bank" option. This modal allows the administrator to configure the search parameters for AI question suggestions. The administrator must then define the desired Question Quantity (up to 50 per search) and filter the results by Difficulty Level (Easy, Medium, Hard, or Any). The modal also provides a Test Bank Overview, showing the total available questions and their current difficulty distribution, before the administrator executes the search by clicking "Search Test Bank".

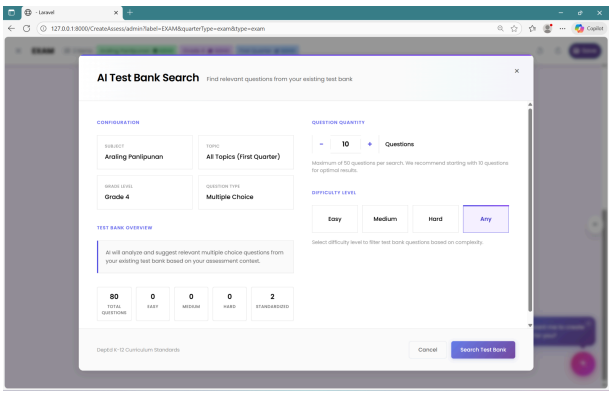


Figure 82. AsseSync Admin Assessment Editing Interface AI Suggest Questions from Test Bank

Figure 83 depicts the AI Test Bank Search Loading State, which appears immediately after the administrator initiates a search for relevant questions. This screen provides visual feedback with a central animation and a text message indicating that the system is actively analyzing the test bank using AI to find the most relevant questions. It also includes an estimated duration for the task ("This may take 10-30 seconds"), effectively managing the user's expectation while the background processing is completed.

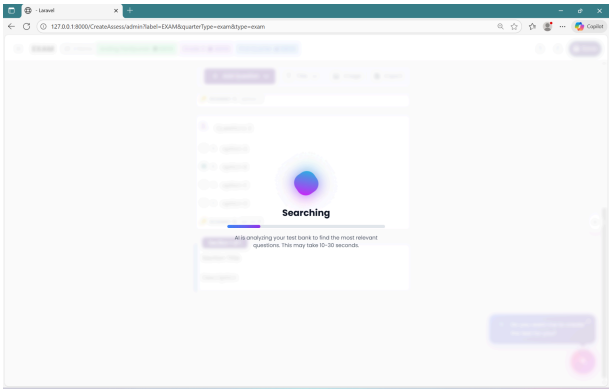


Figure 83. AsseSync Admin Assessment Editing Interface AI Suggest Questions from Test Bank - Loading State

Figure 84 illustrates the Assessment Creation Interface after the AI Test Bank Search is complete, presenting the AI Suggested Questions for review. A toast notification confirms that the questions are ready (e.g., "10 AI-generated questions ready for review"). The system inserts a dedicated block containing the suggested questions, allowing the administrator to preview each one. The administrator is provided with bulk action buttons: "Accept All" to instantly add all suggested questions to the assessment, or "Reject All" to discard the entire batch. This feature ensures human oversight while leveraging AI for efficient content suggestion.

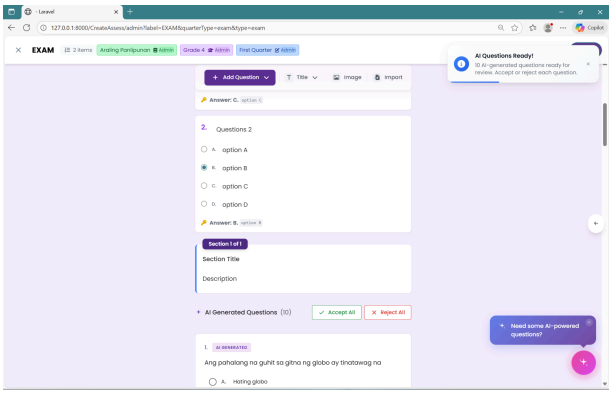


Figure 84. AsseSync Admin Assessment Editing Interface AI Suggest Questions from Test Bank - Questions Imported Interface

Figure 85 illustrates the ability of the administrator to individually review and manage the questions suggested by the AI Test Bank Search. For each "AI GENERATED" question, the administrator can examine the question prompt and the suggested options, including the pre-selected correct answer (marked with a green checkmark, e.g., Option A). Beneath each question block, the administrator has granular control via the "Accept" and "Reject" buttons.

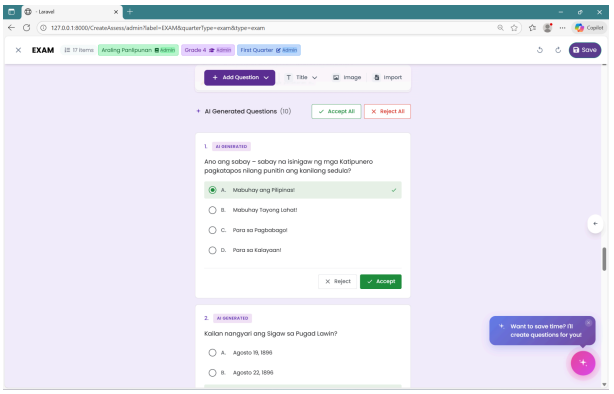


Figure 85. AsseSync Admin Assessment Editing Interface AI Suggest Questions from Test Bank - Individual Review Interface

Figure 86 displays the Assessment Question Generator Modal, which is used when the administrator selects the "AI Create Questions" option. This modal allows for the configuration and generation of entirely new assessment questions using generative AI. Similar to the search function, it configures parameters like the Subject (Araling Panlipunan), Grade Level (Grade 4), and Question Type (Multiple Choice). The administrator sets the Question Quantity (up to 10 per generation) and the desired Cognitive Complexity (Easy, Medium, or Hard), based on Bloom's Taxonomy. The process is initiated by clicking "Generate Questions."

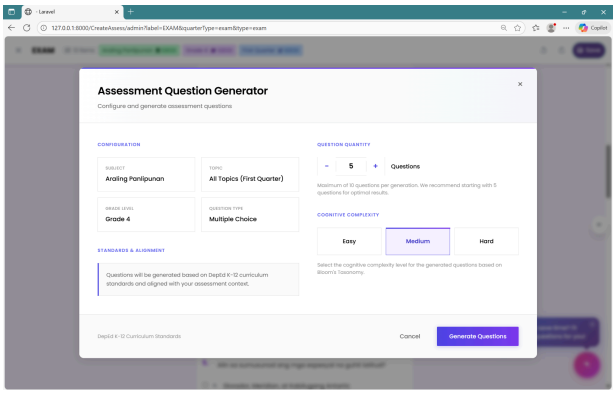


Figure 86. AsseSync Admin Assessment Interface AI Question Generator Modal

Figure 87 depicts the AI Question Generator Loading State, which appears immediately after the administrator clicks the "Generate Questions" button. This screen provides visual feedback with a central animation and a text message indicating that the system is actively generating new questions using AI based on the specified configurations. The message, "AI is creating questions based on your specifications," is displayed, along with an estimated duration ("This may take 10-30 seconds"). This screen effectively informs the user that the system is processing the request and manages expectations regarding the wait time for the generative process.

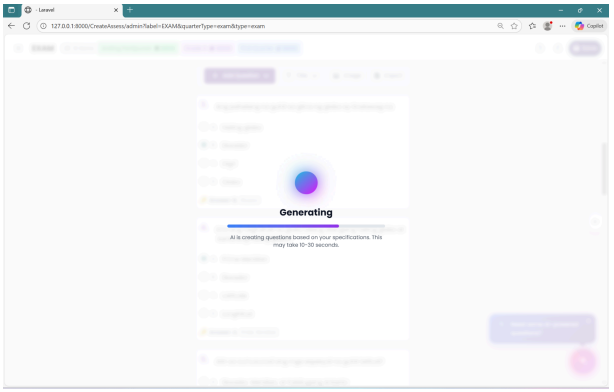


Figure 87. AsseSync Admin Assessment Interface AI Question Generator - Loading State

Figure 88 displays the Assessment Creation Interface after the AI Question Generator process is completed. The new questions are presented in a dedicated review block, clearly labeled "AI Generated Questions" (e.g., 5 generated questions). A toast notification confirms readiness: "AI-generated questions ready for review." For each generated question, the administrator can view the prompt, the multiple-choice options, and the AI-determined correct answer (highlighted in green). The administrator uses the "Accept All" and "Reject All" buttons for batch processing or the individual "Accept/Reject" controls to curate the newly created assessment content.

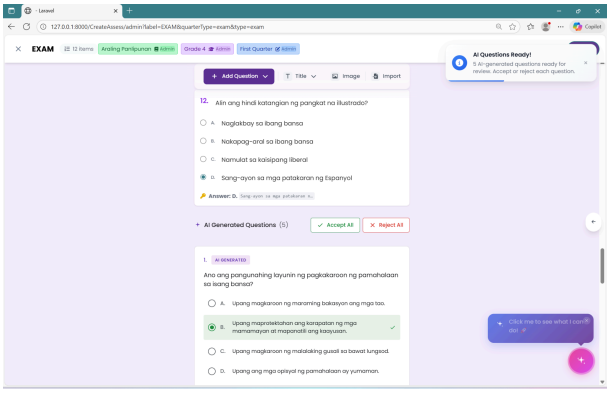


Figure 88. AsseSync Admin Assessment Interface AI Question Generator - Review Interface

Figure 89 further illustrates the administrator's curation capabilities over the AI-generated questions. For each question, which is distinctly labeled "AI GENERATED," the administrator can verify the AI-selected correct answer (marked with a green checkmark, e.g., Option B). Crucially, the interface provides dedicated "Accept" and "Reject" buttons beneath every question block. This design enables the administrator to perform a meticulous, question-by-question review, ensuring that only content that aligns perfectly with the assessment objectives and curriculum standards is ultimately integrated into the final examination document.

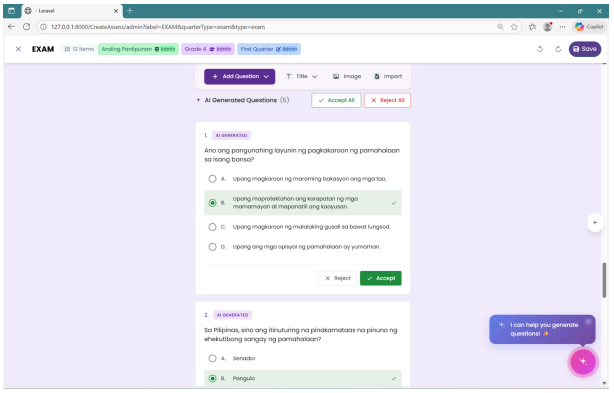


Figure 89. AsseSync Assessment Interface AI Question Generator - Individual Review Interface

Figure 90 displays the Official Document Modal, which is required when an administrator intends to mark an assessment as a standardized document. The modal is specifically titled "Official Document: Required for standardized assessment." It features a clear drag-and-drop or click-to-browse upload area where the final, official version of the assessment document is submitted. Uploading this document is a mandatory step for assessments that need official record-keeping, and the process is completed by clicking "Upload & Validate."

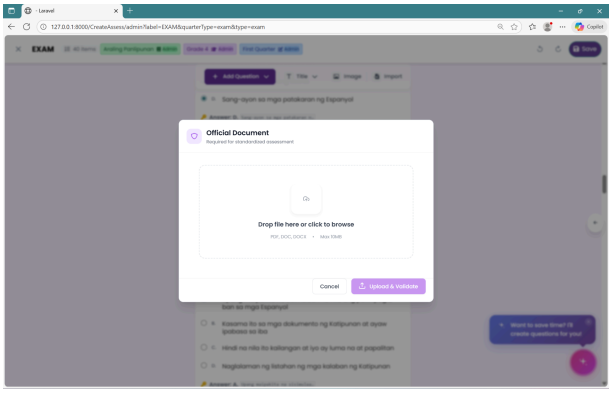


Figure 90. AsseSync Admin Official Document Upload Modal

Figure 91 illustrates the Official Document Modal after a file has been successfully selected by the administrator. The interface confirms the file selection by displaying the document name (e.g., "PT_ARALING PANLIPUNAN 6_Q1.docx") within the upload area. Concurrently, a toast notification appears in the top-right corner, verifying the readiness of the file: "File Selected: [filename] is ready to upload. Click 'Confirm' to proceed." This state confirms the successful local selection, and the administrator must then click "Upload & Validate" to finalize the submission of the official document to the system.

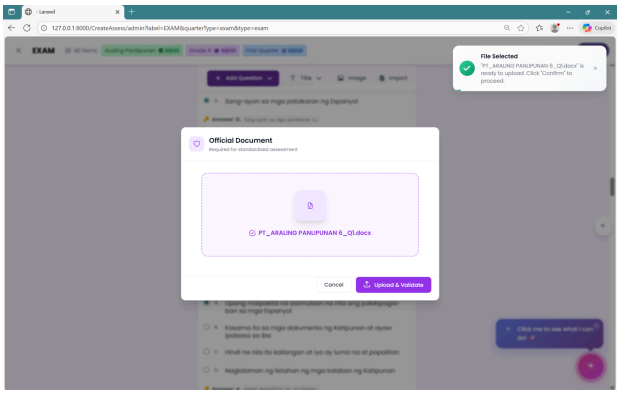


Figure 91. AsseSync Admin Official Document Upload Confirmation State

Figure 92 depicts the loading state of the Official Document Modal after the administrator clicks "Upload & Validate." The modal indicates that the system is currently processing and validating the uploaded official assessment document. The button text changes to "Validating..." and may be accompanied by a subtle loading animation within the upload area. This state confirms that the file transfer is underway and the system is performing the necessary checks to ensure the document meets the requirements for a standardized assessment.

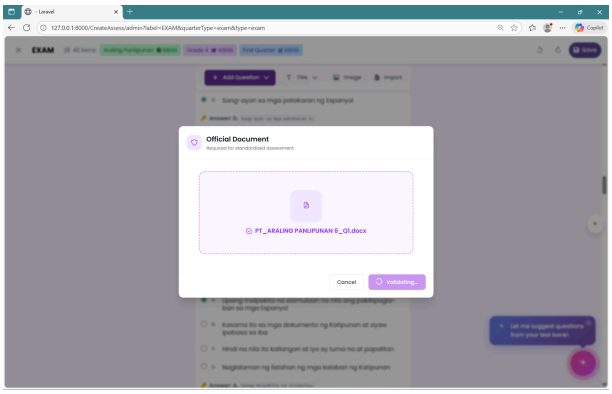


Figure 92. AsseSync Admin Official Document Upload Confirmation State

Figure 93 illustrates the successful completion of the assessment creation process. A central success modal appears with the prominent message: "Assessment Saved Successfully! Redirecting you back..." confirming the finalized creation of the new assessment. Simultaneously, a toast notification confirms the status in the upper right corner: "Assessment Saved: Your standardized assessment has been created successfully." This screen provides the final, positive user feedback before the system automatically redirects the administrator back to the main Admin Dashboard, where the new assessment is available for tracking and management.

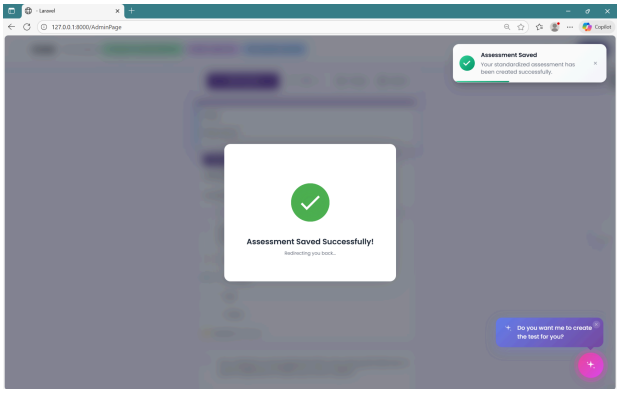


Figure 93. AsseSync Admin Assessment Saved Success Confirmation

Figure 94 displays the Detailed Assessment Saving Progress Modal, which appears when the system is processing the final save action. Unlike simple loading screens, this modal provides the administrator with a step-by-step progress checklist, confirming the successful completion of critical tasks: "Document Uploaded," "AI Validation Passed," and "Saving to Database." The final ongoing step, "Creating Assessment," is shown with a loading bar and percentage indicator (e.g., 75%), assuring the user of the system's progress and transparency as it finalizes the creation of the new assessment record.

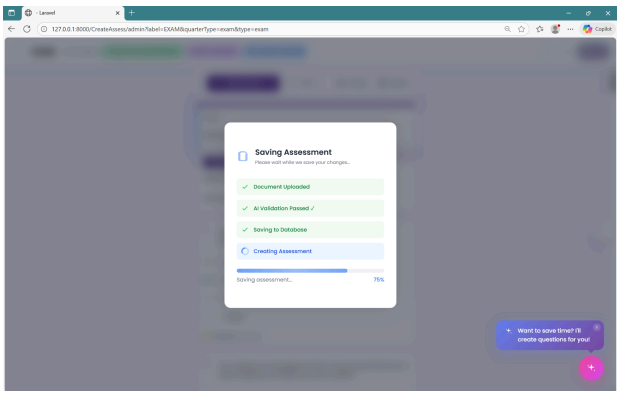


Figure 94. AsseSync Admin Detailed Assessment Saving Progress Modal

Figure 95 displays the Missing Required Content Validation Modal, which appears when the administrator attempts to save an assessment that is incomplete. The modal clearly warns the user about the "Missing Required Content" and specifically identifies the missing item (e.g., "Assessment title is required. Please enter a title for your assessment."). This crucial validation step prevents saving incomplete records to the database. The administrator must click "Review and Fix" to dismiss the modal and return to the editing interface to provide the necessary information before attempting to save the assessment again.

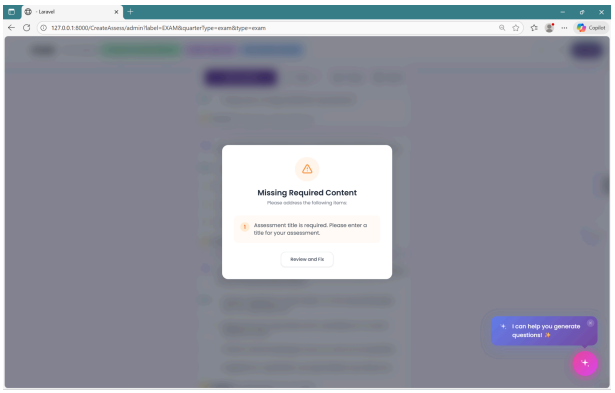


Figure 95. AsseSync Admin Missing Required Content Validation Modal

Figure 96 displays the Official Document Validation Failed Modal, which appears when the system detects a mismatch between the assessment created in the interface and the official document uploaded. The primary error is highlighted both within the modal and via a top-right toast notification ("Document does not match..."). The core failure message states: "Document validation failed" because the uploaded document contains a different number of questions than the assessment form. Crucially, the system provides a remediation option: "Admin override - action will be logged." The administrator can choose to "Override & Save" to proceed despite the mismatch (with the action recorded for accountability) or "Cancel" the operation.

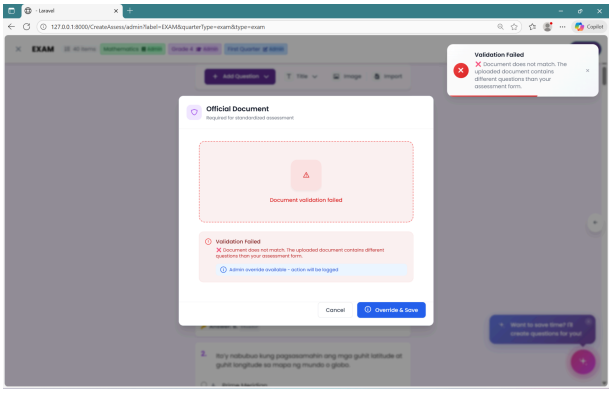


Figure 96. AsseSync Admin Official Document Validation Failed Modal

Figure 97 illustrates the primary actions available for an existing assessment directly from the Assessment List table on the Admin Dashboard. When the administrator interacts with an assessment row, the Action column reveals a set of management buttons. The highlighted "Open Assessment" button is the primary control used to launch the dedicated Assessment Creation and Editing Interface. Clicking this button allows the administrator to review, edit, or modify the content of the selected assessment, including its questions, titles, descriptions, and official documents.

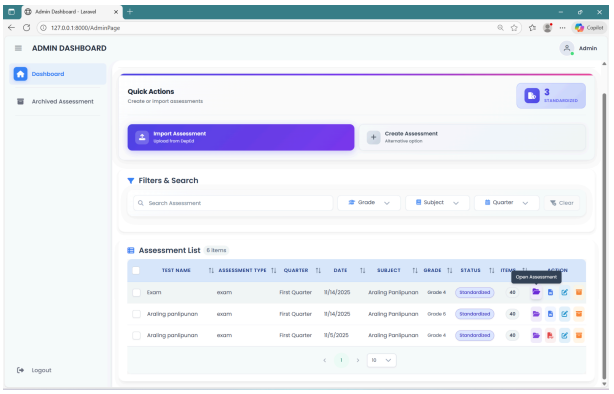


Figure 97. AsseSync Admin Dashboard - Open Assessment Action

Figure 98 illustrates another crucial action available directly from the Assessment List on the Admin Dashboard: the "View Original File" button. This functionality allows the administrator to retrieve and view the original source document (PDF, DOCX, etc.) that was uploaded during the assessment creation process. This feature is vital for verification and archival purposes, ensuring administrators can always reference the authenticated source material used to generate the standardized assessment within the ASSESYNC system.

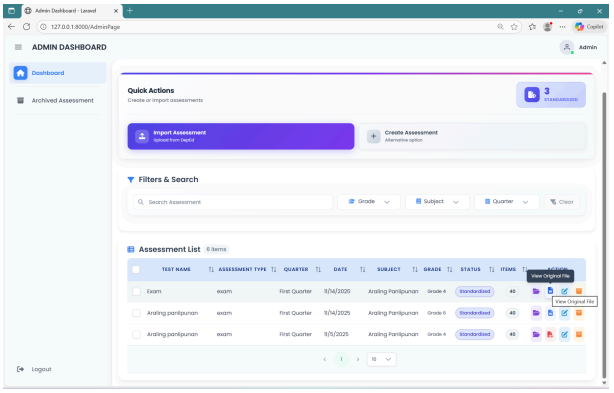


Figure 98. AsseSync Admin Dashboard - View Original File Action

Figure 99 illustrates the "Edit" action available within the Assessment List on the Admin Dashboard. This function is typically associated with modifying the metadata and categorization of an existing assessment, such as updating the Test Name, Quarter, Subject, or Grade level. Clicking the Edit button launches a dedicated modal or interface that allows the administrator to quickly make necessary administrative changes to the assessment's details without altering the question content itself.

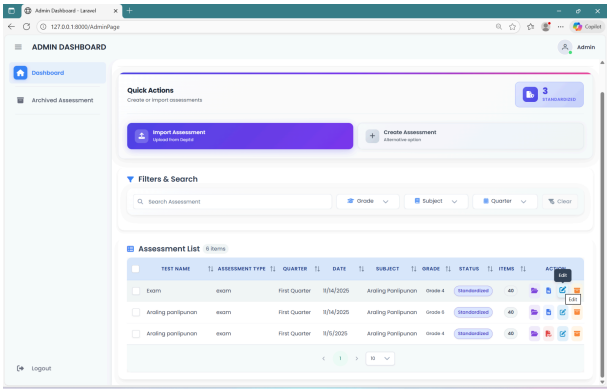


Figure 99. AsseSync Admin Dashboard - Edit Assessment Details Action

Figure 100 displays the Edit Assessment Details Modal, which appears when the administrator clicks the "Edit" action button on an assessment row in the Admin Dashboard. This modal is strictly for modifying the administrative metadata of the assessment. It presents four editable fields: Assessment Name, Quarter Type, Subject, and Grade. These fields use standard input and dropdown menus to allow the administrator to correct or update the categorization of the assessment. Once the necessary changes are made, the administrator clicks "Review Changes" to proceed with saving the updated details, or "Cancel" to discard the modifications.

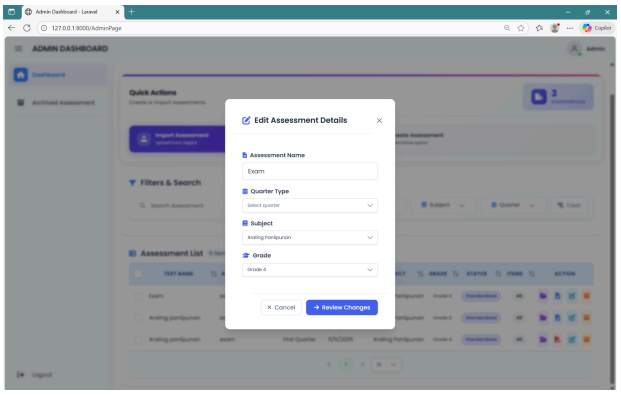


Figure 100. AsseSync Admin Dashboard - Edit Assessment Details Modal

Figure 101 illustrates the "Archive" action available to the administrator directly from the Assessment List on the Admin Dashboard. The Archive function is represented by a dedicated icon within the Action column. Clicking this button initiates the process of moving the selected assessment record from the active list to the Archived Assessment section (visible in the sidebar). This action is essential for maintaining a clean, relevant list of current assessments while retaining historical records that may be needed for future reference or data analysis.

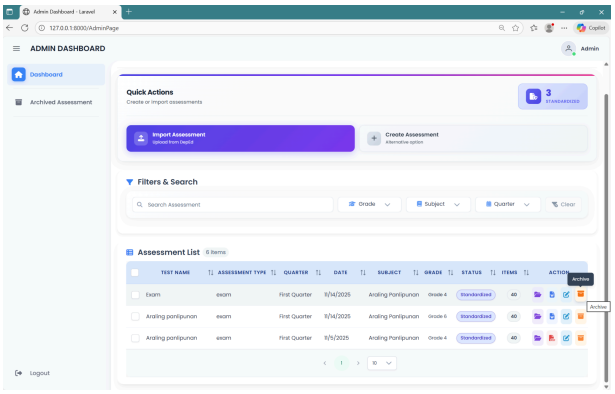


Figure 101. AsseSync Admin Dashboard - Archive Assessment Action

Figure 102 displays the Archive Assessment Confirmation Modal, which appears when the administrator initiates the "Archive" action on an assessment record. This modal requires final authorization to proceed with the action. It asks for confirmation ("Are you sure you want to archive Exam?") and provides an Important Notice clarifying the consequences: "This assessment will be moved to the archive. You can restore it later from the archive section if needed." The administrator must click the orange "Archive" button to proceed with the action or "Cancel" to abort the archiving process.

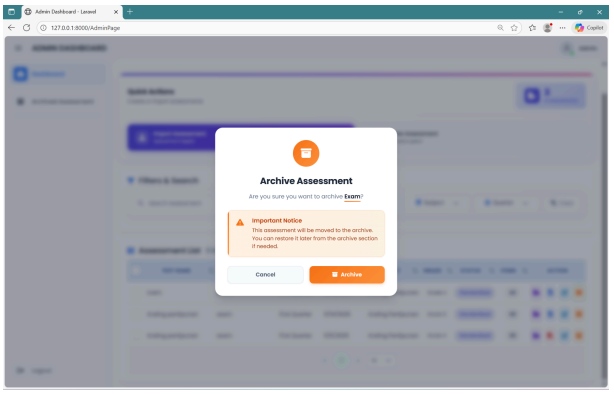


Figure 102. AsseSync Admin Dashboard - Archive Assessment Confirmation Modal

Figure 103 displays the Archive Multiple Assessments Confirmation Modal, which appears when an administrator selects multiple assessments from the dashboard list and initiates the bulk archive action. It provides a warning that archiving is reversible and that all items can be restored later from the Archive section, ensuring the administrator understands the action is safe and temporary. The process is confirmed by clicking the dedicated button, "Archive 2 Assessments," or aborted by clicking "Cancel."

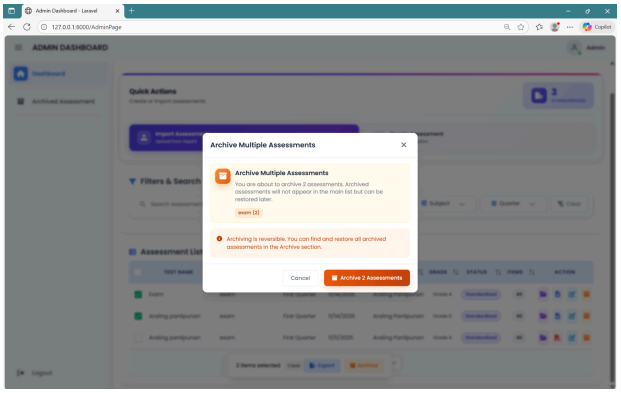


Figure 103. AsseSync Admin Dashboard - Archive Multiple Assessments Confirmation Modal

Figure 104 displays the Archive Multiple Assessments Progress Modal, which appears after the administrator confirms the bulk archiving action. This modal provides clear, real-time feedback on the ongoing process of moving the selected assessments to the archive. It displays the "Archiving Assessments" progress using a progress bar and numerical counter (e.g., "Archiving assessment 1 of 2" at "50%"), allowing the administrator to track the operation's completion status. The button text changes to "Archiving..." to indicate the active state, and the modal retains the reminder that the action is reversible and the assessments can be restored later.

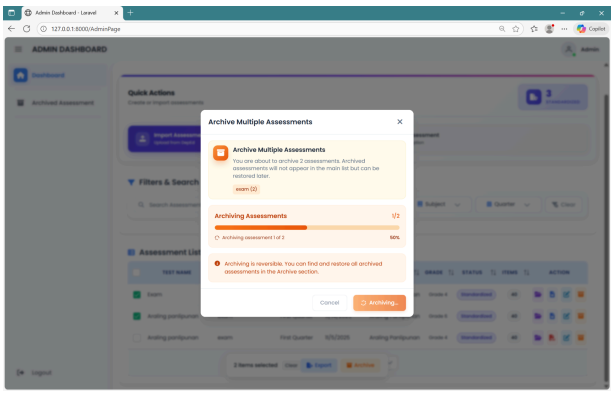


Figure 104. AsseSync Admin Dashboard - Archive Multiple Assessments Progress Modal

Figure 105 displays the Export Assessments Modal, which is used to generate downloadable copies of selected assessments. The modal first provides a summary of the selected content, detailing the number of assessments and the total number of questions being exported. Under Export Options, the administrator can choose the output Format, with Word Document currently available and PDF noted as "Coming Soon." The Inclusions section offers customizable options: toggles allow the user to choose whether to include Bubble Sheets (with bubble options) and Answer Keys (at the end of each document). A notice clarifies that each assessment will be exported as a separate document. The process is initiated by clicking "Export Assessment."

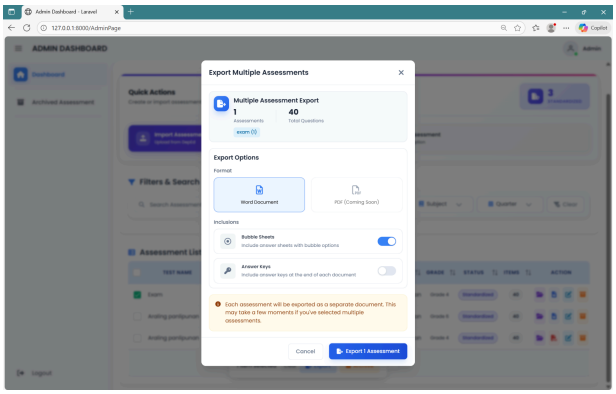


Figure 105. AsseSync Admin Dashboard - Export Assessments Modal

Figure 106 displays the Export Multiple Assessments Loading State, which appears after the administrator clicks the "Export Assessment" button. This modal provides real-time progress feedback using a loading bar to track the conversion and preparation of the documents.The administrator is informed that the generated documents will be downloaded sequentially ("Your documents will be downloaded one by one"). The loading bar tracks the progress of the operation, ensuring the user is informed throughout the export finalization process.

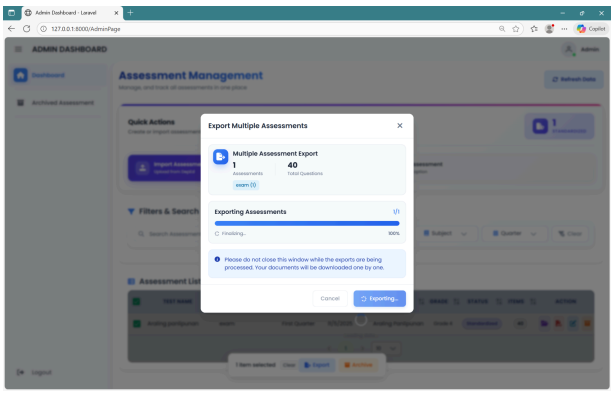


Figure 106. AsseSync Admin Dashboard - Export Assessments Loading State

Figure 107 displays the dedicated Archived Assessment Page, which allows administrators to manage, track, and restore assessments that have been moved out of the active list. This page includes a Filters & Search section, enabling the administrator to easily locate specific archived questionnaires by filtering based on Grade, Subject, or Quarter. The primary function of this interface is executed via the "Restore Selected" button, which allows the administrator to move archived assessments back to the active Admin Dashboard list.

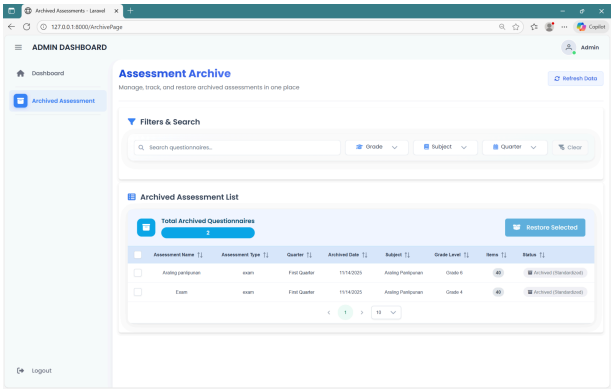


Figure 107. AsseSync Archived Assessment Page

Figure 108 displays the Restore Assessments Confirmation Modal, which appears when the administrator initiates the process of restoring one or more archived assessments. The modal requires confirmation ("Are you sure you want to restore 2 selected assessments?") and clearly explains the consequence of the action under the section "What happens when you restore?". A visual indicator further clarifies the transition from the Archived state to the Active state. The administrator confirms the action by clicking the "Restore" button.

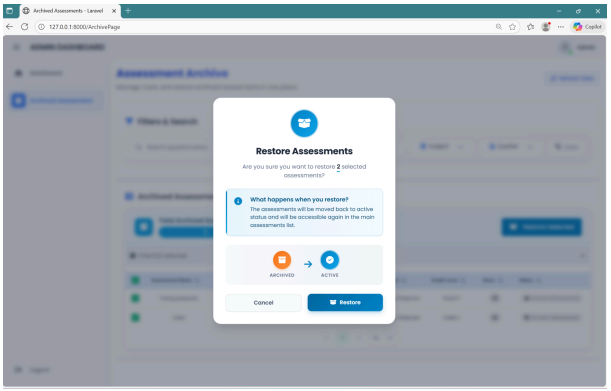


Figure 108. AsseSync Admin Restore Assessments Confirmation Modal

Figure 109 displays the Restore Assessments Loading State Modal, which appears after the administrator confirms the action to restore archived assessments. This modal provides clear, real-time feedback that the system is currently processing the request to change the status of the selected assessments. The button text changes to "Restoring..." and is accompanied by a subtle loading animation, indicating that the system is actively moving the assessment records from the Archived section back to the Active main assessments list.

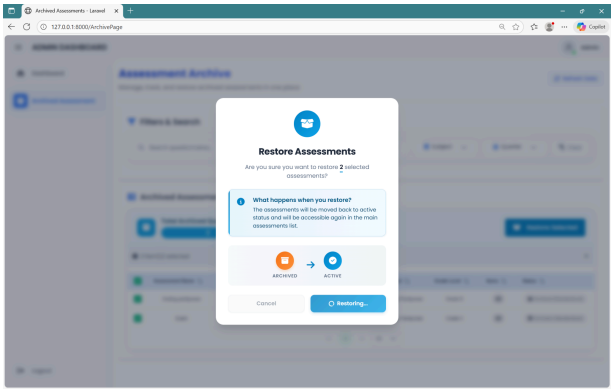


Figure 109. AsseSync Admin Restore Assessments Loading State Modal

Figure 110 displays the Teacher Dashboard of the ASSESYNC system, serving as the main Classes Page where teachers manage their assigned subjects and grade levels. Upon successful login, a toast notification confirms access ("Login Successful"). The interface features a left-hand navigation sidebar that organizes classes by Grade Level (e.g., Grade 4, Grade 5) and includes links to My Classes and Archived Classes. The main content area utilizes a card-based layout to visually represent each assigned class, displaying the Subject (Mathematics, Filipino, AP, etc.) and the School Year (2024-2025), allowing the teacher to easily select and view the details of a specific class.

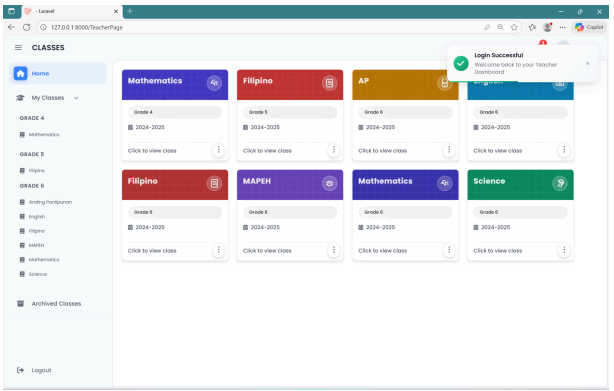


Figure 110. AsseSync Teacher Dashboard - Classes Page

Figure 111 displays the Notifications Pop-up accessible from the Teacher Dashboard via the bell icon in the top-right corner. This system keeps teachers informed of significant updates, such as the creation of new teaching materials. The pop-up shows a list of recent notifications, indicating the count of unread messages and providing an option to "Mark all as read." The example notification, "New Assessment Available," informs the teacher that an assessment has been added to the test bank for a specific grade and subject (Grade 6 Araling Panlipunan). This feature is essential for maintaining awareness of available resources within the system.

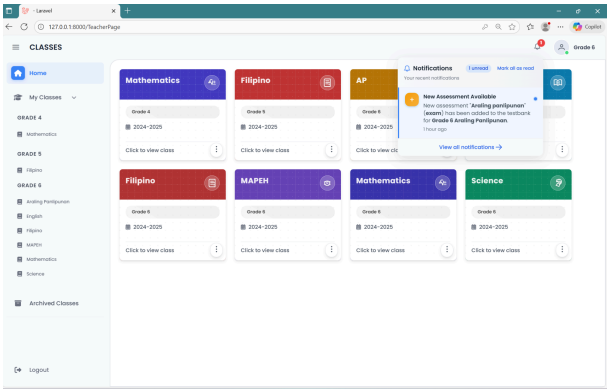


Figure 111. AsseSync Teacher Dashboard - Notifications Pop-up

Figure 112 displays the dedicated Notifications Page, accessed by clicking the "View all notifications" link from the pop-up or the navigation path. This interface serves as a central hub for the teacher to stay updated with all assessment activities. The page lists all notifications, providing details such as the event (e.g., New Assessment Available), the specific assessment involved (Araling Panlipunan exam), and the time elapsed since the alert. The teacher can manage their notifications by viewing the unread count and using the "Mark all as read" action, ensuring they maintain awareness of all relevant system updates.

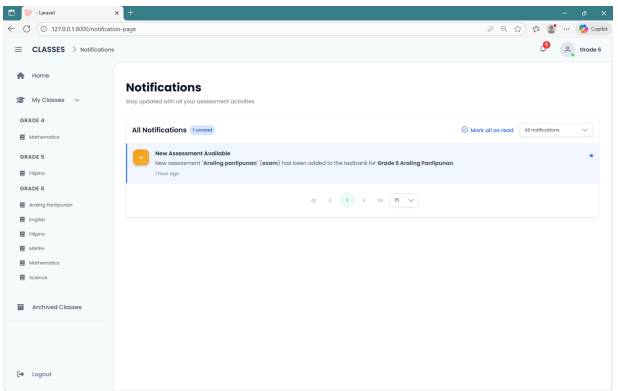


Figure 112. AsseSync Teacher Dashboard - Notifications Pop-up

Figure 113 illustrates the "Archive Class" action available to the teacher on the main Dashboard. By clicking the three-dot ellipsis menu on any class card (such as the MAPEH class shown), the teacher can select the "Archive Class" option. This function allows the teacher to move a class that is no longer active out of the main My Classes list and into the Archived Classes section (visible in the sidebar). Archiving is an essential administrative function for maintaining a focused view of the current teaching load while preserving historical data.

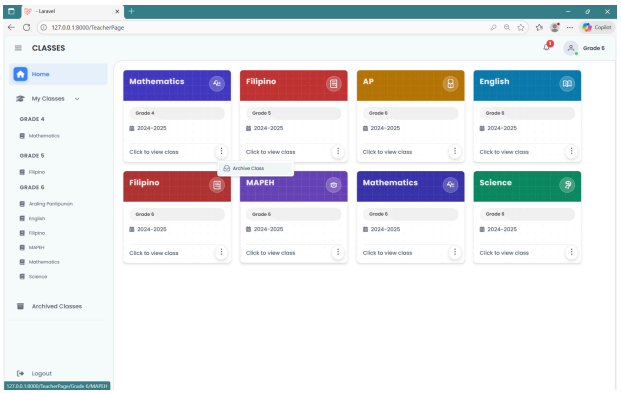


Figure 113. AsseSync Teacher Dashboard - Archive Class Action

Figure 114 displays the Archive Class Confirmation Modal, which appears when a teacher initiates the "Archive Class" action. This modal serves as a crucial warning and confirmation step. It identifies the class being archived (e.g., Grade 4 - Mathematics) and provides an Important Notice warning that archiving the class will make all associated assessments, student data, and class materials temporarily inaccessible. The notice also assures the user that "Your data is safe and can be restored." A visual indicator reinforces the transition from the Active state to the Archived state. The teacher proceeds with the action by clicking "Next".

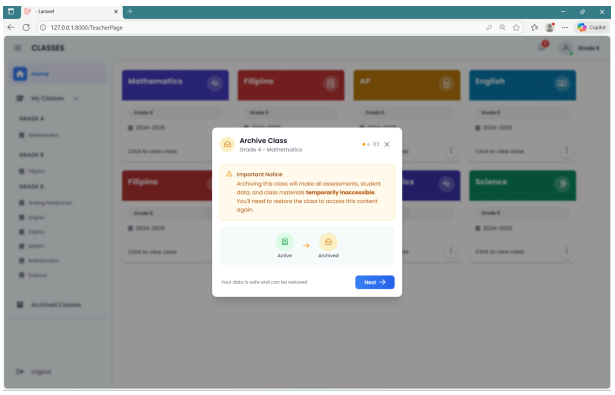


Figure 114. AsseSync Teacher Dashboard - Archive Class Confirmation Modal

Figure 115 displays the second stage of the Archive Class workflow, providing final confirmation and options for logging the reason for the action. The teacher can input an optional "Reason for archiving" (e.g., "End of school year"), which will be saved with the archived class record. A final, prominent red warning requires the teacher to explicitly "Confirm Archive Action," reiterating that the class will become temporarily inaccessible. After reviewing the details and confirming, the teacher clicks the orange "Archive Class" button to finalize the process, or clicks "Back" to return to the previous screen.

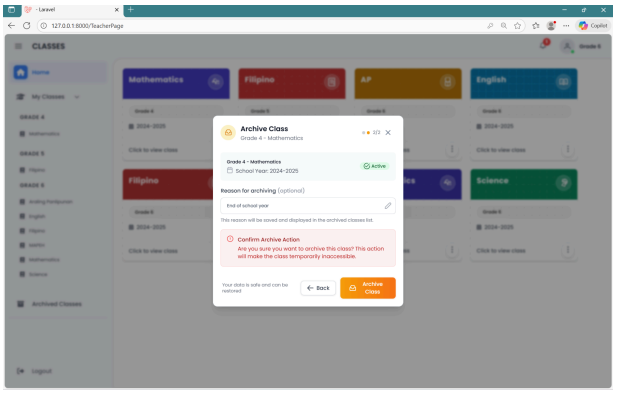


Figure 115. AsseSync Teacher Dashboard - Archive Class Confirmation and Reason Modal

Figure 116 displays the Archive Class Loading State Modal, which appears after the teacher confirms the archive action. This modal provides visual feedback that the system is currently processing the request to move the class data (including assessments and student records) to the archive. The button text changes to "Archiving..." and is accompanied by a subtle loading indicator (e.g., "2/2" progress count at the top), signifying that the system is actively moving the class record from the Active list to the Archived Classes section. The modal retains the reassurance that the data is safe and can be restored later.

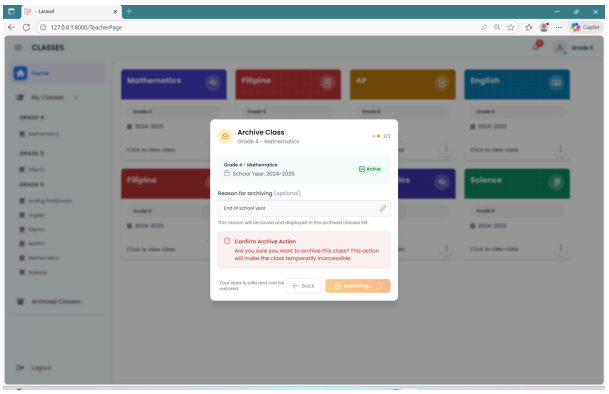


Figure 116. AsseSync Teacher Dashboard - Archive Class Loading State Modal

Figure 117 displays the dedicated Archived Classes Page for the teacher, which is used to view and manage classes that have been archived. This interface features a search bar to quickly locate specific archived classes and a sorting option (e.g., "Recently archived"). The archived classes are displayed in a card-based layout, similar to the main dashboard, but each card is clearly marked with an "Archived" badge. Key information shown includes the subject, grade level, school year, the date the class was archived, and the reason (e.g., "End of school year"). The teacher can restore these classes by viewing the class details.

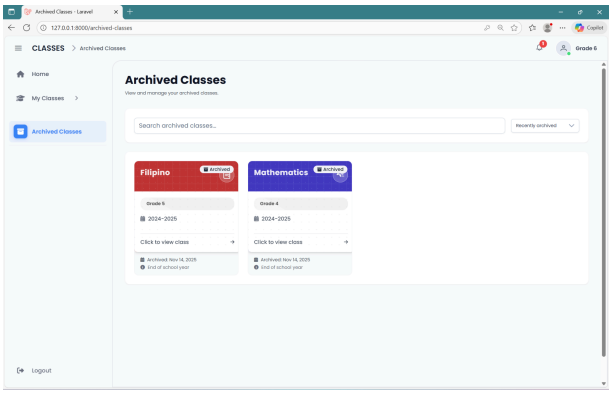


Figure 117. AsseSync Teacher Archived Classes Page

Figure 118 illustrates the Class Archived Success Toast notification, which appears on the Teacher Dashboard immediately after a class has been successfully archived. The toast provides positive feedback with a green checkmark and the message: "Class Archived: [Class Name] has been archived successfully." Concurrently, the archived class card (e.g., Grade 5 - Filipino) is removed from the main My Classes list, confirming the successful operation and updating the teacher's active class view. The archived class is now accessible via the Archived Classes link in the sidebar.

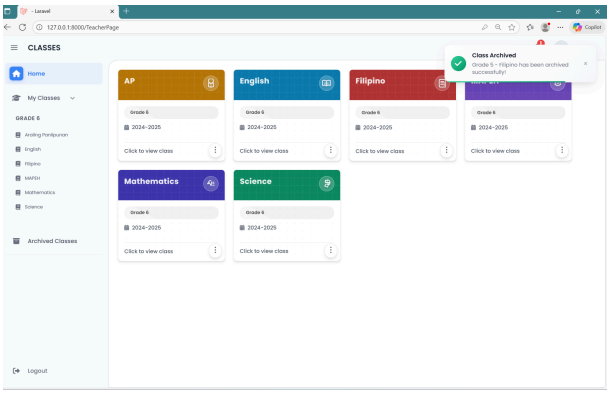


Figure 118. AsseSync Teacher Class Archived Success Toast

Figure 119 displays the Restore Class Action available on the Archived Classes Page. When a teacher hovers over an archived class card, a restoration option appears, prompting the teacher to "Restore this class to active status." This functionality is the primary method for retrieving an archived class and making its associated data (assessments, student records) accessible once again on the main Teacher Dashboard. This ensures the archiving action is reversible and allows teachers to bring back previous year's classes as needed for reference.

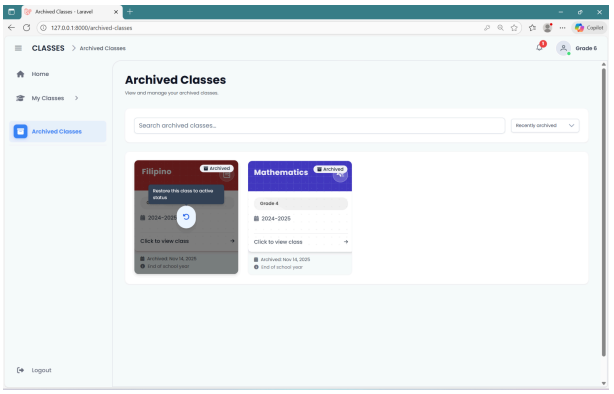


Figure 119. AsseSync Teacher Archived Classes Page - Restore Class Action

Figure 120 displays the Restore Class Confirmation Modal, which appears when the teacher initiates the restoration of an archived class. The modal clearly confirms the class being restored (e.g., Grade 5 - Filipino) and provides a green "Ready to Restore" notification. The text confirms the action's positive outcome: "This class will be restored to active status and all assessments, student data, and materials will become accessible again." A visual indicator reinforces the transition from Archived back to Active status. The teacher completes the process by clicking the "Restore Class" button, making the class immediately available on the main Teacher Dashboard.

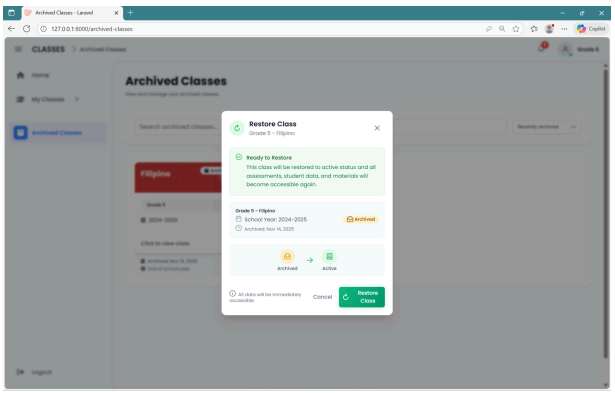


Figure 120. AsseSync Teacher Archived Classes Page - Restore Class Action

Figure 121 displays the Restore Class Loading State Modal, which appears as the system executes the command to restore an archived class. The modal indicates that the restoration process is actively underway. The button text changes to "Restoring..." and is accompanied by a subtle loading animation, signifying that the system is currently moving the selected class record (e.g., Grade 4 - Mathematics) from the Archived section back to the Active main classes list. The modal retains the assurance that the class data and materials will soon be immediately accessible on the Teacher Dashboard.

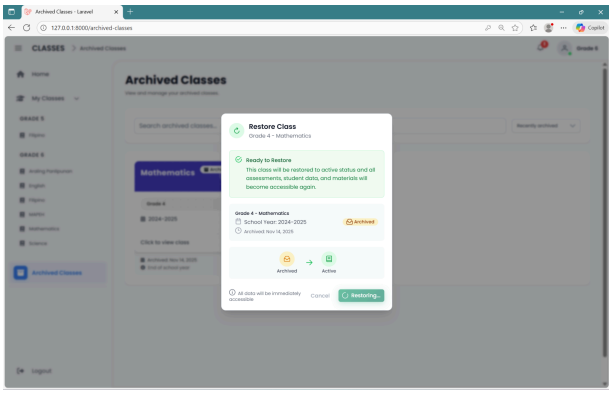


Figure 121. AsseSync Teacher Archived Classes Page - Restore Class Loading State Modal

Figure 122 illustrates the successful completion of the class restoration process. A toast notification appears in the top-right corner, confirming the operation with a green checkmark and the message: "Class Restored: (Class Name) has been restored successfully!" Concurrently, the restored class card (e.g., Grade 5 - Filipino) is removed from the Archived Classes list, updating the view to show only the remaining archived items. This confirms that the class has been successfully moved back to the My Classes section of the Teacher Dashboard and is fully accessible to the teacher.

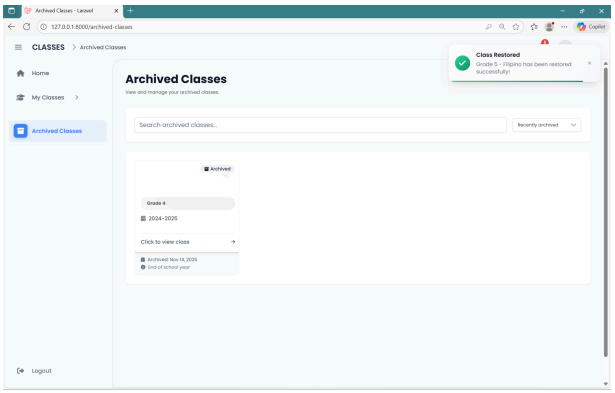


Figure 122. AsseSync Teacher Archived Classes Page - Class Restored Success Toast and Archive Update

Figure 123 A displays the Teacher Subject/Classwork Page, which appears after a teacher clicks on a specific class card (e.g., Grade 4 - Mathematics) from the main dashboard. This serves as the primary workspace for managing course content. The header prominently displays the class details, including the number of Assessments. The interface is structured with tabs for different modules: Classwork, Testbank, Item Analysis Results, and Class List. The Classwork tab is further organized by Quarter, allowing the teacher to view and manage assessments for the First, Second, Third, and Fourth Quarters. The page provides key actions: "Create Topic" and "Create Assessment" to populate the class content.

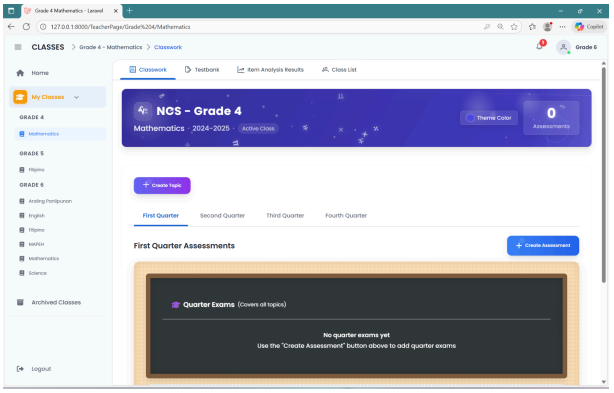


Figure 123.A. AsseSync Teacher Subject/Classwork Page

Figure 124 B displays the initial, empty state of the Teacher Subject/Classwork Page for a selected quarter (First Quarter). This interface indicates that no content has been created yet. Prominent messages guide the teacher to the next steps: the First Quarter Assessments section displays a notice stating "No quarter exams yet," prompting the use of the "Create Assessment" button. Similarly, the topics section indicates "No Topics Yet," instructing the teacher to "Create your first topic" using the "Create Topic" button located at the top of the interface. This design effectively encourages the teacher to begin populating the course with content.

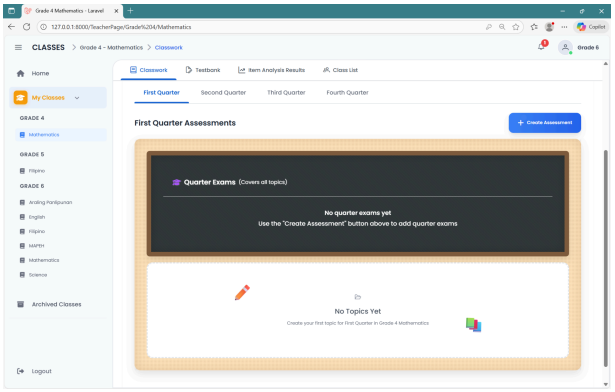


Figure 124.B. AsseSync Teacher Subject/Classwork Page

Figure 125 displays the Create New Topic Modal, which appears when the teacher clicks the "Create Topic" button on the Subject/Classwork Page. This modal is the interface for establishing a new instructional unit within the class. It requires the teacher to input a Topic Title (a descriptive name) and select the corresponding Quarter from a dropdown menu, ensuring the topic is correctly organized within the course structure. An optional Description field is provided for additional context. A key instruction confirms the context: "This topic will be created for Grade 4 Mathematics." The process is finalized by clicking the "Create Topic" button.

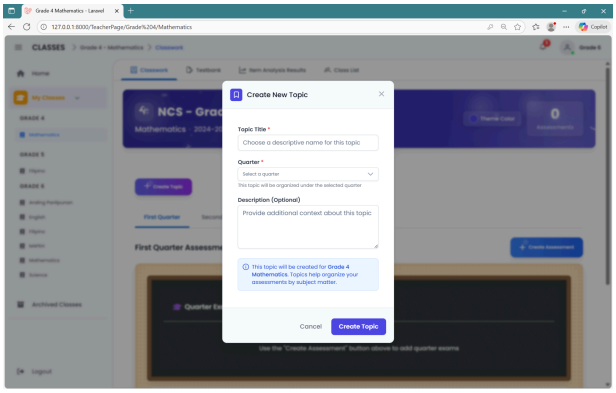


Figure 125. AsseSync Teacher Create New Topic Modal

Figure 126 illustrates the Select Assessment Type Menu that appears when the teacher clicks the "Create Assessment" button on the Classwork Page. This menu prompts the teacher to choose the format of the new assessment being created. The available options allow the teacher to define different types of examinations, including Quiz, Test, and Quarter Exam. This selection defines the nature and scope of the assessment, guiding the teacher to the appropriate subsequent creation interface for the chosen assessment type.

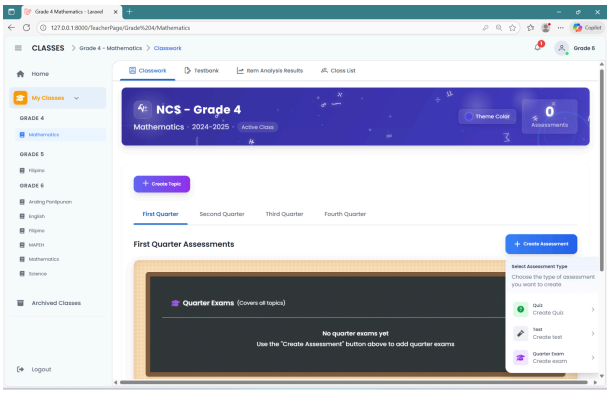


Figure 126. AsseSync Teacher Select Assessment Type Menu

Figure 127 displays the Pick a Topic Modal, which appears during the assessment creation process. This modal requires the teacher to link the assessment to a previously created instructional topic (e.g., "Rizal"), ensuring organizational clarity. The interface features a Search bar to quickly locate a specific topic, and it allows filtering between All Topics and the currently selected Quarter (First Quarter). Once the desired topic is selected, the assessment is categorized under that unit, and the teacher proceeds by clicking the "Select" button.

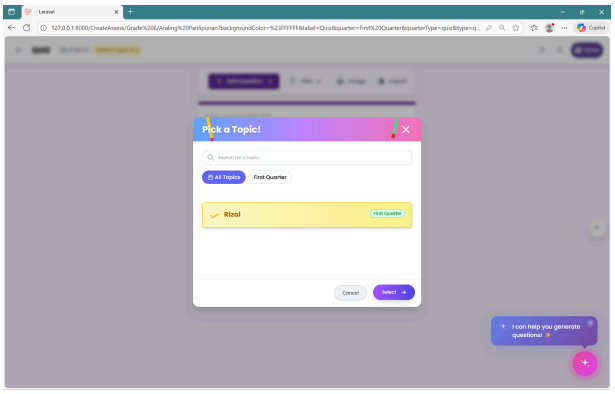


Figure 127. AsseSync Teacher Assessment Creation Interface - Pick a Topic Modal

Figure 128 displays the main interface for the teacher when creating a new assessment (labeled as QUIZ in the header). This editing screen allows the teacher to build the questionnaire content, providing fields for the overall assessment Title and Description. Crucially, the interface for each question includes a robust rich-text editor toolbar. This toolbar provides standard formatting options such as Bold (B), Italic (I), Underline (U), text alignment, and advanced scientific/mathematical options. This allows teachers to precisely format the question prompt and answer options for clarity before defining the Multiple Choice answers.

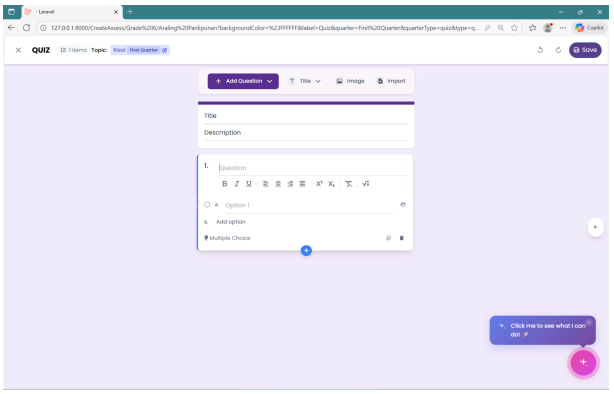


Figure 128. AsseSync Teacher Assessment Creation Interface - Question Editor

Figure 129 illustrates the "Add Question" dropdown menu within the teacher's assessment creation interface (for a QUIZ linked to the Rizal topic). By clicking this button, the teacher can choose the format of the question they wish to insert into the assessment. The primary option visible is "Multiple Choice," which adds a question block requiring the definition of a prompt, multiple options, and the designation of a single correct answer. This feature is fundamental for manually building the assessment content item by item.

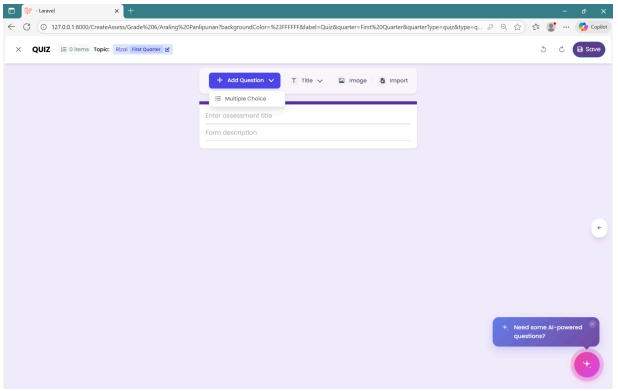


Figure 129. AsseSync Teacher Assessment Creation Interface - Add Question Menu

Figure 130 displays the assessment editing interface immediately after the teacher selects to add a Multiple Choice question. A dedicated block is inserted into the document, numbered sequentially (e.g., "1."). This block provides the necessary structure for the teacher to input the Question prompt and the answer options (starting with "Option 1"). The selected answer type, "Multiple Choice," is labeled clearly at the bottom of the question block. This element forms the basic unit for building the main content of the quiz or exam.

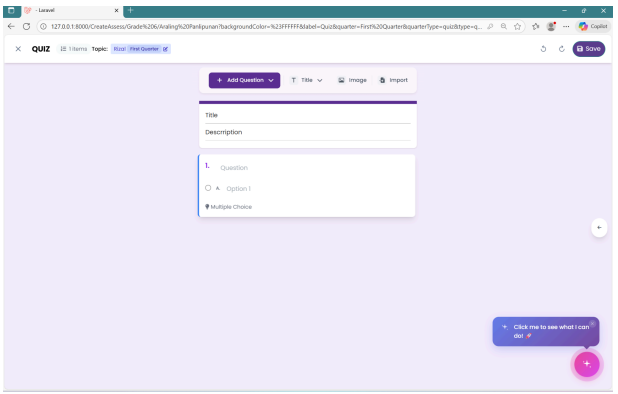


Figure 130. AsseSync Teacher Assessment Creation Interface - Multiple Choice Question Block

Figure 131 illustrates the options available to the teacher when clicking the "Title" button on the assessment creation toolbar. This functionality is used to add structural headers to the quiz or exam. The dropdown menu provides two distinct organizational elements: "Title" (for major section headers) and "Section Title" (for sub-headings). This allows the teacher to effectively divide the assessment into logical parts and provide specific instructions or context for different sets of questions, enhancing the test's structure and clarity.

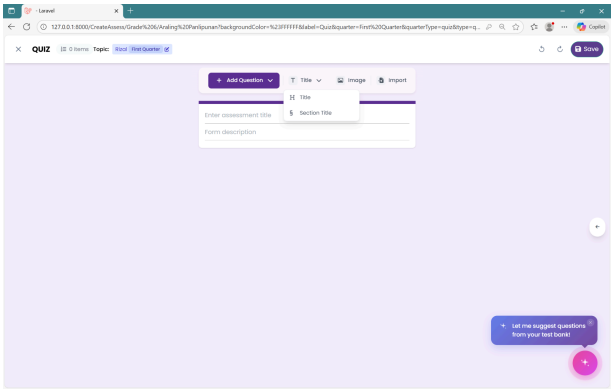


Figure 131. AsseSync Teacher Assessment Creation Interface - Title Menu

Figure 132 illustrates the result of inserting both the Title and Section Title elements into the teacher's assessment document. Both elements provide dedicated input fields for a title (initially labelled "Untitled Title") and an optional Description. The Section Title element is clearly labeled and numbered (e.g., "Section 1 of 1"), indicating its subordinate role to the main title. This dual-level structuring capability allows the teacher to organize the quiz or exam into hierarchical sections, ensuring comprehensive and easy-to-follow instructions for test-takers.

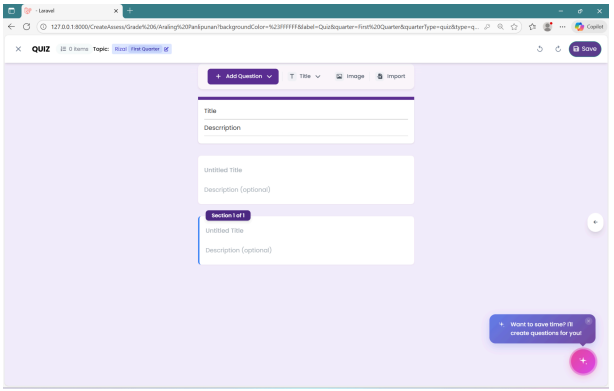


Figure 132. AsseSync Teacher Assessment Creation Interface - Title and Section Title Elements

Figure 133 illustrates the Add Image feature within the teacher's Assessment Creation Interface, accessed by clicking the "Image" dropdown button and selecting "Add Image." This functionality inserts a dedicated image block into the assessment, providing fields for an Image Title and a drag-and-drop or click-to-upload area. The system supports standard formats (PNG, JPG, GIF) and specifies a maximum file size limit of up to 10MB. This feature allows teachers to easily embed visual aids, charts, maps, or diagrams into their quizzes or exams to support question prompts or provide supplementary context.

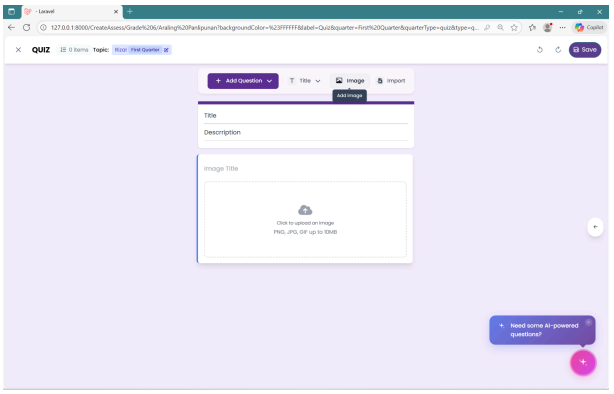


Figure 133. AsseSync Teacher Assessment Creation Interface - Add Image Feature

Figure 134 illustrates the Import feature within the teacher's assessment creation interface, which opens the Test Bank sidebar. This functionality allows the teacher to efficiently reuse approved questions from standardized assessments available in the institution's repository. The sidebar enables the teacher to search assessments and displays a list of available examinations, showing key contextual data such as the subject (Araling Panlipunan), grade level (Grade 6), and the number of questions. This tool significantly accelerates the process of building high-quality, pre-validated assessment content by importing entire exams or selected questions.

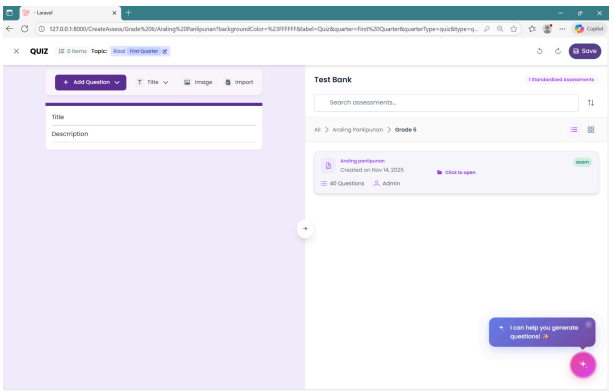


Figure 134. AsseSync Teacher Assessment Creation Interface - Import from Test Bank

Figure 135 displays the Question Selection Modal, which appears when the teacher selects an assessment from the Test Bank sidebar. This modal allows for the selective importing of questions into the current assessment document. It presents a detailed preview of all questions within the source assessment, including the question number, type (Multiple Choice), and the correct answer key (highlighted in green). The teacher can choose questions individually or use the "Select All" function. The total number of selected items is displayed, and the chosen questions are added to the quiz by clicking the "Import Selected" button.

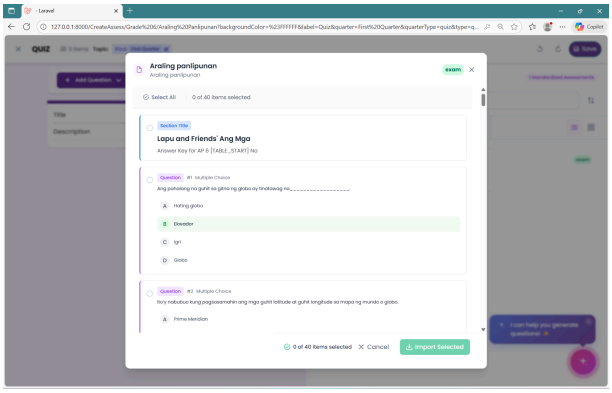


Figure 135. AsseSync Teacher Import Assessment Question Selection Modal

Figure 136 showcases the Artificial Intelligence (AI) features accessible to the teacher via the floating action button in the Assessment Creation Interface. This functionality offers two distinct options to accelerate content creation: "AI Create Questions" and "AI Suggest Questions from Test Bank." The "AI Create Questions" option initiates the generative process, creating entirely new questions based on the assessment's subject and grade level. Conversely, the "AI Suggest Questions from Test Bank" option employs AI to intelligently filter and propose relevant questions from the institution's existing Test Bank.

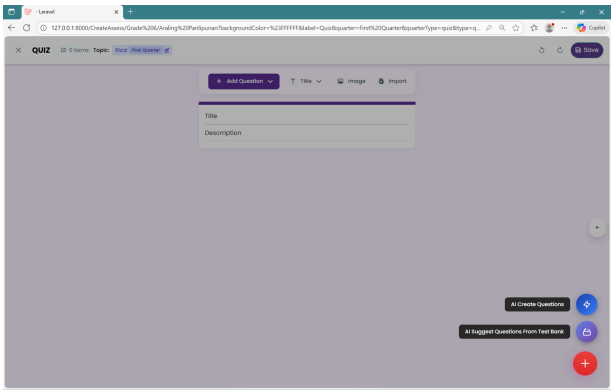


Figure 136. AsseSync Teacher Assessment Creation Interface - AI Question Generation Options

Figure 137 displays the AI Test Bank Search Modal, which is activated when the teacher selects the "AI Suggest Questions from Test Bank" option. This modal allows the teacher to configure the search parameters for AI question suggestions. The teacher must then define the desired Question Quantity (up to 50 per search) and filter the results by Difficulty Level (Easy, Medium, Hard, or Any). The modal also provides a Test Bank Overview, showing the inventory of available questions based on complexity, before the teacher initiates the search by clicking "Search Test Bank."

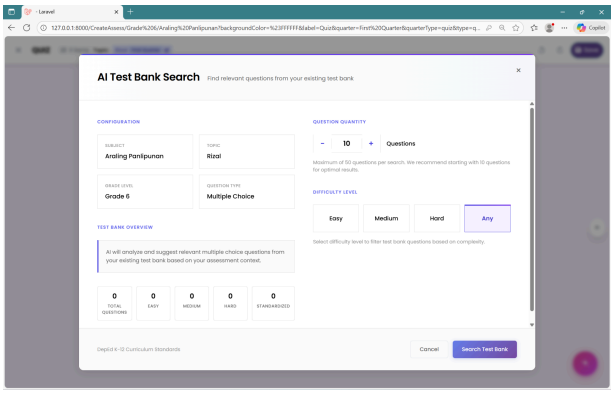


Figure 137. AsseSync Teacher Assessment Creation Interface - AI Test Bank Search Modal

Figure 138 displays the Assessment Question Generator Modal, which is used when the teacher selects the "AI Create Questions" option. This modal is the interface for configuring the parameters of the generative AI to create entirely new questions. The teacher defines the desired Question Quantity (up to 10 per generation) and the required Cognitive Complexity (Easy, Medium, or Hard), based on Bloom's Taxonomy. The modal reassures the user that generation is aligned with DepEd K-12 Curriculum Standards. The process is initiated by clicking the "Generate Questions" button.

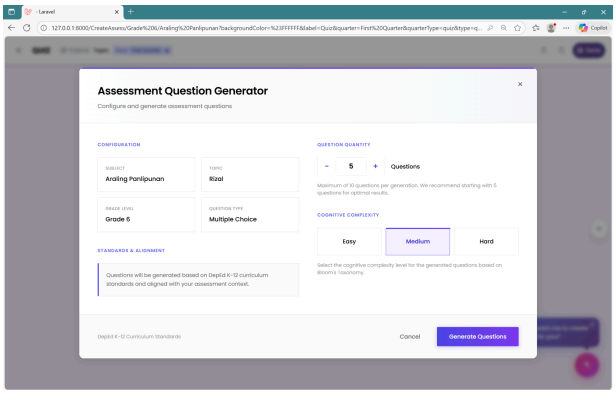


Figure 138. AsseSync Teacher Assessment Creation Interface - Question Generator Modal

Figure 139 illustrates the AI Generated Questions Review Interface after the teacher utilizes the generative AI tool. The interface presents a dedicated block of questions, clearly labeled "AI Generated Questions," ready for acceptance or rejection. For each question, the teacher can view the question prompt, the multiple-choice options, and the AI-determined correct answer (highlighted in green, e.g., Option C, "Jose Rizal"). The teacher has granular control through the "Accept" and "Reject" buttons beneath each question block, allowing for precise curation. Bulk actions are also available via the "Accept All" and "Reject All" buttons at the top of the block.

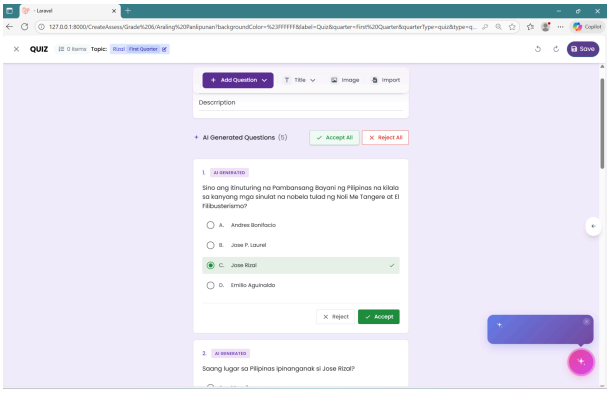


Figure 139. AsseSync Teacher Assessment Creation Interface - AI Generated Questions Review

Figure 140 displays the Open Assessment Loading Progress Modal, which appears when a user attempts to open an existing assessment from the dashboard or classwork page. This modal provides a transparent, step-by-step view of the system's process for preparing the assessment interface. The progress list details the sequence of required operations: "Fetching assessment," "Loading questions," "Preparing components," and "Finalizing." This detailed progress bar and checklist effectively informs the user that the system is actively retrieving the data and setting up the editing workspace.

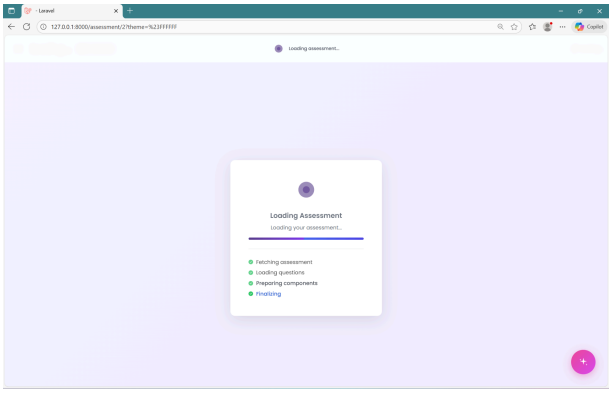


Figure 140. AsseSync Teacher Open Assessment Loading Progress Modal

Figure 141 displays the Version History Modal, which allows users to manage and switch between different versions of a questionnaire. This feature is vital for auditing and controlling assessment integrity. The modal clearly lists the "Main Assessment" (which is currently Active) and displays previous versions (e.g., Version 2, Version 1), along with key details such as creation context and the number of items. For any previous version, the user is provided with two powerful actions: "Switch to vX" to load that version for temporary review, and the orange "Use as Main Version" button, which allows the user to officially revert the active assessment to a selected prior state.

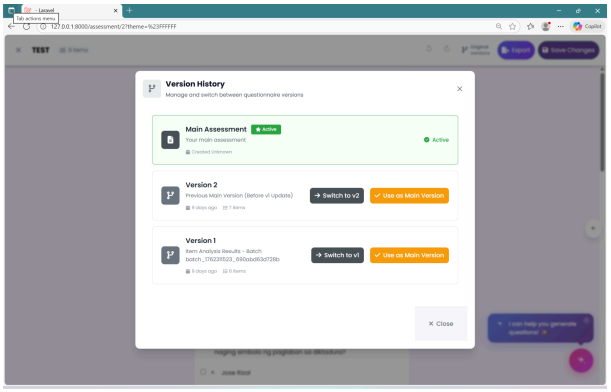


Figure 141. AsseSync Teacher Assessment Version History Modal

Figure 142 displays the Assessment Use Version Confirmation Modal, which appears when a user clicks the "Use as Main Version" button within the Version History modal. This final confirmation step is crucial due to the severity of the action. The modal clearly outlines "What will happen".

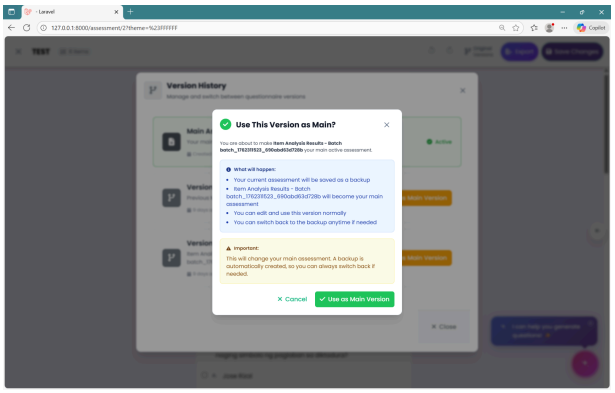


Figure 142. AsseSync Teacher Assessment Use Version Confirmation Modal

Figure 143 displays the Delete Topic Warning Modal, which appears when a teacher initiates the deletion of an instructional topic (e.g., "Addition"). This modal serves as a critical final safeguard, requiring confirmation before executing a permanent action. The warning explicitly states: "This topic will be permanently deleted and cannot be recovered. You can't undo this action." This ensures the teacher is fully aware of the irreversible nature of the deletion. The teacher must confirm the action by clicking the red "Delete topic" button, or can abort the operation by clicking "Cancel."

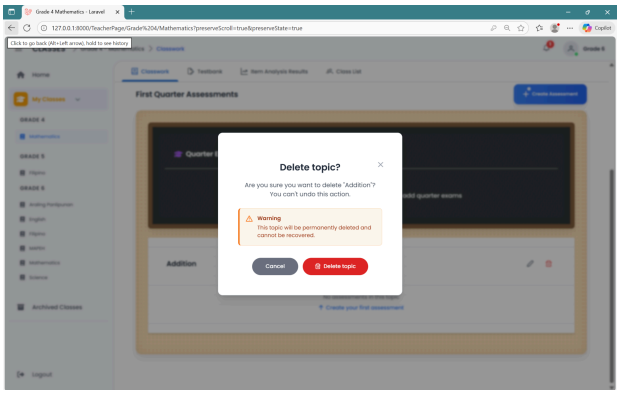


Figure 143. AsseSync Teacher Delete Topic Warning Modal

Figure 144 displays a critical Warning Toast Notification that appears when a teacher attempts to delete the last remaining topic within a quarter (e.g., "Addition" from First Quarter). The warning informs the user that this action will have a consequence for assessment creation: "Deleting '[Topic]' will remove the last topic from First Quarter. You won't be able to create assessments until you add topics to all quarters again." This system safeguard prevents the teacher from inadvertently removing the structural basis required for organizing new assessments, ensuring that all subsequent exams or quizzes are properly linked to an instructional topic.

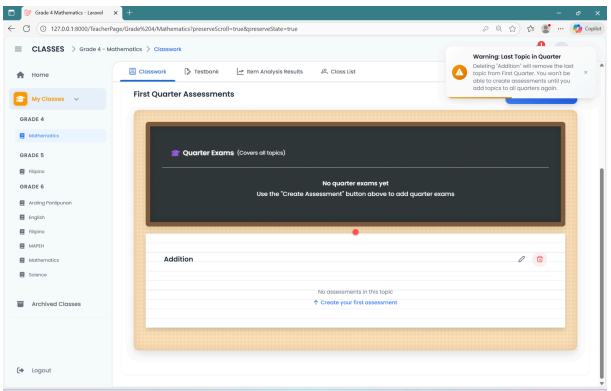


Figure 144. AsseSync Teacher Delete Topic Warning Toast Notification

Figure 145 displays the Choose Evaluation Method Modal, which prompts the teacher to select how student responses will be processed and scored. The modal presents two distinct methods which is the “Scan Bubble Sheets” as the main feature and “Manual Input” as the alternative feature.

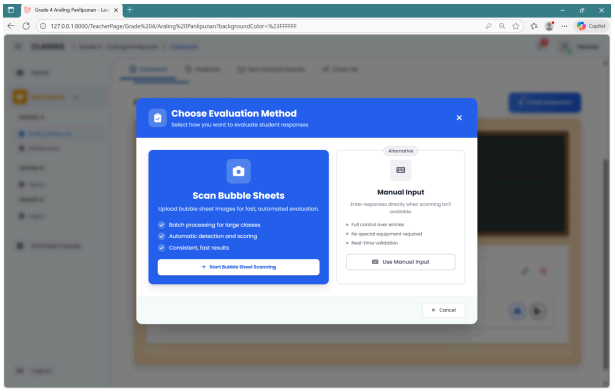


Figure 145. AsseSync Teacher Choose Evaluation Method Modal

Figure 146 displays the Upload Responses - Bubble Sheet Scanning Modal, which appears when the teacher selects the automated evaluation method. This modal is the interface for uploading and scanning student bubble sheet images. The header confirms the target Questionnaire details, including the subject (Araling Panlipunan), grade, and item count (40 Item(s)). The teacher must first select the corresponding Class Section from a dropdown menu. The core functionality is the Upload bubble sheet images area, which supports PNG, JPG, and JPEG formats. Once the images are selected, the teacher initiates the automated scoring process by clicking "Upload Bubble Sheets".

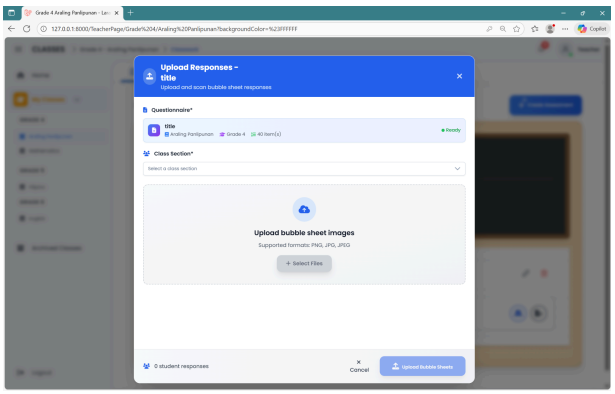


Figure 146. AsseSync Teacher Upload Responses - Bubble Sheet Scanning Modal

Figure 147 displays the Bubble Sheet Image Validation Loading State, which appears after the teacher initiates the bubble sheet scanning process. This modal provides transparent, real-time feedback that the system is actively processing the uploaded images. The loading screen displays a message indicating that AI is analyzing the bubble sheets to ensure image quality and compatibility for scoring. A progress bar and numerical counter track the status (e.g., "0/2 Images validated" at "33% complete"), managing the teacher's expectation. The teacher can choose to stop the process using the "Cancel Validation" button.

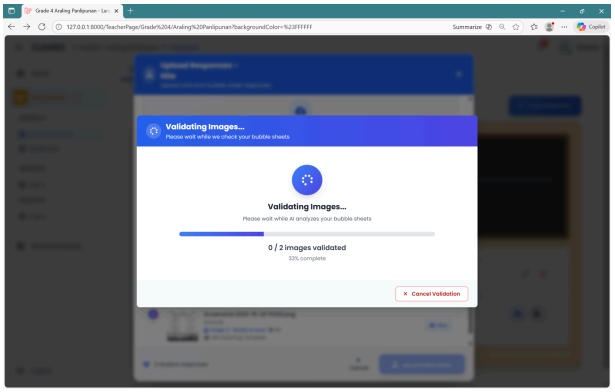


Figure 147. AsseSync Teacher Upload Responses - Bubble Sheet Image Validation Loading State

Figure 148 illustrates the Successful Image Validation Modal after the AI processing is complete. The modal displays a checklist for each file, showing specific confirmation points such as "valid bubble sheet," "Quality: excellent," and confirmation of the question count. Once validated, the teacher proceeds to the next stage of the evaluation process by clicking the "Continue to Student Detection" button.

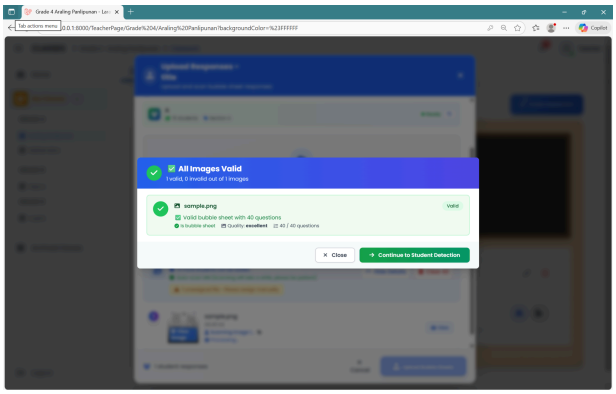


Figure 148. AsseSync Teacher Upload Responses - Successful Bubble Sheet Validation Modal

Figure 149 displays the Invalid Image Validation Modal, which appears when one or more uploaded bubble sheets fail the AI processing checks. A warning banner alerts the user: "Some Images Need Attention," and an "Action Required" notice instructs the teacher to please remove or replace the invalid images. The specific issue for the failed image is detailed in the "Issues found" section (e.g., "Barcode mismatch"), providing the necessary information for remediation. The teacher can choose to "Remove invalid images to continue" and then click "Go Back & Remove Invalid" to proceed with only the valid sheets.

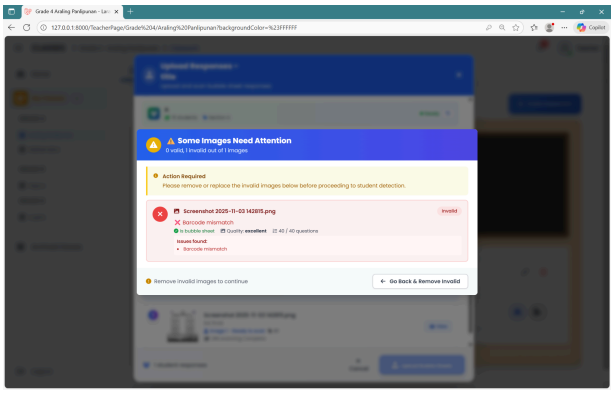


Figure 149. AsseSync Teacher Upload Responses - Successful Bubble Sheet Validation Modal

Figure 150 illustrates the Upload Responses Modal after an image has been successfully uploaded and automatically processed for student assignment. The modal confirms the successful operation, showing "1 image selected (1/15 students)" and displaying the status "All files assigned - Ready to upload." The auto-scan feature successfully identified the student associated with the file (e.g., "Caroline Jacobson Word") using the unique identifier (such as the LRN or barcode) printed on the bubble sheet, confirming the scanning is complete. The teacher can now review the assigned file before clicking "Upload Bubble Sheets" to finalize the scoring process.

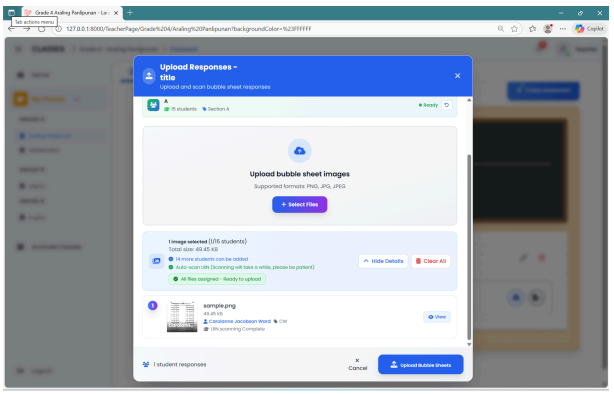


Figure 150. AsseSync Teacher Upload Responses - Automatic Student Assignment State

Figure 151 displays the Finalizing Results Loading Modal, which represents the last phase of the automated evaluation and scoring process. This modal informs the teacher that the system is currently preparing the detailed analysis report. It provides a clear progress breakdown, showing the current state as "Phase 5 of 6" and displaying a percentage completion on the progress bar. The text "Finalizing results... Preparing detailed analysis report" confirms that the scoring is complete and the system is generating the comprehensive data required for the Item Analysis module. The teacher has the option to "Cancel Upload" if necessary, though the process is near completion.

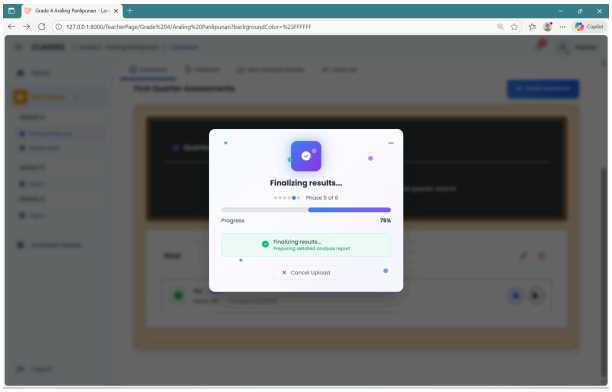


Figure 151. AsseSync Teacher Upload Responses - Finalizing Results and Item Analysis Loading Modal

Figure 152 displays the Scanning Complete Modal, which appears after the automated scoring process (including validation and analysis) is finished. The modal confirms the successful operation with the message "Bubble sheet has been successfully processed." It provides a Processing Summary, confirming the number of students scanned and the number of questions scored. The teacher selects "Yes, Analyze" to view the report or "No, Skip Analysis" to return to the Classwork page.

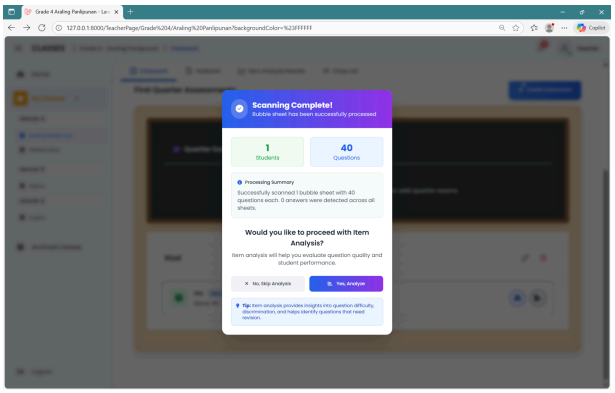


Figure 152. AsseSync Teacher Upload Responses - Scanning Complete and Item Analysis Prompt Modal

Figure 153 displays the initial interface for the Manual Input Evaluation method, which is selected as an alternative to automated bubble sheet scanning. This page allows the teacher to directly enter student responses into the system. The header confirms the assessment details, including the Grade, Subject (Araling Panlipunan), and 40 Items. The primary step required to begin is to Select Class Section from the dropdown menu. A prompt confirms the empty state: "No Section Selected," guiding the teacher to choose a section from the dropdown above to start inputting student responses. This page also includes a "Load Saved Progress" button for retrieving previously started evaluation sessions.

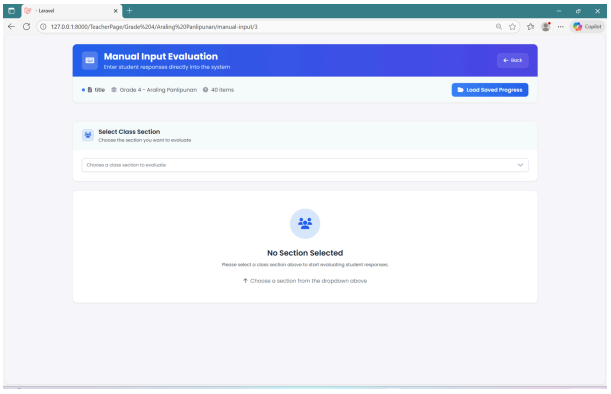


Figure 153. AsseSync Teacher Manual Input Evaluation Page

Figure 154 displays the Manual Input Evaluation - Student Answer Entry Modal, which opens after the teacher selects a class section. This interface is the main tool for directly recording a student's responses into the system. The modal clearly identifies the student being evaluated and indicates the progress . The interface presents a grid layout of all questions, where the teacher clicks the corresponding multiple-choice option or "No Answer." Shortcut keys are provided for efficiency. The interface also features "Auto-saving responses" and buttons to navigate between students.

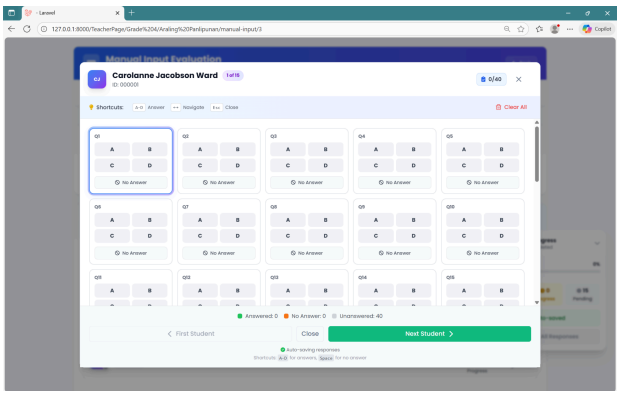


Figure 154. AsseSync Teacher Manual Input Evaluation - Student Answer Entry Modal

Figure 155 displays the All Questions Answered Confirmation Modal, which appears when the teacher has finished manually inputting all responses for a student. The modal confirms the status with the message "All Questions Answered!" and shows the completion count (40/40 Completed). A crucial prompt instructs the teacher to double-check their inputs and confirms that edits can still be made before proceeding. The teacher is given two options: "Review Responses" to re-examine the answers, or "Proceed to Next Student" to save the current student's data and move on to the next entry in the class list.

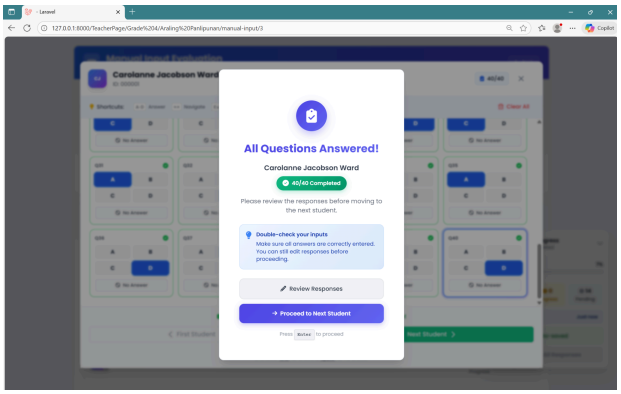


Figure 155. AsseSync Teacher Manual Input Evaluation - All Questions Answered Confirmation Modal

Figure 156 displays the Submit All Responses Confirmation Modal, which appears when the teacher is ready to finalize and submit all manually evaluated student responses for analysis. The modal provides a Progress Summary, confirming the Total Students and the number Completed, with a 100% Completion Rate. Item analysis will be generated automatically, and the teacher will be redirected to the results page. A final note confirms that all pending changes will be saved automatically before submission. The teacher confirms the action by clicking the blue "Submit All Responses" button.

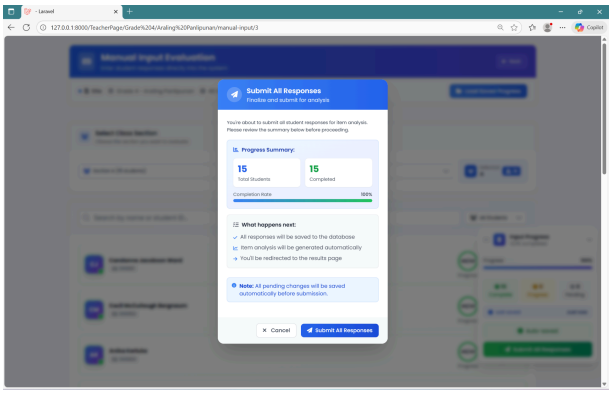


Figure 156. AsseSync Teacher Manual Input Evaluation - Submit All Responses Confirmation Modal

Figure 157 displays the Export Assessment Options Modal, which appears when a teacher initiates the export action for an assessment. This modal allows the teacher to configure the output settings before downloading the document(s). The teacher first selects the target Class Section; the system confirms that student names will be assigned automatically to the exported documents. Under Export Options, the teacher chooses the output Format, with Word Document currently available. The Inclusions section offers two important toggles: Bubble Sheets and Answer Keys. The export process is initiated by clicking the blue "Export Assessment" button, and the files will be downloaded automatically.

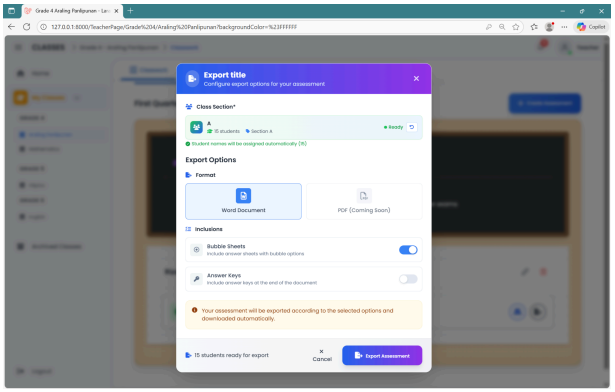


Figure 157. AsseSync Teacher Export Assessment Options Modal

Figure 158 displays the Export Assessment Generation Loading State, which appears after the teacher initiates the export action. This modal provides transparent, step-by-step feedback on the process of creating the downloadable document. The progress bar at the bottom tracks the overall completion, while a sequence of steps: Prepare, Generate, Finalize, and Download is illuminated as the process advances. A real-time message indicates the current sub-task, assuring the user that the system is actively generating the document based on the configured export options. The button text changes to "Generating..." to indicate the active state.

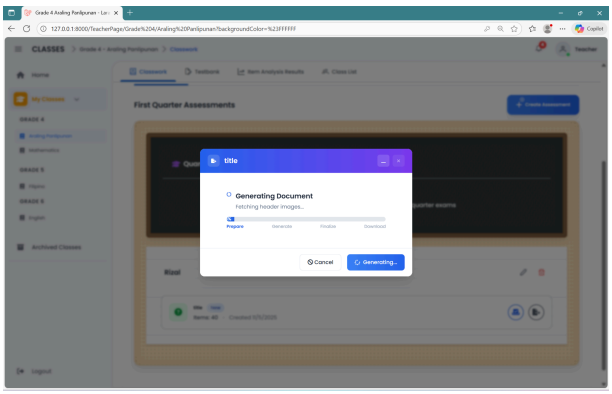


Figure 158. AsseSync Teacher Export Assessment Generation Loading State

Figure 159 displays the Item Analysis Results Page, which serves as the central hub for teachers to access and review Assessment Insights for their class. The interface allows teachers to filter and search analysis reports based on Status, Quarter, or Date. A comprehensive Overview section tracks the status of all analyses, detailing the number of reports that are Completed, In Progress, Pending, and the Total assessments requiring analysis. Below the overview, a list of processed assessments is displayed, showing the Topic/Title, Quarter, date, number of students, and number of items. Teachers can easily identify completed analyses and access the detailed insights for each specific assessment.

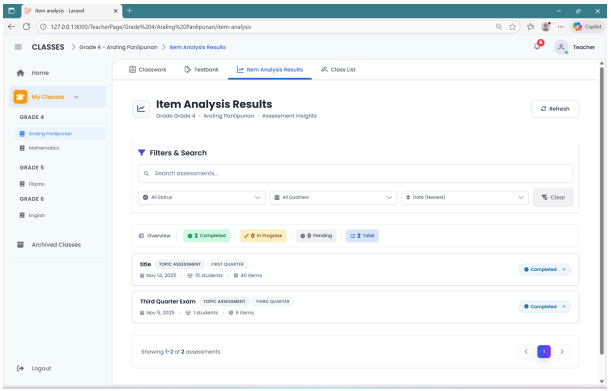


Figure 159. AsseSync Teacher Item Analysis Results Page

Figure 160 displays the Assessment Analysis Page, which provides a comprehensive scan of results and item analysis after student responses have been processed. The interface is organized into tabs: Student Scores and Item Analysis. The Analysis Overview confirms that analysis is completed for all 40 questions.

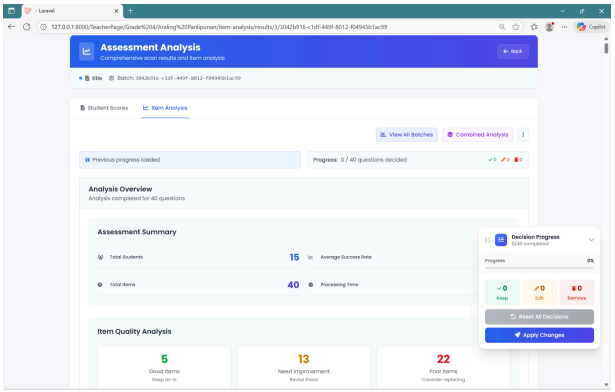


Figure 160. AsseSync Teacher Assessment Analysis - Item Analysis Page

Figure 161 displays the Item Quality Analysis section of the detailed results page. This interface provides an immediate statistical overview of the entire assessment's item quality. The system classifies items into three color-coded categories based on performance and statistical metrics.

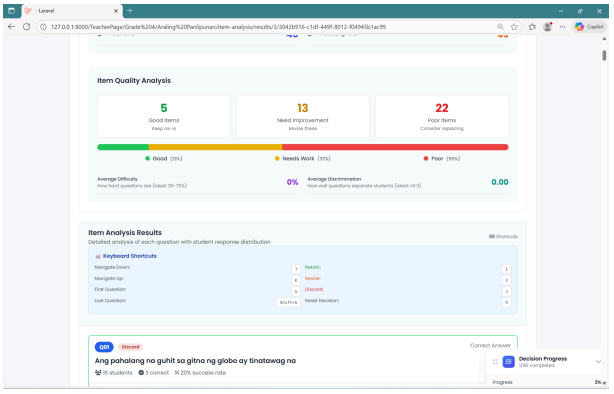


Figure 161. AsseSync Teacher Assessment Analysis - Item Quality Analysis Summary

Figure 162 illustrates the Detailed Question Analysis section, providing metrics for a single item (e.g., Q01). This view allows the teacher to make specific, data-driven decisions on each question.

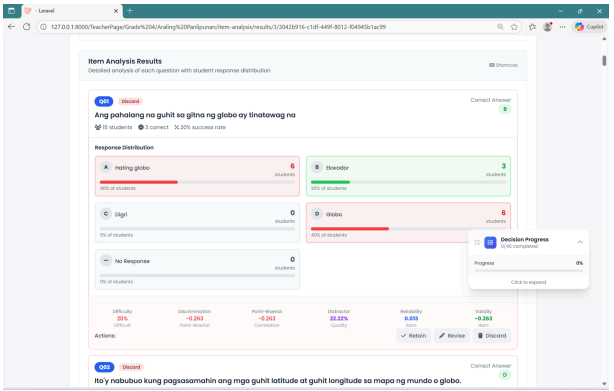


Figure 162. AsseSync Teacher Assessment Analysis - Detailed Question Response and Statistical Analysis

Figure 163 illustrates the Retain action taken on a specific question within the Item Analysis Results. When the teacher clicks the green "Retain" button, it signifies the decision to keep the question in the assessment without any changes. A confirmation message, "Retained as-is," appears below the action buttons. This decision is typically made for questions classified as Good Items based on favorable Difficulty and Discrimination statistics, confirming their validity for future use in the assessment. The action is then logged in the Decision Progress panel.

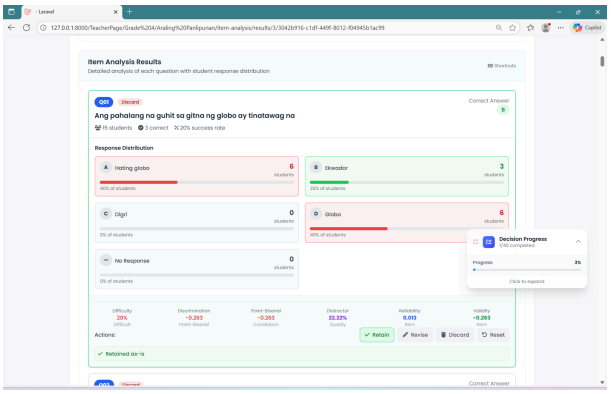


Figure 163. AsseSync Teacher Assessment Analysis - Retain Action

Figure 164 depicts the Revise action, used when the item analysis suggests a question needs improvement. When the teacher clicks the orange "Revise" button, the question block expands to activate the editing interface. This allows the teacher to directly modify the question text, correct the answer, or adjust the distractor options (as shown by the visible rich-text editor toolbar). A "Pending revision confirmation" message appears, and the teacher must click "Confirm Revision" to finalize the changes to the question. This action is used to correct statistical flaws like poor distractor quality or highly imbalanced difficulty.

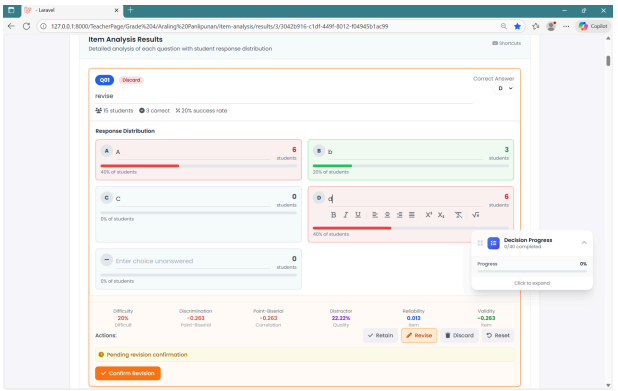


Figure 164. AsseSync Teacher Assessment Analysis - Revise Action

Figure 165 illustrates the Discard action, which is used when the item analysis indicates a question is poor and should be replaced. When the teacher clicks the red "Discard" button, the interface presents the option to "Create Replacement Question." This feature provides input fields for Question Text and four answer choices (A-D) with a designated Correct Answer dropdown. This action allows the teacher to immediately replace the discarded item with a new, manually created question, ensuring the assessment maintains the required number of items. The replacement is finalized by clicking "Confirm Replacement Question."

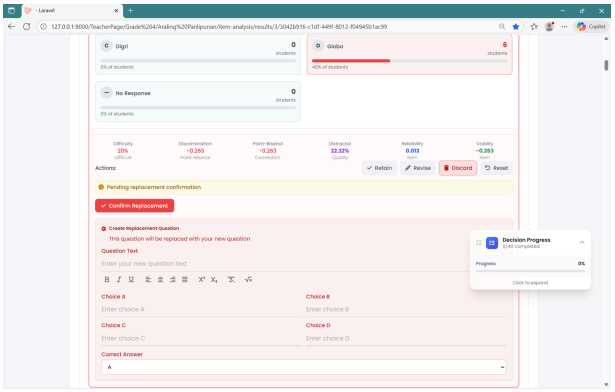


Figure 165. AsseSync Teacher Assessment Analysis - Discard Action

Figure 166 shows the Reset action, which is a utility function available during the item review process. When the teacher clicks "Reset," it immediately undoes any pending Retain, Revise, or Discard decisions made on that specific question. A brief confirmation prompt, "Reset decision," is visible. This feature allows the teacher to quickly revert the question to its original state before making a new, final decision, providing flexibility during the critical item curation workflow.

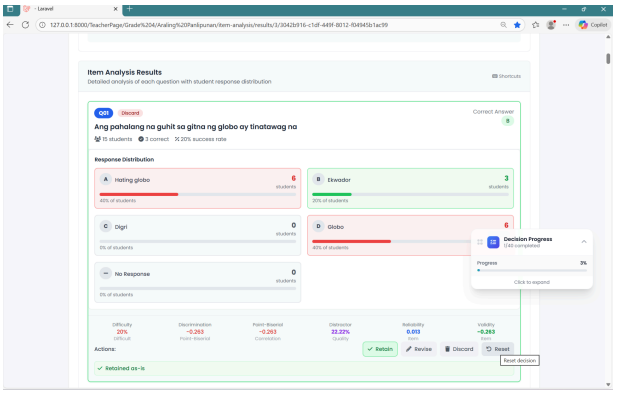


Figure 166. AsseSync Teacher Assessment Analysis - Reset Action

Figure 167 displays the Incomplete Analysis Validation Modal, which appears if the teacher attempts to finalize changes without making a decision (Retain, Revise, or Discard) on all reviewed questions. The modal warns of "Incomplete Analysis" and lists the "Questions Awaiting Decision" (e.g., 39 questions). This safeguard ensures that the data-driven item curation is thoroughly completed. The list shows the question number and a "Pending" status for each item that still requires a decision. The teacher must complete all pending decisions before clicking "Continue" to save and apply the changes to the assessment.

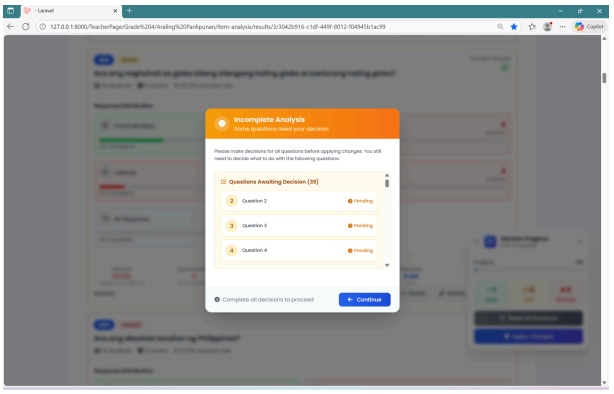


Figure 167. AsseSync Teacher Incomplete Analysis Validation Modal

Figure 168 illustrates the final state of the Item Analysis review process, indicating that all necessary decisions have been made on every question. The Decision Progress panel confirms that the progress is 100% completed (40/40 completed). The summary shows the final tallies of items designated to Keep (e.g., 39 items), Edit (e.g., 1 item), and Remove (e.g., 0 items). This state signifies that the teacher has reviewed the data, curated the assessment, and is ready to save the revisions. The final action is taken by clicking the blue "Apply Changes" button, which commits the item-level decisions (Retain, Revise, Discard) to the assessment record.

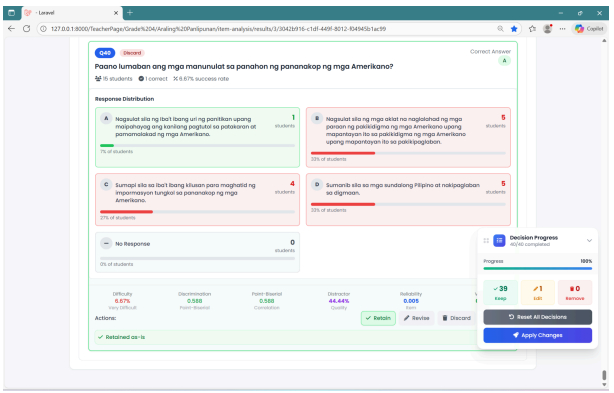


Figure 168. AsseSync Teacher Item Analysis - Ready to Apply Changes State

Figure 169 illustrates the Applying Changes Progress State which appears immediately after the teacher clicks the "Apply Changes" button to finalize the Item Analysis decisions. The Decision Progress panel displays a loading indicator and the button text changes to "Processing..." to confirm that the system is actively working. During this state, the system is performing the necessary tasks to implement the teacher's decisions creating replacement questions, deleting items, and marking retained/revised items before saving the new version of the assessment to the database.

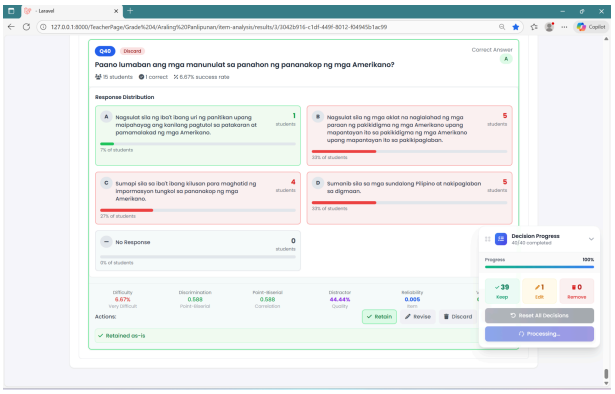


Figure 169. AsseSync Teacher Item Analysis - Applying Changes Progress

Figure 170 displays the successful conclusion of the Item Analysis workflow. A success toast notification appears in the top-right corner with the message: "Version 1 Created Successfully: Your item analysis decisions have been saved as a new questionnaire version." This confirms that the system has implemented all the Keep, Edit, and Remove decisions made by the teacher and saved the curated assessment as a new, stable version. This ensures the original assessment remains intact in the Version History while the improved questionnaire becomes the new active version for future use.

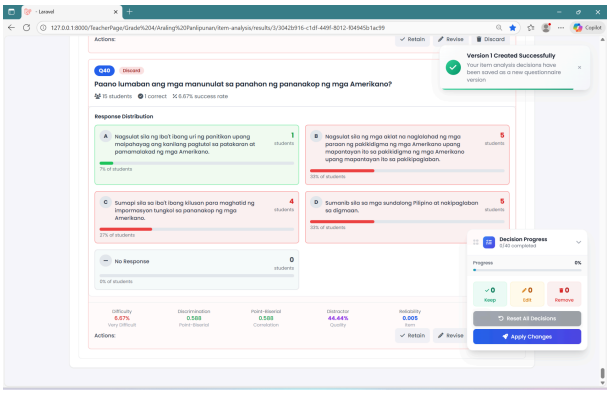


Figure 170. AsseSync Teacher Item Analysis - Version Created Successfully

Figure 171 displays the Student Scores tab of the Assessment Analysis interface, which provides a comprehensive overview of the class performance after scoring is complete. The page includes Summary Statistics, detailing the Total Students (e.g., 15), Total Questions (e.g., 40), the Average Score (e.g., 26%), and the Scan Date. The Summary Statistics also feature a color-coded quality assessment showing the number of scores that are Perfect, Good, or Need improvement. The main section, Student Score Results, lists each student and shows their detailed response matrix, score (e.g., 16/40), and overall status, allowing teachers to quickly identify performance levels.

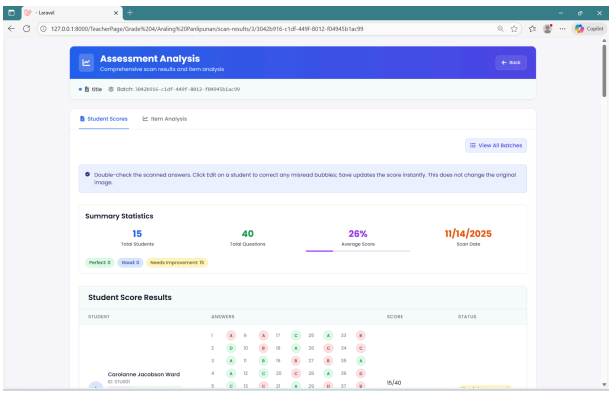


Figure 171. AsseSync Teacher Assessment Analysis - Student Scores Page Overview

Figure 172 illustrates the Manual Edit Interface for student scores, activated when the teacher clicks the "Edit" button next to a student's record (e.g., Carolanne Jacobson Ward). This function allows the teacher to manually correct misread bubbles or scoring errors made by the automatic scanner. When active, the answer matrix becomes fully editable, allowing the teacher to click on individual bubbles to change the recorded response (Correct, Incorrect, or Blank). After making adjustments, the teacher must click "Save" to commit the manual changes, or "Cancel" to discard them. This manual correction capability ensures the final student data is accurate.

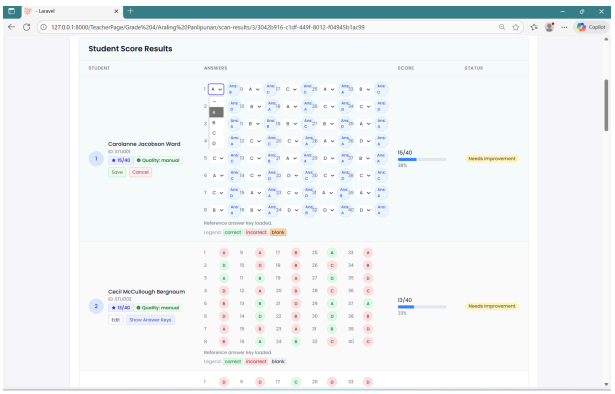


Figure 172. AsseSync Teacher Assessment Analysis - Manual Edit Interface

Figure 173 displays the Student Response Updated Confirmation state following a successful manual score correction. After the teacher saves the edited responses, a success toast notification appears in the top-right corner with the message: "Student Response Updated: Successfully updated answers for [Student Name]." This confirms that the manually corrected scores have been saved to the database. The student's record (e.g., Carolanne Jacobson Ward) immediately updates to show the new score and status (e.g., "Quality: manual"), reflecting the necessary adjustments and maintaining data integrity.

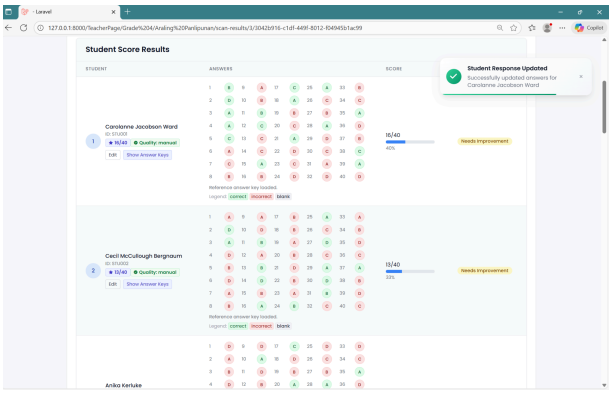


Figure 173. AsseSync Teacher Assessment Analysis - Edit Saved Confirmation

Figure 174 illustrates the Show Answer Keys feature available within the Student Score Results. When the teacher clicks the "Show Answer Keys" button next to a student's record, the interface overlays the Reference Answer Key onto the student's response matrix. This overlay visually indicates the correct choice for every question, allowing the teacher to immediately verify which questions the student answered correctly (green circle), incorrectly (red circle), or left blank. The legend below the matrix clarifies the color-coding system. The teacher can hide the key at any time by clicking "Hide Answer Keys."

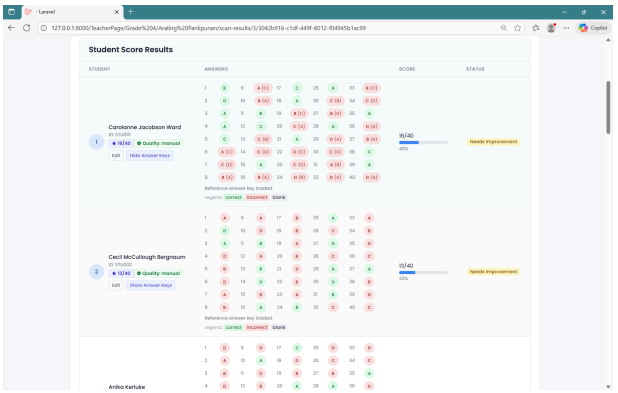


Figure 174. AsseSync Teacher Assessment Analysis - Show Answer Keys

Figure 175 displays the All Scan Results Modal, which is accessed by clicking the "View All Batches" button on the Assessment Analysis page. This modal is essential for managing assessments that have been scored or evaluated multiple times (e.g., in different class sections or through different methods). The modal lists all available scan batches for the current assessment, categorized by method (e.g., Manual Input or Scan Session). Each batch entry shows the date and time of the scan, the class section, and the number of students processed in that batch. The current batch being viewed is clearly labeled ("Currently Viewing"), and the teacher can select any other batch to view its detailed results and analysis.

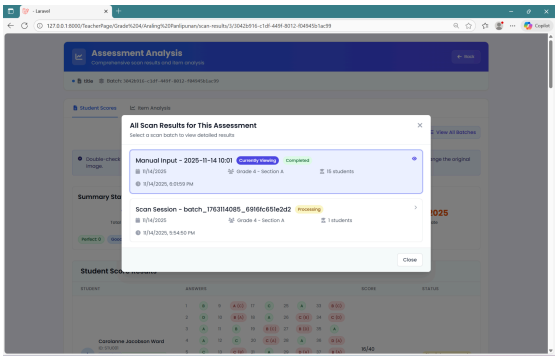


Figure 175. AsseSync Teacher Assessment Analysis - All Scan Results Modal

Figure 176 displays the Scan Batch Results Modal, which allows the user to easily view and switch between different item analysis results generated from separate evaluation batches (e.g., scoring done at different times or via different methods). The modal lists all available batches, clearly showing the processing method (Manual Input or Scan Session), the date and time of the process, and the Section evaluated (e.g., Section A). The currently active batch is highlighted with the "Current" badge, and other batches may show a "Processing" status. This functionality provides flexibility for teachers who score assessments intermittently or use mixed evaluation methods.

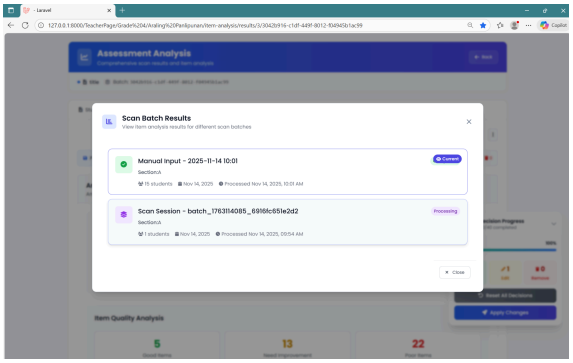


Figure 176. AsseSync Teacher Assessment Analysis - Scan Batch Results Modal

Figure 177 illustrates the Combined Item Analysis Modal, which enables the teacher to merge data from multiple batches to generate a single, comprehensive analysis report. This is particularly useful for combining results from different class sections or evaluation methods into one view. The modal lists all available batches and provides checkboxes for the user to select the desired batches to combine. Information displayed for each batch includes the method, section, student count, and processing time. The teacher can use "Select All" or "Clear All" for efficiency and initiates the combined report generation by clicking "Perform Analysis."

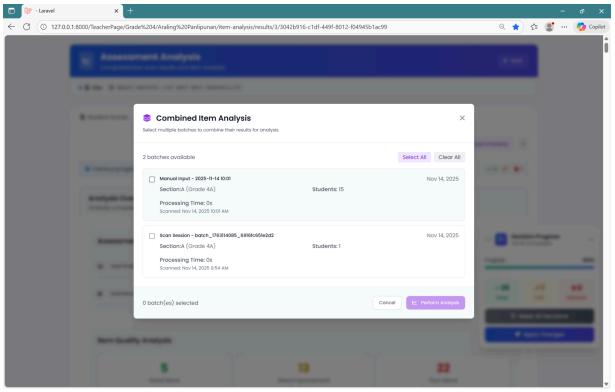


Figure 177. AsseSync Teacher Assessment Analysis - Combined Item Analysis Modal

Figure 178 illustrates the Export Results functionality available on the Item Analysis tab of the Assessment Analysis page. This feature is accessed via the three-dot ellipsis menu located near the "Combined Analysis" button. Clicking "Export Results" allows the teacher to download the comprehensive statistical data generated by the item analysis (including Item Quality Analysis, Response Distribution, and psychometric metrics) in a transferable format, such as Excel or CSV. This enables the teacher to conduct further external analysis or include the data in administrative reports.

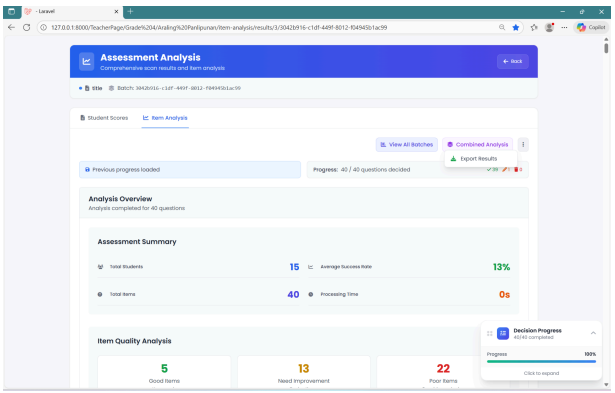


Figure 178. AsseSync Teacher Assessment Analysis - Export Results Option

Figure 179 displays the Test Bank Page for the teacher's selected subject (Grade 6 Araling Panlipunan). This interface allows the teacher to manage and organize all assessments relevant to that course. The page features a Filters & Search section, enabling the teacher to quickly locate specific assessments by filtering based on Quarter or searching by name. The main Assessments list displays key details for each available examination, including the Assessment Name, Type, Quarter, Date, Status (Standardized), and number of Items. This central repository ensures teachers have easy access to all approved and available assessment materials for their class.

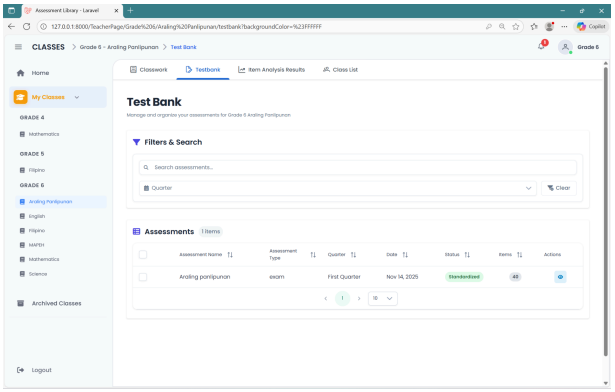


Figure 179. AsseSync Teacher Test Bank Page

Figure 180 illustrates the Class List Page, which is accessed via the tab on the teacher's subject interface. This page is dedicated to managing and viewing all students enrolled across the class sections. The header provides an immediate summary of the Total Students and Active Sections (e.g., 78 students across 4 sections). The interface is organized by section tabs (Section A, Section B, etc.), allowing the teacher to easily switch views. The main display area, Section A Students, presents a searchable list for that specific section, showing essential student data including the LRN, Name, and Phone Number. This page is crucial for managing enrollment records and contact information.

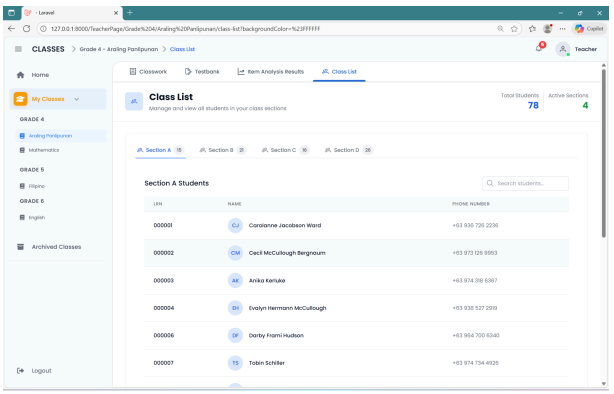


Figure 180. AsseSync Teacher Class List Page

5.2. Development Results

5.2.1. Functionality Test of the AsseSync: Assessment Management System

Developers created two task scenarios, the System Administrator and Teacher to assess the unique functions associated with each role. Participants had the chance to provide feedback while testing these scenarios. Subsequently, developers identified the system's weaknesses requiring further improvement based on the collected participant feedback.

5.2.1.1. System Administrator Functionality Test Results

Table 5 shows the task scenarios set by the developers for the system administrator. The system admin user has successfully navigated the platform and accomplished all assigned tasks.

Table 5. System Administrator User Task Scenarios

Participants	Actions			
	Import Assessment	Archive Assessment	Restore Archived Assessment	Create Assessment
P1	✔	✔	✔	✔

5.2.1.2. Teacher Functionality Test Results

Table 6 showcases the task scenarios formulated by the developers for the students, signaling a positive user experience and the effectiveness of the system's design. User feedback aids in pinpointing areas for enhancement, facilitating further optimization of the system. This marks a positive advancement for the Library Management System and its developers.

Table 6. Teacher User Task Scenarios

Participants	Actions				
	Create Assessment	Item Analysis	Export Item Analysis Result	Test Bank	View Student Scores
P2	✔	✔	✔	✔	✔
P3	✔	✔	✔	✔	✔
P4	✔	✔	✔	✔	✔
P5	✔	✔	✔	✔	✔
P6	✔	✔	✔	✔	✔
P7	✔	✔	✔	✔	✔
P8	✔	✔	✔	✔	✔
P9	✔	✔	✔	✔	✔
P10	✔	✔	✔	✔	✔
P11	✔	✔	✔	✔	✔
P12	✔	✔	✔	✔	✔
P13	✔	✔	✔	✔	✔
P14	✔	✔	✔	✔	✔
P15	✔	✔	✔	✔	✔
P16	✔	✔	✔	✔	✔
P17	✔	✔	✔	✔	✔
P18	✔	✔	✔	✔	✔
P19	✔	✔	✔	✔	✔
P20	✔	✔	✔	✔	✔
P21	✔	✔	✔	✔	✔
P22	✔	✔	✔	✔	✔
P23	✔	✔	✔	✔	✔
P24	✔	✔	✔	✔	✔
P25	✔	✔	✔	✔	✔
P26	✔	✔	✔	✔	✔
P27	✔	✔	✔	✔	✔
P28	✔	✔	✔	✔	✔
P29	✔	✔	✔	✔	✔
P30	✔	✔	✔	✔	✔

5.3.1.4. Summary of Functionality Test

The testing phase for the AsseSync: Assessment Management System in Naawan Central School has concluded, involving a diverse user group including the system administrator and teachers. The feedback was positive, with users commending the system for its user-friendly design and its convenience.

However, the System Administrator provided a minor suggestion on the ”Import Assessment”. Instead of putting the details of the questionnaire manually, he suggested a feature to automate data entry. When the admin attaches an assessment, the system will automatically extract the required details and populate the corresponding input fields. The developers have taken the feedback into consideration and implemented the suggestion from the System Admin User. Moreover, all of the participants have successfully completed all of the tasks assigned to them without any issues and did not provide any other comments on the function list. The testing phase has been a positive advancement, ensuring the effectiveness and user-friendliness of the AsseSync: Assessment Management System across every user.

5.3.1. Usability Test of the AsseSync: Assessment Management System

5.3.1.1. The Test Participants

The system administrator, responsible for Importing Assessments from the DepEd in the desktop application of the AsseSync: Assessment Management System, was one of the 30 available testers. The remaining 29 testers were teachers assigned to Create an Assessment, Item analysis an assessment, and use the testbank for creation of assessment in the teacher module on the website application of the AsseSync: Assessment Management System.

A. The Usability Test Setup

The developers provided a brief yet clear orientation to the system participants on the testing day. After obtaining consent through the signed form (refer to Appendix A), the testing commenced, with developers setting up the necessary tools and assigning specific task scenarios to each participant.

Distinct task scenarios were created for the system administrator and teachers by the developers. Once participants independently complete their assigned tasks, developers use the System Usability Scale (SUS) form (refer to Appendix C) to evaluate and score each participant, assessing their system usage experience. The testing concluded with developers offering a brief concluding comment.

B. The Usability Test Results

B.1 System Admin Usability Test Results

The table 7 shows a mixed but strong feedback from Participant 1. While the overall usability score 92.5 indicates an excellent user experience according to Adjective Rating Scale.

Table 7. Admin User Task Scenarios

Participants	Questions										
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	
P1	3	2	5	1	5	1	5	1	5	1	92.5

B.2 Teacher Usability Test Results

Table 8 displays the Teacher Usability Test Results for 29 participants P2 through P30 across ten questions, with individual question ratings on a scale of 1 to 5, and a final calculated overall score. Most scores fall in the 70s and 80s, suggesting the system is generally perceived as Good to Excellent. The high scores (90s) indicate some teachers found the system outstanding, while the scores near 60 suggest a segment of the user base struggles with it. Lower scores indicate a transitional learning curve, suggesting that additional user onboarding would further enhance faculty adoption.

Table 8. Teacher User Task Scenarios

Participants	Question										
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	
P2	5	2	5	1	5	1	5	1	5	1	97.5
P3	1	5	2	2	5	2	5	1	3	1	62.5
P4	4	3	5	1	5	2	5	1	5	1	90
P5	2	2	5	2	5	1	5	1	4	1	85
P6	1	1	5	3	5	1	5	1	4	1	82.5
P7	4	1	4	4	4	2	5	2	4	1	77.5
P8	2	2	5	2	5	2	5	3	5	1	80
P9	5	2	4	1	5	2	5	3	5	1	87.5
P10	4	4	3	5	5	2	5	3	5	1	67.5
P11	1	4	5	5	3	2	4	2	5	1	60
P12	3	4	5	5	5	2	4	2	5	1	70
P13	5	1	3	4	5	3	5	2	5	1	80
P14	4	5	4	3	5	1	5	2	4	1	75
P15	1	2	2	5	5	1	5	2	3	1	62.5
P16	1	3	5	4	5	1	4	1	5	1	75

Participants	Question										
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	
P2	5	2	5	1	5	1	5	1	5	1	97.5
P3	1	5	2	2	5	2	5	1	3	1	62.5
P4	4	3	5	1	5	2	5	1	5	1	90
P17	3	5	3	3	4	1	2	1	5	1	65
P18	2	4	4	3	5	1	3	1	4	1	70
P19	2	1	5	4	5	1	3	1	4	1	77.5
P20	1	5	3	4	5	1	5	1	4	1	65
P21	2	1	3	4	5	1	5	1	4	2	75
P22	1	4	5	5	5	1	5	1	4	1	70
P23	2	5	5	5	5	1	4	1	5	1	70
P24	2	2	5	5	4	2	4	1	5	3	67.5
P25	1	1	4	4	4	2	4	2	5	1	70
P26	4	3	4	4	4	2	4	5	5	3	60
P27	2	5	5	4	5	1	5	1	5	3	70
P28	3	2	5	5	5	1	4	1	5	3	75
P29	3	1	5	5	5	2	5	1	5	1	82.5
P30	3	2	5	5	4	1	5	2	4	1	75

C. Summary of the Usability Test Results

In summary, the result of the gathered data from the SUS form (Appendix C) of the 30 test participants is that 1 is the System Administrator, participants 2 to 30 are the teachers. As shown in Table It suggests that 8 of participants, out of 30, have good to very good perceptions of the system's usability, as indicated by the system's "Excellent" acceptability score of 80 to 97.5 on the SUS and ARS Rubric. However, the majority of the participants from the teachers who reported lower scores (Three (3) C and One (1) D grades), indicating that there may be areas of the system that are not meeting the expectations of some users or causing difficulty for them. The questions presented in the survey suggest that participants find the system to be simple, easy to use, intuitive, and well-integrated, with a sense of confidence using it without support or needing to learn.

Table 9. Usability Test Results

Participants	Question										
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	
P1	3	2	5	1	5	1	5	1	5	1	92.5
P2	5	2	5	1	5	1	5	1	5	1	97.5
P3	1	5	2	2	5	2	5	1	3	1	62.5
P4	4	3	5	1	5	2	5	1	5	1	90
P5	2	2	5	2	5	1	5	1	4	1	85

P6	1	1	5	3	5	1	5	1	4	1	82.5
P7	4	1	4	4	4	2	5	2	4	1	77.5
P8	2	2	5	2	5	2	5	3	5	1	80
P9	5	2	4	1	5	2	5	3	5	1	87.5
P10	4	4	3	5	5	2	5	3	5	1	67.5
P11	1	4	5	5	3	2	4	2	5	1	60
P12	3	4	5	5	5	2	4	2	5	1	70
P13	5	1	3	4	5	3	5	2	5	1	80
P14	4	5	4	3	5	1	5	2	4	1	75
P15	1	2	2	5	5	1	5	2	3	1	62.5
P16	1	3	5	4	5	1	4	1	5	1	75
P17	3	5	3	3	4	1	2	1	5	1	65
P18	2	4	4	3	5	1	3	1	4	1	70
P19	2	1	5	4	5	1	3	1	4	1	77.5
P20	1	5	3	4	5	1	5	1	4	1	65
P21	2	1	3	4	5	1	5	1	4	2	75
P22	1	4	5	5	5	1	5	1	4	1	70
P23	2	5	5	5	5	1	4	1	5	1	70
P24	2	2	5	5	4	2	4	1	5	3	67.5
P25	1	1	4	4	4	2	4	2	5	1	70
P26	4	3	4	4	4	2	4	5	5	3	60
P27	2	5	5	4	5	1	5	1	5	3	70
P28	3	2	5	5	5	1	4	1	5	3	75
P29	3	1	5	5	5	2	5	1	5	1	82.5
P30	3	2	5	5	4	1	5	2	4	1	75

5.4. Summary of the Chapter

The SUS data, as highlighted in this chapter, for the AsseSync: Assessment Management System suggests that most participants seem to have an unfavourable impression of the system's usability. This is evident in the high SUS scores, which are mostly in different ranges. However, there are a number of participants who scored higher, which suggests that certain aspects of the system may not meet certain user expectations or present challenges. In conclusion, while the AsseSync: Assessment Management system generally received positive SUS Average score, the identification of lower scores underscores potential areas for improvement; nevertheless, users positive feedback on the system's ease of navigation, simplicity, logic, integration, and independent usability provides valuable insights for enhancing overall user satisfaction.

CHAPTER 6
CONCLUSION AND RECOMMENDATIONS

6.1. Conclusion

The development of AsseSync: Assessment Management System for Naawan Central School successfully addressed the need for a faster, accurate, and standardized assessment process. Through its system-assisted workflow ranging from test creation or import, bubble sheet generation, automated checking, and item analysis, the system significantly reduces manual work while improving assessment reliability. Features such as automatic computation of item difficulty, distractor quality, and discrimination index allow teachers to make evidence-based decisions in refining their assessments. Evaluated and imported assessments from DepEd can then be stored in the system's test bank, promoting consistency and long-term enhancement of the school's testing materials.

The System Usability Scale (SUS) results presented in chapter 5 indicate that users generally found the system usable and beneficial. Although a small subset of participants (specifically those with lower computer literacy) gave lower scores, reflecting some usability challenges, the overall SUS average remained positive. While the system is generally useful and easy to navigate, some teachers, particularly those who are not computer literate still require initial guidance or training before they can use the system independently. These strengths confirm AsseSync's effectiveness, while the identified areas of improvement provide directions for refining user experience in future iterations.

Overall, AsseSync has demonstrated its potential as a practical and valuable solution for modernizing assessment practices at Naawan Central School. It not only enhances accuracy and efficiency but also contributes to continuous test improvement, making it a reliable tool for both teachers and the institution.

6.2. Recommendations

To further improve this project, here are the following recommendations:

Based on the findings and user feedback, several recommendations are proposed to further enhance the capabilities and usability of AsseSync. First, the integration of online examinations and quizzes is encouraged to allow teachers to administer assessments digitally, thereby reducing reliance on printed materials and increasing flexibility and accessibility. It is also recommended that the system support a wider variety of question types such as identification, matching type, true or false, and short-answer items to make the platform more versatile and applicable to diverse assessment needs. Furthermore, improvements to the user interface and overall user experience should be prioritized, particularly by refining specific UI elements, enhancing visual clarity, and increasing system responsiveness in order to address concerns identified in the lower SUS scores.

Strengthening error handling and feedback mechanisms, including clearer prompts, notifications, and user guidance, would also help minimize user difficulties and support smoother navigation.

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DEFINITION OF TERMS

AsseSync - a software application that automates and facilitates the process of creating, administering, scoring, and reporting on educational or corporate assessments.

Assessment Management - refers to the processes, procedures, and strategies used to plan, develop, deliver, score, analyze, and store all forms of educational or corporate assessments.

Item Analysis - is a statistical process used to evaluate the quality and effectiveness of individual test questions (items) in an assessment. It helps determine how well each item is performing in measuring the intended knowledge or skill.

Test Bank - is a large, organized repository or database of pre-written, categorized, and tagged assessment questions (items). These items are usually associated with specific learning objectives, difficulty levels, and have undergone Item Analysis to ensure quality.

APPENDIX A

(Consent Form)

Mindanao State University at Naawan
College of Business and Information Technology
Department of Information Technology

Consent Form

Part I: Information Sheet

Title of the Project
AsseSync: Assessment Management System

Main Purpose of the Project
The purpose of *AsseSync* is to automate the assessment process at Naawan Central School by helping teachers efficiently manage, evaluate, and record student assessments. It aims to reduce manual work, improve accuracy, and enhance data management through automated checking, item analysis, and AI-powered insights, ultimately supporting data-driven teaching and better learning outcomes.

Duration/Period: From Novemeber 3, 2025 to Novemeber 7, 2025

Name of Developers: John Robert G. Adlawan

Jaymei Rey D. Cortez

Lloyd S. Gonzaga
Contact number of Developers: 09100796597

09363275590

09633407873

Part II: Certificate of Consent
Written Consent Form for Research Participants/ Informants or Parents/Guardians of Children (below 18 years old who are respondents) *To be signed at the beginning of the interview/survey.

Direction: please read the following statements carefully and put a check (/) in the corresponding box if you agree:

☐	I fully understand the nature and purpose of this research project.
☐	I give my voluntary consent to participate in this research.
☐	I give my consent for the researchers to record the related activities.
☐	I give my consent for the researcher to record the conduct of the Activities (video, audio, photos)
☐	I can stop participating in the study at any time, for any reason if I so decide.
☐	If I have questions about the research in general or about my role in the study, I can Freely ask the researcher about my concerns.

_____ _____
Name and Signature of Participant/Respondent Date (Day/Month/Year)

APPENDIX B

(Task Scenarios)

1 Test Scenarios

1.1 Admin Login

No.	Test Scenario	Expected Results	Pass / Fail	Tested by Name & Signature	Remarks
	Access the login page.	The user shall be able to access the login page.			
	Actual Result: The system successfully displays the login page when accessed.				
	Input a valid email address and password then click the login button.	The user shall be able to log in to the AsseSync and see the Dashboard.			
	Actual Result: The system successfully logs in the user after entering a valid email address and password. The user is redirected to the AsseSync Dashboard, where all main features and modules are displayed properly.				

1.2 Admin Module

No.	Test Scenario	Expected Results	Pass / Fail	Tested by Name & Signature	Remarks
	View Admin Dashboard and Assessment Data List	The user shall be able to view the Admin Dashboard and see the list of all assessment data, including details such as title, date created, and status.			
	Actual Result: The user was able to view the Admin Dashboard and see the list of all assessment data, including details such as title, date created, and status.				
	Create Assessment	The user shall be able to create a new assessment by filling out the required fields (e.g., title, description, items, choices, images and answer keys).			

		The new assessment successfully saves it to the system.			
	Actual Result: The user was able to create a new assessment by filling out the required fields and successfully saved it to the system.				
	Admin Dashboard Import Assessment	The user shall be able to import an assessment file (e.g., PDF or DOCX format) into the system, and the imported assessment data shall be displayed correctly in the assessment list.			
	Actual Result: The user was able to import an assessment file into the system, and the imported assessment data was displayed correctly in the assessment list.				
	Search Assessment	The user shall be able to search for an assessment using keywords or titles. The system shall display matching assessment records based on the search criteria.			
	Actual Result: The user was able to search for an assessment using keywords or titles. The system displayed matching assessment records based on the search criteria.				
	Assessment List Filter by Grade Level	The user shall be able to filter assessments by grade level. The system shall display assessment records that match the selected grade level.			
	Actual Result: The user was able to filter assessments by grade level. The system displayed assessment records that matched the selected grade level.				
	Assessment List Filter by Subject	The user shall be able to filter the assessment list by subject. The system shall display assessment records that match the selected subject.			

	Actual Result: The user was able to filter the assessment list by subject. The system displayed assessment records that matched the selected subject.				
	Assessment List Filter by Quarter	The user shall be able to filter the assessment list by quarter. The system shall display assessment records that match the selected quarter.			
	Actual Result: The user was able to filter the assessment list by quarter. The system displayed assessment records that matched the selected quarter.				
	Assessment List Clear Filter	The user shall be able to clear all applied filters in the assessment list. The system shall display the complete list of assessments after the filters are cleared.			
	Actual Result: The user was able to clear all applied filters in the assessment list. The system displayed the complete list of assessments after the filters were cleared.				
	Assessment List Open Assessment	The user shall be able to open a selected assessment from the list. The system shall display the complete details of the selected assessment			
	Actual Result: The user was able to open a selected assessment from the list. The system displayed the complete details of the selected assessment.				
0	View Original Document of the Assessment	The user shall be able to view the original document of the selected assessment. The system shall display the original document clearly and accurately.			
	Actual Result: The user was able to view the original document of the selected assessment. The system displayed the original document clearly and accurately.				
1	Assessment List Edit	The user shall be able to edit the details of a			

	Assessment Details	selected assessment from the list. The system shall save and display the updated assessment details successfully.			
	Actual Result: The user was able to edit the details of a selected assessment from the list. The system saved and displayed the updated assessment details successfully.				
2	Assessment List Archive Assessment	The user shall be able to archive a selected assessment from the list. The system shall move the archived assessment to the archive section and remove it from the active list.			
	Actual Result: The user was able to archive a selected assessment from the list. The system moved the archived assessment to the archive section and removed it from the active list.				
3	View Archive Assessment List	The user shall be able to view the list of archived assessments. The system shall display all assessments that have been archived with their corresponding details.			
	Actual Result: The user was able to view the list of archived assessments. The system displayed all assessments that had been archived with their corresponding details.				
4	Search Archived Assessment	The user shall be able to search for an archived assessment using keywords or titles. The system shall display matching archived assessment records based on the search criteria.			

	Actual Result: The user was able to search for an archived assessment using keywords or titles. The system displayed matching archived assessment records based on the search criteria.				
5	Archive Assessment List Filter by Grade	The user shall be able to filter the archived assessment list by grade level. The system shall display archived assessment records that match the selected grade level.			
	Actual Result: The user was able to filter the archived assessment list by grade level. The system displayed only the archived assessment records that matched the selected grade level.				
6	Archive Assessment List Filter by Subject	Archived assessments are filtered by subject.			
	Actual Result: The user was able to filter the archived assessment list by grade level. The system displayed archived assessment records that matched the selected grade level.				
7	Archive Assessment List Filter by Quarter	The user shall be able to filter the archived assessment list by quarter. The system shall display archived assessment records that match the selected quarter.			
	Actual Result: The user was able to filter the archived assessment list by quarter. The system displayed archived assessment records that matched the selected quarter.				
8	Archive Assessment List Clear Filter	The user shall be able to clear all applied filters in the archived assessment list. The system shall display the complete list of archived assessments after the filters are cleared.			
	Actual Result: The user was able to clear all applied filters in the archived assessment list. The system displayed the complete list of archived assessments after the filters were cleared.				

Overall Comment(s) / Suggestion(s)

1 Approval

The undersigned acknowledge that they have reviewed the contents of the **User Acceptance Test Plan** and agree with the information presented within this document.

Development Team
JOHN ROBERT G. ADLAWAN ***Front-end Developer*** **LLOYD S. GONZAGA** ***Back-end Developer*** **JAYMEI REY D. CORTEZ** ***System Analyst*** _____ Date

END-USER
_____ _____ Date

1 Test Scenarios

1.1 Teacher Login

No.	Test Scenario	Expected Results	Pass / Fail	Tested by Name & Signature	Remarks
	Access the login page.	The user shall be able to access the login page.			
	Actual Result: The system successfully displays the login page when accessed.				
	Input user credentials then click the login button.	The user shall be able to log in to the AsseSync and see the Dashboard.			
	Actual Result: The system successfully logs in the user after entering the user credentials. The user is redirected to the AsseSync Dashboard, where all main features and modules are displayed properly.				

1.2 Teacher Module

No.	Test Scenario	Expected Results	Pass / Fail	Tested by Name & Signature	Remarks
	View Subject List	The user shall be able to view the list of all subjects in the system that has been assigned to the teacher. The system shall display			

		each subject with its corresponding details such as subject name and assigned grade level.			
	Actual Result: The user was able to view the list of all subjects in the system that had been assigned to the teacher. The system displayed each subject with its corresponding details such as subject name and assigned grade level.				
	View Subject Page	The user shall be able to view the subject page assigned to them. The system shall display the classwork, test bank, Item analysis result and class list for the selected class.			
	Actual Result: The user was able to view the subject page assigned to them. The system displayed the classwork, test bank, item analysis result, and class lists for the selected class.				
	Create Topic	The user shall be able to create a new topic under a selected subject or class. The system shall save the new topic and display it in the topic list.			
	Actual Result: The user was able to create a new topic under a selected subject or class. The system saved the new topic and displayed it in the topic list.				
	Create Assessment	The user shall be able to create a new assessment under a selected topic. The system shall save the created assessment and display it in the topic assessment list.			
	Actual Result: The user was able to create a new assessment under a selected topic. The system saved the created assessment and displayed it in the topic assessment list.				
	View Assessments by Quarter Exams	The user shall be able to view assessments categorized under Quarter			

	and Topic Assessments	Exams and Topic Assessments. The system shall display the list of assessments grouped accordingly under each category.			
	Actual Result: The user was able to view assessments categorized under Quarter Exams and Topic Assessments. The system displayed the list of assessments grouped accordingly under each category.				
	Classwork Open Assessment	The user shall be able to open a selected assessment from the classwork section. The system shall display the complete assessment details, including title, description, and list of items. The user shall be able to edit the assessment or individual items as needed.			
	Actual Result: The user was able to open a selected assessment from the classwork section. The system displayed complete assessment details, including title, description, and list of items. The user was able to edit the assessment or individual items as needed.				
	Switch Assessment Version Control	The user shall be able to switch between different versions of an assessment. The system shall display the selected version's details, including its items, settings, and results.			
	Actual Result: The user was able to switch between assessment versions, and the system successfully displayed the corresponding details of the selected version, including its items, settings, and results.				
	Evaluate Assessment Bubble sheet Scanning	The user shall be able to evaluate an assessment using the bubble sheet scanning feature. The system shall scan the			

		submitted bubble sheets, automatically check the answers, compute the scores, and record the results accurately.			
	Actual Result: The user was able to evaluate an assessment using the bubble sheet scanning feature. The system scanned the submitted bubble sheets, automatically checked the answers, computed the scores, and recorded the results accurately.				
	Evaluate Assessment Manual Input	The user shall be able to manually input students' scores for a selected assessment. The system shall save the manually entered scores and update the evaluation results accordingly.			
	Actual Result: The user was able to manually input students' scores for a selected assessment. The system saved the manually entered scores and updated the evaluation results accordingly				
0	View Evaluation Results	The user shall be able to view the evaluation results of a selected assessment. The system shall display each student's score, item analysis, and overall performance summary.			
	Actual Result: The user was able to view the evaluation results of a selected assessment. The system displayed each student's score, item analysis, and overall performance summary.				
1	View Test bank	The user shall be able to view the test bank containing all saved standardized assessment. The system shall display the list of questions with the correct answer.			
	Actual Result: The user was able to view the test bank containing all standardized assessments saved. The system displayed the list of questions with the correct answers.				
2	Search Test Bank Assessment	The user shall be able to search for a test bank			

		assessment using keywords or titles. The system shall display matching standardized assessments based on the search criteria.			
	Actual Result: The user was able to search for a test bank assessment using keywords or titles. The system displayed the matching standardized assessments based on the search criteria				
3	Filter Test bank by Assessment Type	The user shall be able to filter the test bank by assessment type. The system shall display the list of standardized assessments that match the selected assessment type.			
	Actual Result: The user was able to filter the test bank by assessment type. The system displayed the list of standardized assessments that matched the selected assessment type.				
4	Clear Test bank Filter	The user shall be able to clear all applied filters in the test bank. The system shall display the complete list of standardized assessments after the filters are cleared.			
	Actual Result: The user was able to clear all applied filters in the test bank. The system displayed the complete list of standardized assessments after the filters were cleared.				
5	View Item Analysis Result	The user shall be able to view the item analysis results of a selected assessment. The system shall display details such as item difficulty, discrimination index, distractor, reliability and validity, number of correct responses, and overall item performance.			

	Actual Result: The user was able to view the item analysis results of a selected assessment. The system displayed details such as item difficulty, discrimination index, distractors, reliability and validity, number of correct responses, and overall item performance.				
6	Search Item Analysis Result	The user shall be able to search for an item analysis result using keywords or assessment titles. The system shall display the matching item analysis results based on the search criteria.			
	Actual Result: The user was able to search for an item analysis result using keywords or assessment titles. The system displayed the matching item analysis results based on the search criteria.				
7	Filter Item Analysis Result by Status	The user shall be able to filter item analysis results by status (e.g., completed, pending, or in progress). The system shall display only the item analysis results that match the selected status.			
	Actual Result: The user was able to filter item analysis results by status (e.g., completed, pending, or in progress). The system displayed only the item analysis results that matched the selected status.				
8	Filter Item Analysis Result by Quarter	The user shall be able to filter item analysis results by quarter. The system shall display only the item analysis results that correspond to the selected quarter.			
	Actual Result: The user was able to filter item analysis results by quarter. The system displayed only the item analysis results that corresponded to the selected quarter.				
9	Sort Item Analysis Result by Date	The user shall be able to sort item analysis results by date (newest or oldest). The system shall display the			

		item analysis results arranged according to the selected date order.			
	Actual Result: The user was able to sort item analysis results by date (newest or oldest). The system displayed the item analysis results arranged according to the selected date order.				
0	Clear Item Analysis Result Filter	The user shall be able to clear all applied filters in the item analysis results. The system shall display the complete list of item analysis results after the filters are cleared.			
	Actual Result: The user was able to clear all applied filters in the item analysis results. The system displayed the complete list of item analysis results after the filters were cleared.				
1	View Item Analysis Result Performance Metrics	The user shall be able to view the performance metrics of a selected item analysis result. The system shall display metrics such as item difficulty, discrimination index, distractors, reliability, validity, and overall student performance.			
	Actual Result: The user was able to view the performance metrics of a selected item analysis result. The system displayed metrics such as item difficulty, discrimination index, distractors, reliability, validity, and overall student performance.				
2	Combine Results	The user shall be able to combine multiple item analysis results into a single report. The system shall merge the selected results and display a consolidated report with all relevant metrics.			
	Actual Result: The user was able to combine multiple item analysis results into a single report. The system merged the selected results and displayed a consolidated report with all relevant metrics.				

3	View Student Scores	The user shall be able to view the scores of students for a selected assessment. The system shall display each student's score, total possible score, and performance summary.			
	Actual Result: The user was able to view the scores of students for a selected assessment. The system displayed each student's score, total possible score, and performance summary.				
4	Edit Student Answers	The user shall be able to edit the answers submitted by a student for a selected assessment. The system shall update the student's responses and recalculate the score accordingly.			
	Actual Result: The user was able to edit the answers submitted by a student for a selected assessment. The system updated the student's responses and recalculated the score accordingly.				
5	Retain Items	The user shall be able to retain specific items from the item analysis results when generating or updating assessments. The system shall preserve the retained items.			
	Actual Result: The user was able to retain specific items from the item analysis results when generating or updating assessments. The system preserved the retained items.				
6	Revise Items	The user shall be able to revise items from the item analysis results. The system shall save the revised items.			
	Actual Result: The user was able to revise items from the item analysis results. The system saved the revised items.				
7	Discard Items	The user shall be able to discard or remove selected			

		items from the item analysis results. The system shall delete the discarded items and update the assessment or item analysis accordingly.			
	Actual Result: The user was able to discard or remove selected items from the item analysis results. The system deleted the discarded items and updated the assessment or item analysis accordingly.				
8	Reset Items	The user shall be able to reset selected items from the item analysis results to their original state. The system shall restore the items to their original state.			
	Actual Result: The user was able to reset selected items from the item analysis results to their original state. The system restored the items to their original state.				
9	Apply Changes / Create New Version of the Assessment	The user shall be able to apply changes made to items in the item analysis results or create a new version of the assessment. The system shall save the changes or generate a new version of the assessment.			
	Actual Result: The user was able to apply changes made to items in the item analysis results or create a new version of the assessment. The system saved the changes or generated a new version of the assessment.				
0	View Archive Classes	The user shall be able to view the list of archived classes. The system shall display all archived classes with their corresponding details, such as class section and grade level.			

	Actual Result: The user was able to view the list of archived classes. The system displayed all archived classes with their corresponding details, such as class section and grade level.				
1	Restore Archived Class/Subject by Grade Level	The user shall be able to restore an archived class or subject by selecting the grade level. The system shall move the restored class or subject back to the active list and make it available for use.			
	Actual Result: The user was able to restore an archived class or subject by selecting the grade level. The system moved the restored class or subject back to the active list and made it available for use.				
2	View All Notifications	The user shall be able to view all notifications in the system. The system shall display a list of notifications with details such as notification type, date, and related action or item.			
	Actual Result: The user was able to view all notifications in the system. The system displayed a list of notifications with details such as notification type, date, and related action or item.				
3	Mark as Read	The user shall be able to mark notifications as read. The system shall update the status of the selected notifications and remove the unread indicator**.**			
	Actual Result: The user was able to mark notifications as read. The system updated the status of the selected notifications and removed the unread indicator.				

Overall Comment(s) / Suggestion(s)

1 Approval

The undersigned acknowledge that they have reviewed the contents of the **User Acceptance Test Plan** and agree with the information presented within this document.

Development Team
JOHN ROBERT G. ADLAWAN ***Front-end Developer***
LLOYD S. GONZAGA ***Back-end Developer***
JAYMEI REY D. CORTEZ ***System Analyst***
_____ Date

END-USER
_____
_____ Date

APPENDIX C

(SUS Form)

SUS Form

Instruction
Based on your experience today, check the box that reflects your immediate response to each statement. Make sure you respond to every statement.

		Strongly Disagree 1	2	3	4	Strongly Agree 5
1.	I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
2.	I found the system unnecessarily complex.	☐	☐	☐	☐	☐
3.	I thought the system was easy to use.	☐	☐	☐	☐	☐
4.	I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5.	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
6.	I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
7.	I would imagine that most people would learn to use this system very quickly	☐	☐	☐	☐	☐
8.	I found the system very cumbersome/awkward to use.	☐	☐	☐	☐	☐
9.	I felt very confident using the system.	☐	☐	☐	☐	☐
10.	I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

Additional Feedback (Optional): Please provide additional feedbacks or comments about your experience with the system,

Signature: _____

Thank you for you time and participating in this usability assessment. Your feedback is valuable in helping us to improve the system's Usability.

APPENDIX D

(Orientation Script)

“Good Day, everyone. Thank you for agreeing to take part in our study. My name is John Robert G. Adlawan, and these are the other members of the group namely Jaymei Rey D. Cortez, and Lloyd S. Gonzaga"

"During the rest of the session, I'll be working from a script to ensure that our instructions to everyone who participates in the study are the same. We're here to learn about how the users experience the use of the AMS, a Assessment Management System for Naawan Central School. Our objective today is to observe your usage on the web-based system."

"By the way, we are from MSU-Naawan and are currently taking up Bachelor of Science in Information Technology major in Database Management. We are working on our capstone entitled "AsseSync: Assessment Management System", and we are here to let you explore the system. So please be honest in your feedback—we need to know exactly what you think, not what you think we want to hear. We are also here to learn and take notes. The whole session will not go beyond 1 hour. Do you have any questions before we begin? [Answer any questions]."

APPENDIX E

(Data Dictionary)

schema_name	table_name	obj_ty	order	is_key	column_name	data_type	nullable	column_descr
public	answer_keys	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	answer_keys	TBL	2	FK,UK	question_id	bigint(64)	NOT NULL	NULL
public	answer_keys	TBL	3		correct_choice	character varying(1)	NULL	NULL
public	answer_keys	TBL	4		text_answer	text	NULL	NULL
public	answer_keys	TBL	5		answer_type	character varying(50)	NULL	NULL
public	answer_keys	TBL	6		created_at	timestamp without time zone(6)	NULL	NULL
public	answer_keys	TBL	7		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessment_content_order	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	assessment_content_order	TBL	2	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	assessment_content_order	TBL	3		content_type	character varying(50)	NULL	NULL
public	assessment_content_order	TBL	4		content_id	bigint(64)	NOT NULL	NULL
public	assessment_content_order	TBL	5		display_order	integer(32)	NOT NULL	NULL
public	assessment_content_order	TBL	6		created_at	timestamp without time zone(6)	NULL	NULL

public	assessment_content_order	TBL	7		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessment_images	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	assessment_images	TBL	2	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	assessment_images	TBL	3		title	character varying(255)	NULL	NULL
public	assessment_images	TBL	4		image_path	character varying(500)	NOT NULL	NULL
public	assessment_images	TBL	5		display_order	integer(32)	NULL	NULL
public	assessment_images	TBL	6		created_at	timestamp without time zone(6)	NULL	NULL
public	assessment_images	TBL	7		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessment_questions	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	assessment_questions	TBL	2	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	assessment_questions	TBL	3	FK	question_id	bigint(64)	NOT NULL	NULL
public	assessment_questions	TBL	4		question_number	integer(32)	NOT NULL	NULL
public	assessment_questions	TBL	5		created_at	timestamp without time zone(6)	NULL	NULL
public	assessment_questions	TBL	6		updated_at	timestamp without time zone(6)	NULL	NULL

public	assessment_versions	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	assessment_versions	TBL	2	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	assessment_versions	TBL	3	UK	version_number	integer(32)	NOT NULL	NULL
public	assessment_versions	TBL	4		label	character varying(255)	NULL	NULL
public	assessment_versions	TBL	5		notes	text	NULL	NULL
public	assessment_versions	TBL	6		derived_from_version_id	bigint(64)	NULL	NULL
public	assessment_versions	TBL	7		snapshot_json	jsonb	NULL	NULL
public	assessment_versions	TBL	8		total_items	integer(32)	NULL	NULL
public	assessment_versions	TBL	9		avg_difficulty	numeric(5,2)	NULL	NULL
public	assessment_versions	TBL	10		avg_discrimination	numeric(5,2)	NULL	NULL
public	assessment_versions	TBL	11		is_published	boolean	NULL	NULL
public	assessment_versions	TBL	12		is_current	boolean	NULL	NULL
public	assessment_versions	TBL	13		version_type	character varying(50)	NULL	NULL
public	assessment_versions	TBL	14		change_summary	text	NULL	NULL
public	assessment_versions	TBL	15		decision_metadata	jsonb	NULL	NULL

public	assessment_versions	TBL	16		activated_at	timestamp without time zone(6)	NULL	NULL
public	assessment_versions	TBL	17		restore_reason	text	NULL	NULL
public	assessment_versions	TBL	18		created_at	timestamp without time zone(6)	NULL	NULL
public	assessment_versions	TBL	19		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessments	TBL	1	PK	assessment_id	bigint(64)	NOT NULL	Formerly questionnaire_id - primary key
public	assessments	TBL	2	FK	user_id	bigint(64)	NOT NULL	NULL
public	assessments	TBL	3	FK	subject_id	bigint(64)	NOT NULL	NULL
public	assessments	TBL	4		assessment_name	character varying(255)	NOT NULL	NULL
public	assessments	TBL	5		assessment_description	text	NULL	NULL
public	assessments	TBL	6		assessment_type	character varying(100)	NULL	NULL
public	assessments	TBL	7	FK	topic_id	bigint(64)	NULL	NULL
public	assessments	TBL	8	FK	exam_id	bigint(64)	NULL	NULL
public	assessments	TBL	9		total_items	integer(32)	NULL	NULL
public	assessments	TBL	10		difficulty	character varying(50)	NULL	NULL
public	assessments	TBL	11		status	character varying(50)	NULL	NULL
public	assessments	TBL	12		analysis_status	character varying(50)	NULL	NULL

public	assessments	TBL	13		original_file_path	character varying(500)	NULL	NULL
public	assessments	TBL	14		original_file_name	character varying(255)	NULL	NULL
public	assessments	TBL	15		original_file_type	character varying(50)	NULL	NULL
public	assessments	TBL	16		file_data	bytea	NULL	NULL
public	assessments	TBL	17		file_size	bigint(64)	NULL	NULL
public	assessments	TBL	18		archived_at	timestamp without time zone(6)	NULL	NULL
public	assessments	TBL	19		created_at	timestamp without time zone(6)	NULL	NULL
public	assessments	TBL	20		updated_at	timestamp without time zone(6)	NULL	NULL
public	assessments	TBL	21		deleted_at	timestamp without time zone(6)	NULL	NULL
public	choices	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	choices	TBL	2	FK	question_id	bigint(64)	NOT NULL	NULL
public	choices	TBL	3		choice_letter	character varying(1)	NULL	NULL
public	choices	TBL	4		choice_text	text	NOT NULL	NULL
public	choices	TBL	5		image_path	character varying(500)	NULL	NULL
public	choices	TBL	6		created_at	timestamp without time zone(6)	NULL	NULL

public	choices	TBL	7		updated_at	timestamp without time zone(6)	NULL	NULL
public	class_sections	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	class_sections	TBL	2	FK,UK	class_id	bigint(64)	NOT NULL	NULL
public	class_sections	TBL	3	FK,UK	subject_id	bigint(64)	NOT NULL	NULL
public	class_sections	TBL	4	UK	section_name	character varying(255)	NOT NULL	NULL
public	class_sections	TBL	5		section_code	character varying(50)	NULL	NULL
public	class_sections	TBL	6		max_students	integer(32)	NULL	NULL
public	class_sections	TBL	7		description	text	NULL	NULL
public	class_sections	TBL	8		status	character varying(50)	NULL	NULL
public	class_sections	TBL	9		created_at	timestamp without time zone(6)	NULL	NULL
public	class_sections	TBL	10		updated_at	timestamp without time zone(6)	NULL	NULL
public	classes	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	classes	TBL	2	FK	user_id	bigint(64)	NOT NULL	NULL
public	classes	TBL	3		class_name	character varying(255)	NOT NULL	NULL
public	classes	TBL	4		grade_level	character varying(50)	NULL	NULL
public	classes	TBL	5		school_year	character varying(50)	NULL	NULL
public	classes	TBL	6		status	character varying(50)	NULL	NULL

public	classes	TBL	7		archived_at	timestamp without time zone(6)	NULL	NULL
public	classes	TBL	8		archived_reason	text	NULL	NULL
public	classes	TBL	9		created_at	timestamp without time zone(6)	NULL	NULL
public	classes	TBL	10		updated_at	timestamp without time zone(6)	NULL	NULL
public	enhanced_item_analysis	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	enhanced_item_analysis	TBL	2	FK	scan_batch_id	bigint(64)	NOT NULL	NULL
public	enhanced_item_analysis	TBL	3	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	enhanced_item_analysis	TBL	4		item_analysis_data	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	5		response_distribution	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	6		difficulty_index	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	7		discrimination_index	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	8		distractor_analysis	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	9		educational_insights	jsonb	NULL	NULL
public	enhanced_item_analysis	TBL	10		overall_test_reliability	numeric(5,2)	NULL	NULL

public	enhanced_item_analysis	TBL	11		average_difficulty	numeric(5,2)	NULL	NULL
public	enhanced_item_analysis	TBL	12		average_discrimination	numeric(5,2)	NULL	NULL
public	enhanced_item_analysis	TBL	13		analysis_generated_at	timestamp without time zone(6)	NULL	NULL
public	enhanced_item_analysis	TBL	14		created_at	timestamp without time zone(6)	NULL	NULL
public	enhanced_item_analysis	TBL	15		updated_at	timestamp without time zone(6)	NULL	NULL
public	exams	TBL	1	PK	exam_id	bigint(64)	NOT NULL	NULL
public	exams	TBL	2	FK,UK	topic_id	bigint(64)	NULL	NULL
public	exams	TBL	3	FK,UK	subject_id	bigint(64)	NOT NULL	NULL
public	exams	TBL	4	UK	grade_level	character varying(50)	NULL	NULL
public	exams	TBL	5	UK	quarter	character varying(50)	NULL	NULL
public	exams	TBL	6	UK	exam_type	character varying(50)	NULL	NULL
public	exams	TBL	7		created_at	timestamp without time zone(6)	NULL	NULL
public	exams	TBL	8		updated_at	timestamp without time zone(6)	NULL	NULL
public	exams	TBL	9		deleted_at	timestamp without time zone(6)	NULL	NULL
public	item_analysis	TBL	1	PK	id	bigint(64)	NOT NULL	NULL

public	item_analysis	TBL	2	FK	question_id	bigint(64)	NOT NULL	NULL
public	item_analysis	TBL	3	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	item_analysis	TBL	4		difficulty_level	numeric(5,2)	NULL	NULL
public	item_analysis	TBL	5		difficulty_level_status	character varying(50)	NULL	NULL
public	item_analysis	TBL	6		discrimination_index	numeric(5,2)	NULL	NULL
public	item_analysis	TBL	7		correct_responses	integer(32)	NULL	NULL
public	item_analysis	TBL	8		total_responses	integer(32)	NULL	NULL
public	item_analysis	TBL	9		distractor_quality	numeric(5,2)	NULL	NULL
public	item_analysis	TBL	10		recommendation	character varying(50)	NULL	NULL
public	item_analysis	TBL	11		created_at	timestamp without time zone(6)	NULL	NULL
public	item_analysis	TBL	12		updated_at	timestamp without time zone(6)	NULL	NULL
public	item_analysis_progress	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	item_analysis_progress	TBL	2	FK,UK	assessment_id	bigint(64)	NOT NULL	NULL
public	item_analysis_progress	TBL	3	UK	batch_id	character varying(255)	NOT NULL	NULL
public	item_analysis_progress	TBL	4	UK	question_number	integer(32)	NOT NULL	NULL
public	item_analysis_progress	TBL	5		decision	character varying(50)	NULL	NULL

public	item_analysis_progress	TBL	6		status	character varying(50)	NULL	NULL
public	item_analysis_progress	TBL	7		edits	jsonb	NULL	NULL
public	item_analysis_progress	TBL	8		replacement	jsonb	NULL	NULL
public	item_analysis_progress	TBL	9		created_at	timestamp without time zone(6)	NULL	NULL
public	item_analysis_progress	TBL	10		updated_at	timestamp without time zone(6)	NULL	NULL
public	manual_input_progress	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	manual_input_progress	TBL	2	FK,UK	assessment_id	bigint(64)	NOT NULL	NULL
public	manual_input_progress	TBL	3	FK,UK	section_id	bigint(64)	NOT NULL	NULL
public	manual_input_progress	TBL	4		student_responses	jsonb	NULL	NULL
public	manual_input_progress	TBL	5		total_students	integer(32)	NULL	NULL
public	manual_input_progress	TBL	6		completed_students	integer(32)	NULL	NULL
public	manual_input_progress	TBL	7		in_progress_students	integer(32)	NULL	NULL
public	manual_input_progress	TBL	8	UK	is_submitted	boolean	NULL	NULL
public	manual_input_progress	TBL	9		submitted_at	timestamp without time zone(6)	NULL	NULL

public	manual_input_progress	TBL	10		created_at	timestamp without time zone(6)	NULL	NULL
public	manual_input_progress	TBL	11		updated_at	timestamp without time zone(6)	NULL	NULL
public	notifications	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	notifications	TBL	2	FK	user_id	bigint(64)	NOT NULL	NULL
public	notifications	TBL	3		title	character varying(255)	NOT NULL	NULL
public	notifications	TBL	4		content	text	NOT NULL	NULL
public	notifications	TBL	5		link	character varying(255)	NULL	NULL
public	notifications	TBL	6		type	character varying(50)	NULL	NULL
public	notifications	TBL	7		read_at	timestamp without time zone(6)	NULL	NULL
public	notifications	TBL	8		created_at	timestamp without time zone(6)	NULL	NULL
public	notifications	TBL	9		updated_at	timestamp without time zone(6)	NULL	NULL
public	question_responses	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	question_responses	TBL	2	FK	student_response_id	bigint(64)	NOT NULL	NULL
public	question_responses	TBL	3	FK	question_id	bigint(64)	NOT NULL	NULL
public	question_responses	TBL	4		question_number	integer(32)	NOT NULL	NULL

public	question_responses	TBL	5		student_answer	character varying(10)	NULL	NULL
public	question_responses	TBL	6		is_correct	boolean	NULL	NULL
public	question_responses	TBL	7		confidence_score	numeric(5,2)	NULL	NULL
public	question_responses	TBL	8		question_quality	character varying(50)	NULL	NULL
public	question_responses	TBL	9		choice_analysis	jsonb	NULL	NULL
public	question_responses	TBL	10		created_at	timestamp without time zone(6)	NULL	NULL
public	question_responses	TBL	11		updated_at	timestamp without time zone(6)	NULL	NULL
public	questions	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	questions	TBL	2		is_original	boolean	NULL	NULL
public	questions	TBL	3		usage_count	integer(32)	NULL	NULL
public	questions	TBL	4		question_text	text	NOT NULL	NULL
public	questions	TBL	5		difficulty	character varying(50)	NULL	NULL
public	questions	TBL	6		question_type	character varying(50)	NULL	NULL
public	questions	TBL	7		created_at	timestamp without time zone(6)	NULL	NULL
public	questions	TBL	8		updated_at	timestamp without time zone(6)	NULL	NULL
public	scan_batches	TBL	1	PK	id	bigint(64)	NOT NULL	NULL

public	scan_batches	TBL	2	UK	batch_id	character varying(255)	NOT NULL	NULL
public	scan_batches	TBL	3	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	scan_batches	TBL	4	FK	version_id	bigint(64)	NULL	NULL
public	scan_batches	TBL	5	FK	section_id	bigint(64)	NULL	NULL
public	scan_batches	TBL	6		session_name	character varying(255)	NULL	NULL
public	scan_batches	TBL	7		test_date	date(3)	NULL	NULL
public	scan_batches	TBL	8		status	character varying(50)	NULL	NULL
public	scan_batches	TBL	9		scan_timestamp	timestamp without time zone(6)	NULL	NULL
public	scan_batches	TBL	10		scanner_version	character varying(50)	NULL	NULL
public	scan_batches	TBL	11		processing_stats	jsonb	NULL	NULL
public	scan_batches	TBL	12		detection_accuracy	jsonb	NULL	NULL
public	scan_batches	TBL	13		configuration	jsonb	NULL	NULL
public	scan_batches	TBL	14		processing_time_seconds	integer(32)	NULL	NULL
public	scan_batches	TBL	15		error_message	text	NULL	NULL
public	scan_batches	TBL	16		created_at	timestamp without time zone(6)	NULL	NULL
public	scan_batches	TBL	17		updated_at	timestamp without time zone(6)	NULL	NULL
public	scan_jobs	TBL	1	PK	id	bigint(64)	NOT NULL	NULL

public	scan_jobs	TBL	2	UK	job_id	uuid	NOT NULL	NULL
public	scan_jobs	TBL	3		batch_id	character varying(255)	NULL	NULL
public	scan_jobs	TBL	4	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	scan_jobs	TBL	5		job_type	character varying(50)	NULL	NULL
public	scan_jobs	TBL	6		status	character varying(50)	NULL	NULL
public	scan_jobs	TBL	7		progress_percentage	integer(32)	NULL	NULL
public	scan_jobs	TBL	8		current_phase	character varying(255)	NULL	NULL
public	scan_jobs	TBL	9		job_data	jsonb	NULL	NULL
public	scan_jobs	TBL	10		result_data	jsonb	NULL	NULL
public	scan_jobs	TBL	11		error_message	text	NULL	NULL
public	scan_jobs	TBL	12		started_at	timestamp without time zone(6)	NULL	NULL
public	scan_jobs	TBL	13		completed_at	timestamp without time zone(6)	NULL	NULL
public	scan_jobs	TBL	14		created_at	timestamp without time zone(6)	NULL	NULL
public	scan_jobs	TBL	15		updated_at	timestamp without time zone(6)	NULL	NULL
public	scan_processing_logs	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	scan_processing_logs	TBL	2	FK	scan_batch_id	bigint(64)	NOT NULL	NULL

public	scan_processing_logs	TBL	3		log_level	character varying(50)	NULL	NULL
public	scan_processing_logs	TBL	4		event_type	character varying(100)	NULL	NULL
public	scan_processing_logs	TBL	5		message	text	NULL	NULL
public	scan_processing_logs	TBL	6		context	jsonb	NULL	NULL
public	scan_processing_logs	TBL	7		logged_at	timestamp without time zone(6)	NULL	NULL
public	scan_processing_logs	TBL	8		created_at	timestamp without time zone(6)	NULL	NULL
public	scan_processing_logs	TBL	9		updated_at	timestamp without time zone(6)	NULL	NULL
public	section_titles	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	section_titles	TBL	2	FK	assessment_id	bigint(64)	NOT NULL	NULL
public	section_titles	TBL	3		title	character varying(255)	NOT NULL	NULL
public	section_titles	TBL	4		description	text	NULL	NULL
public	section_titles	TBL	5		type	character varying(50)	NULL	NULL
public	section_titles	TBL	6		display_order	integer(32)	NULL	NULL
public	section_titles	TBL	7		created_at	timestamp without time zone(6)	NULL	NULL
public	section_titles	TBL	8		updated_at	timestamp without time zone(6)	NULL	NULL

public	sessions	TBL	1	PK	id	character varying(255)	NOT NULL	NULL
public	sessions	TBL	2	FK	user_id	bigint(64)	NULL	NULL
public	sessions	TBL	3		ip_address	character varying(45)	NULL	NULL
public	sessions	TBL	4		user_agent	text	NULL	NULL
public	sessions	TBL	5		payload	text	NOT NULL	NULL
public	sessions	TBL	6		last_activity	integer(32)	NOT NULL	NULL
public	student_responses	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	student_responses	TBL	2	FK	scan_batch_id	bigint(64)	NOT NULL	NULL
public	student_responses	TBL	3	FK	student_id	bigint(64)	NOT NULL	NULL
public	student_responses	TBL	4		image_file	character varying(500)	NULL	NULL
public	student_responses	TBL	5		answers	jsonb	NULL	NULL
public	student_responses	TBL	6		detection_method	character varying(50)	NULL	NULL
public	student_responses	TBL	7		total_questions	integer(32)	NULL	NULL
public	student_responses	TBL	8		correct_answers	integer(32)	NULL	NULL
public	student_responses	TBL	9		score_percentage	numeric(5,2)	NULL	NULL
public	student_responses	TBL	10		processing_time_seconds	integer(32)	NULL	NULL
public	student_responses	TBL	11		overall_quality	character varying(50)	NULL	NULL

public	student_responses	TBL	12		quality_grade	character varying(10)	NULL	NULL
public	student_responses	TBL	13		average_confidence	numeric(5,2)	NULL	NULL
public	student_responses	TBL	14		confidence_std	numeric(5,2)	NULL	NULL
public	student_responses	TBL	15		answer_distribution	jsonb	NULL	NULL
public	student_responses	TBL	16		answer_statistics	jsonb	NULL	NULL
public	student_responses	TBL	17		detection_statistics	jsonb	NULL	NULL
public	student_responses	TBL	18		quality_assessment	jsonb	NULL	NULL
public	student_responses	TBL	19		created_at	timestamp without time zone(6)	NULL	NULL
public	student_responses	TBL	20		updated_at	timestamp without time zone(6)	NULL	NULL
public	students	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	students	TBL	2	FK	section_id	bigint(64)	NOT NULL	NULL
public	students	TBL	3	UK	student_id	character varying(255)	NOT NULL	NULL
public	students	TBL	4		first_name	character varying(255)	NOT NULL	NULL
public	students	TBL	5		middle_name	character varying(255)	NULL	NULL
public	students	TBL	6		last_name	character varying(255)	NOT NULL	NULL
public	students	TBL	7		email	character varying(255)	NULL	NULL

public	students	TBL	8		phone	character varying(50)	NULL	NULL
public	students	TBL	9		birth_date	date(3)	NULL	NULL
public	students	TBL	10		gender	character varying(50)	NULL	NULL
public	students	TBL	11		address	text	NULL	NULL
public	students	TBL	12		guardian_name	character varying(255)	NULL	NULL
public	students	TBL	13		guardian_phone	character varying(50)	NULL	NULL
public	students	TBL	14		guardian_email	character varying(255)	NULL	NULL
public	students	TBL	15		status	character varying(50)	NULL	NULL
public	students	TBL	16		enrollment_date	date(3)	NULL	NULL
public	students	TBL	17		created_at	timestamp without time zone(6)	NULL	NULL
public	students	TBL	18		updated_at	timestamp without time zone(6)	NULL	NULL
public	subjects	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	subjects	TBL	2		subject_name	character varying(255)	NOT NULL	NULL
public	subjects	TBL	3		subject_description	text	NULL	NULL
public	subjects	TBL	4		created_at	timestamp without time zone(6)	NULL	NULL

public	subjects	TBL	5		updated_at	timestamp without time zone(6)	NULL	NULL
public	topics	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	topics	TBL	2	FK	subject_id	bigint(64)	NOT NULL	NULL
public	topics	TBL	3		topic_name	character varying(255)	NOT NULL	NULL
public	topics	TBL	4		topic_description	text	NULL	NULL
public	topics	TBL	5		quarter	character varying(50)	NULL	NULL
public	topics	TBL	6		grade_level	character varying(50)	NULL	NULL
public	topics	TBL	7		created_at	timestamp without time zone(6)	NULL	NULL
public	topics	TBL	8		updated_at	timestamp without time zone(6)	NULL	NULL
public	uploaded_images	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	uploaded_images	TBL	2	FK	question_id	bigint(64)	NOT NULL	NULL
public	uploaded_images	TBL	3		image_title	character varying(255)	NULL	NULL
public	uploaded_images	TBL	4		image_path	character varying(500)	NOT NULL	NULL
public	uploaded_images	TBL	5		created_at	timestamp without time zone(6)	NULL	NULL

public	uploaded_images	TBL	6		updated_at	timestamp without time zone(6)	NULL	NULL
public	user_subject_colors	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	user_subject_colors	TBL	2	FK,UK	user_id	bigint(64)	NOT NULL	NULL
public	user_subject_colors	TBL	3	FK,UK	subject_id	bigint(64)	NOT NULL	NULL
public	user_subject_colors	TBL	4		color	character varying(50)	NOT NULL	NULL
public	user_subject_colors	TBL	5		created_at	timestamp without time zone(6)	NULL	NULL
public	user_subject_colors	TBL	6		updated_at	timestamp without time zone(6)	NULL	NULL
public	users	TBL	1	PK	id	bigint(64)	NOT NULL	NULL
public	users	TBL	2		name	character varying(255)	NOT NULL	NULL
public	users	TBL	3		last_name	character varying(255)	NULL	NULL
public	users	TBL	4	UK	email	character varying(255)	NOT NULL	NULL
public	users	TBL	5		email_verified_at	timestamp without time zone(6)	NULL	NULL
public	users	TBL	6		password	character varying(255)	NOT NULL	NULL
public	users	TBL	7		remember_token	character varying(100)	NULL	NULL

public	users	TBL	8		created_at	timestamp without time zone(6)	NULL	NULL
public	users	TBL	9		updated_at	timestamp without time zone(6)	NULL	NULL

APPENDIX F

(Gantt Chart)

CAPSTONE DESIGN GANTT CHART

Project Title	AsseSync: Assessment Management System
Project Members	Adlawan, John Robert G., Cortez, Jaymei Rey D., Gonzaga Lloyd S.
Adviser	Ms. Jehanie May A. Macasawang

ID	Task Name	AUGUST				SEPTEMBER				OCTOBER				NOVEMBER				DECEMBER			
		W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
1	Finalizing Design for AMS																				
2	Meeting with the Client																				
2.1	Showing Final Design																				
2.2	Intel/Material Gathering																				
2.3	Feedback																				
3	Applying Changes																				
4	Building Backend																				
5	Functionality Testing																				
5.6	Applying Suggestions																				
6	Meet with the Adviser																				
7	Meeting with the Client																				
7	UAT Testing																				
7.8	Applying Suggestions																				
8	Meet with the Adviser																				
9	Finalize System and Papers																				
10	Application for Defense																				
11	Final Defense																				